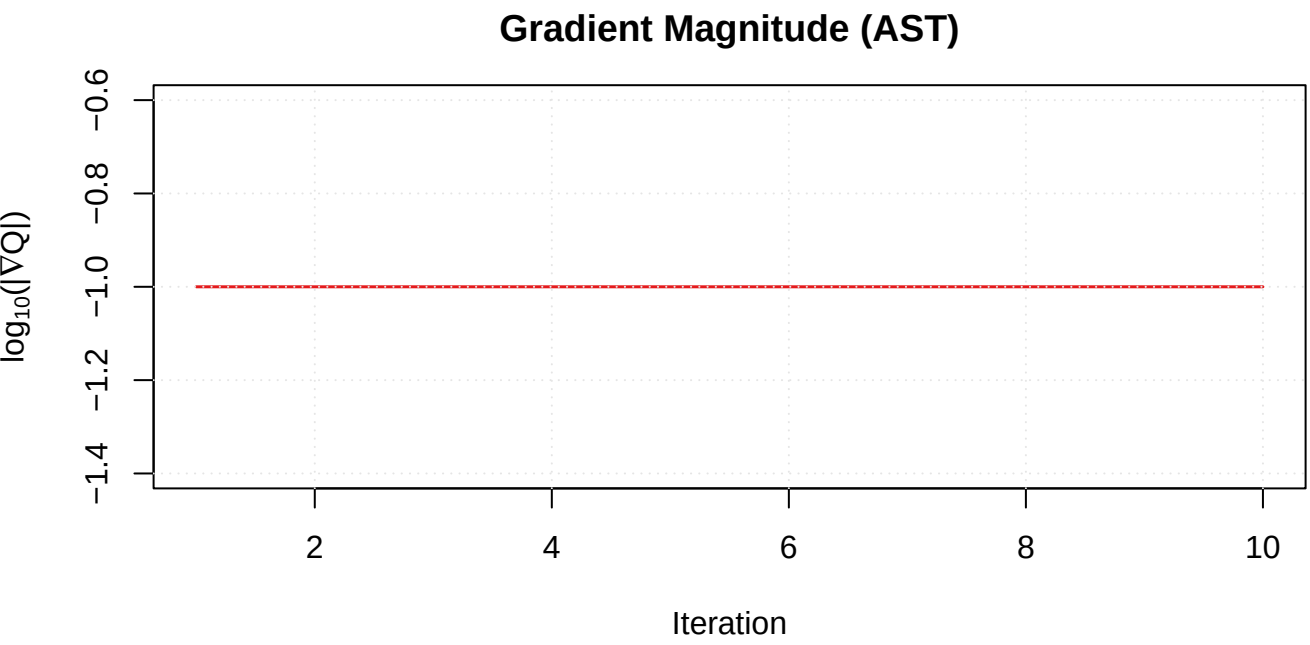
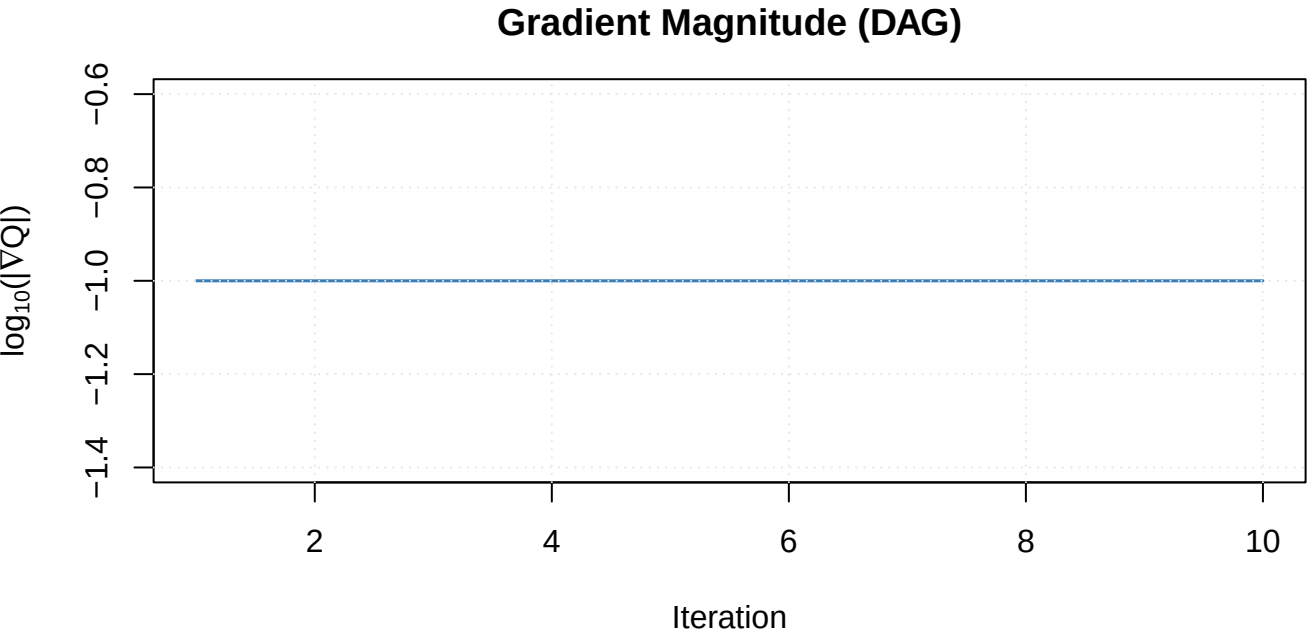
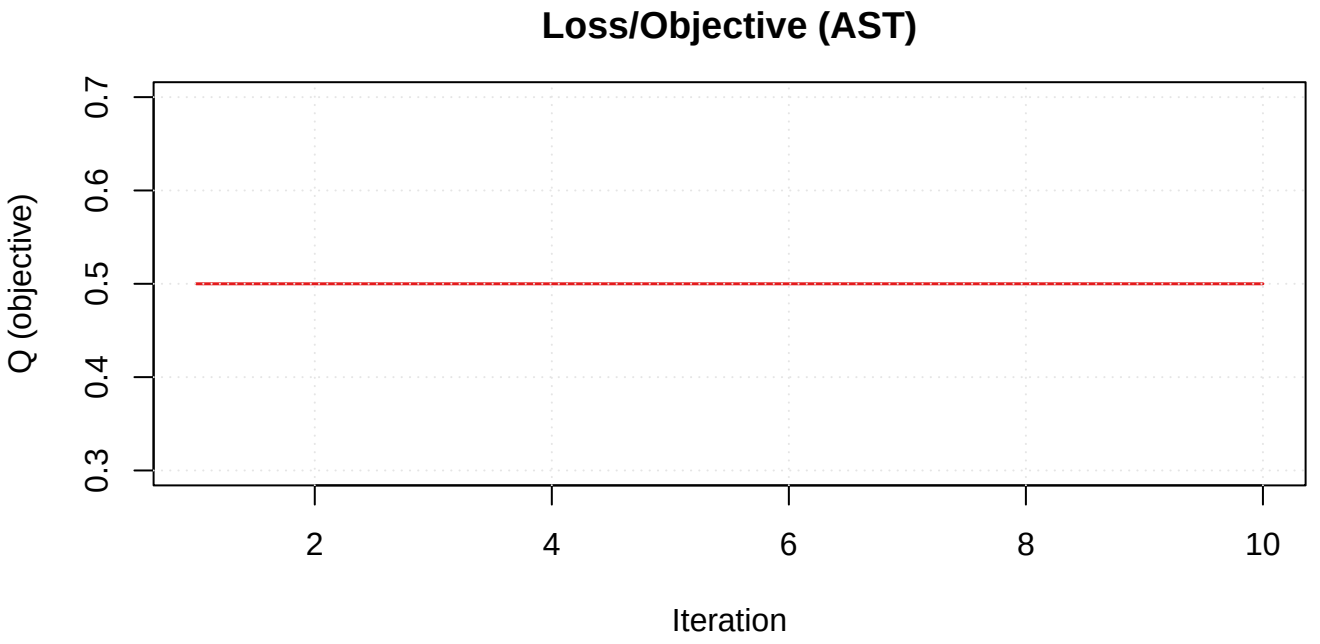
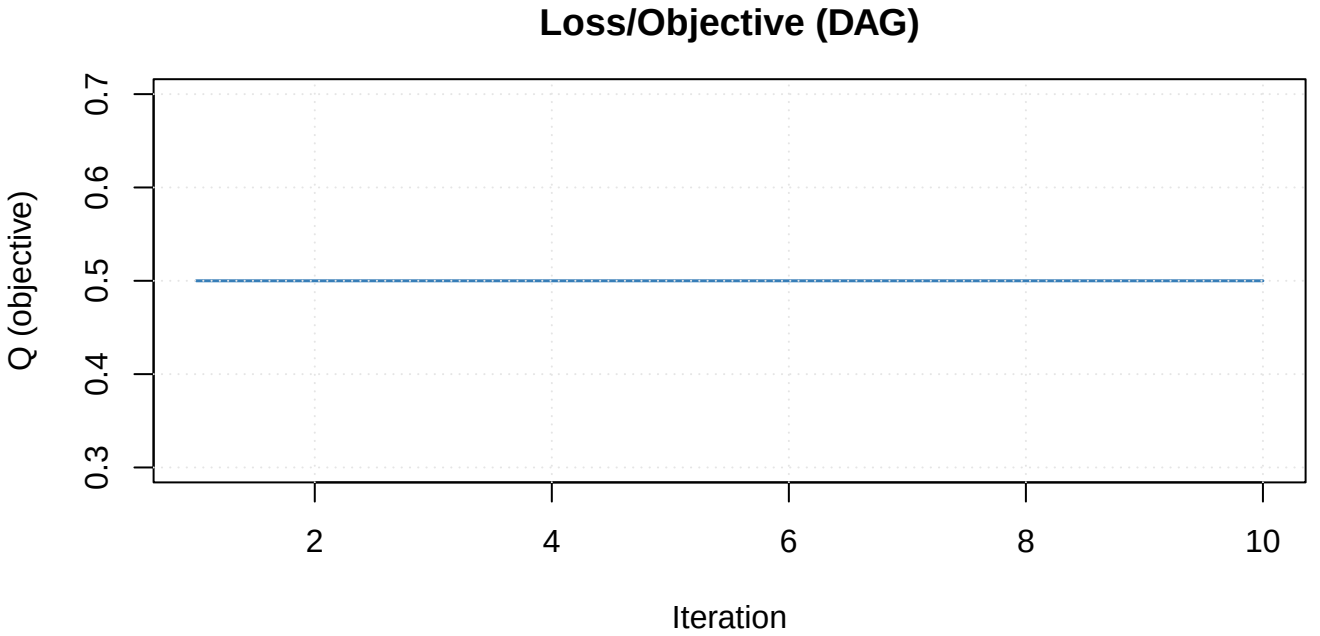
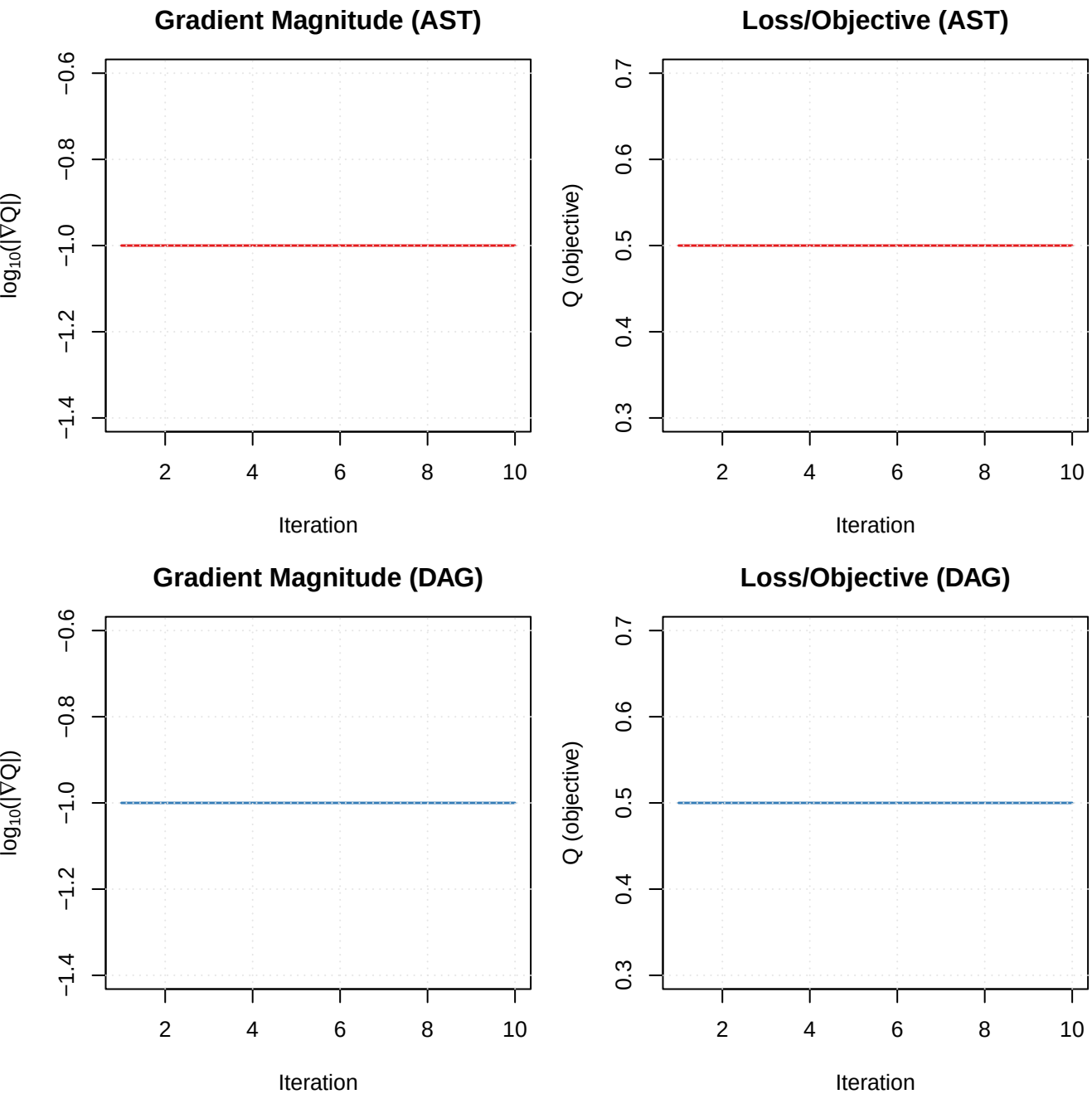
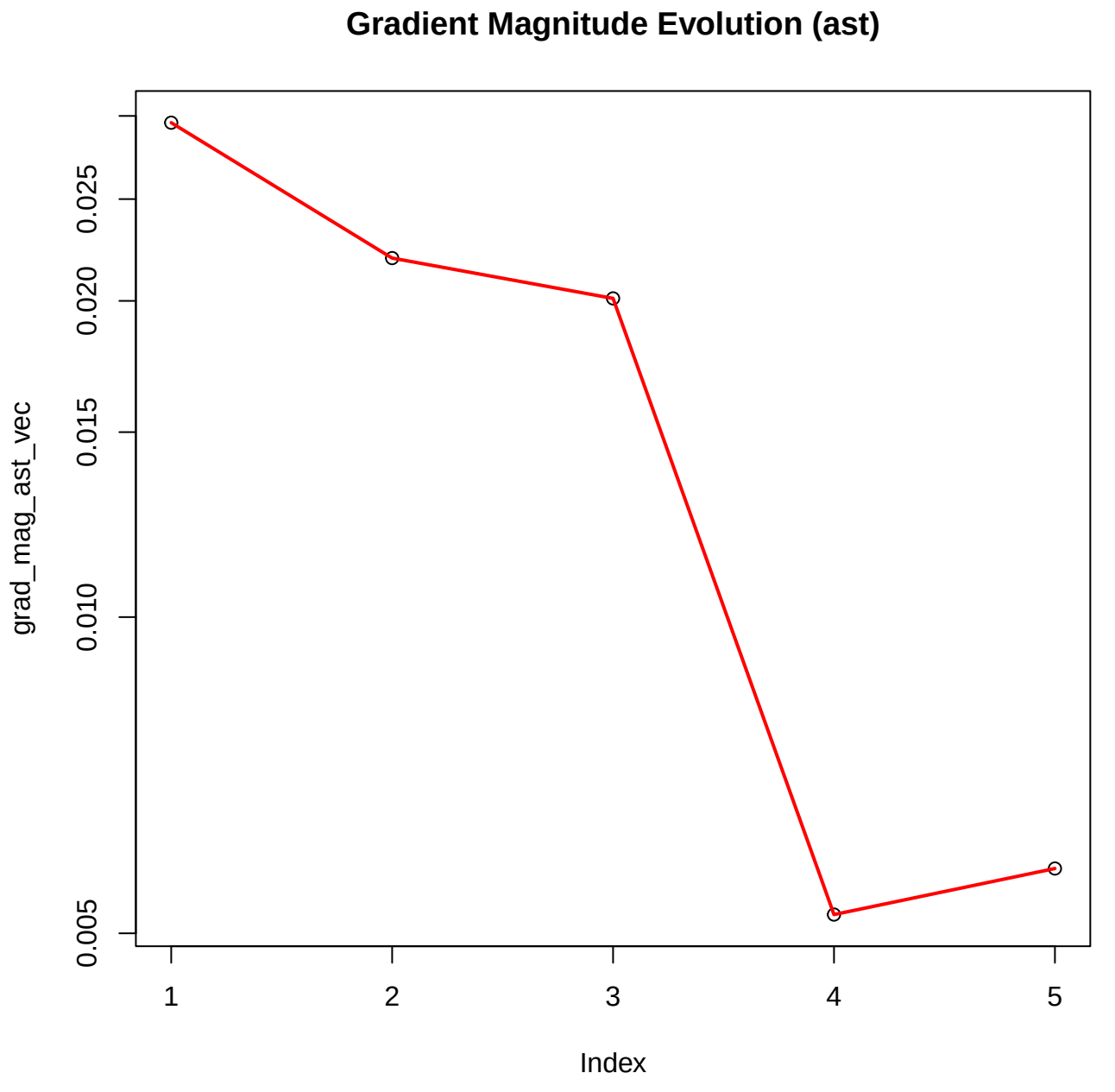

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

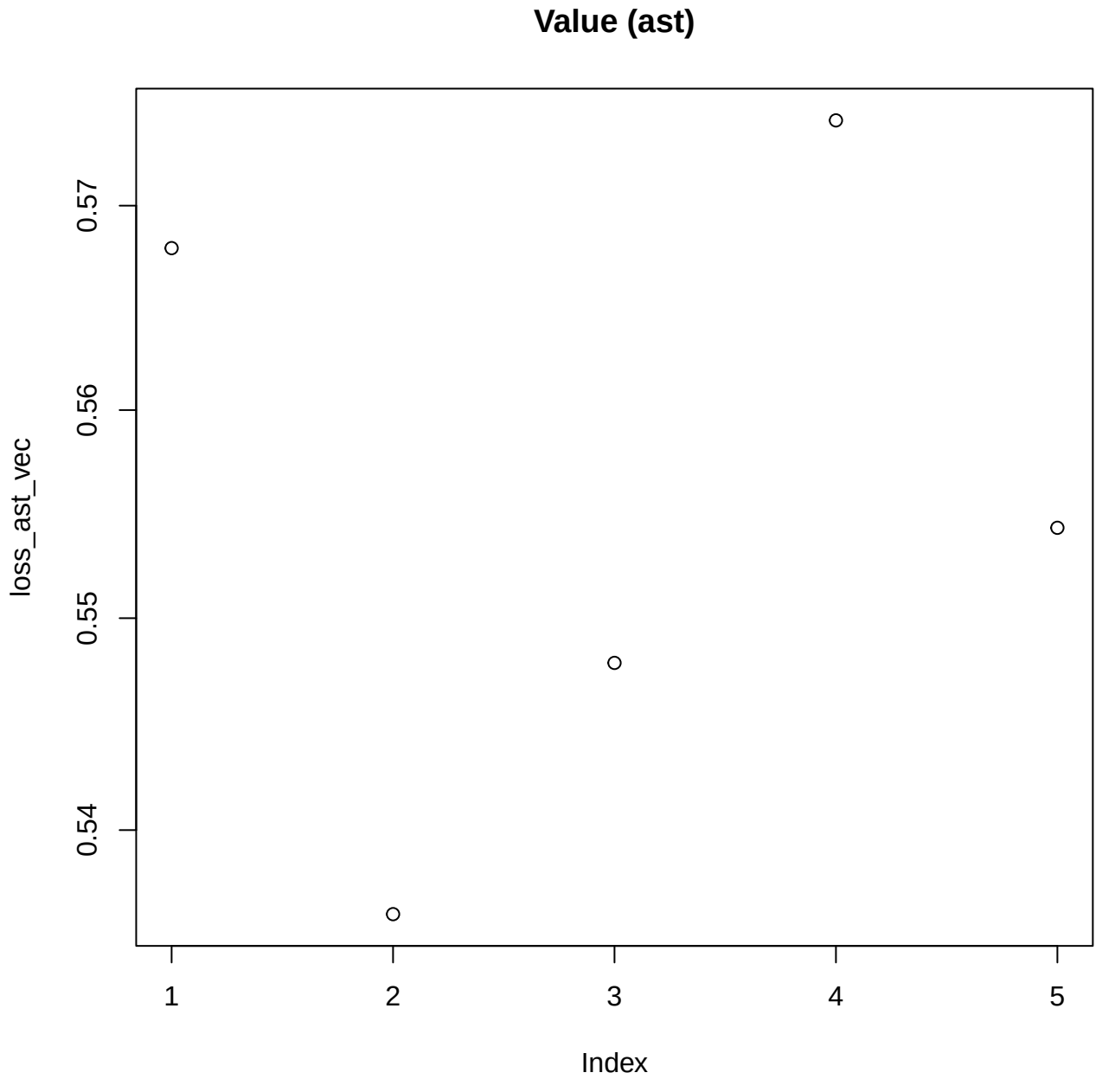

Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration

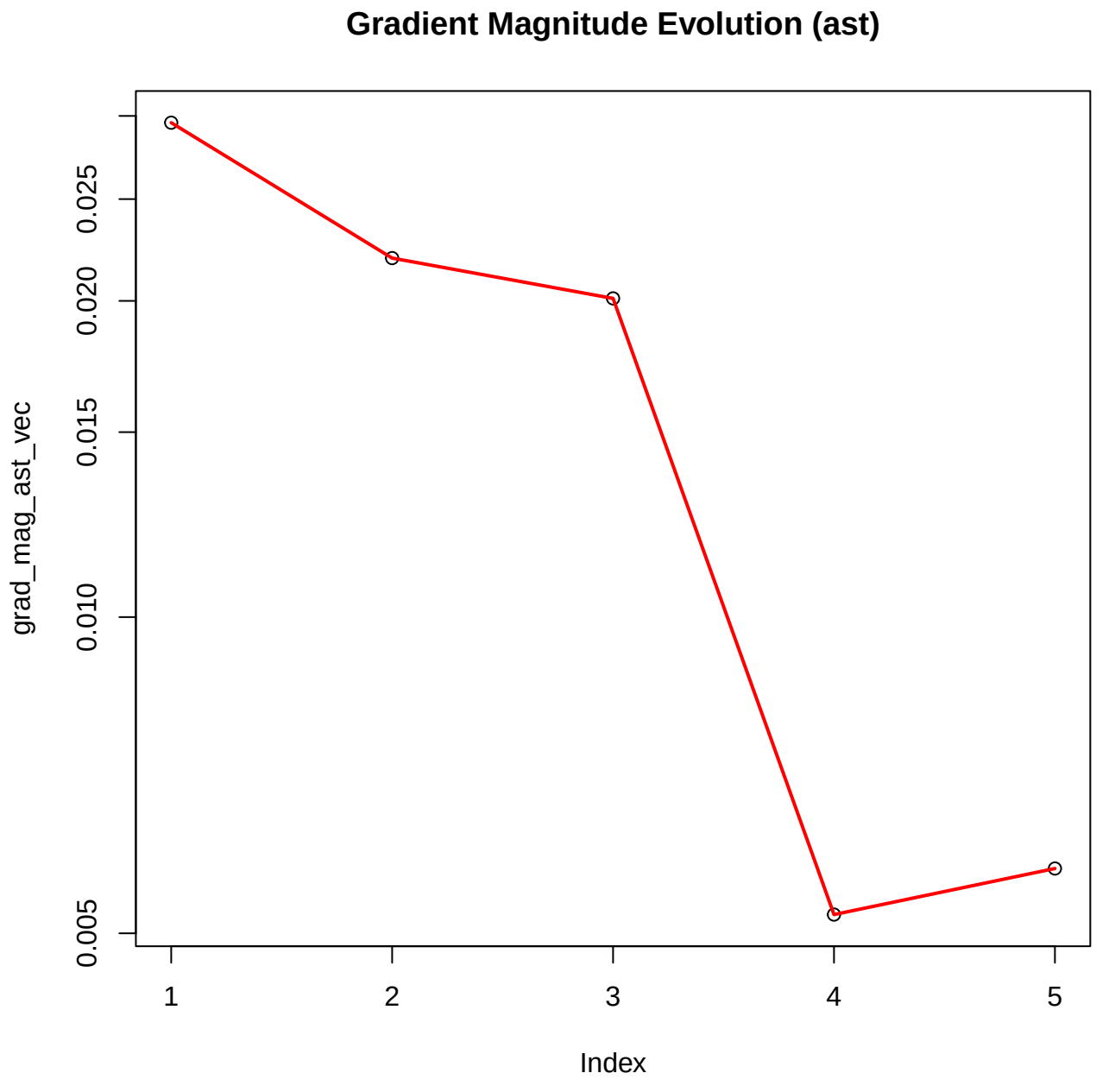

Loss/Objective (AST)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

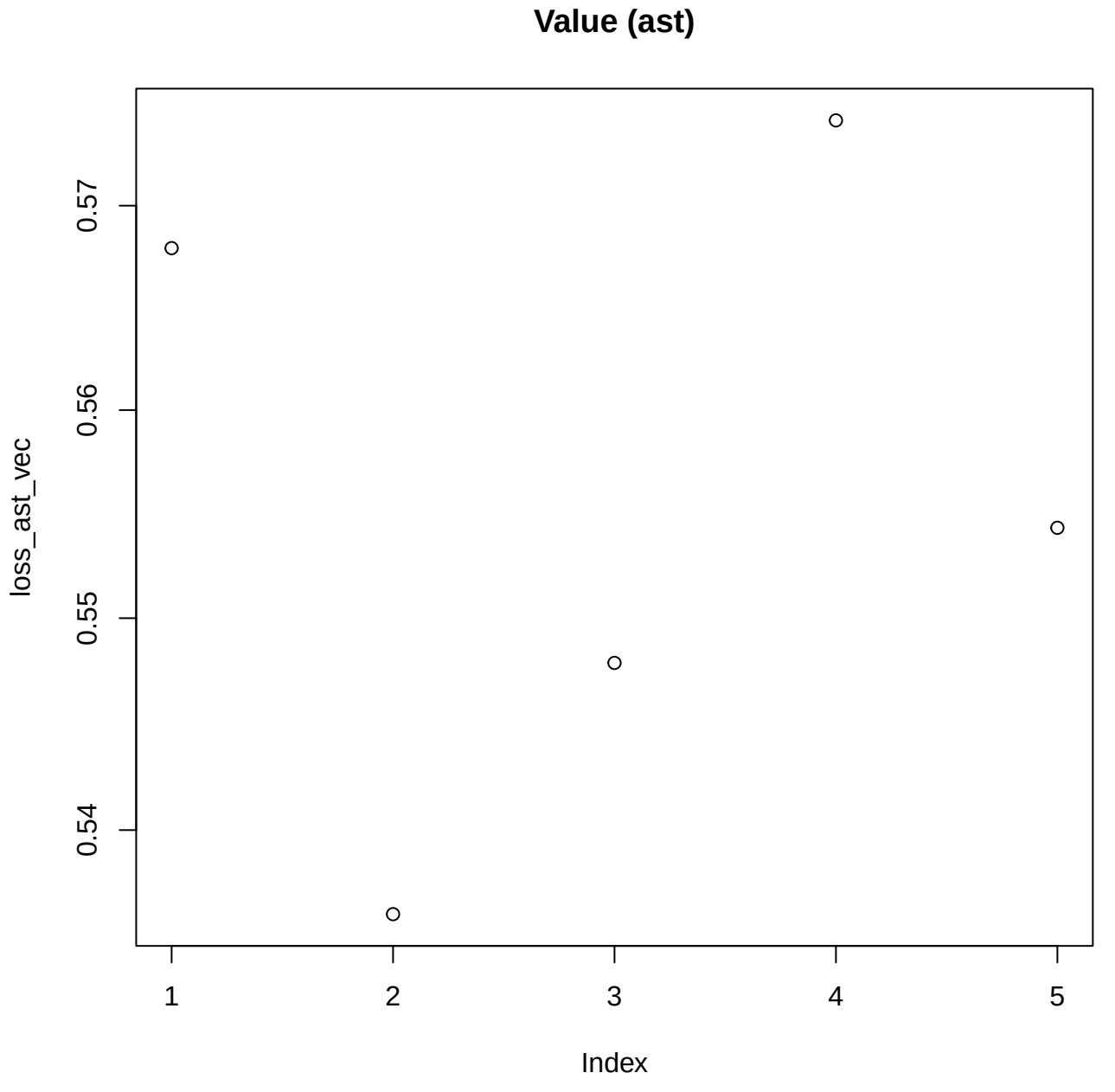

Loss/Objective (DAG)
Q (objective)
0.3
0.4
0.5
0.6
0.7
2
4
6
8
10
Iteration

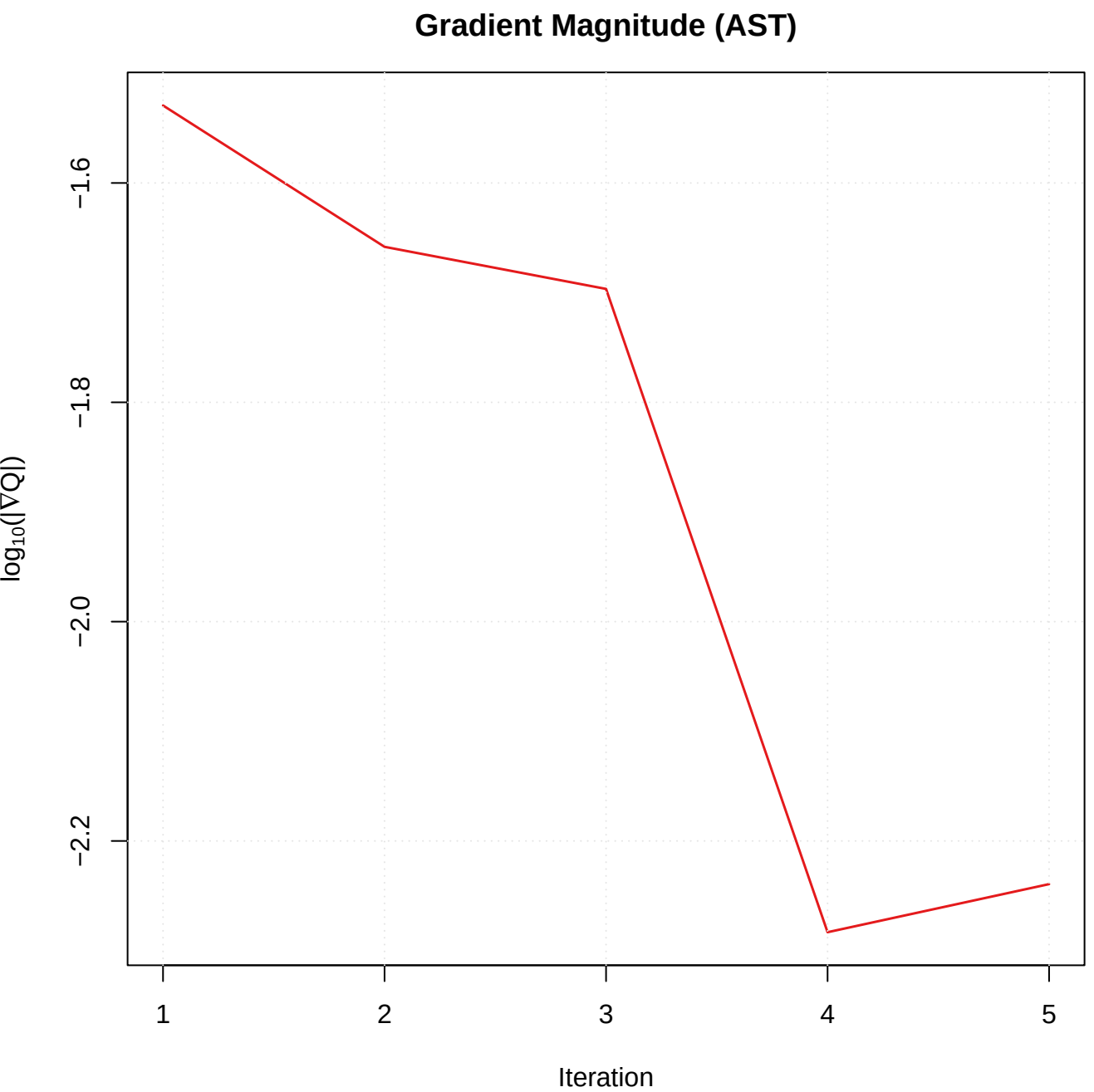

Gradient Magnitude (AST)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (AST)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration
Gradient Magnitude (DAG)
log10(|∇Q|)
−0.6
−0.8
−1.0
−1.2
−1.4
2
4
6
8
10
Iteration
Loss/Objective (DAG)
Q (objective)
0.7
0.6
0.5
0.4
0.3
2
4
6
8
10
Iteration

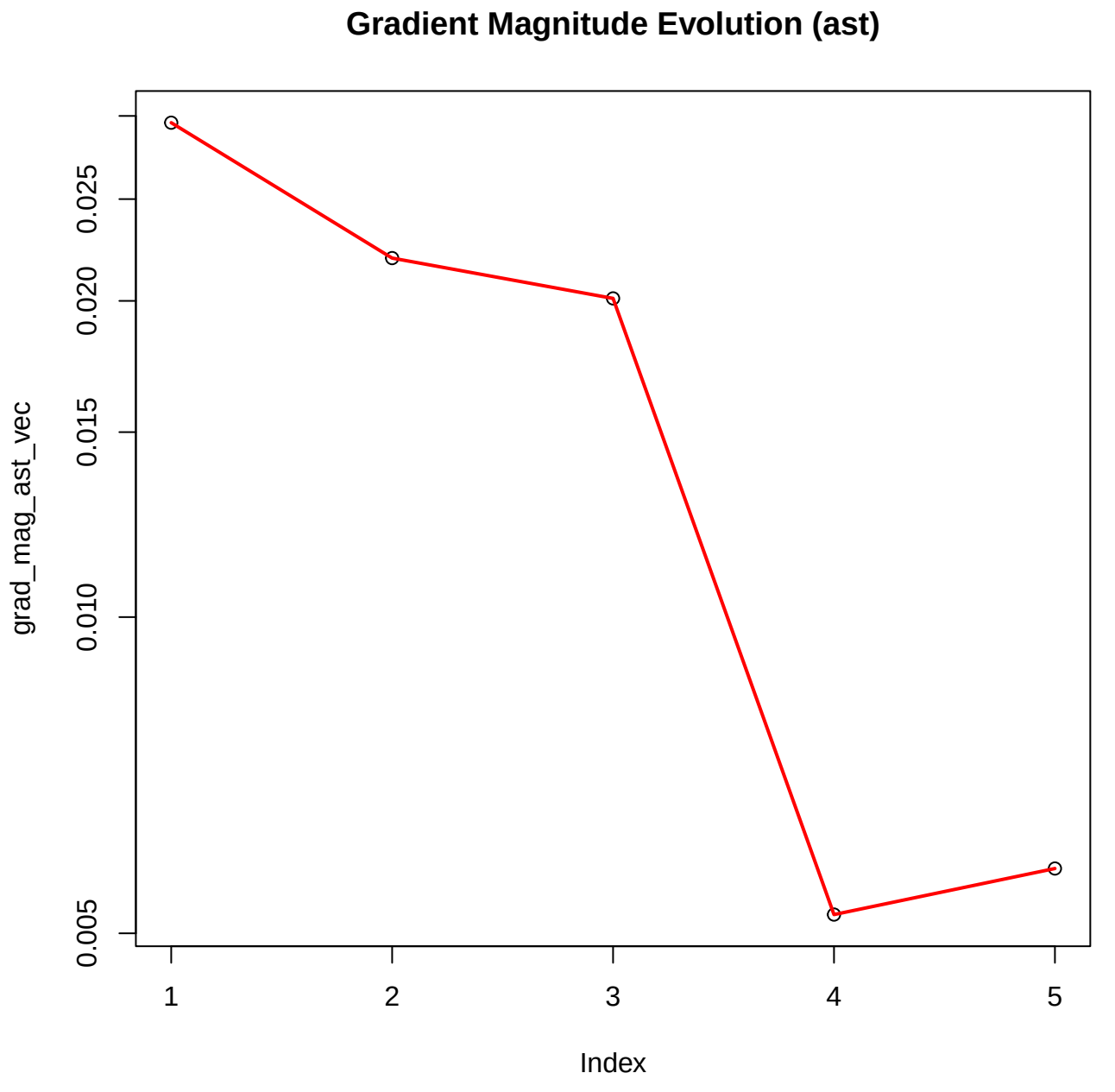

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

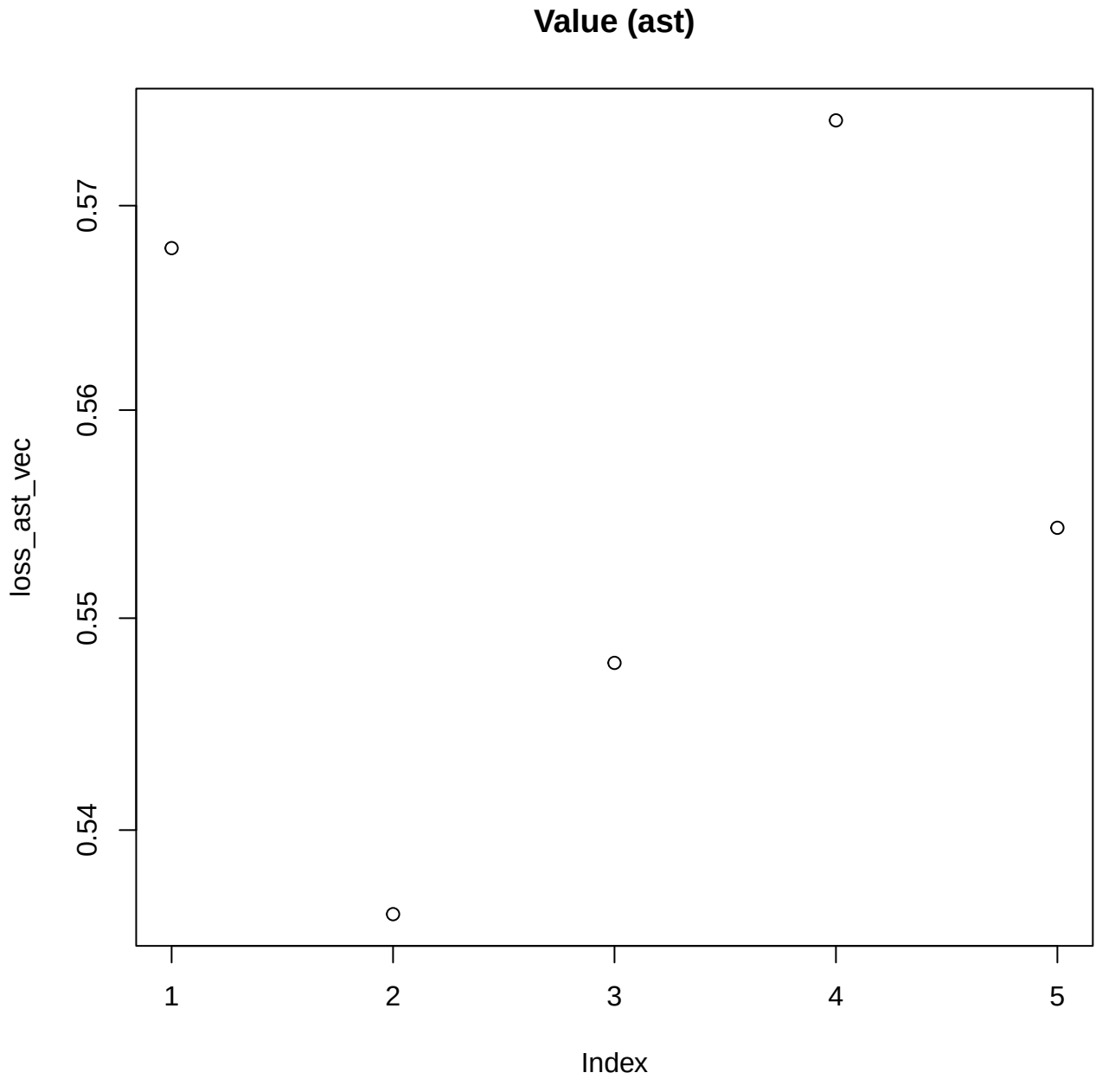

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

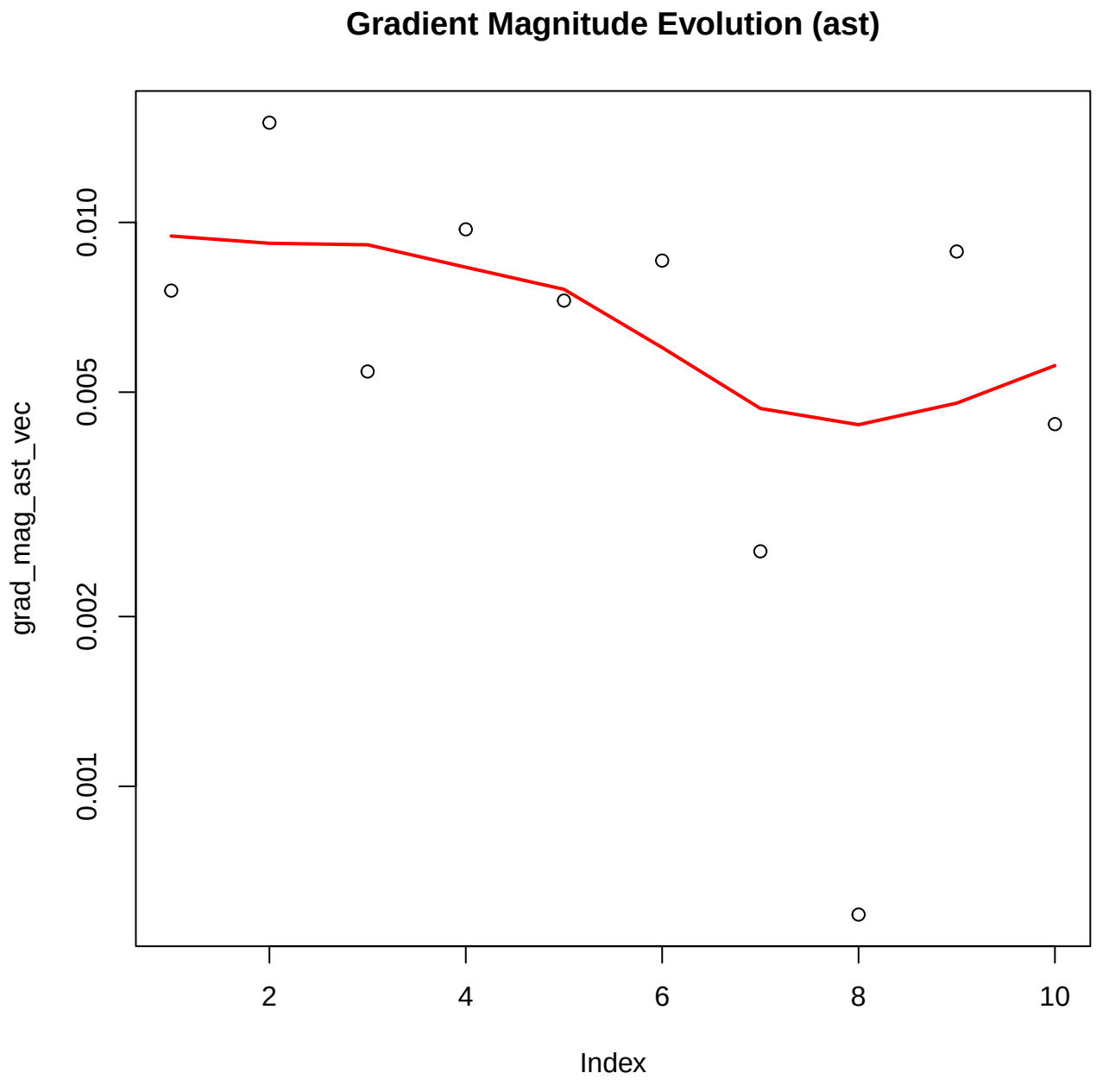

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

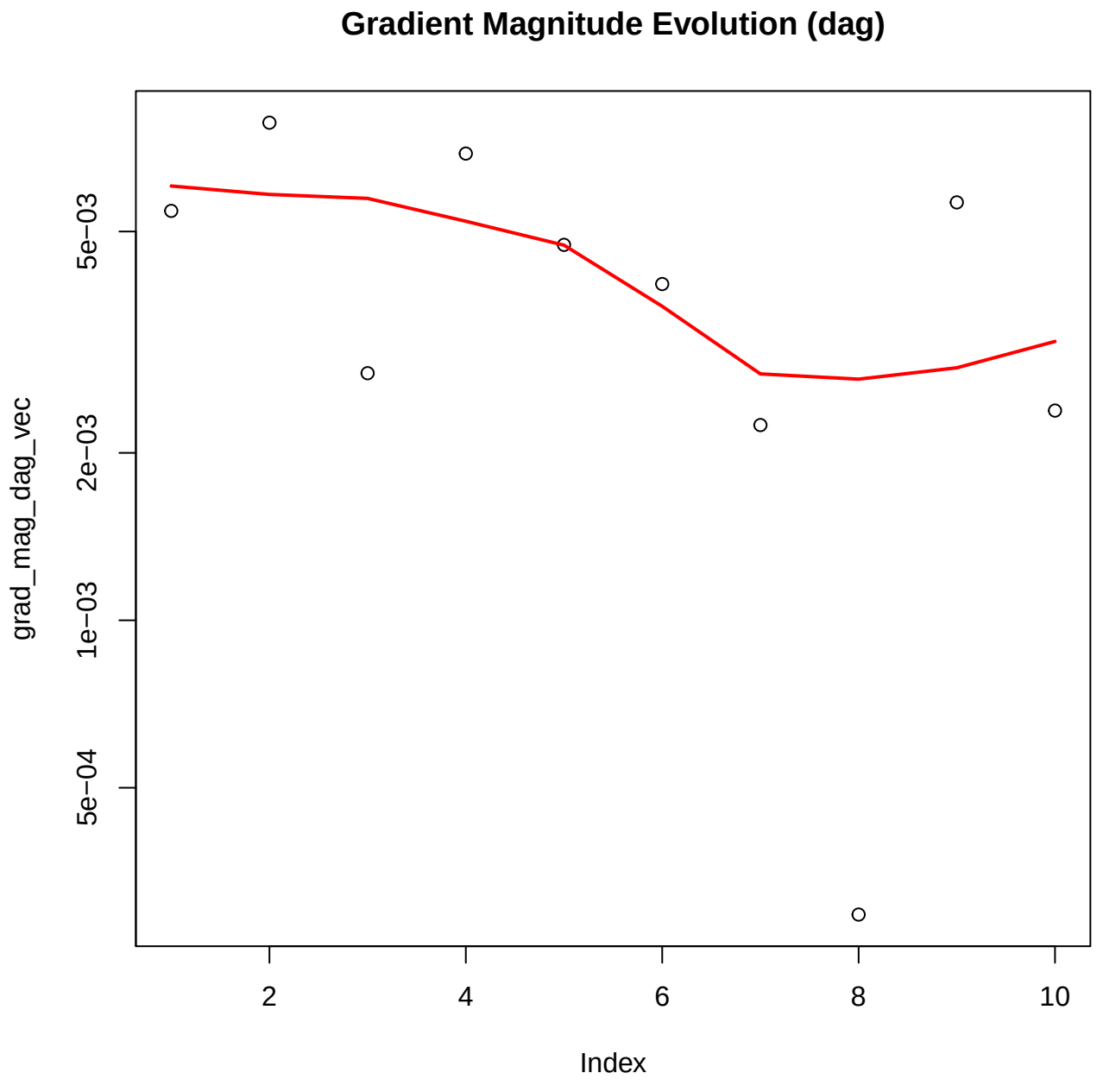

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

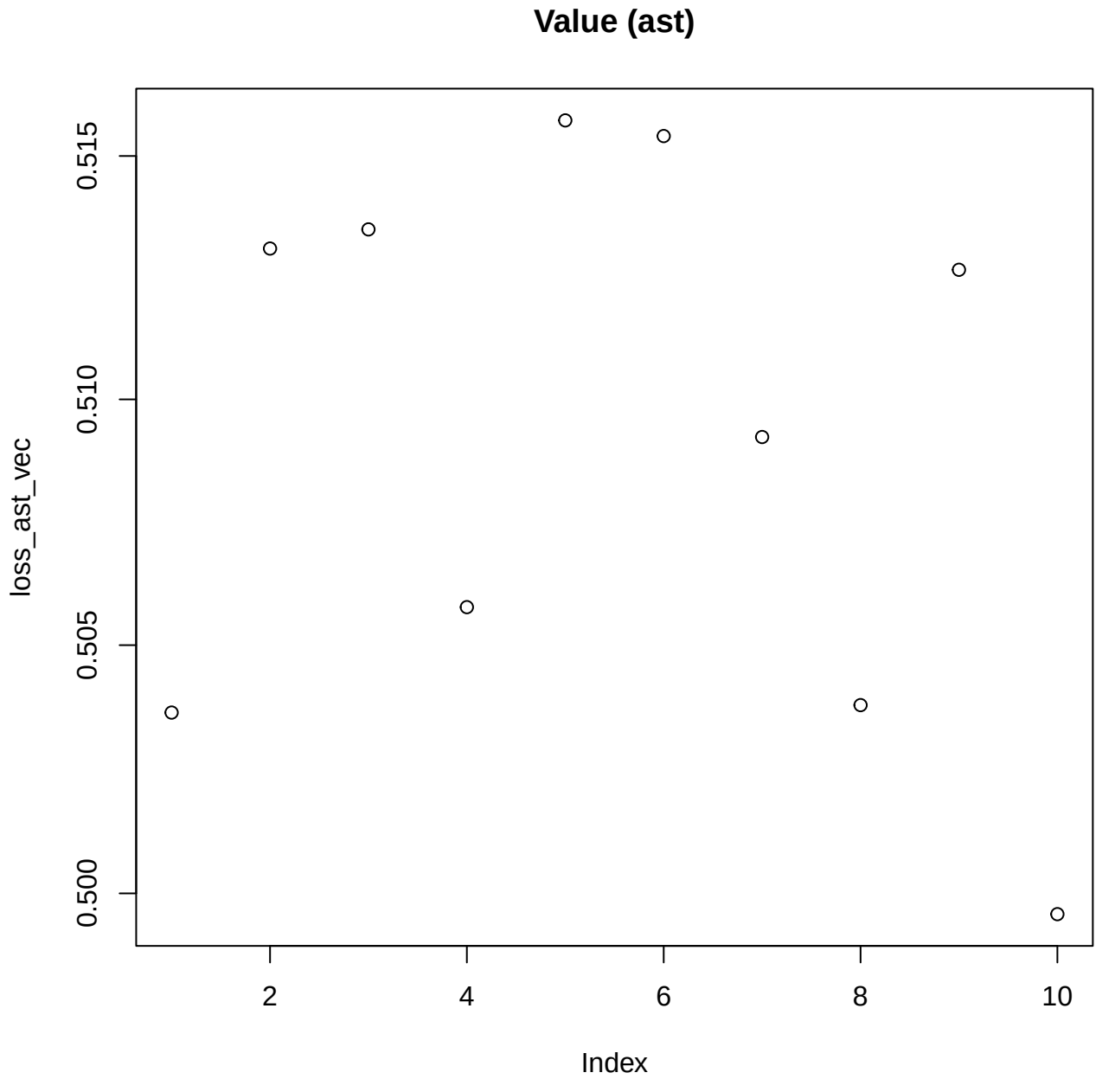

Gradient Magnitude (AST)
−1.6
−1.8
−2.0
−2.2
log10(|∇Q|)
1
2
3
4
5
Iteration

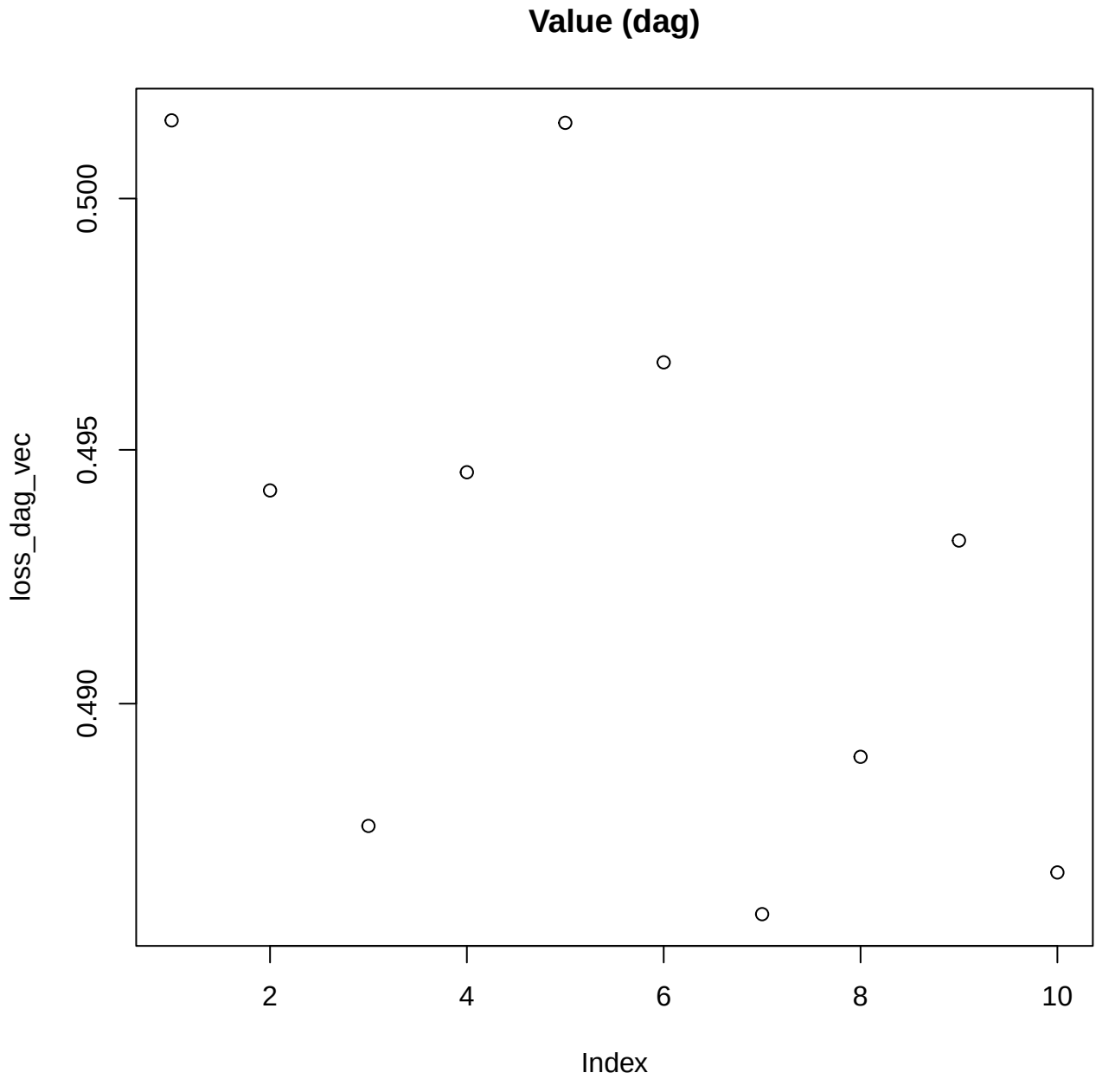

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.005
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

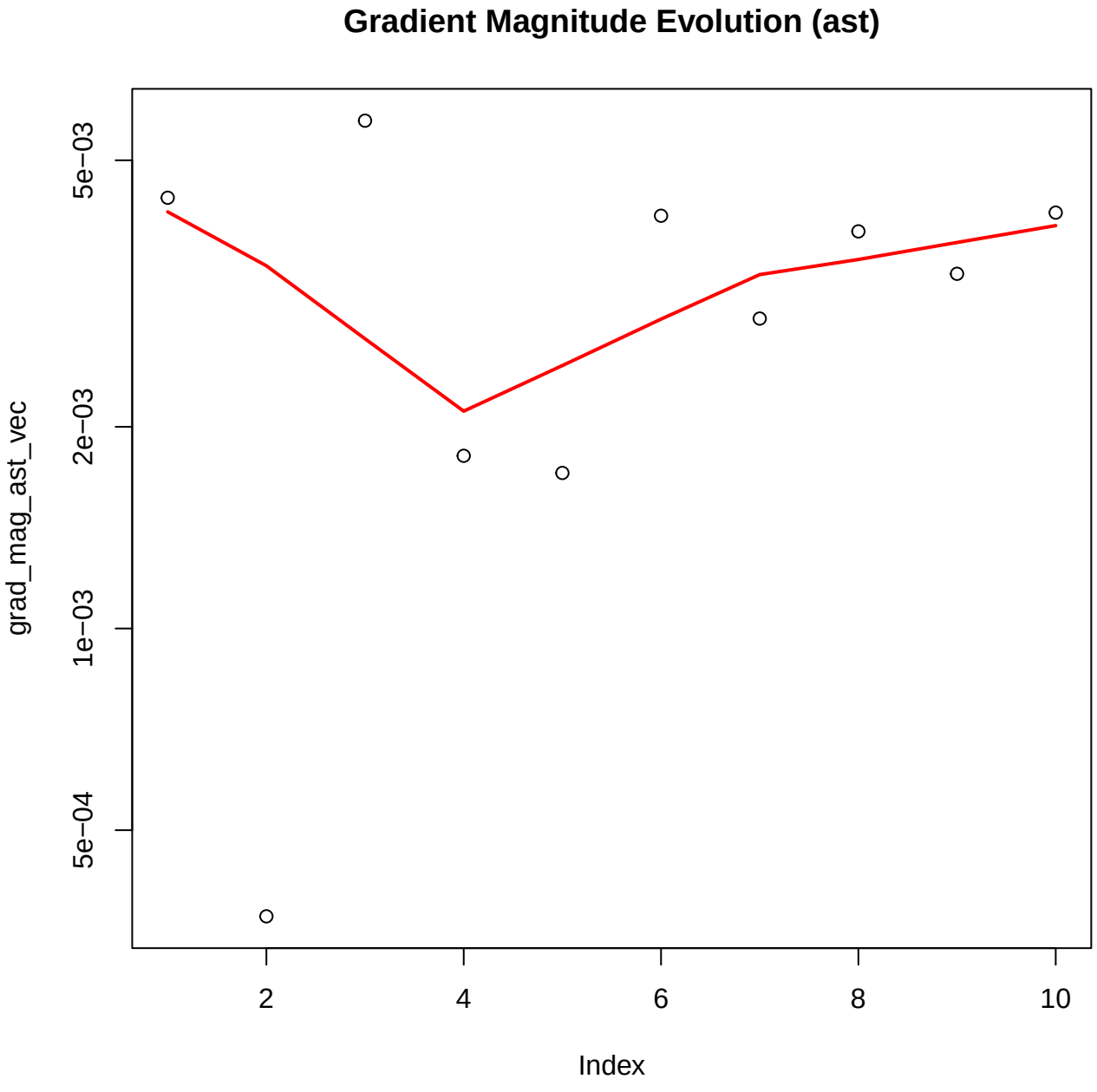

Value (ast)
loss_ast_vec
0.54
0.55
0.56
0.57
1
2
3
4
5
Index

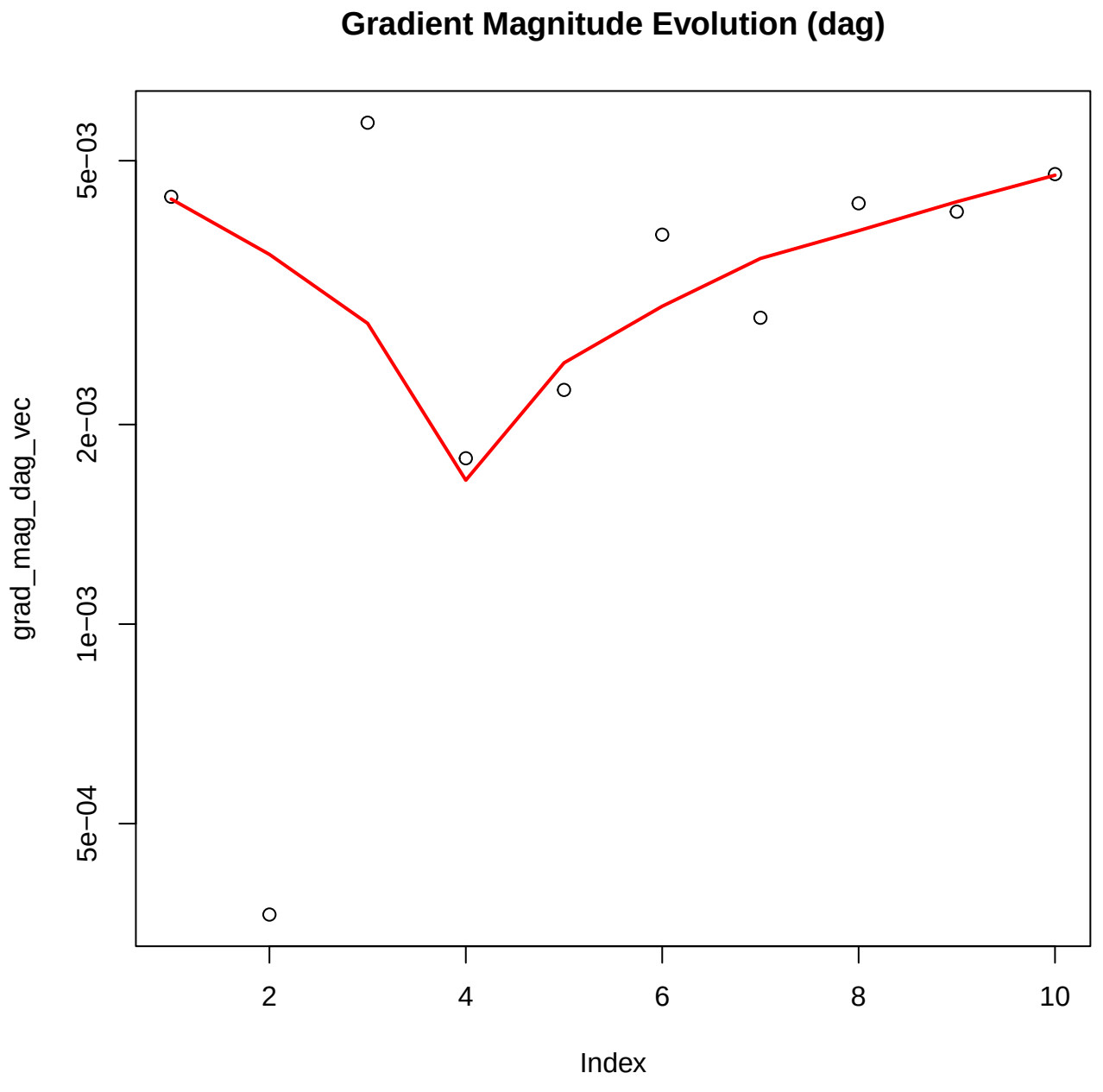

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.001
0.002
0.005
0.010
2
4
6
8
10
Index

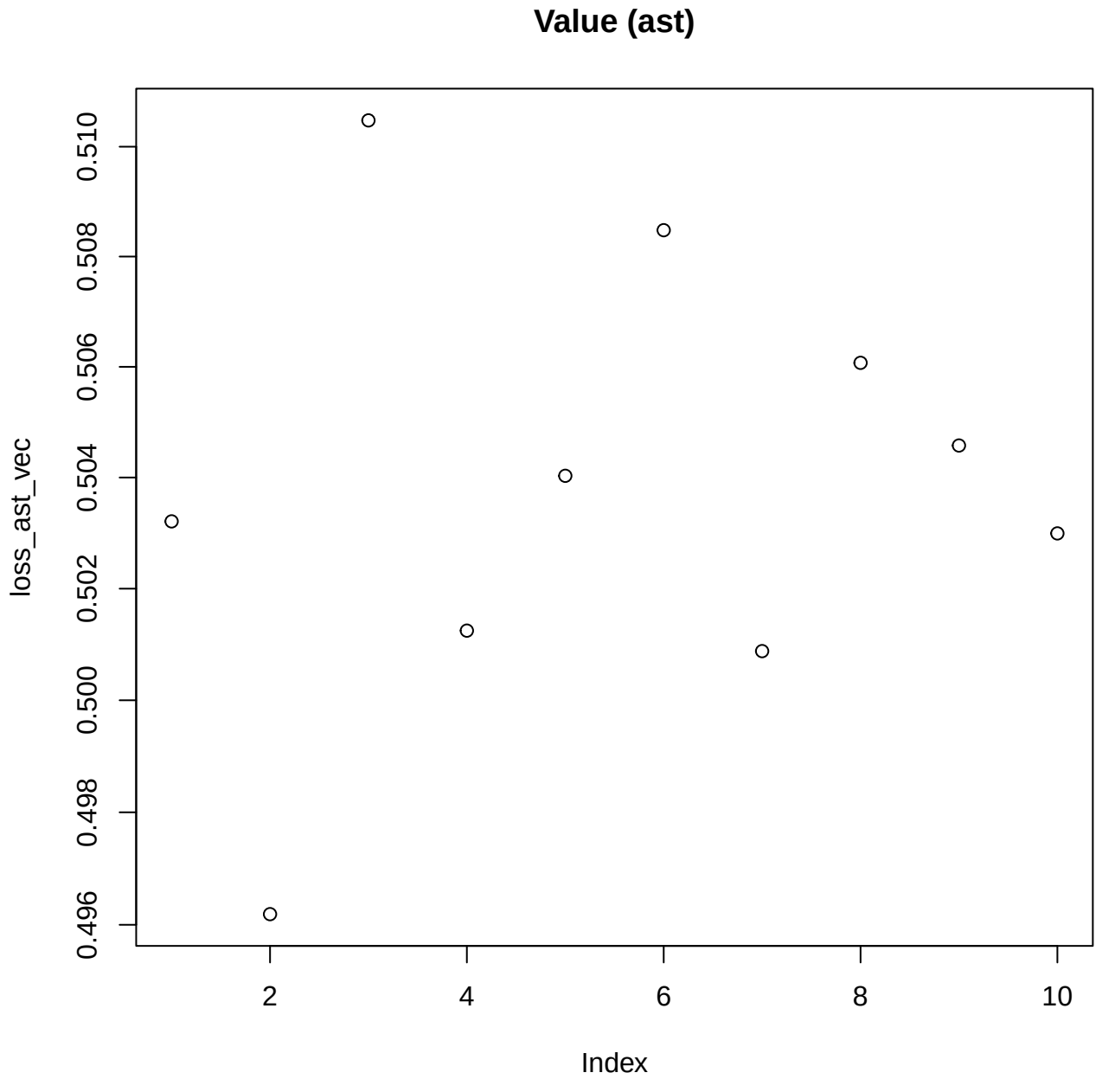

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

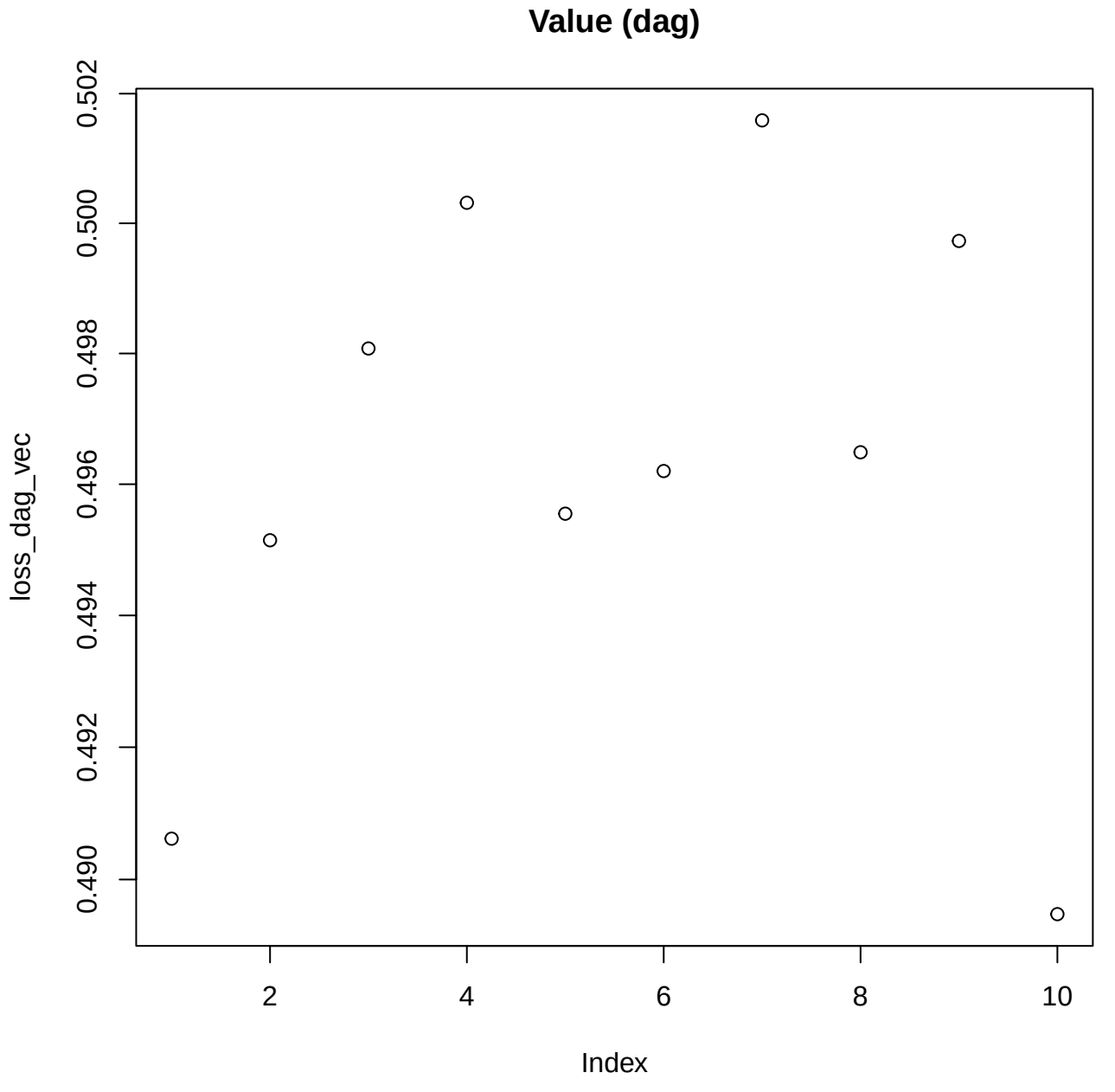

Value (ast)
loss_ast_vec
0.500
0.505
0.510
0.515
2
4
6
8
10
Index

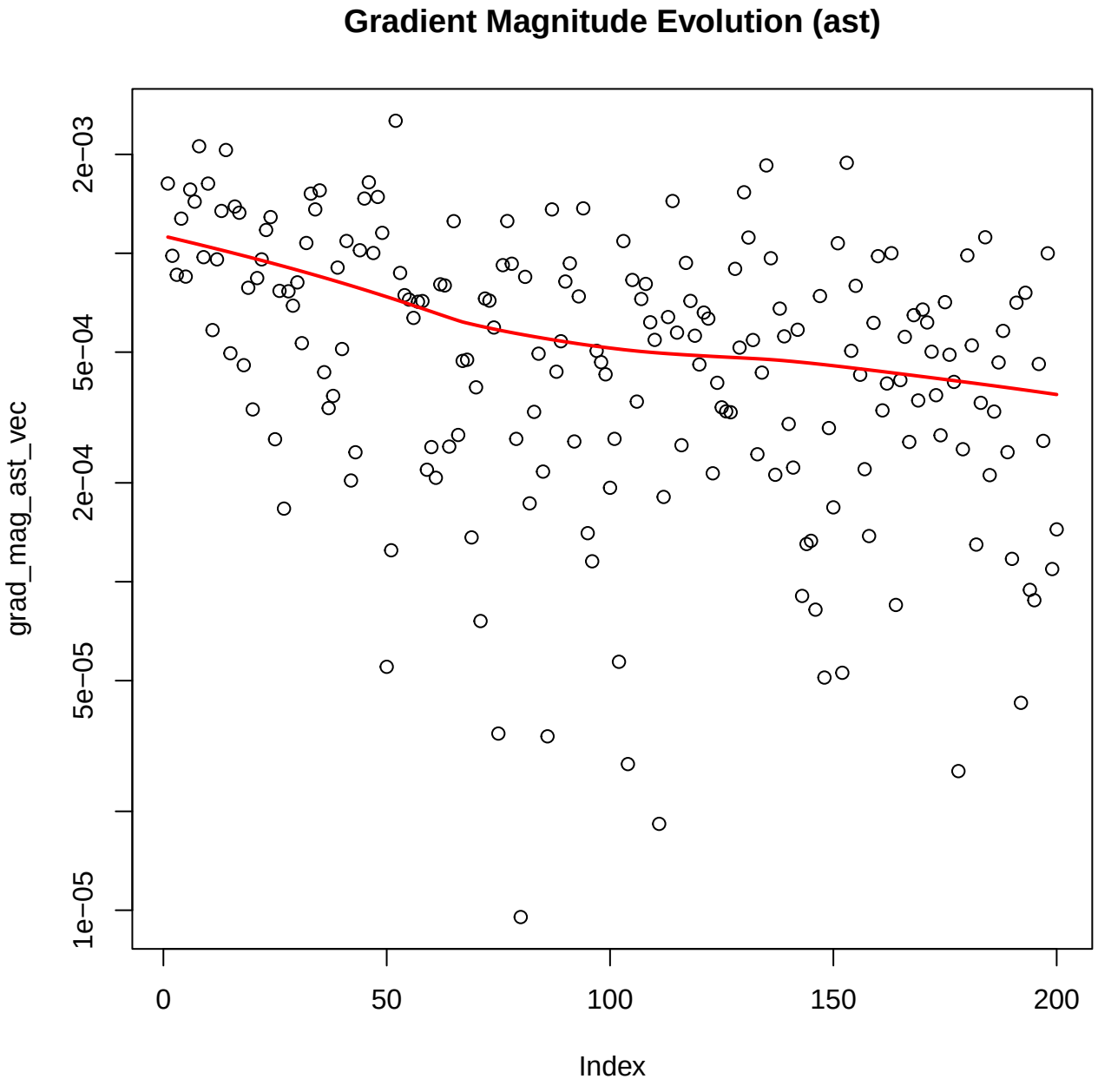

Value (dag)
loss_dag_vec
0.490
0.495
0.500
2
4
6
8
10
Index

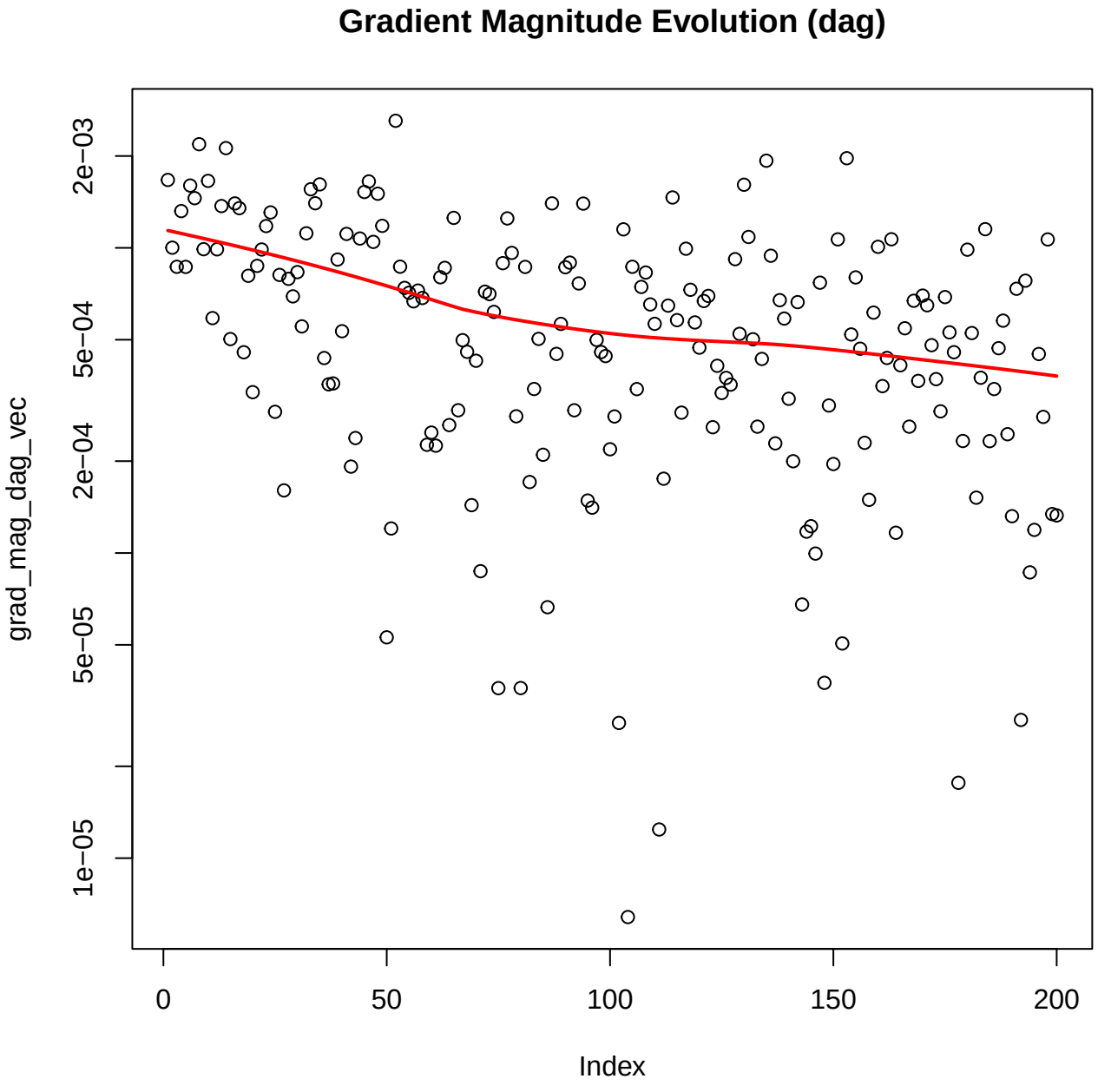

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

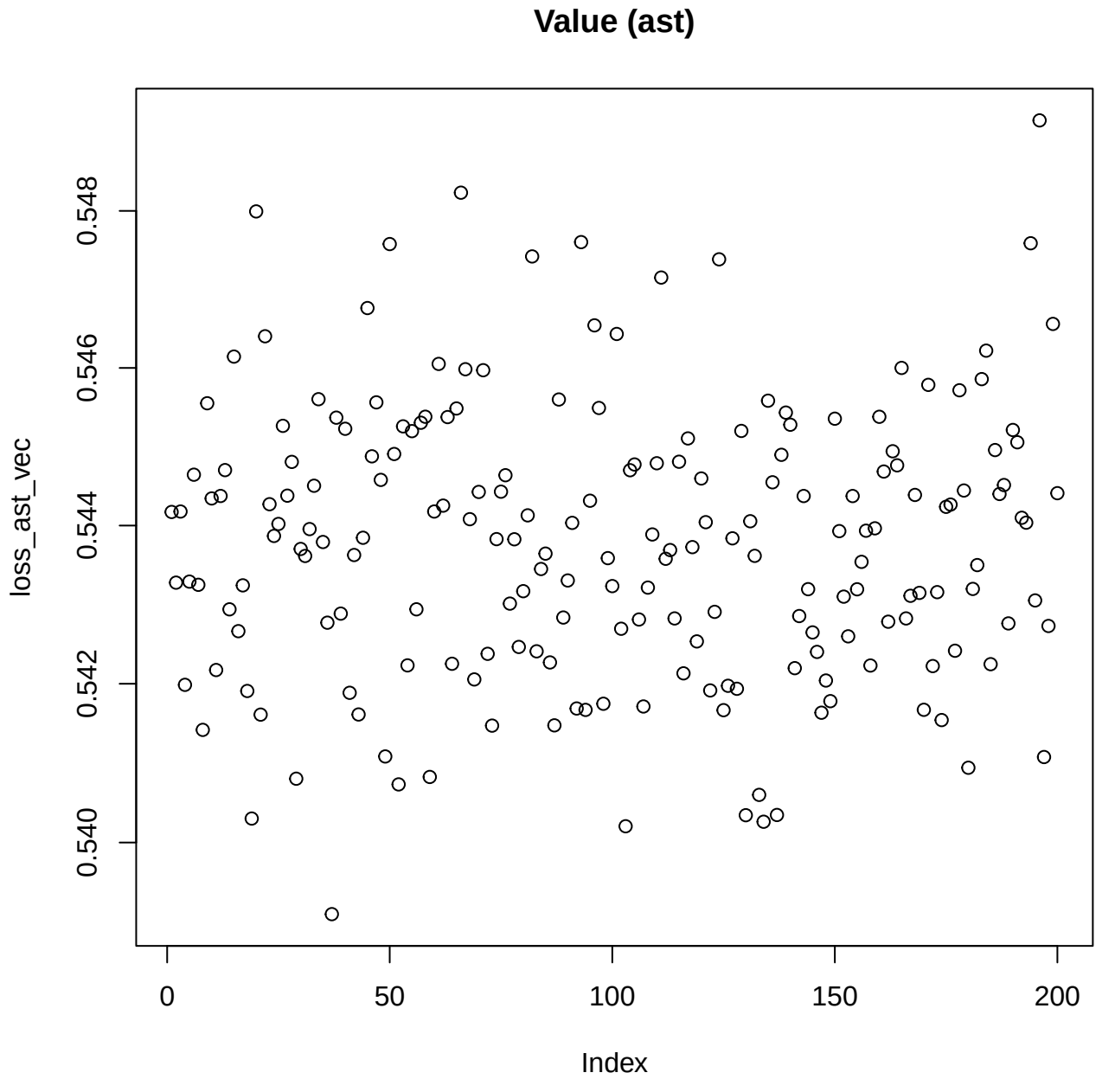

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
5e−03
2e−03
1e−03
5e−04
2
4
6
8
10
Index

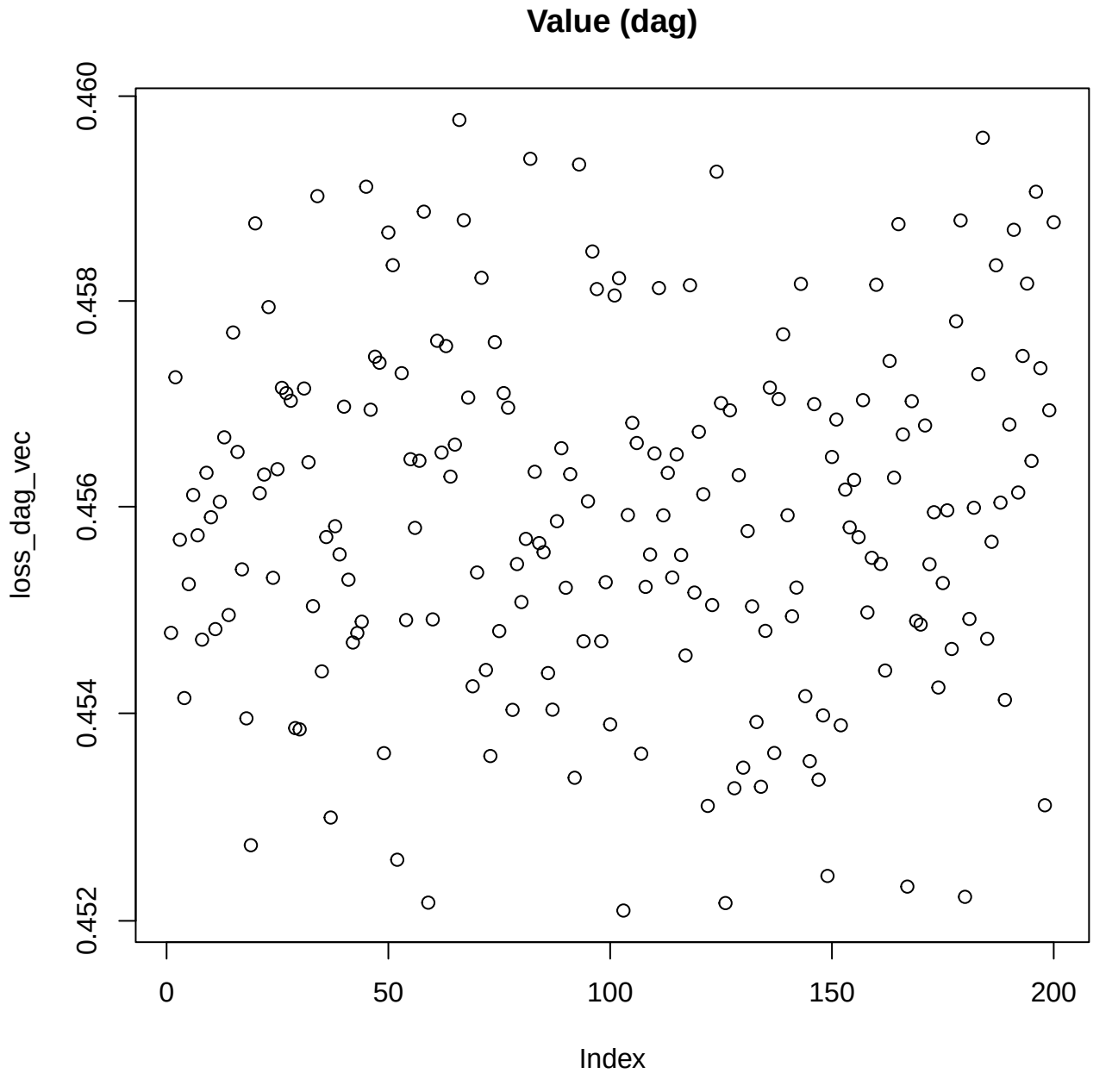

Value (ast)
loss_ast_vec
0.496
0.498
0.500
0.502
0.504
0.506
0.508
0.510
2
4
6
8
10
Index

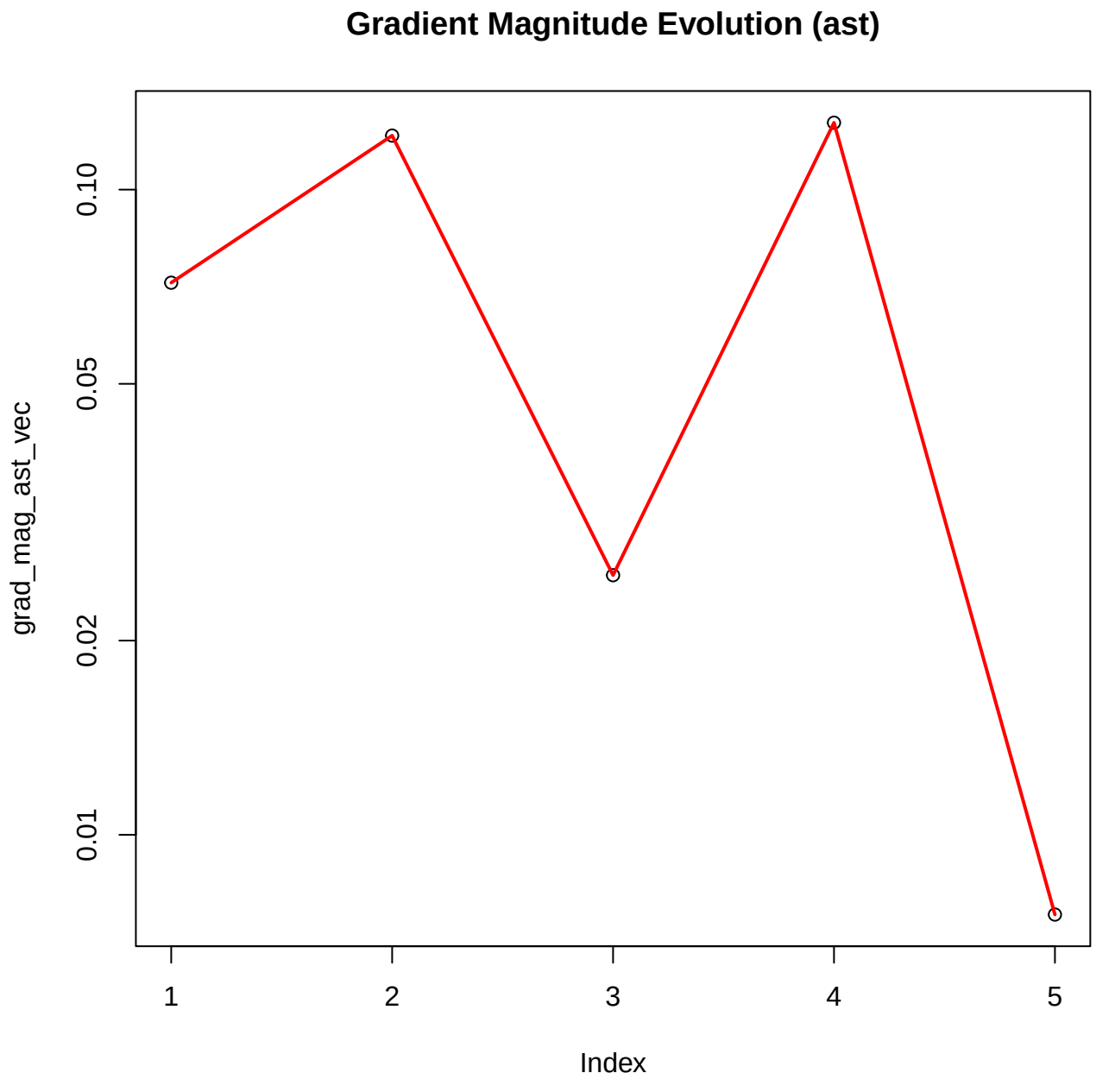

Value (dag)
loss_dag_vec
0.490
0.492
0.494
0.496
0.498
0.500
0.502
2
4
6
8
10
Index

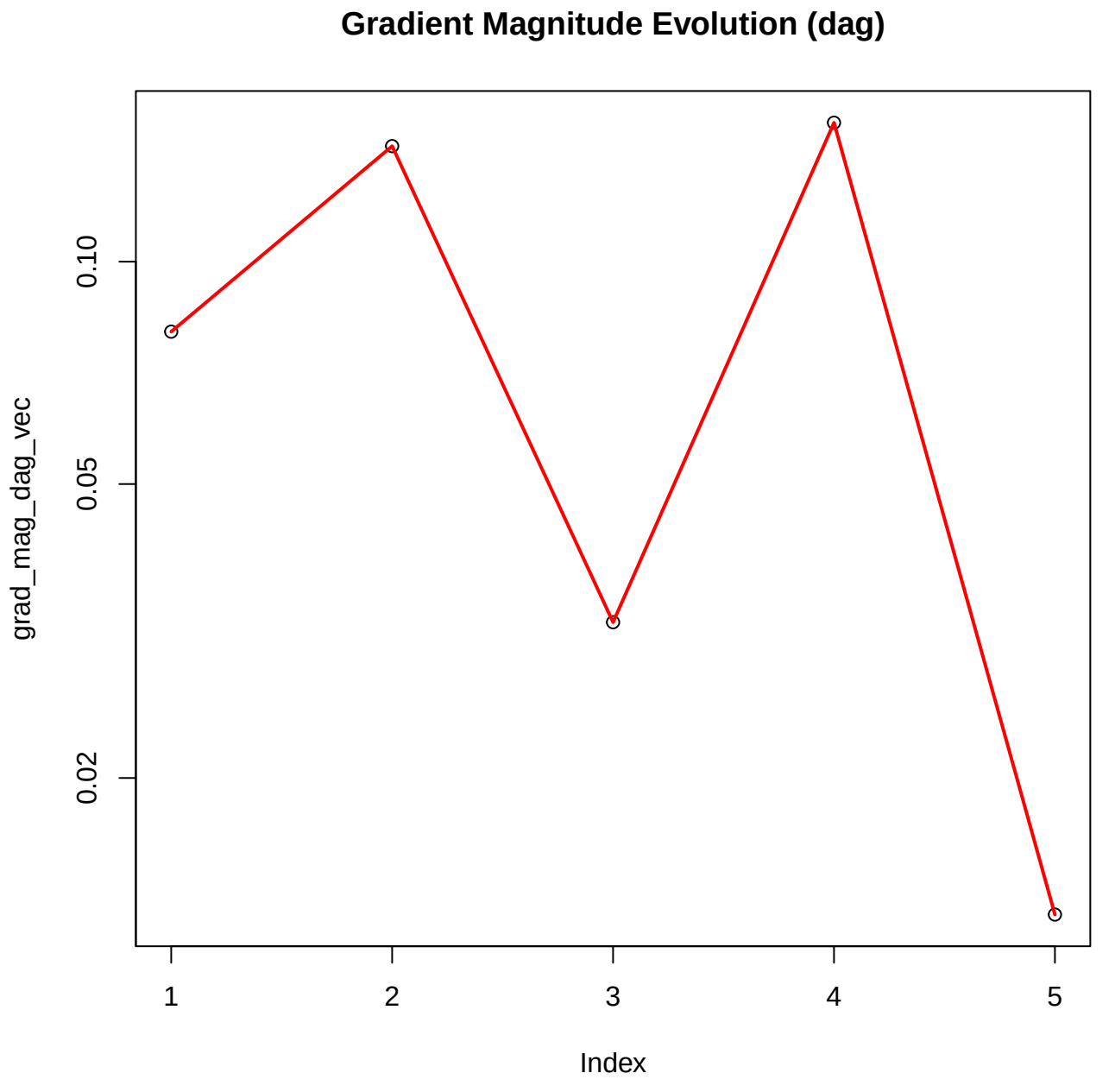

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

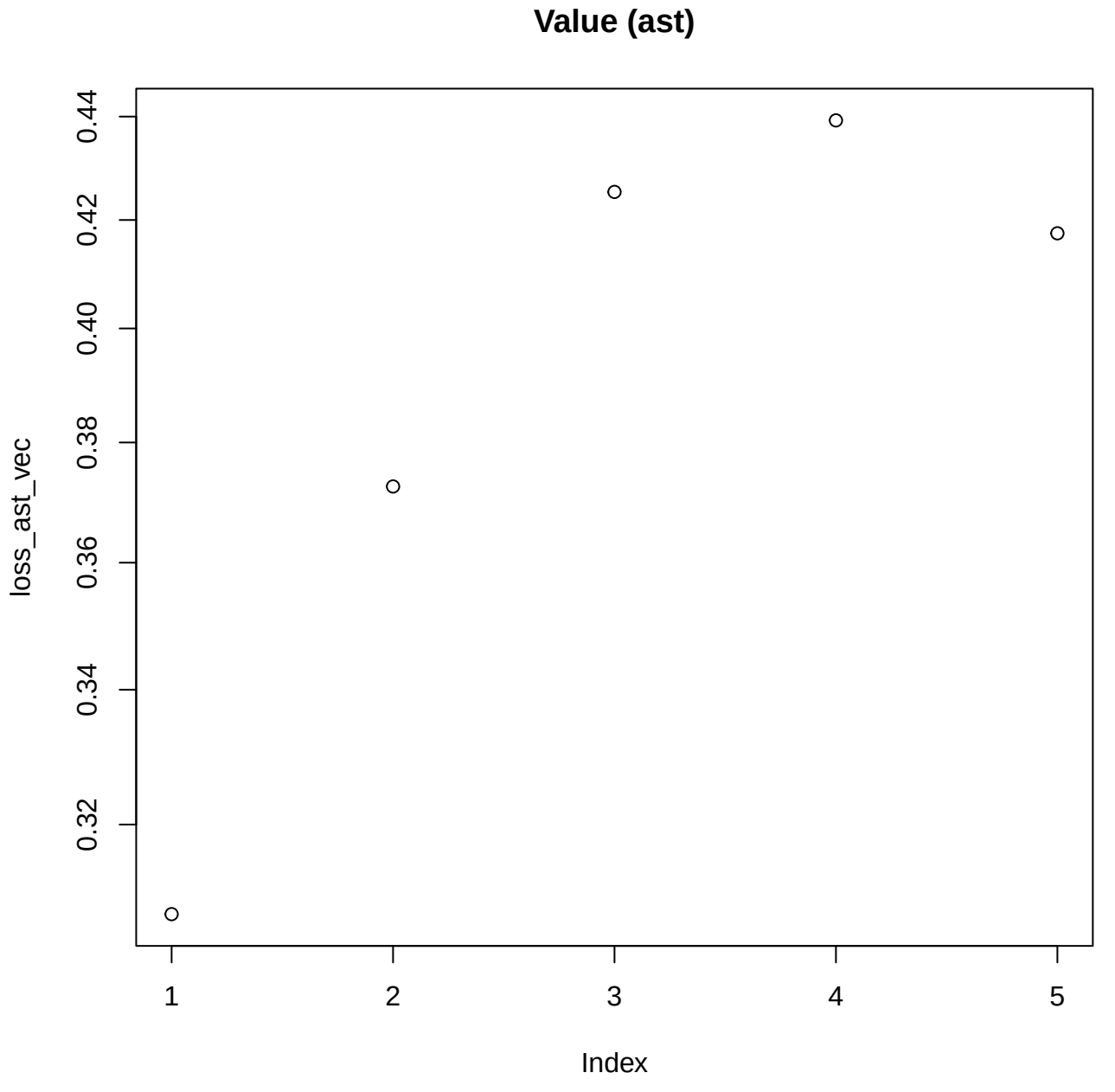

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
2e−03
5e−04
2e−04
5e−05
1e−05
0
50
100
150
200
Index

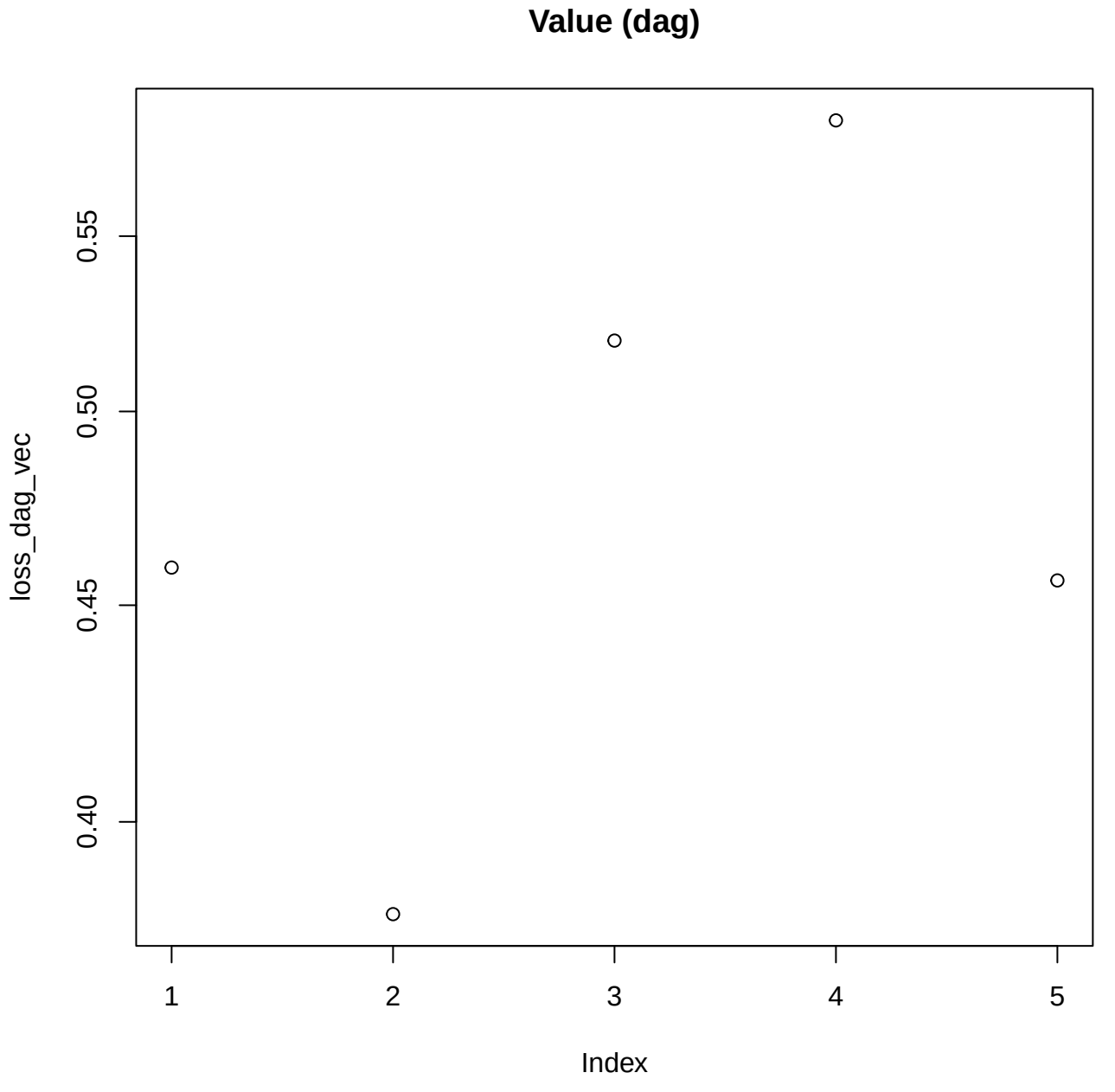

Value (ast)
loss_ast_vec
0.540
0.542
0.544
0.546
0.548
0
50
100
150
200
Index

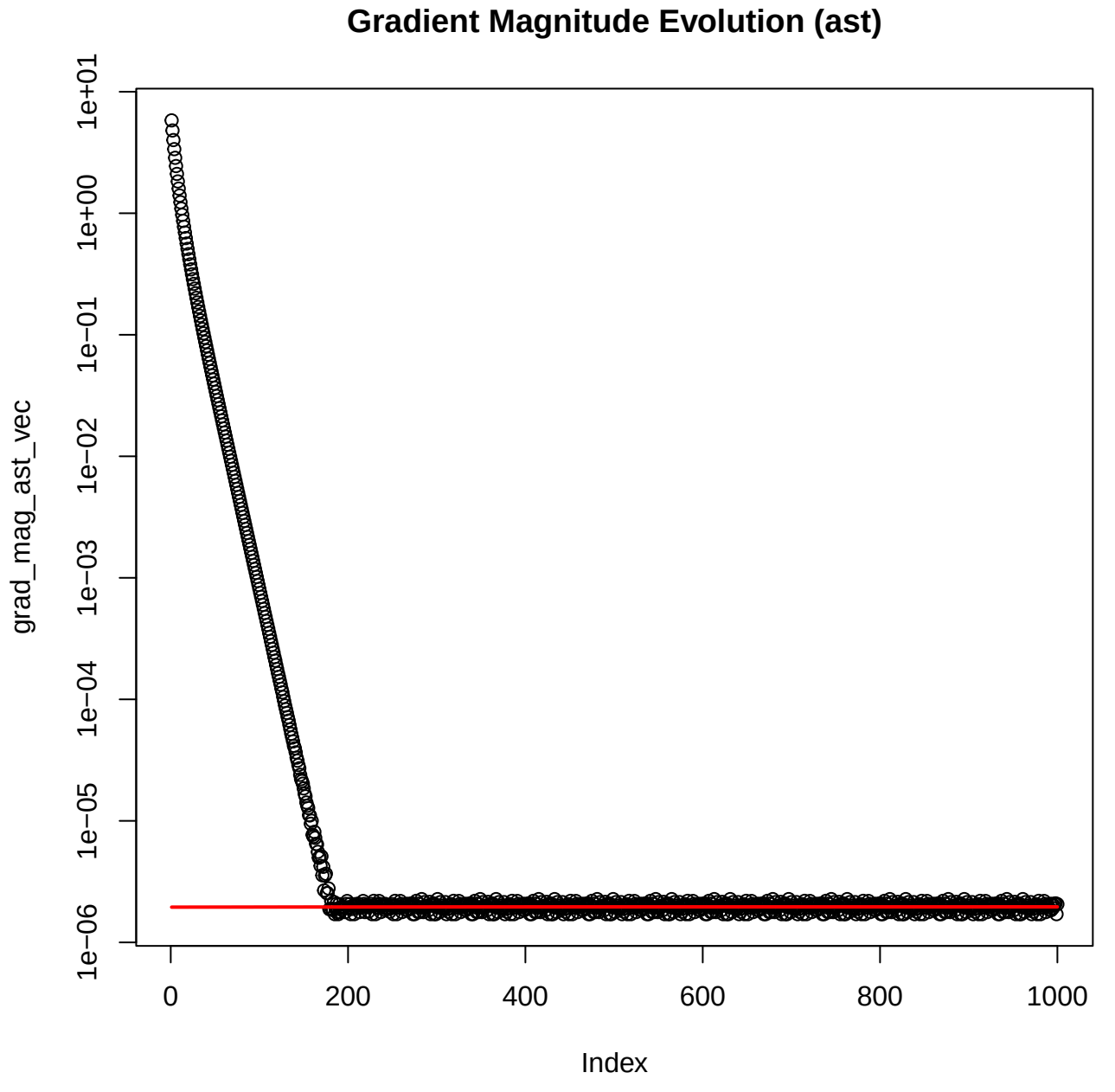

Value (dag)
loss_dag_vec
0.452
0.454
0.456
0.458
0.460
0
50
100
150
200
Index

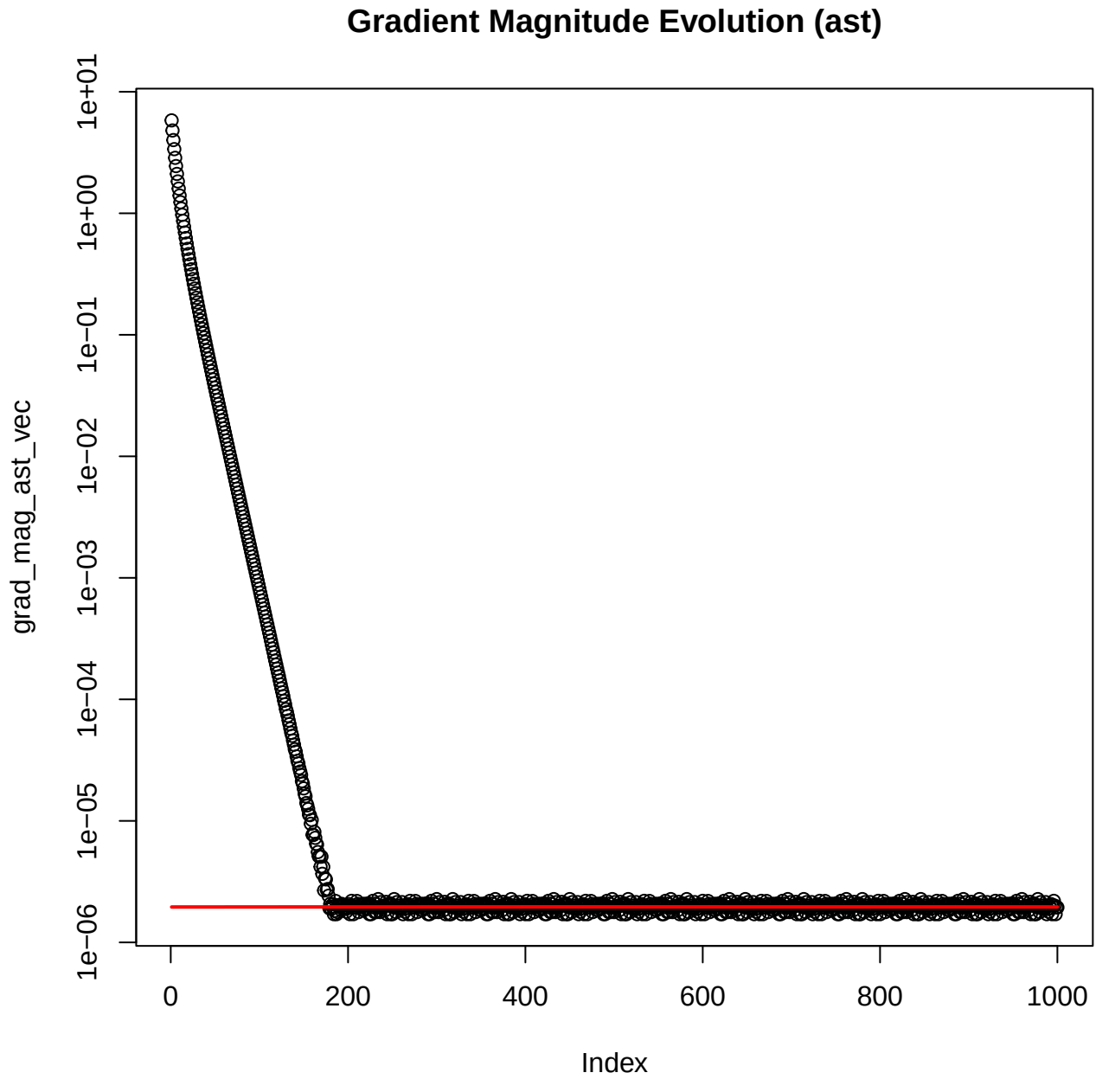

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
1
2
3
4
5
Index

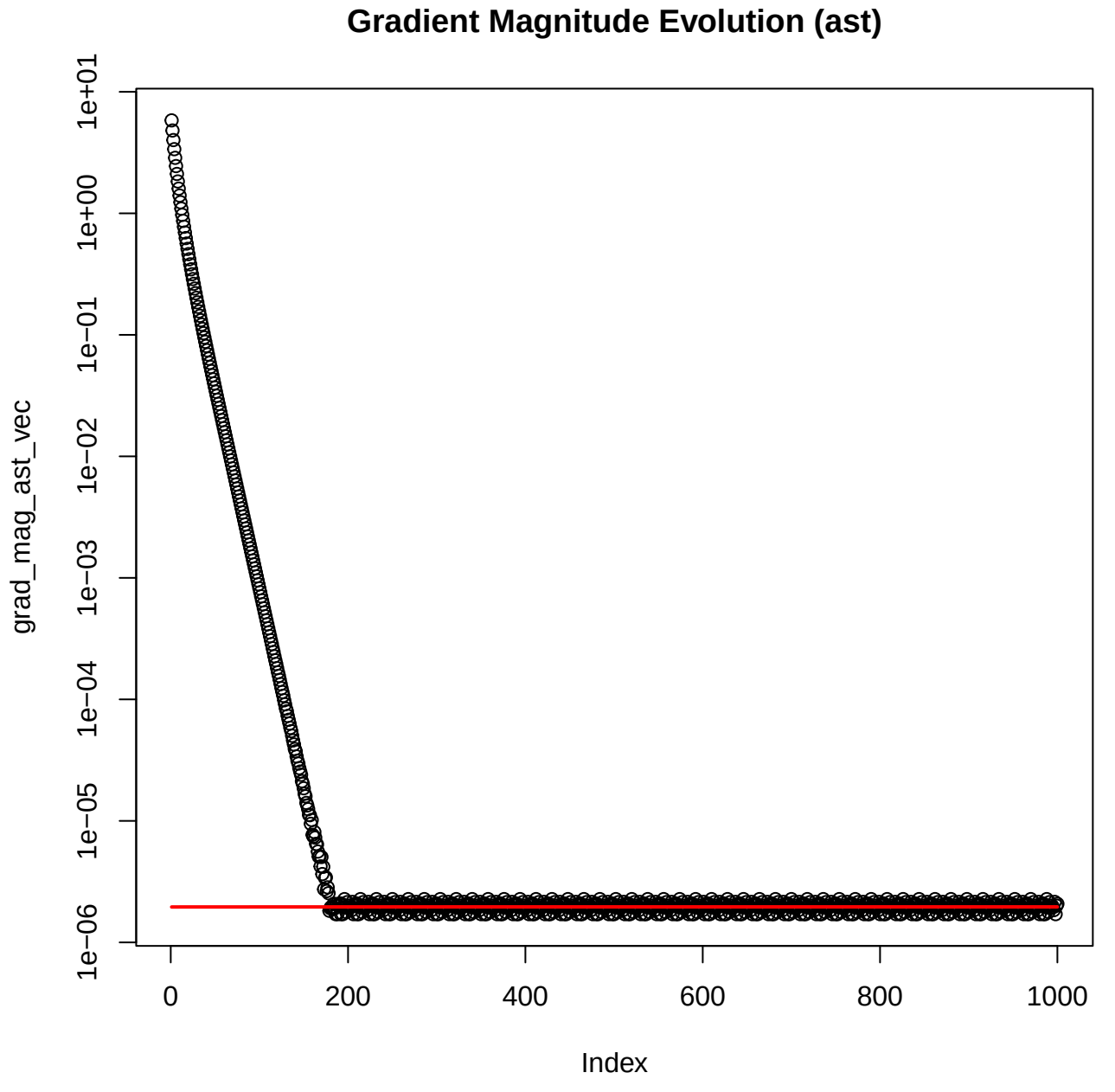

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.02
0.05
0.10
1
2
3
4
5
Index

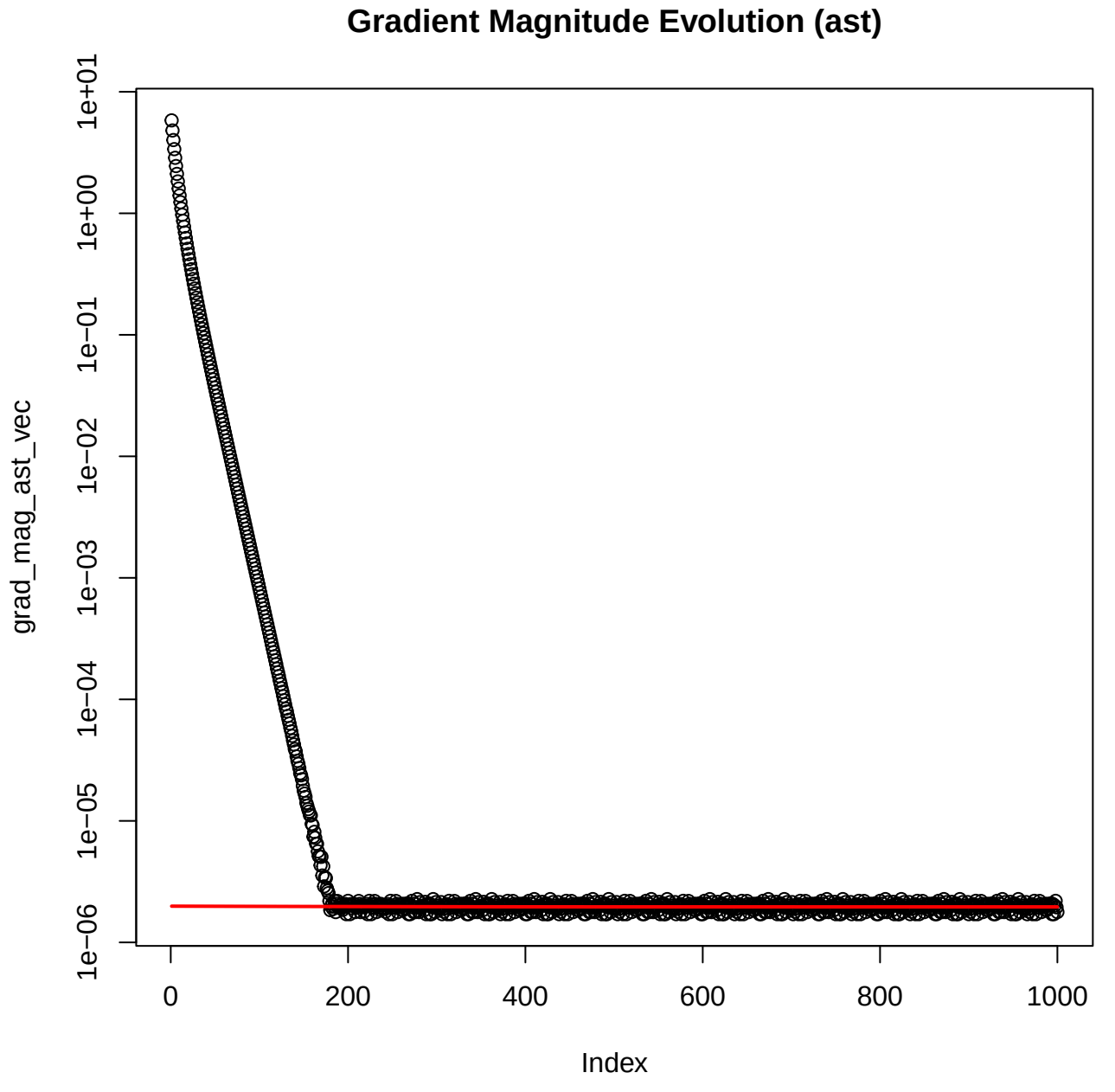

Value (ast)
loss_ast_vec
0.32
0.34
0.36
0.38
0.40
0.42
0.44
1
2
3
4
5
Index

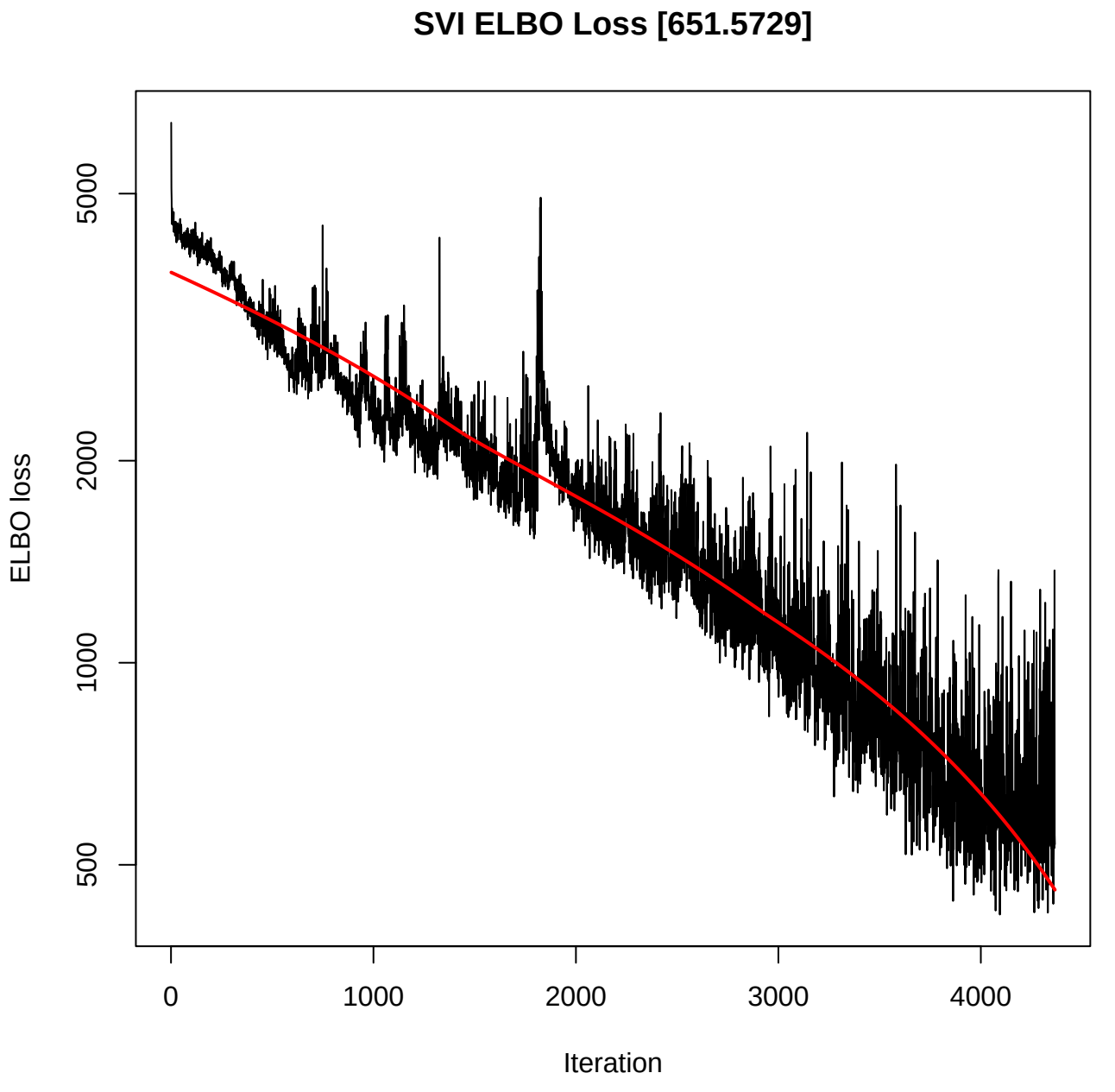

Value (dag)
loss_dag_vec
0.40
0.45
0.50
0.55
1
2
3
4
5
Index

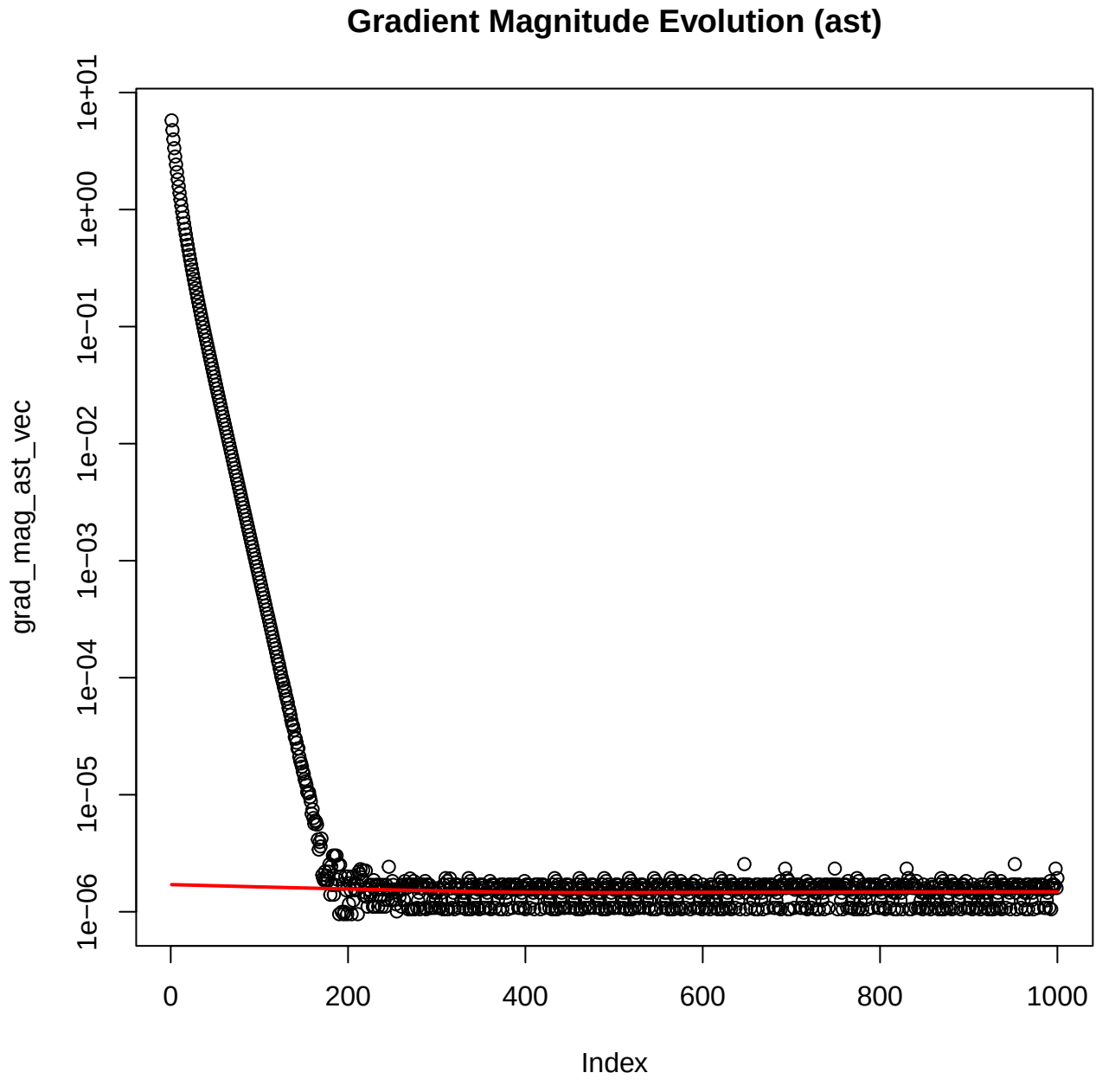

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

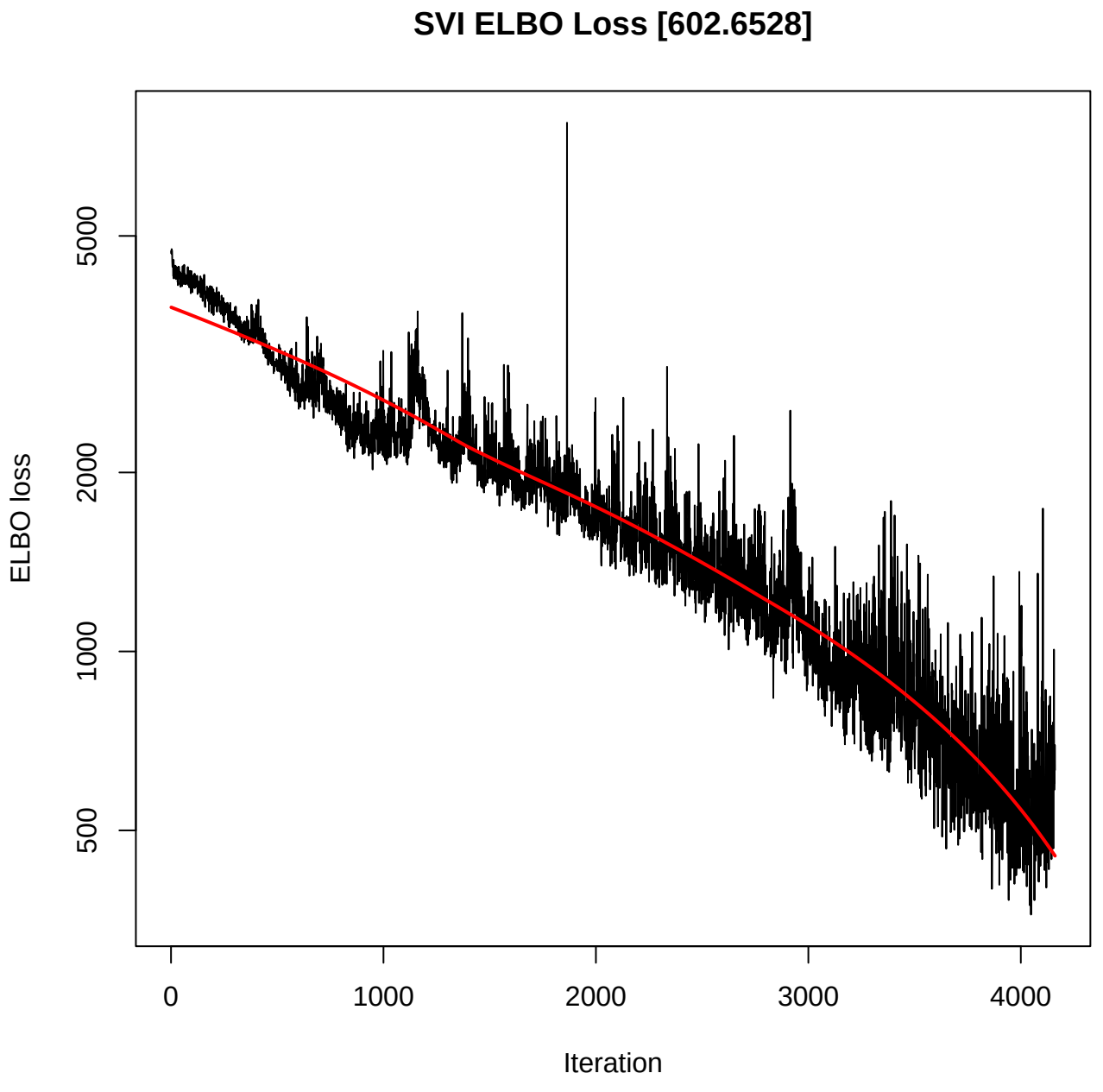

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

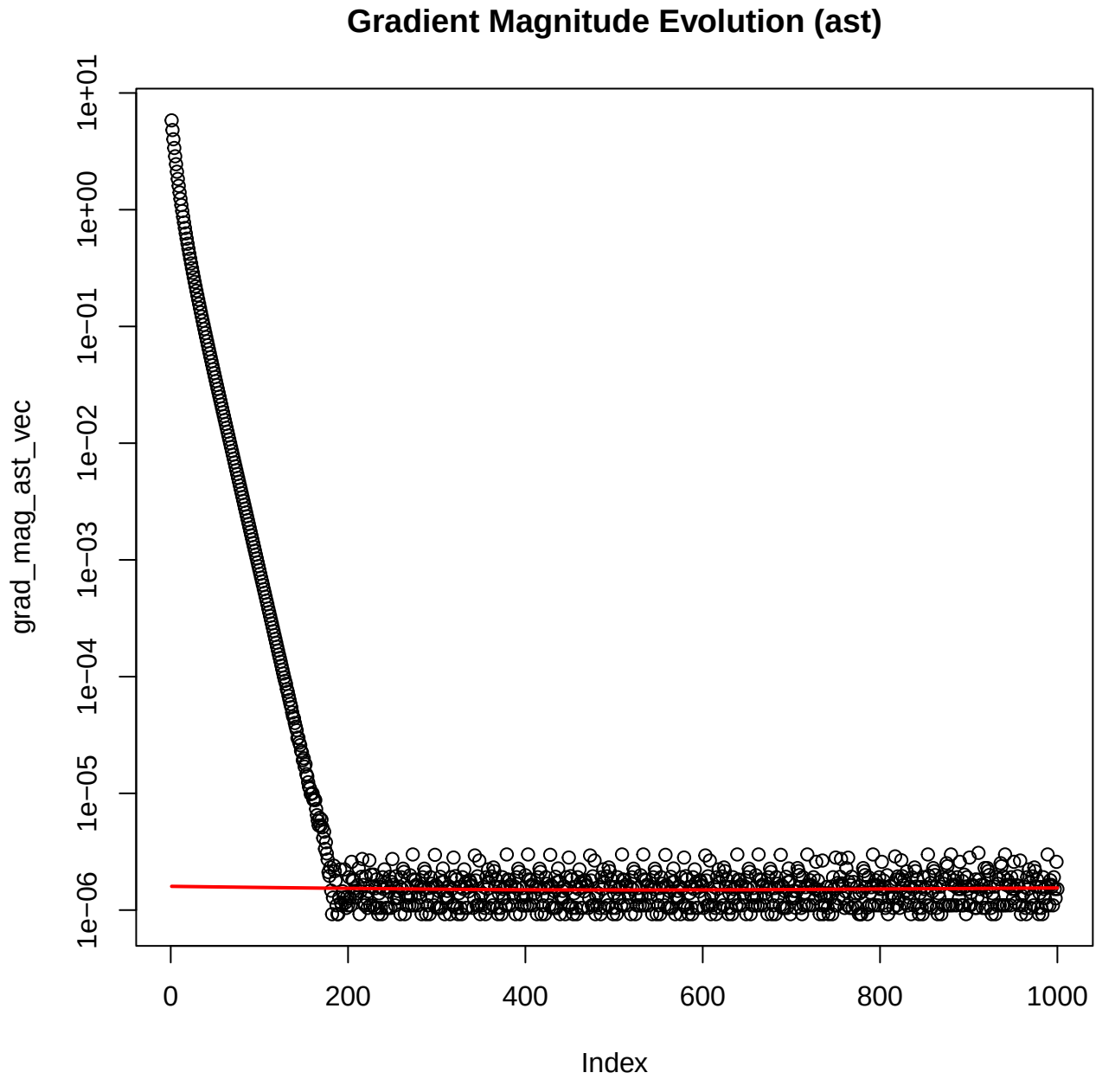

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

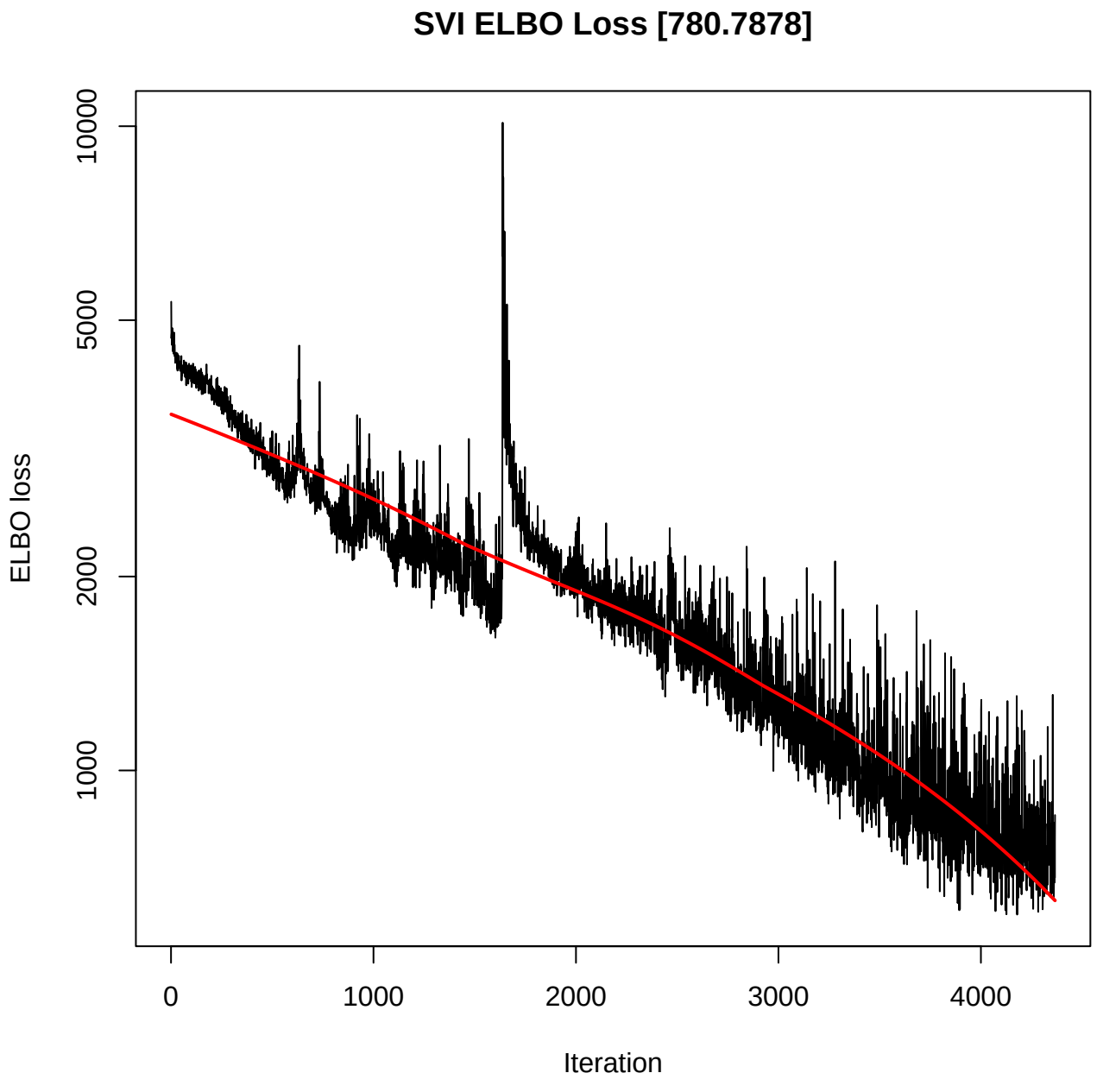

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

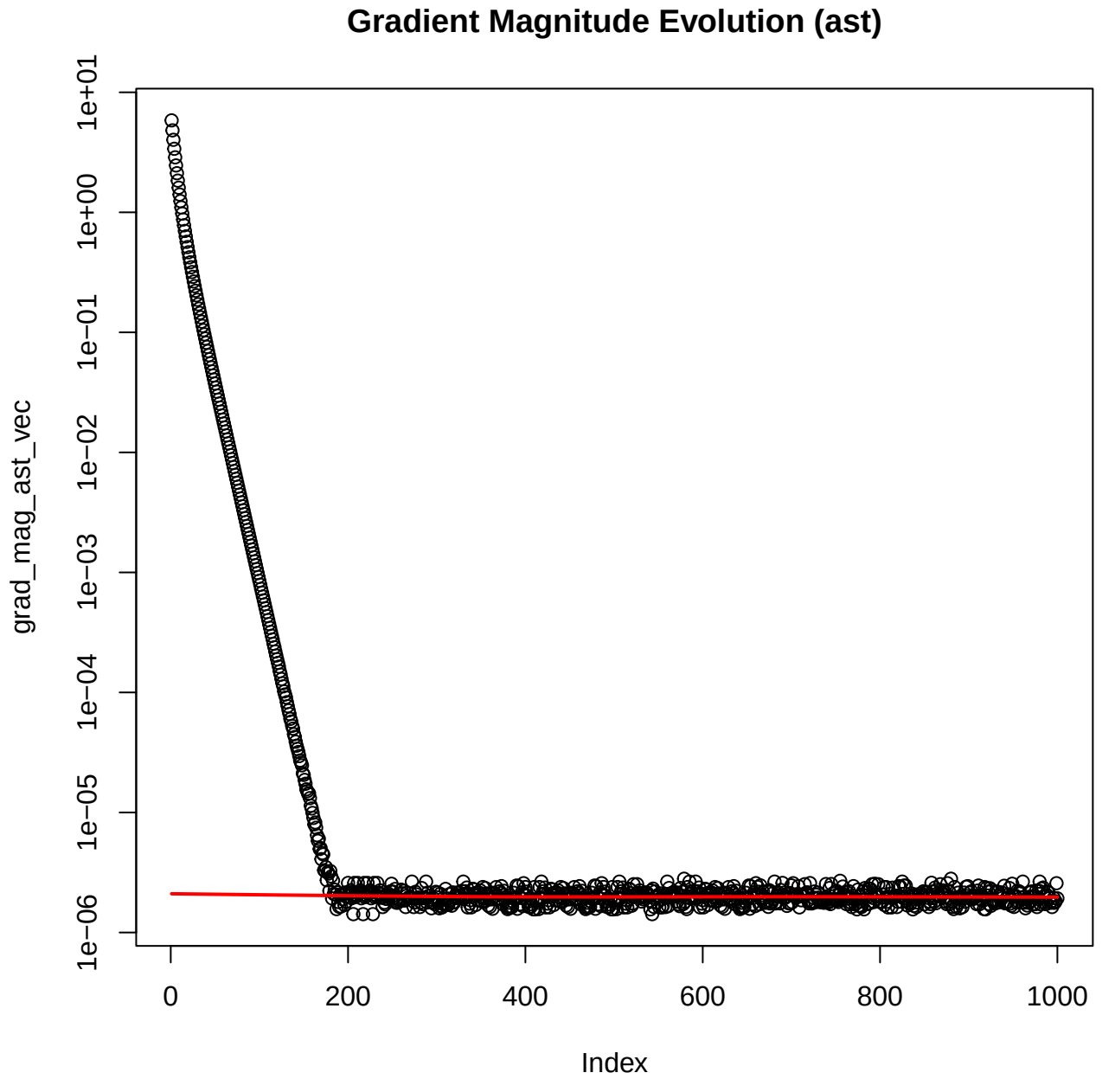

SVI ELBO Loss [651.5729]
ELBO loss
500
1000
2000
5000
0
1000
2000
3000
4000
Iteration

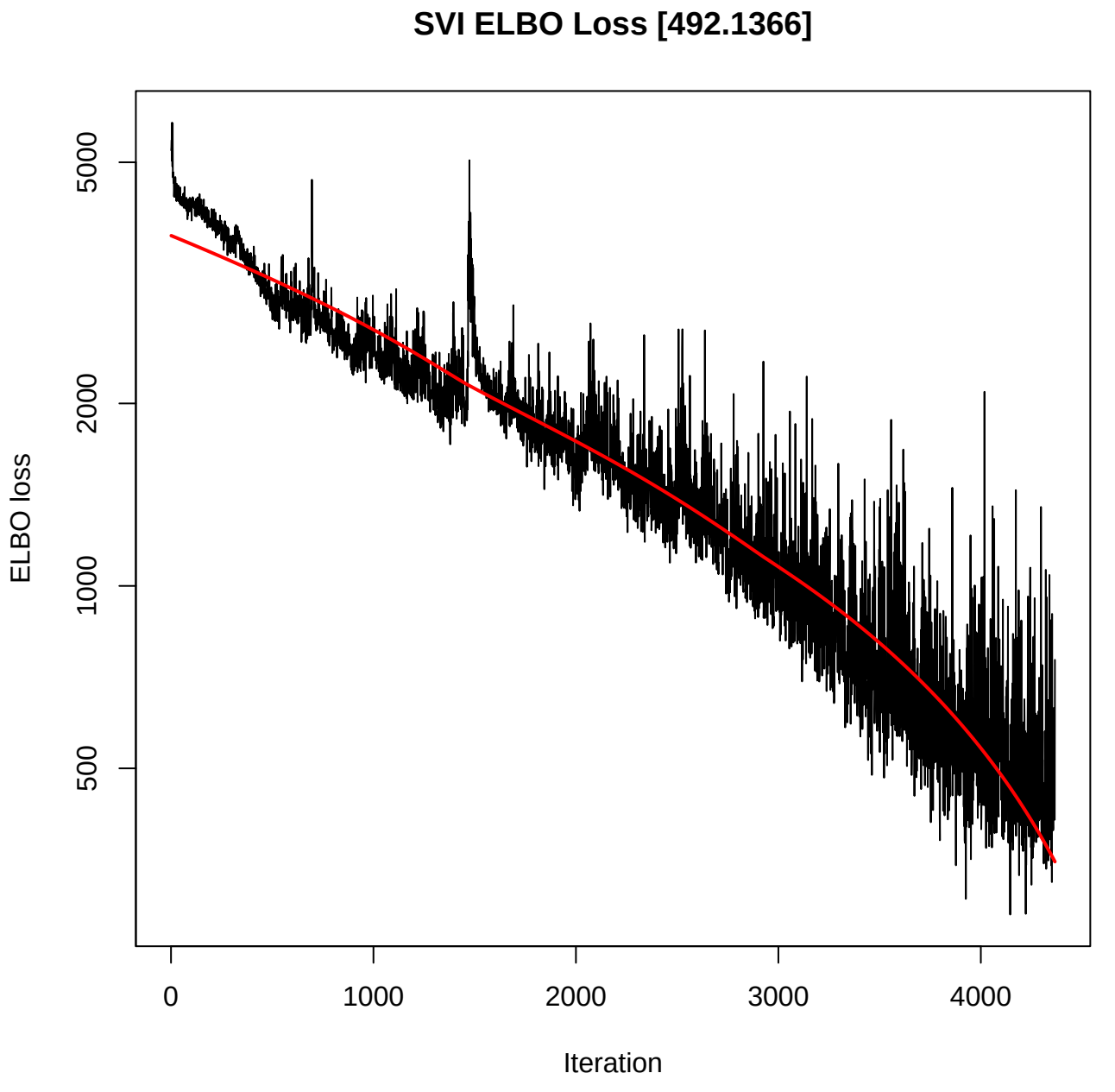

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

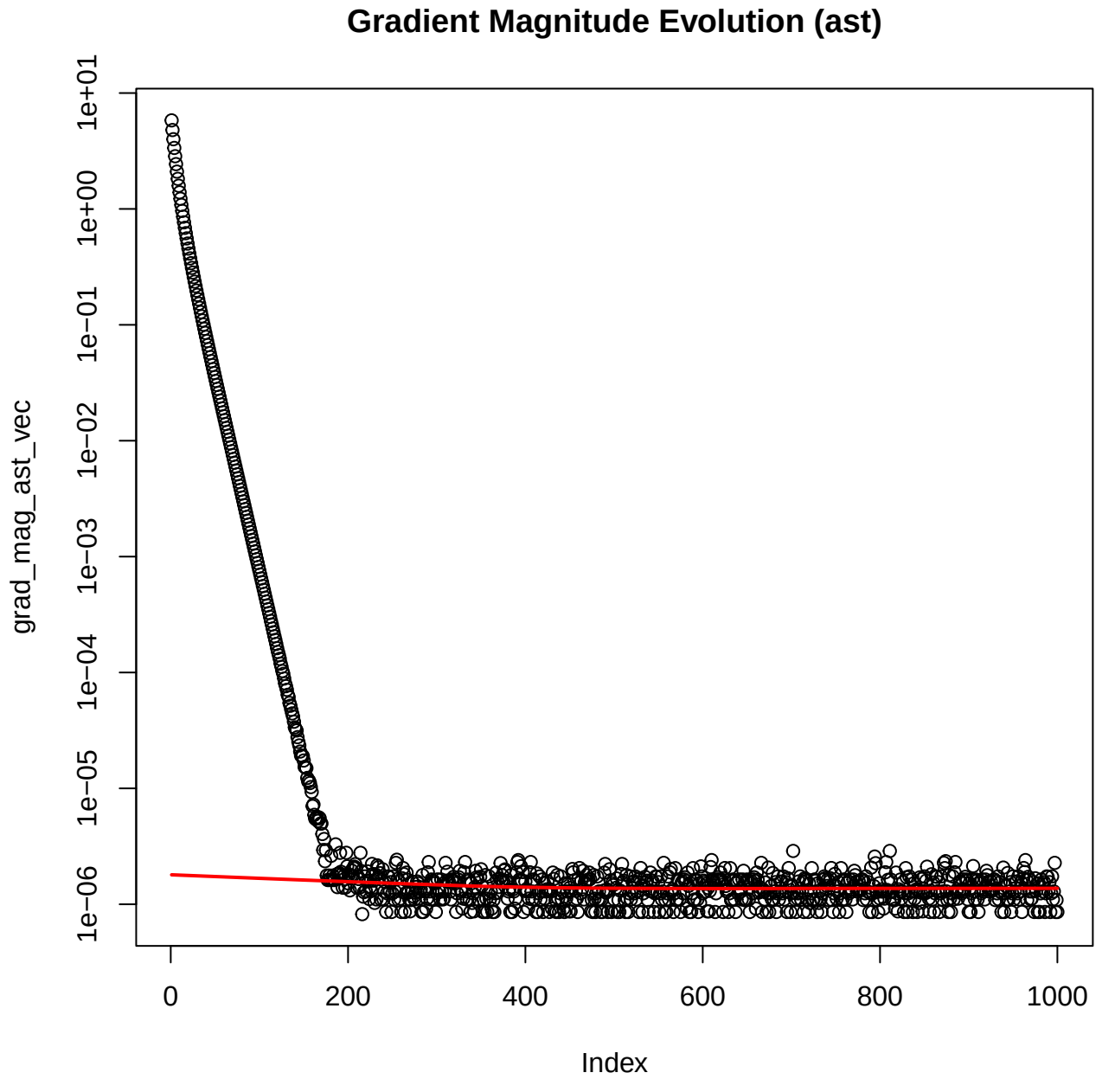

SVI ELBO Loss [602.6528]
ELBO loss
500
1000
2000
5000
0
1000
2000
3000
4000
Iteration

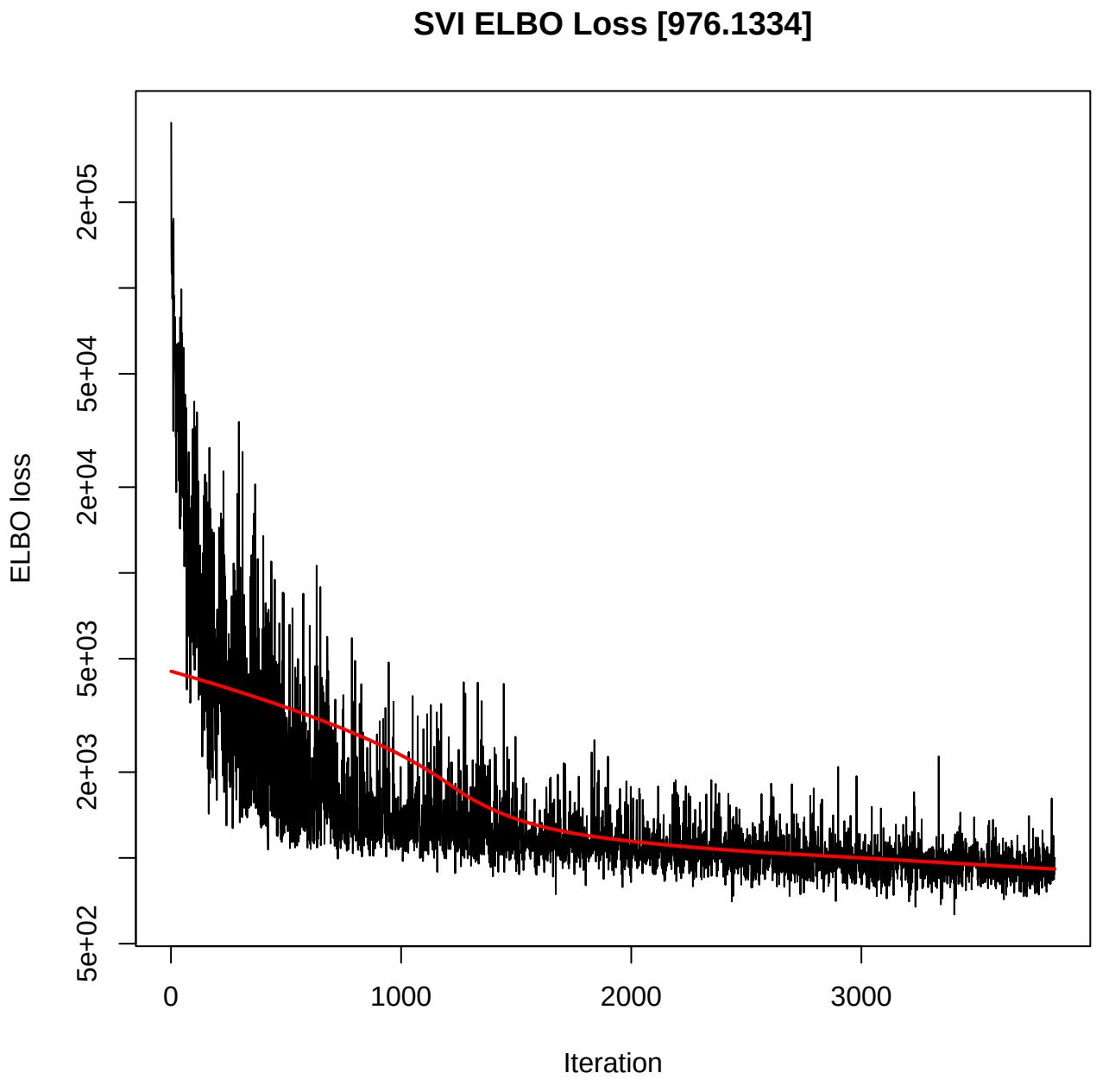

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

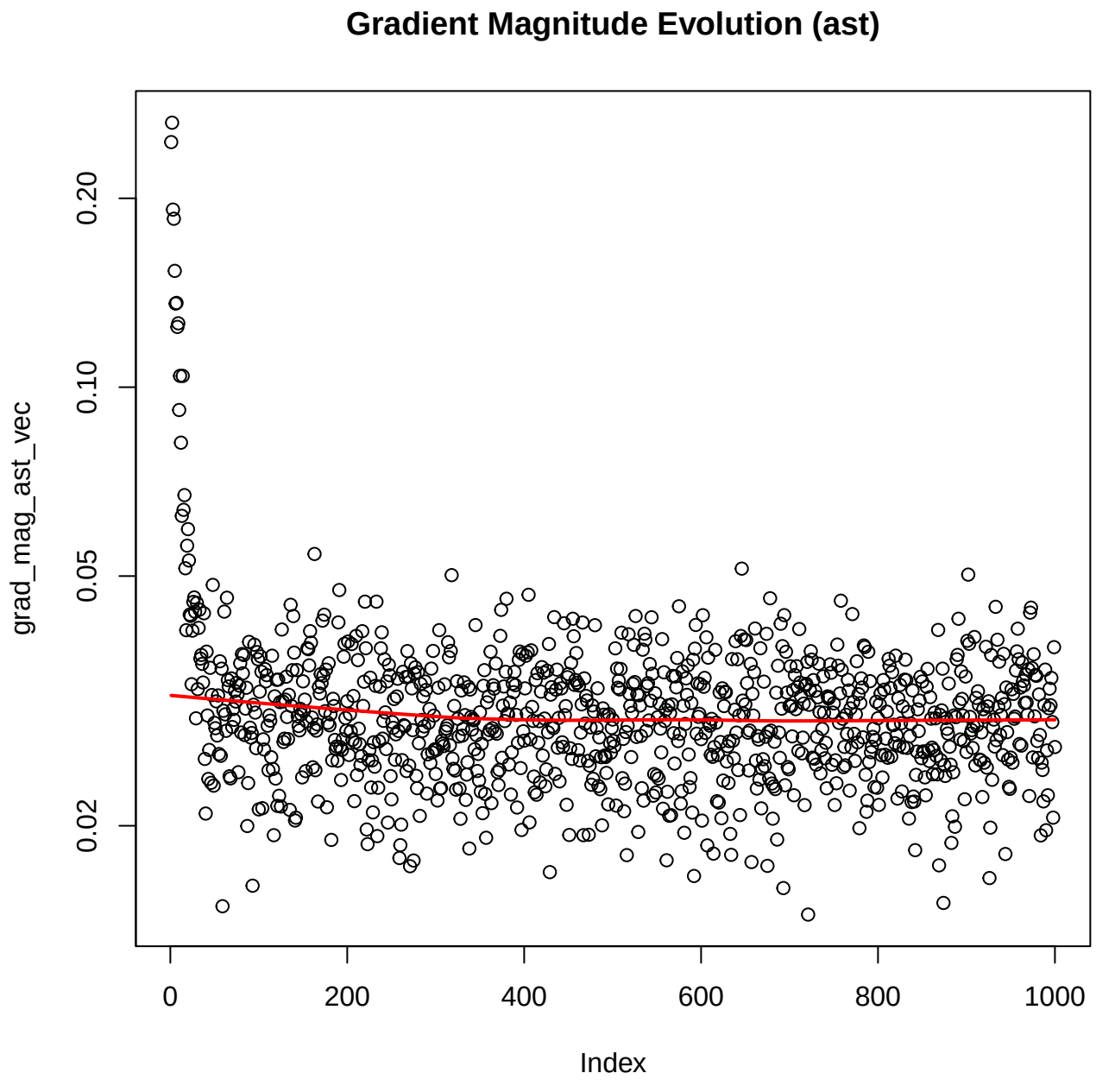

SVI ELBO Loss [780.7878]
ELBO loss
10000
5000
2000
1000
0
1000
2000
3000
4000
Iteration

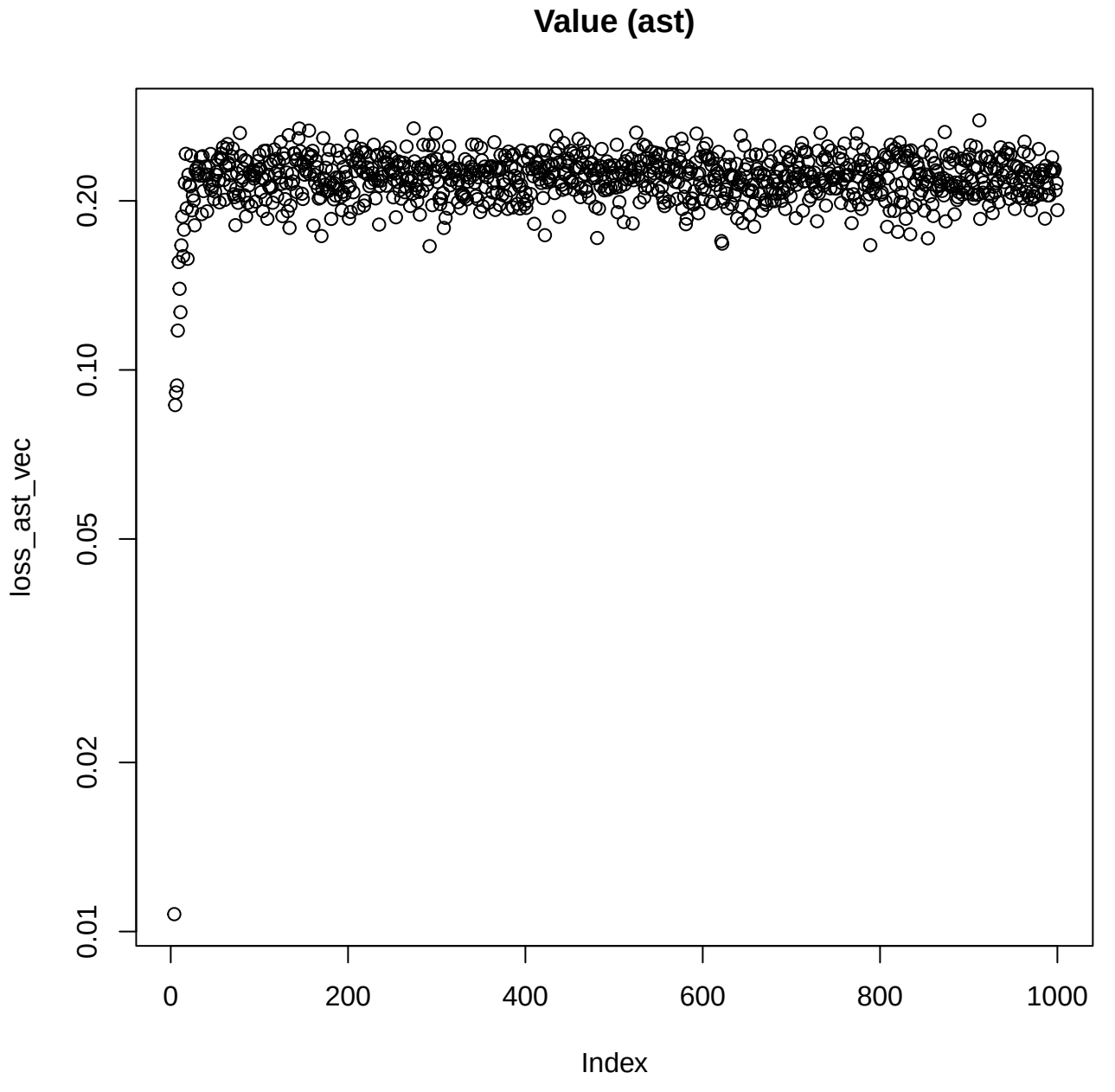

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index

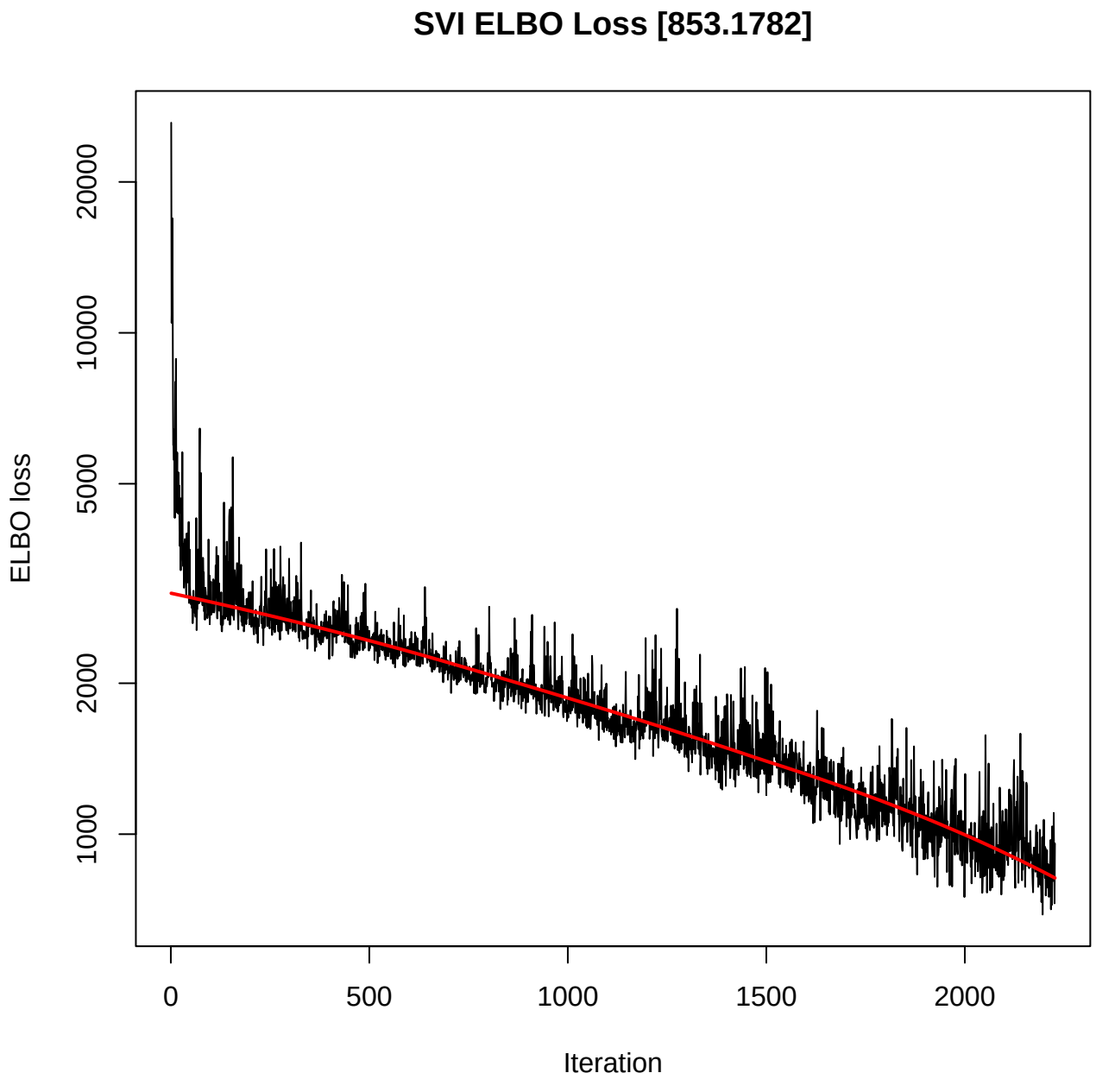

SVI ELBO Loss [492.1366]
ELBO loss
5000
2000
1000
500
0
1000
2000
3000
4000
Iteration

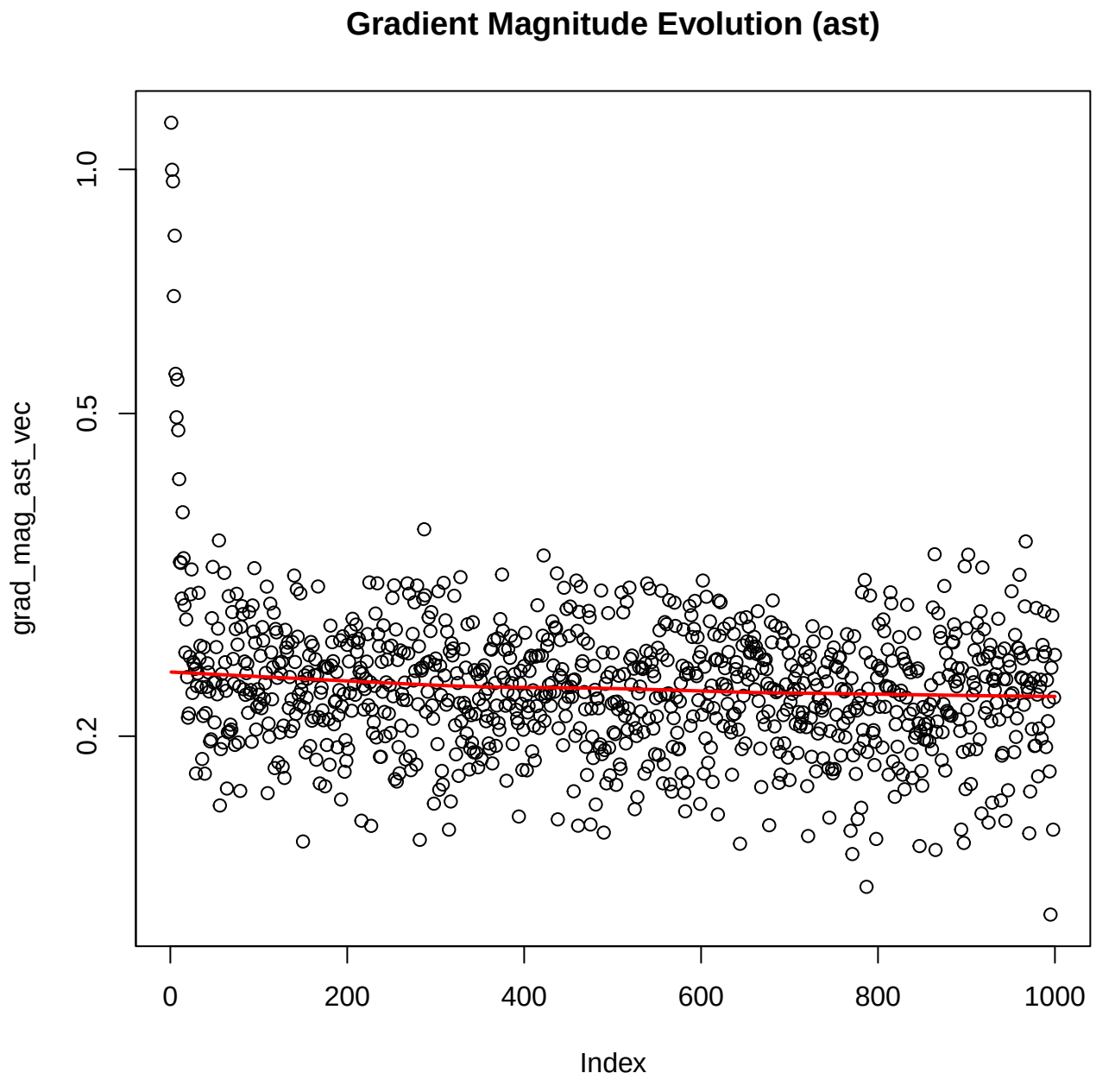

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
1e+01
1e+00
1e−01
1e−02
1e−03
1e−04
1e−05
1e−06
0
200
400
600
800
1000
Index


SVI ELBO Loss [976.1334]
ELBO loss
2e+05
5e+04
2e+04
5e+03
2e+03
5e+02
0
1000
2000
3000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index


Value (ast)
loss_ast_vec
0.01
0.02
0.05
0.10
0.20
0
200
400
600
800
1000
Index


SVI ELBO Loss [853.1782]
ELBO loss
20000
10000
5000
2000
1000
0
500
1000
1500
2000
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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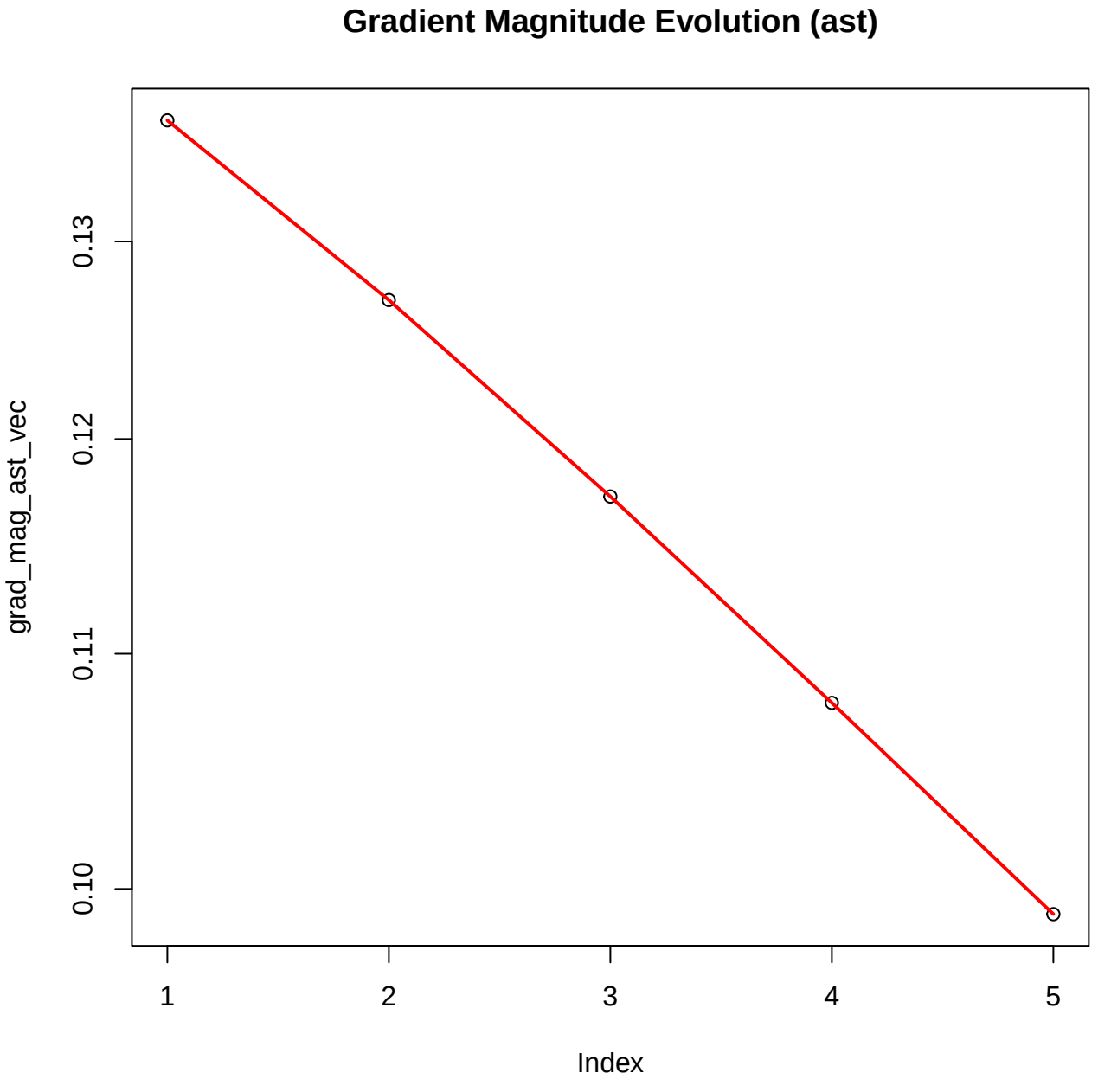
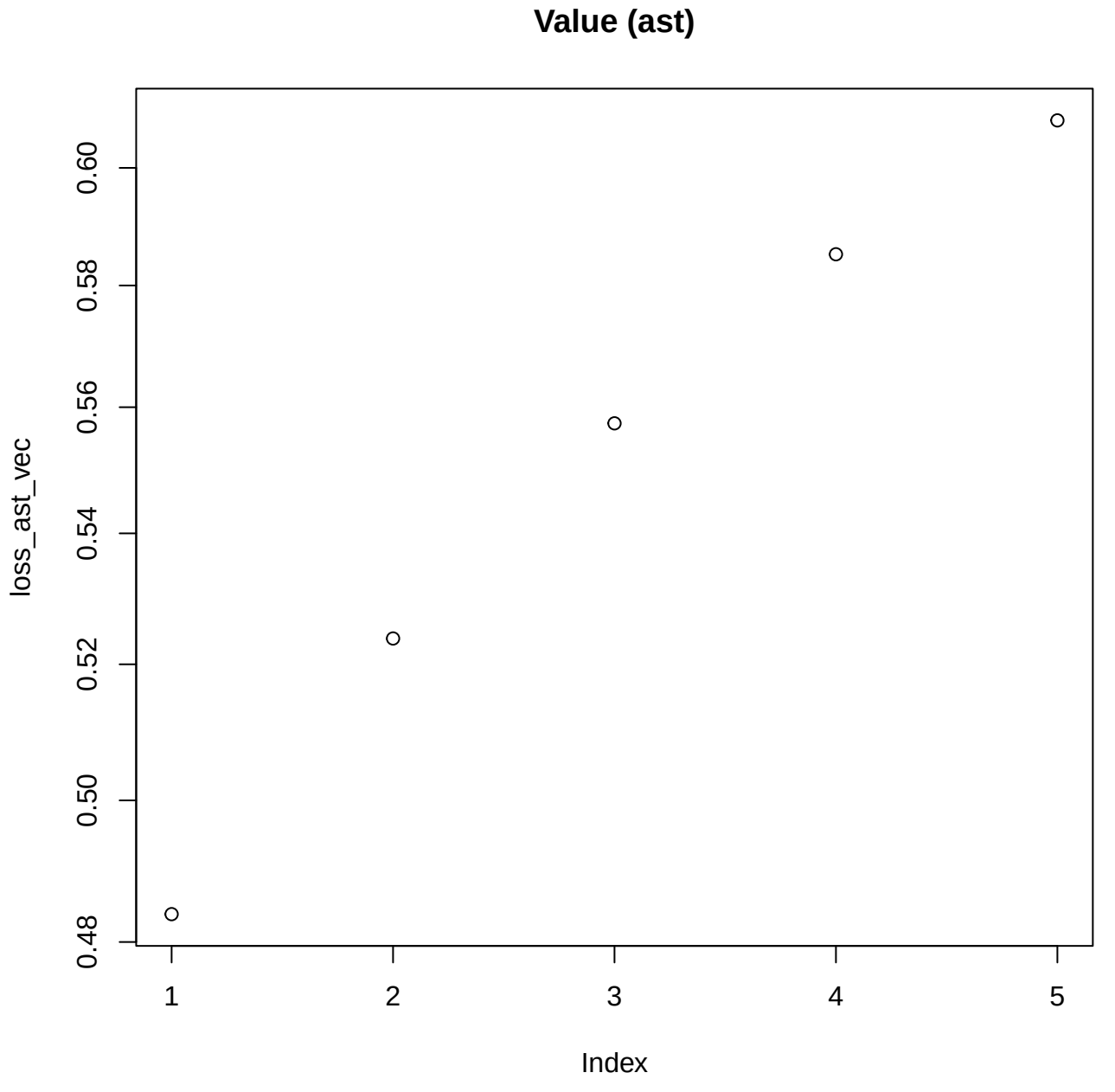
0.5
0.2
0
200
400
600
800
1000
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.10
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
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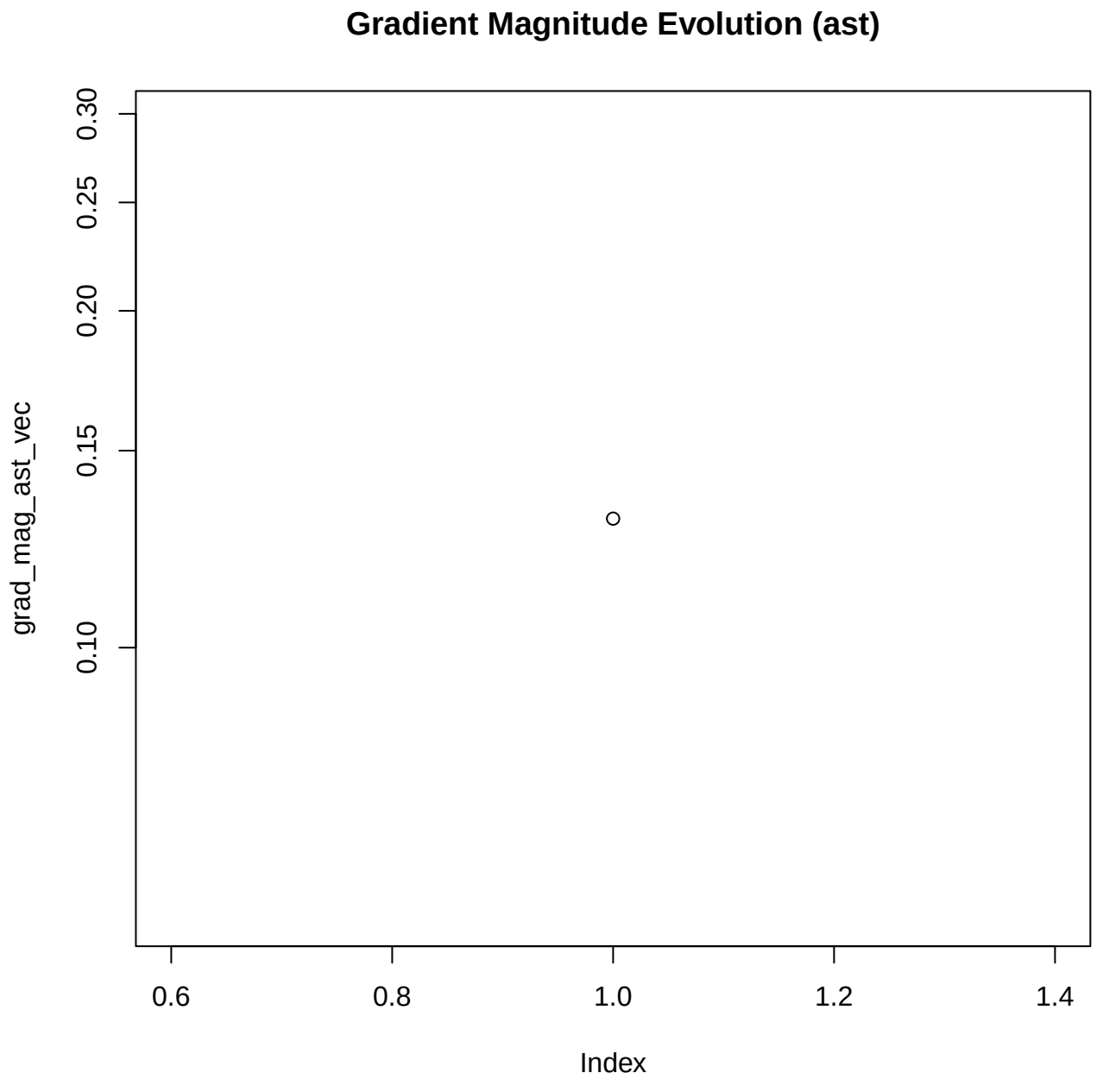
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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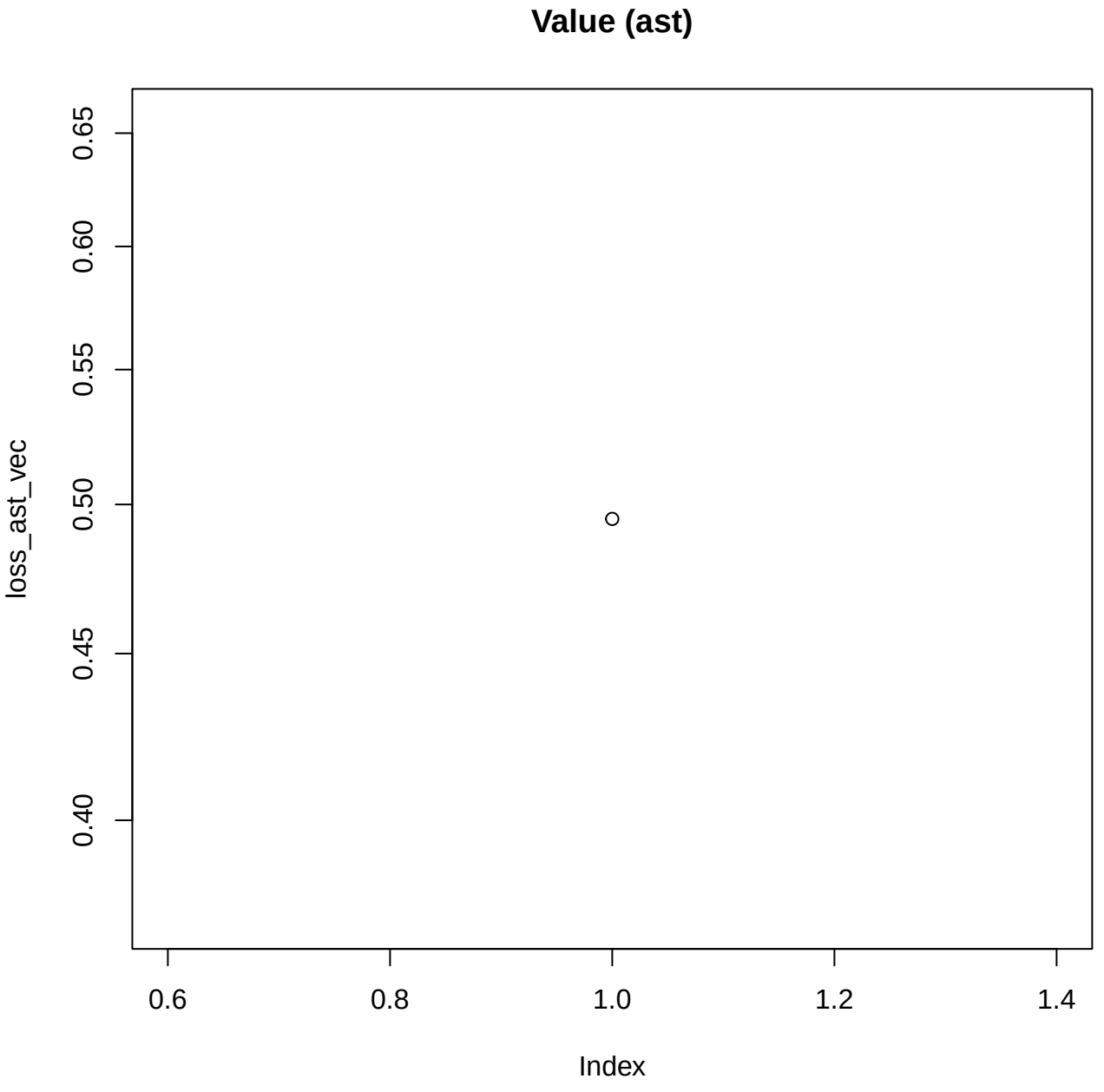
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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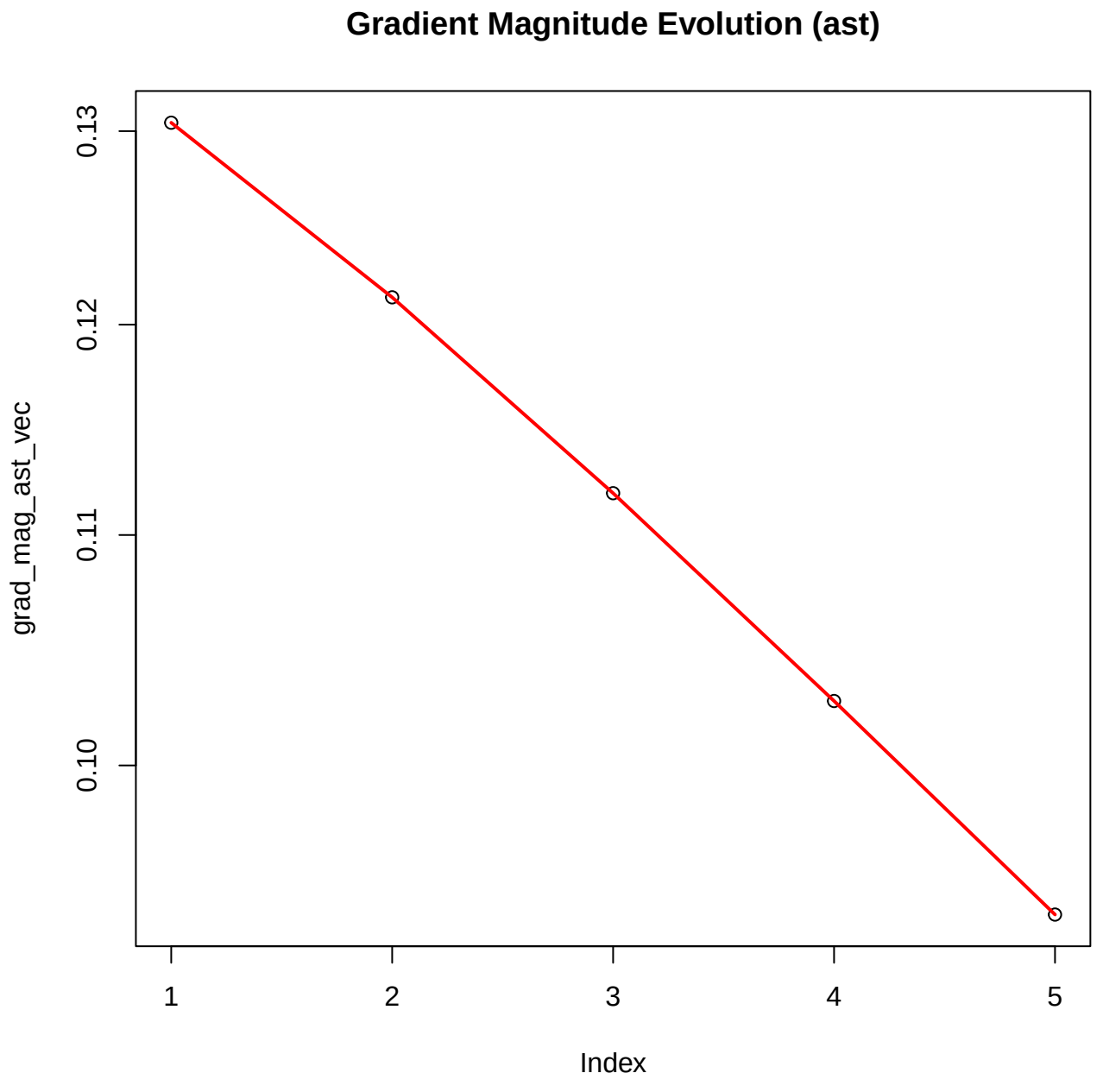
0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.13
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3
4
5
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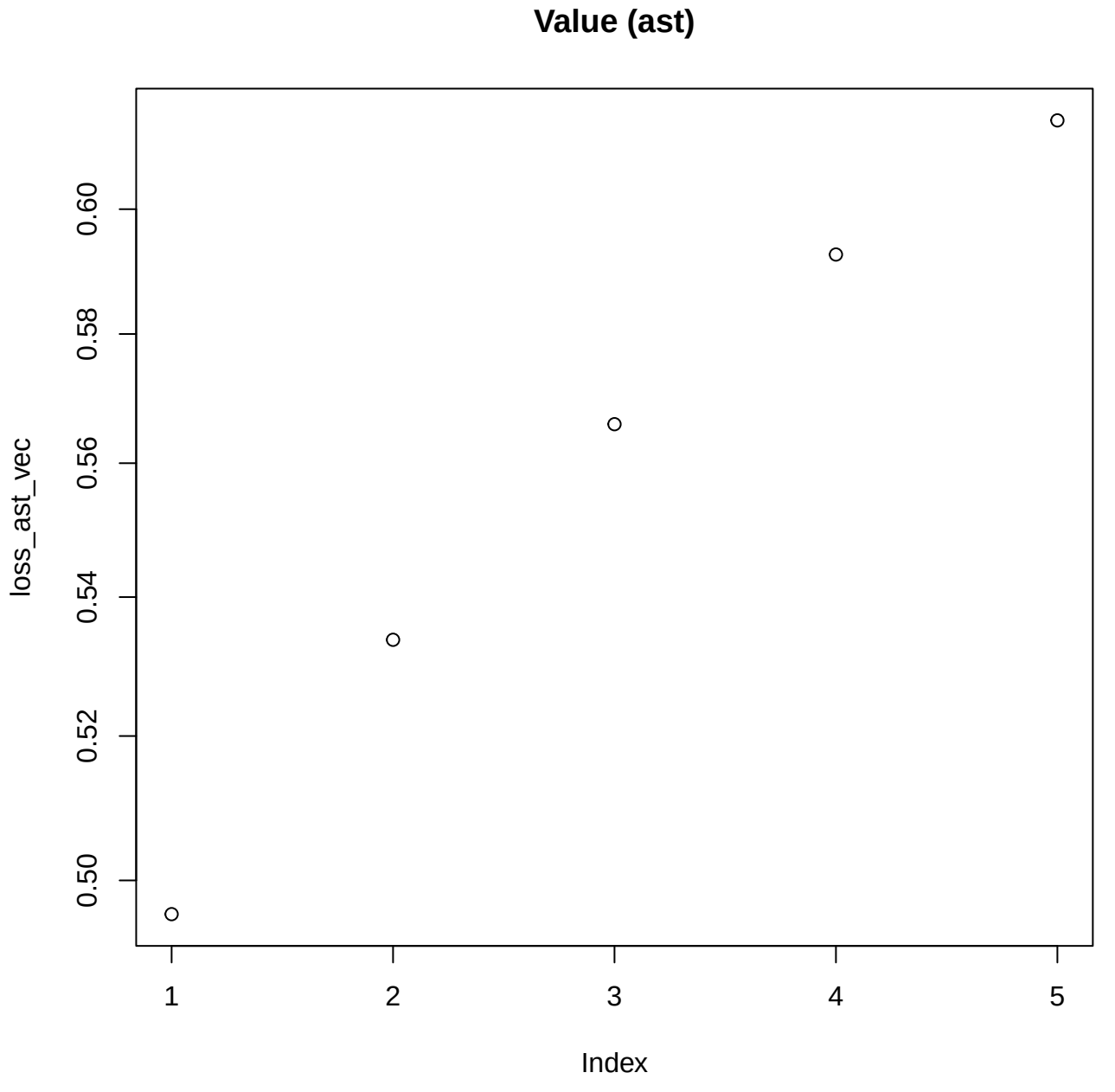
Value (ast)
loss_ast_vec
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0.52
0.54
0.56
0.58
0.60
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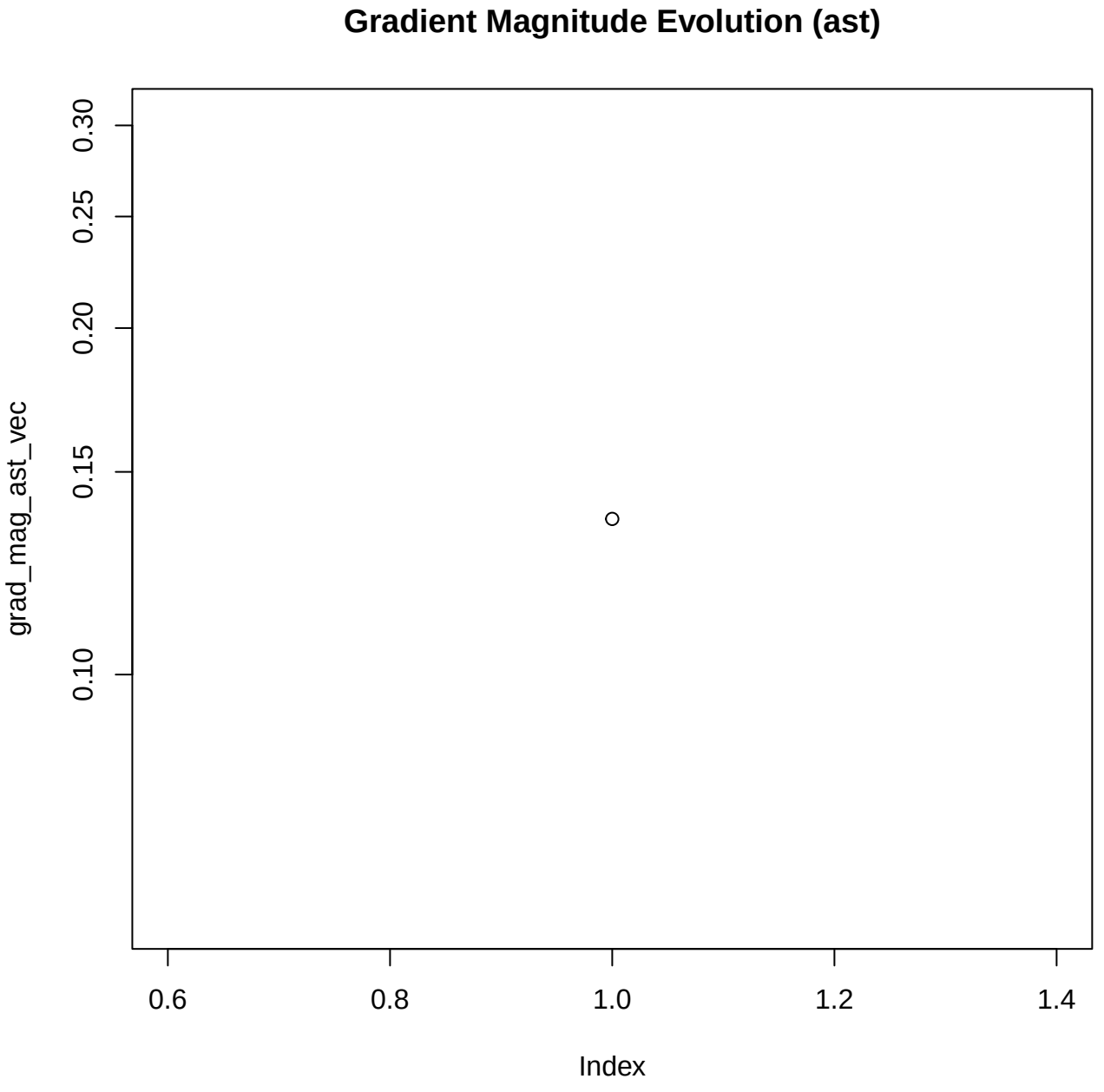
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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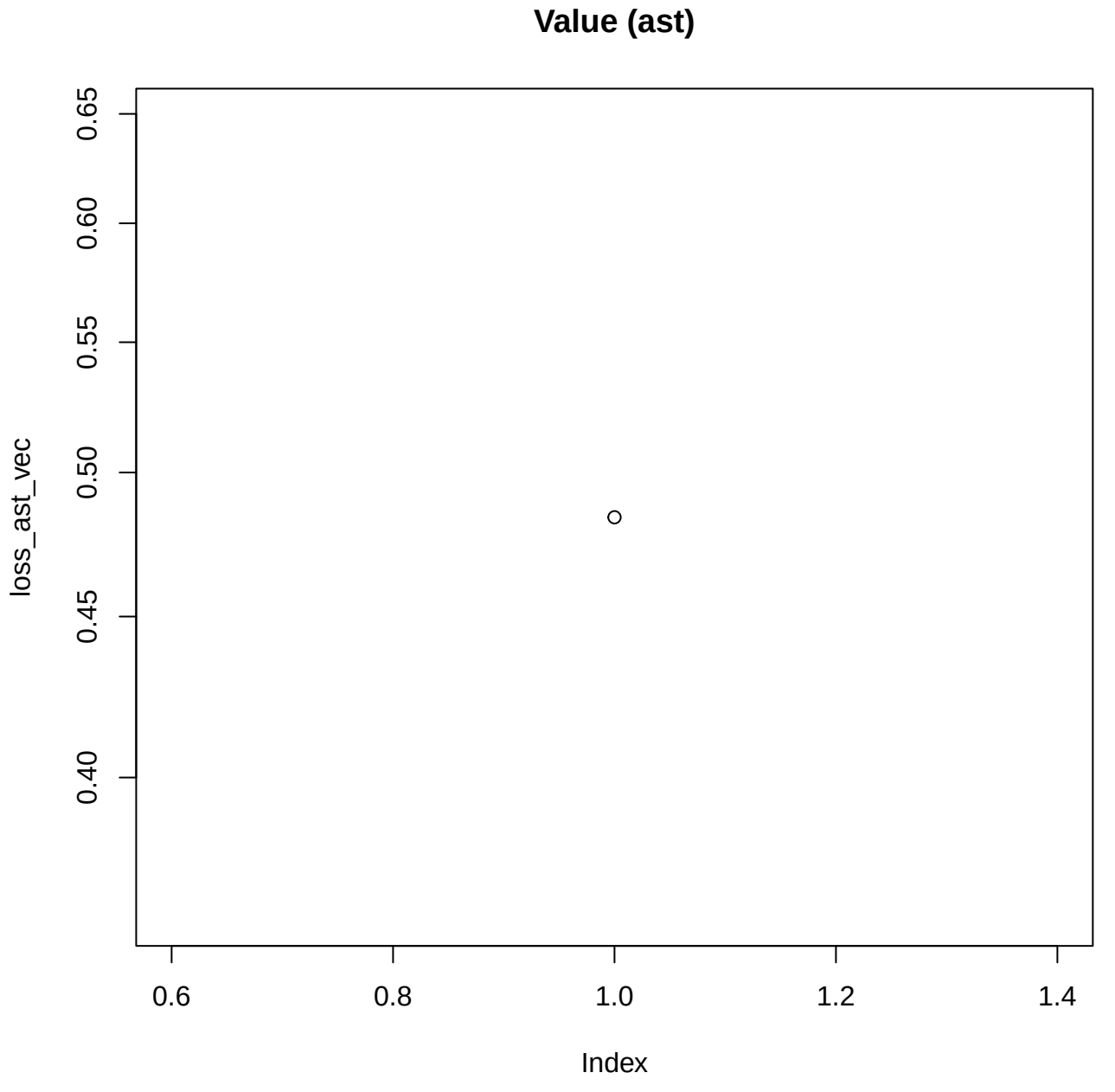
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.60
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
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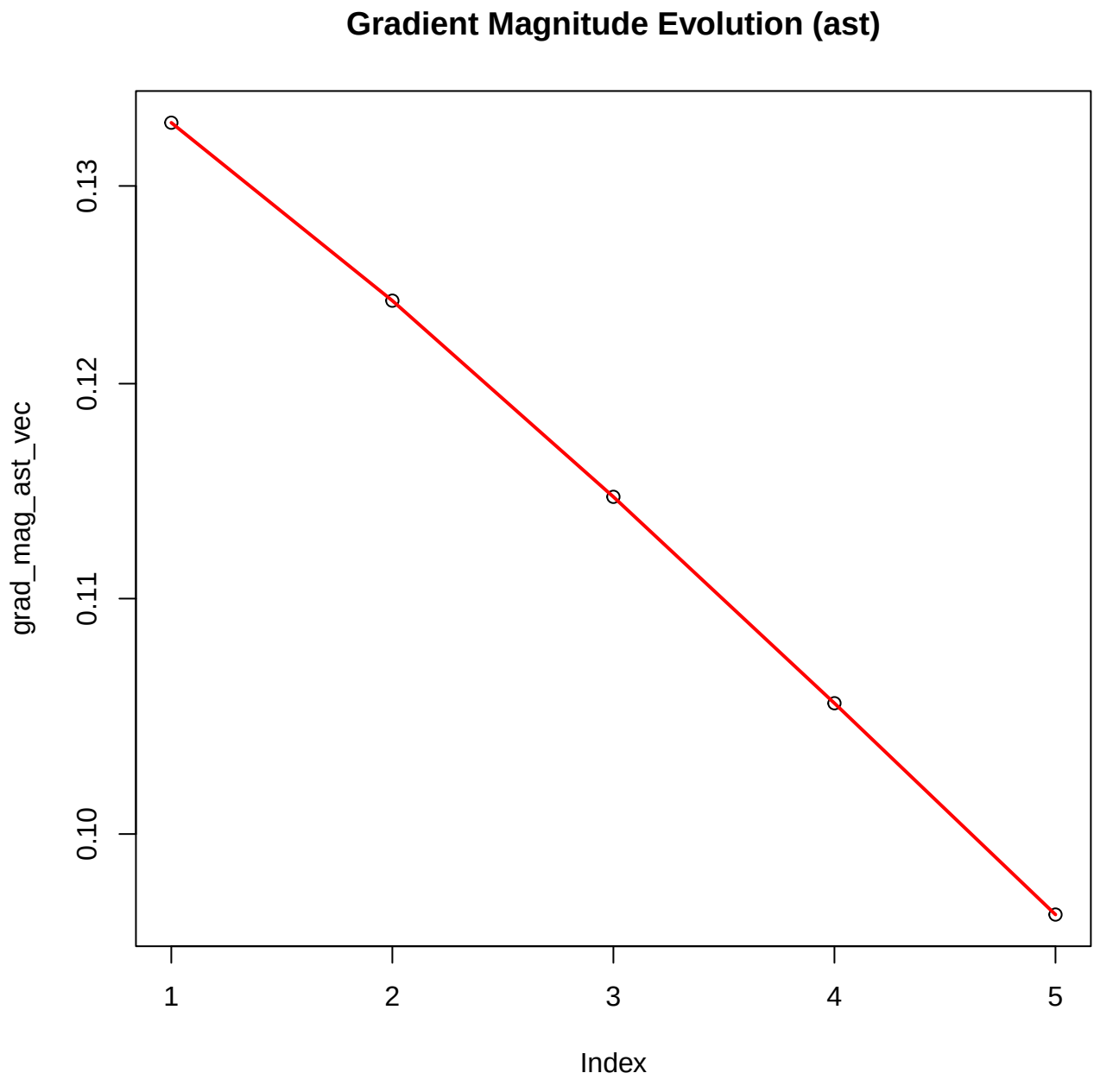
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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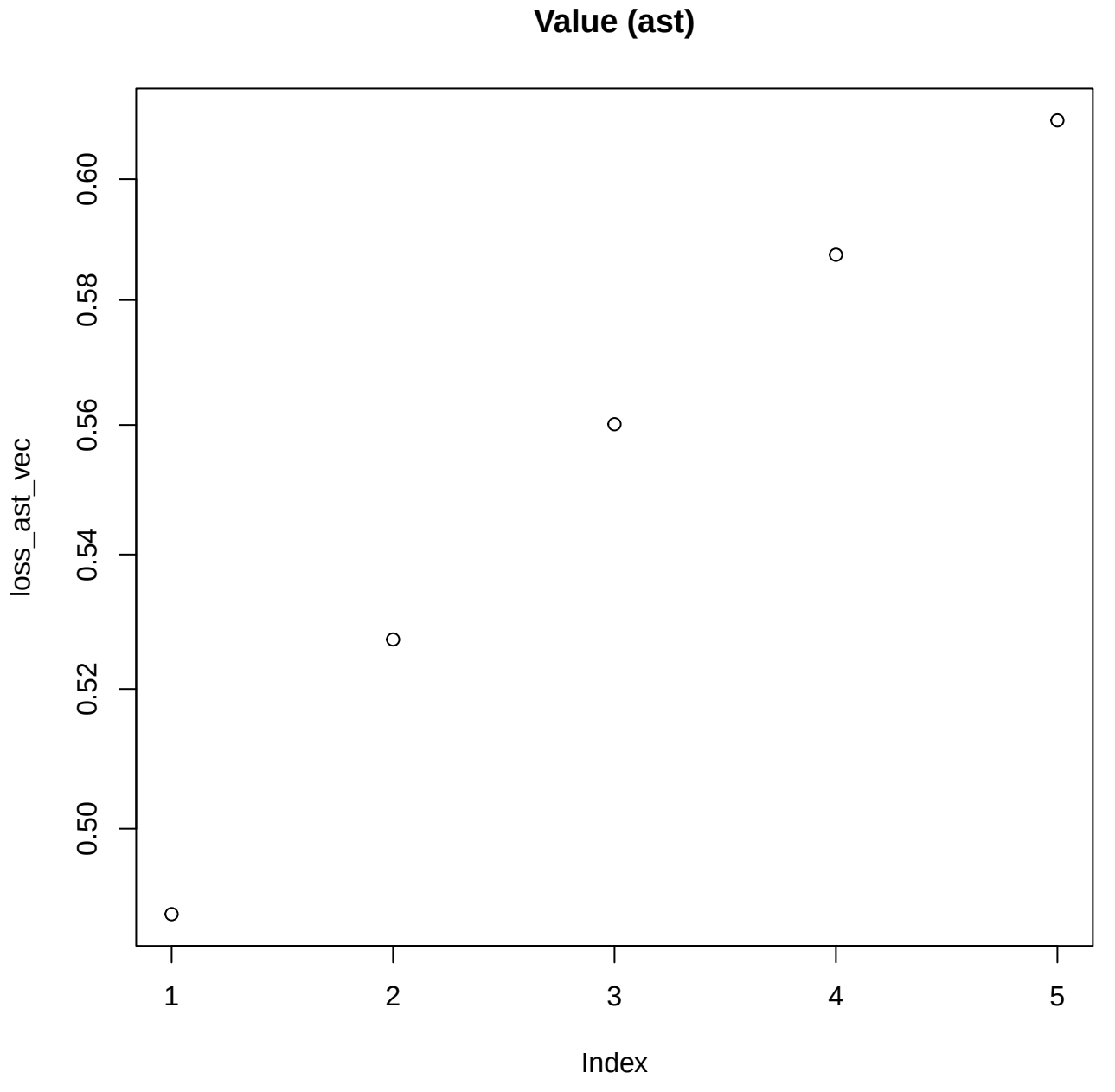
0.11
0.12
0.13
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
0.56
0.58
0.60
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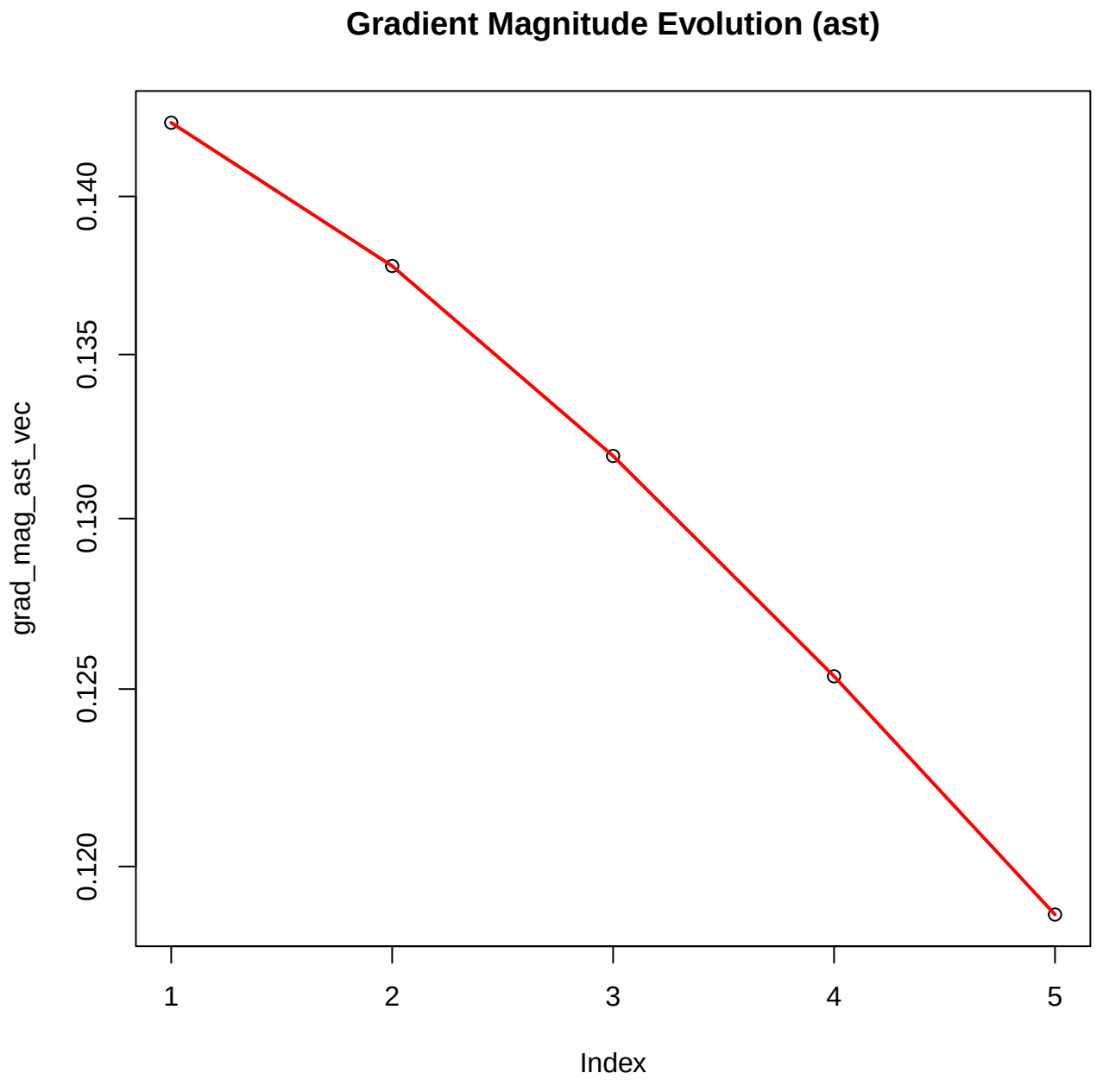
3
4
5
Index


Gradient Magnitude Evolution (ast)
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0.130
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0.140
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2
3
4
5
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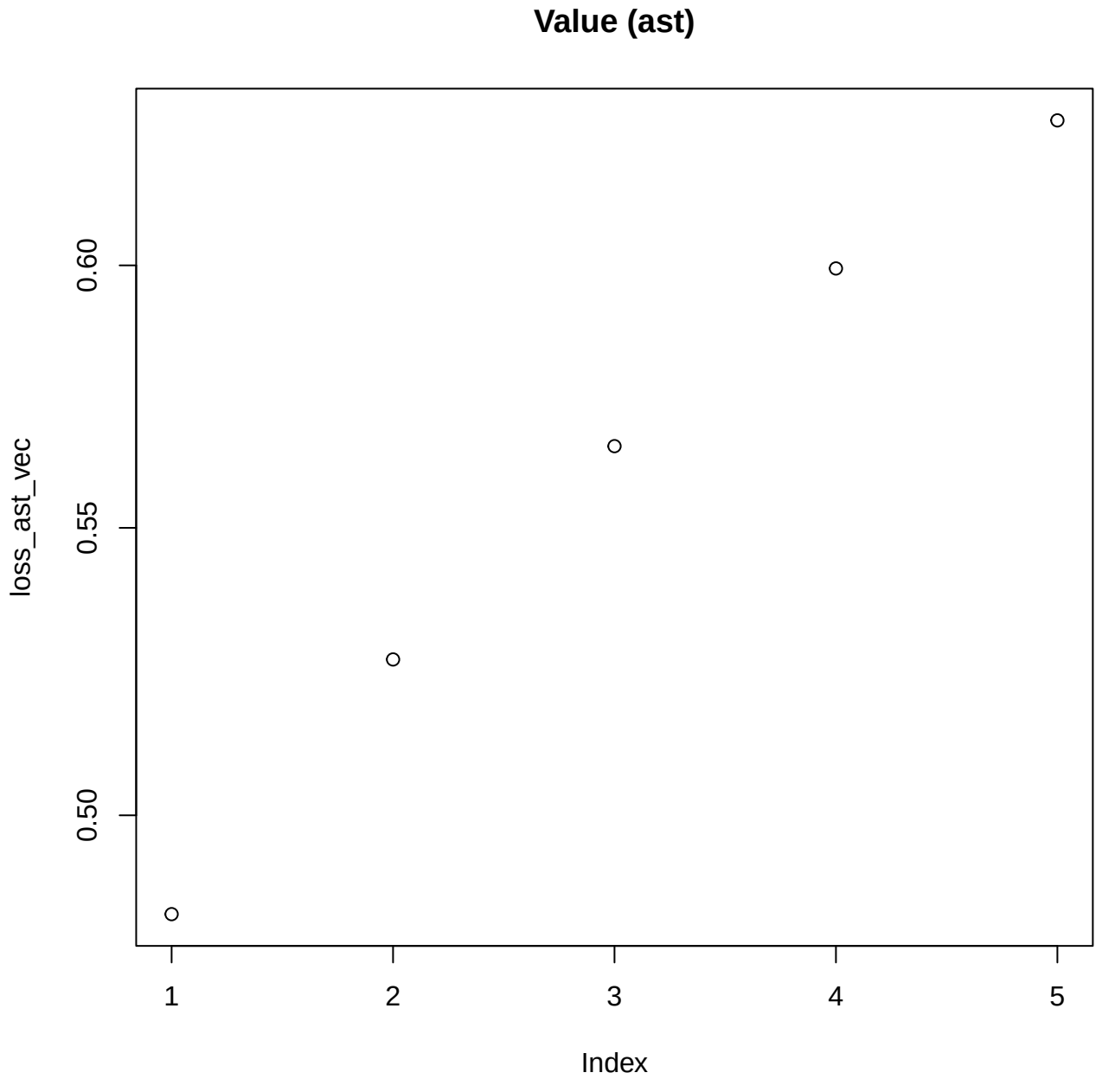
Value (ast)
loss_ast_vec
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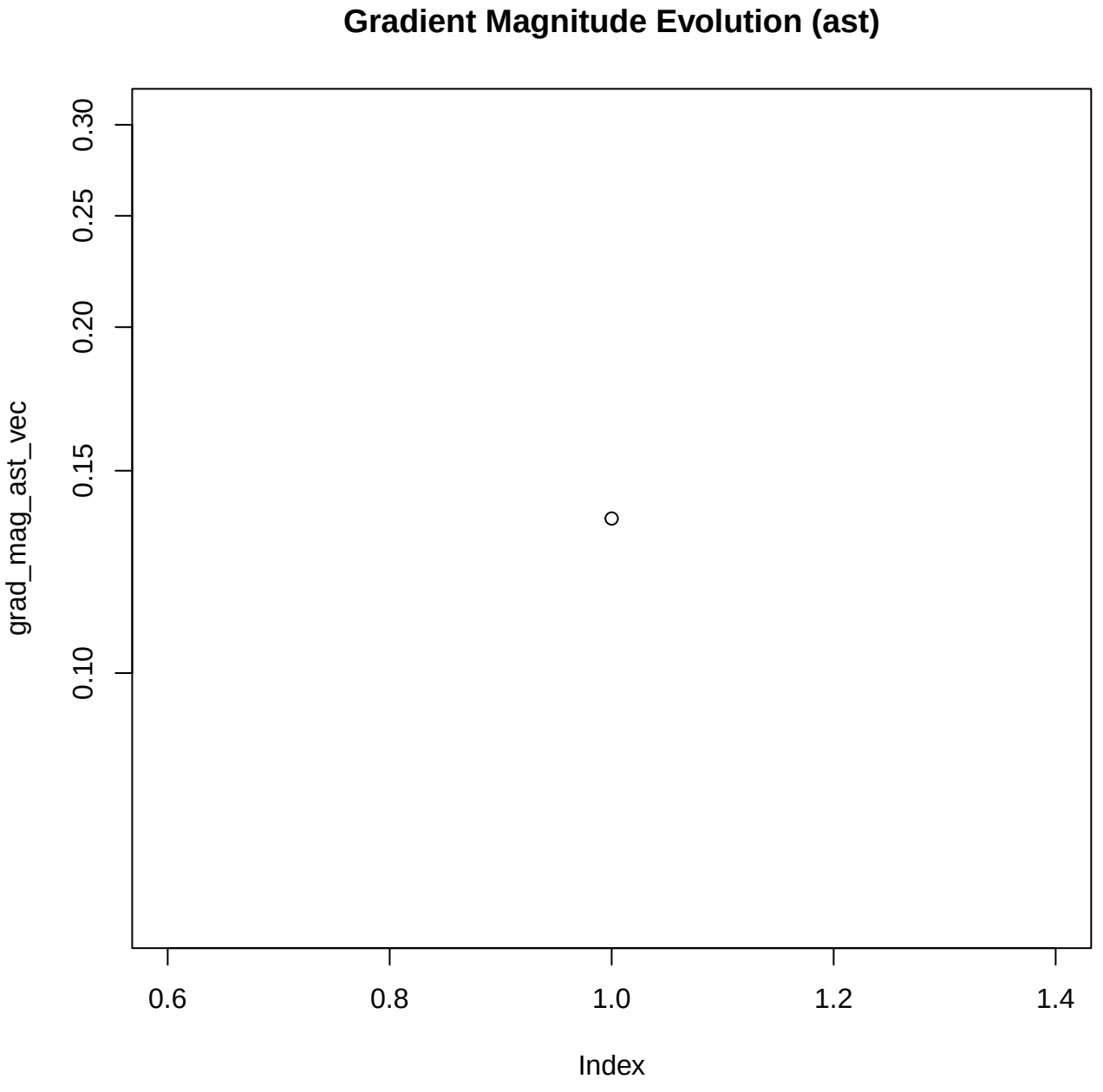
0.60
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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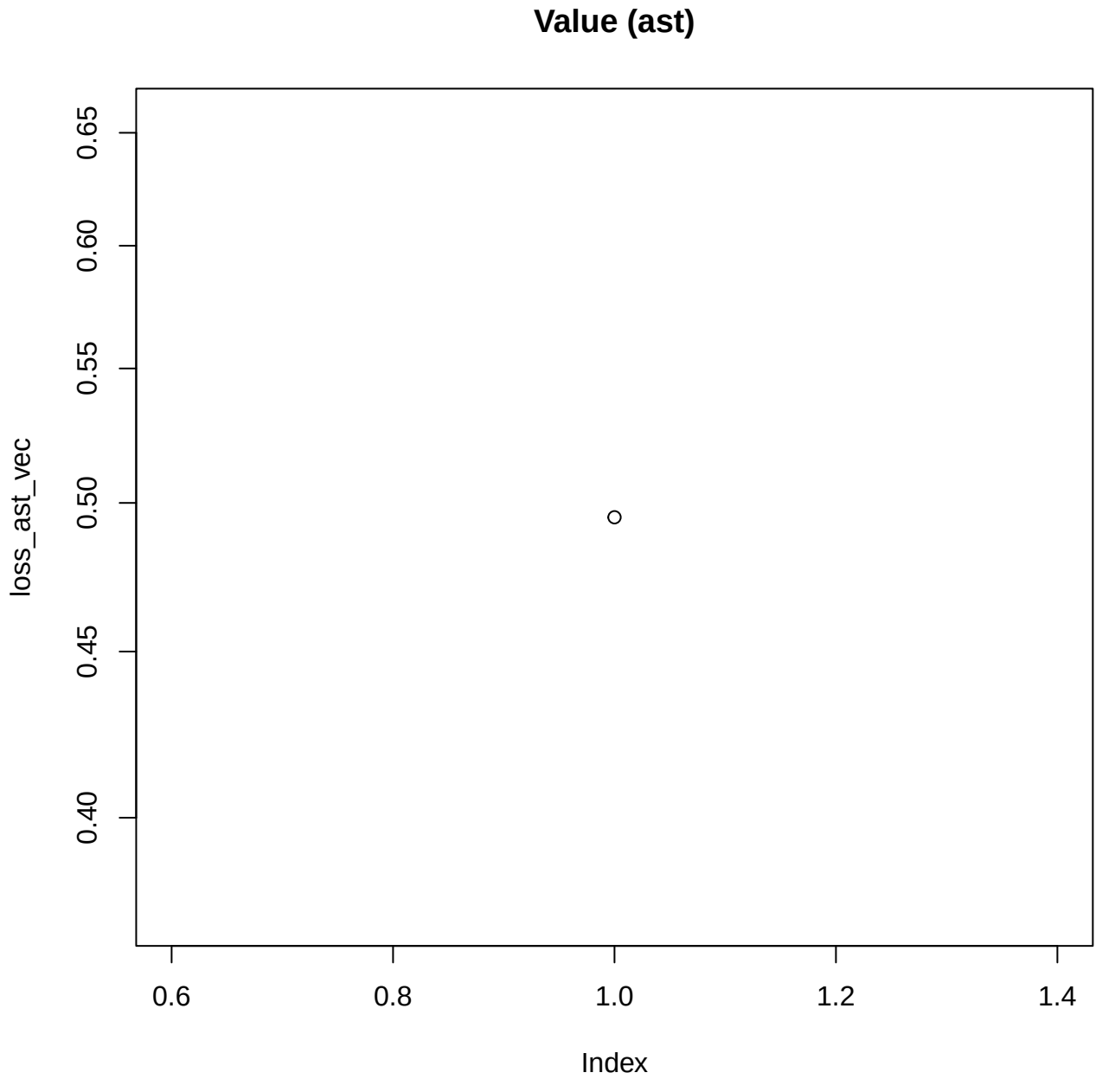
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.55
0.50
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0.40
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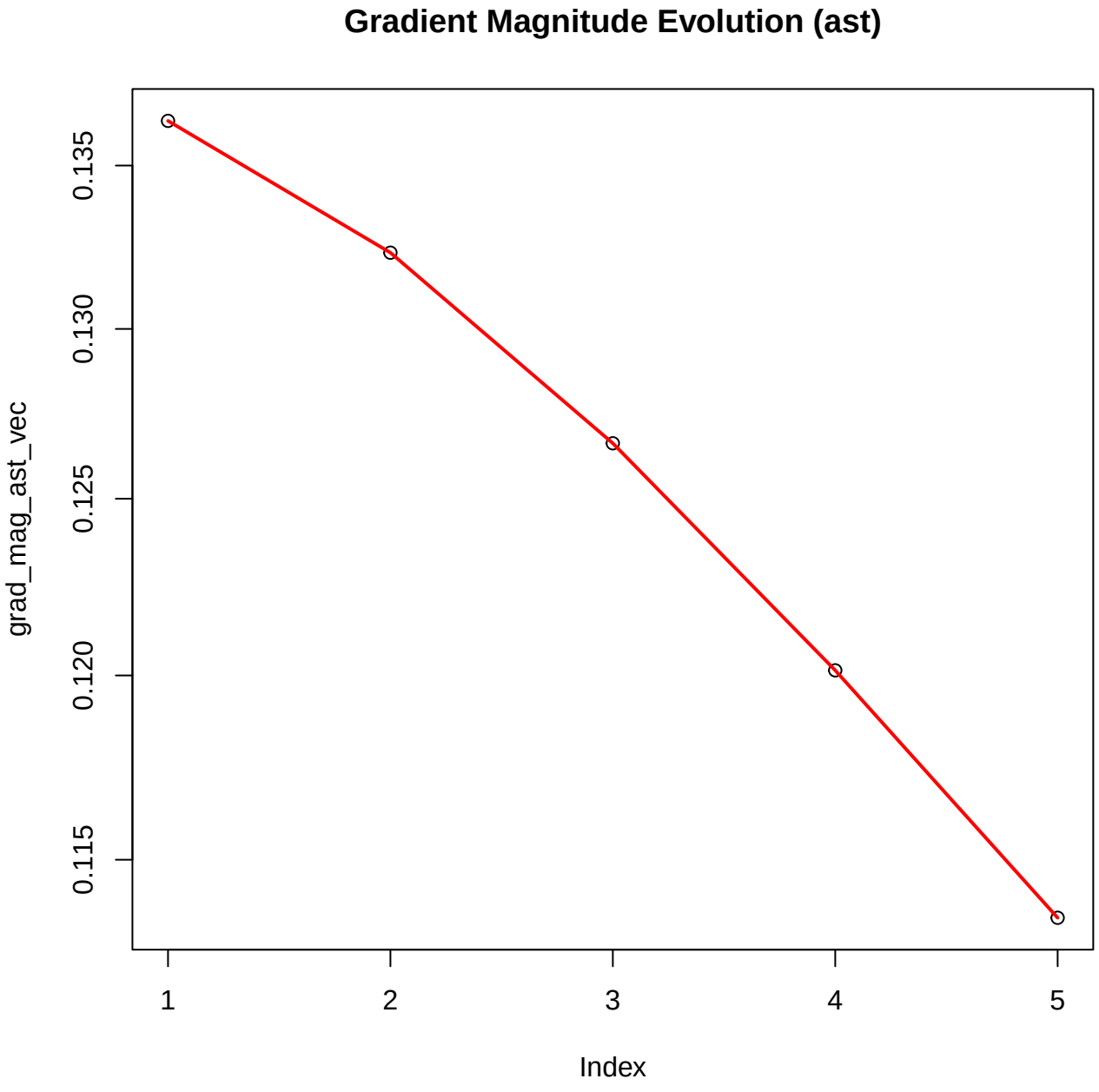
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
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0.120
0.125
0.130
0.135
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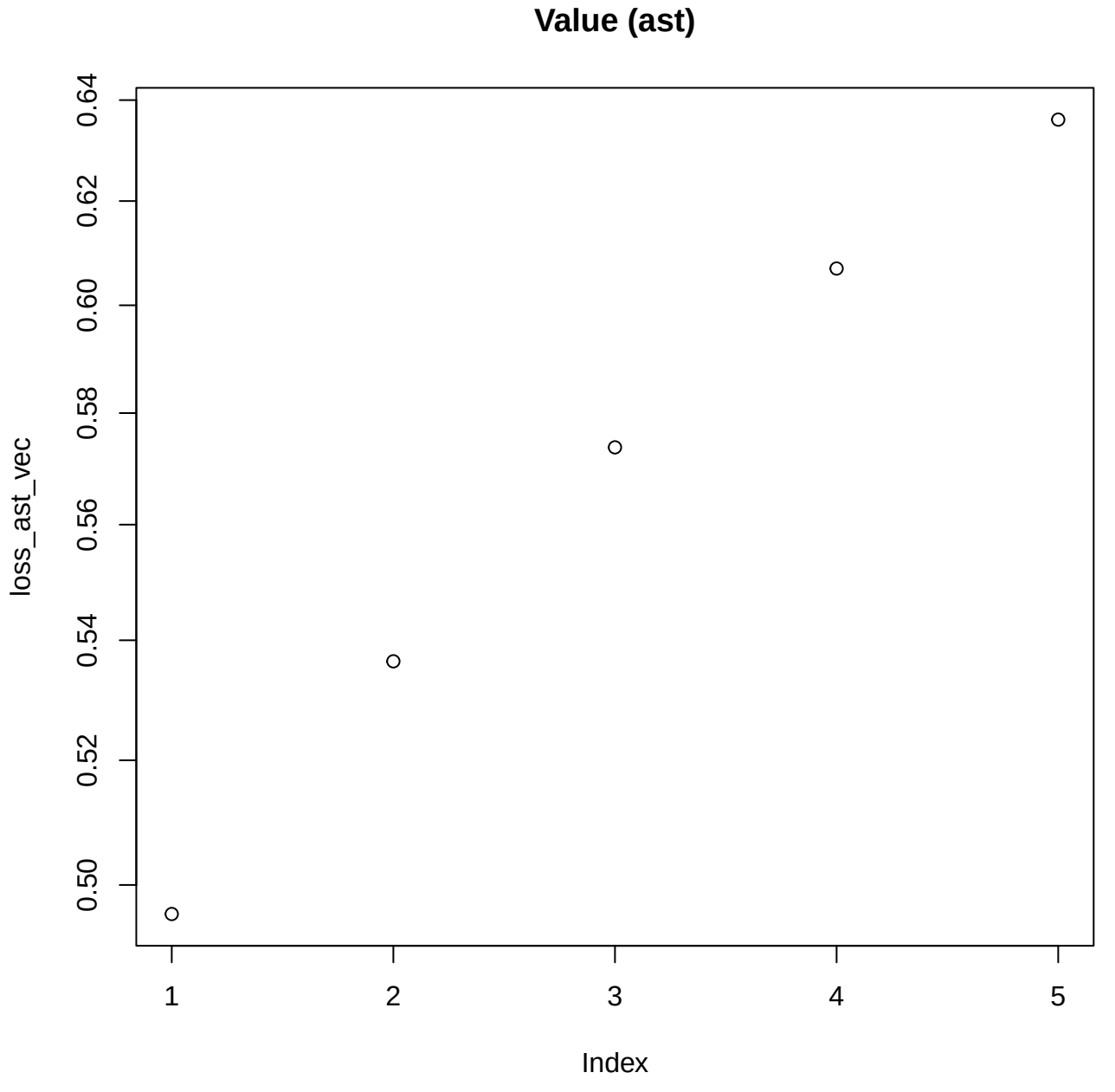
2
3
4
5
Index


Value (ast)
loss_ast_vec
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5
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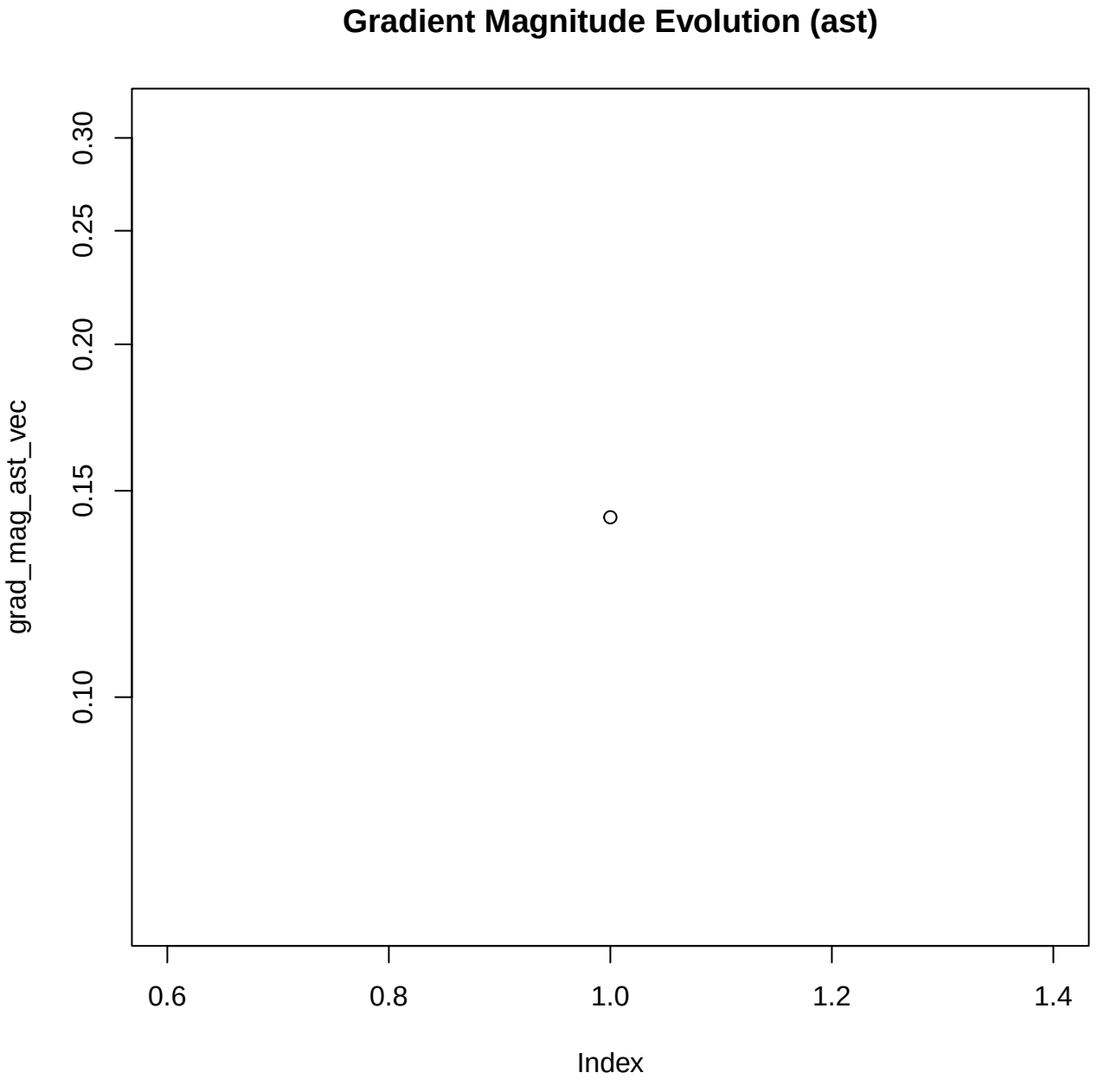
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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1.2
1.4
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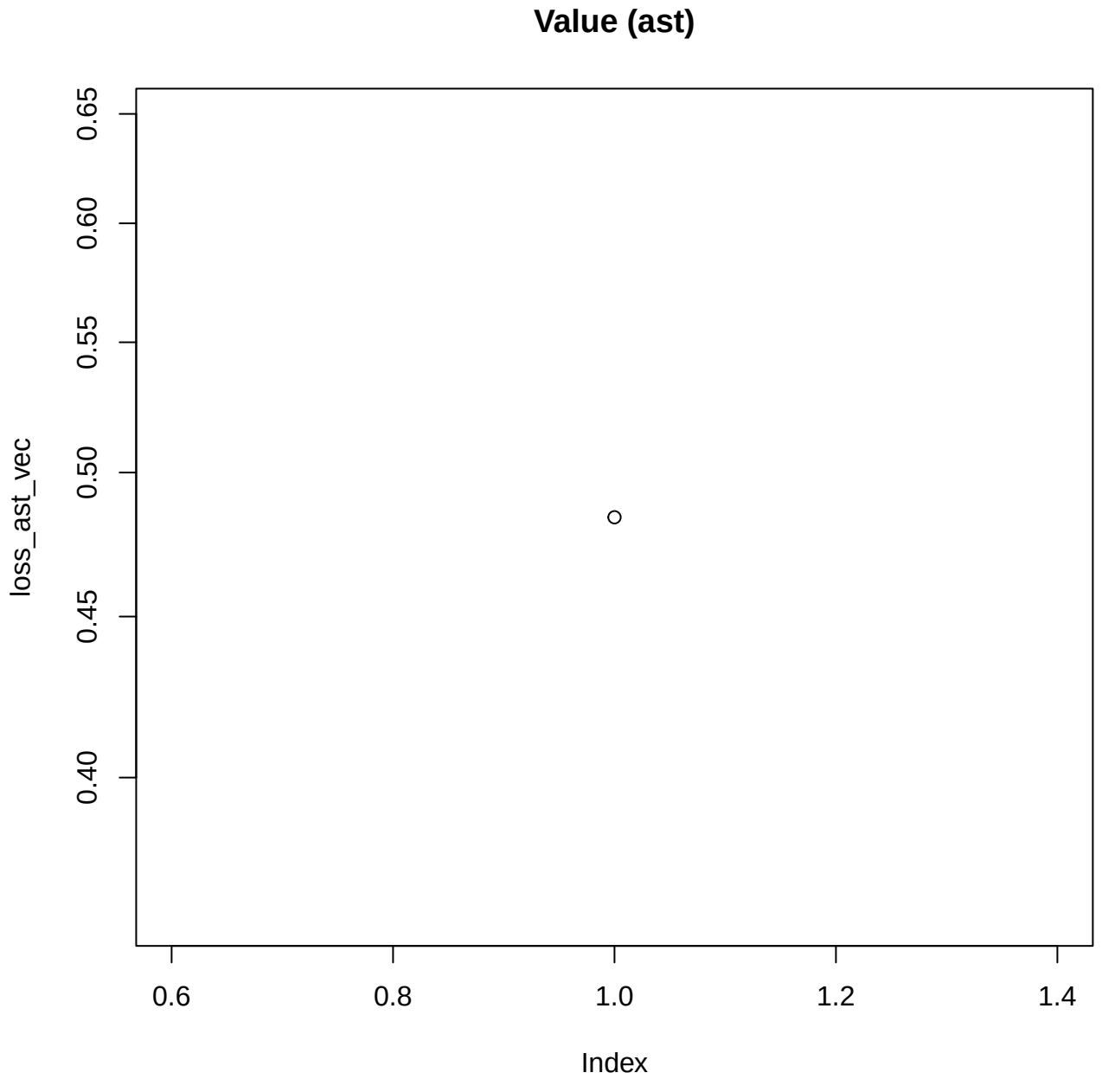
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loss_ast_vec
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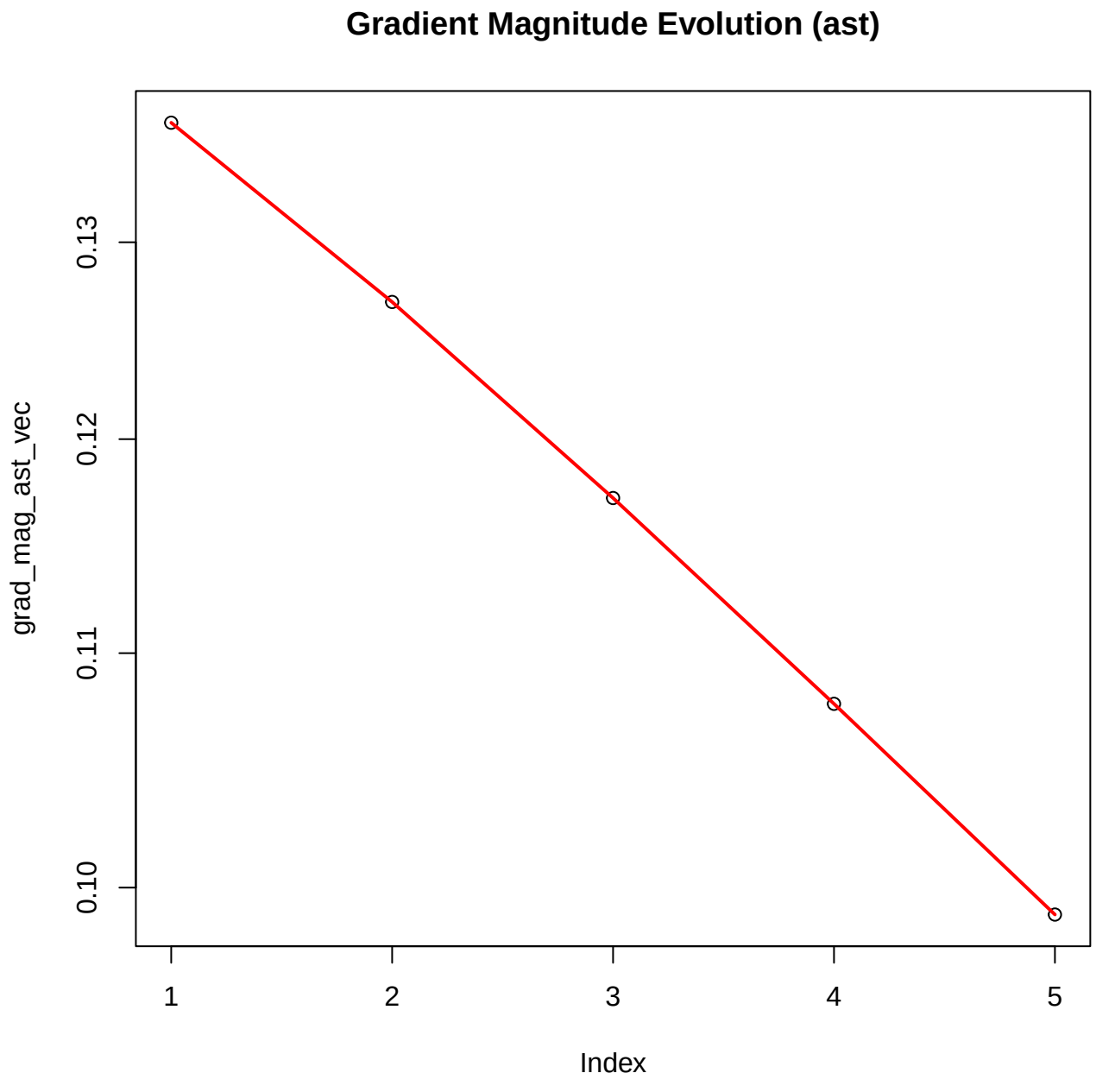
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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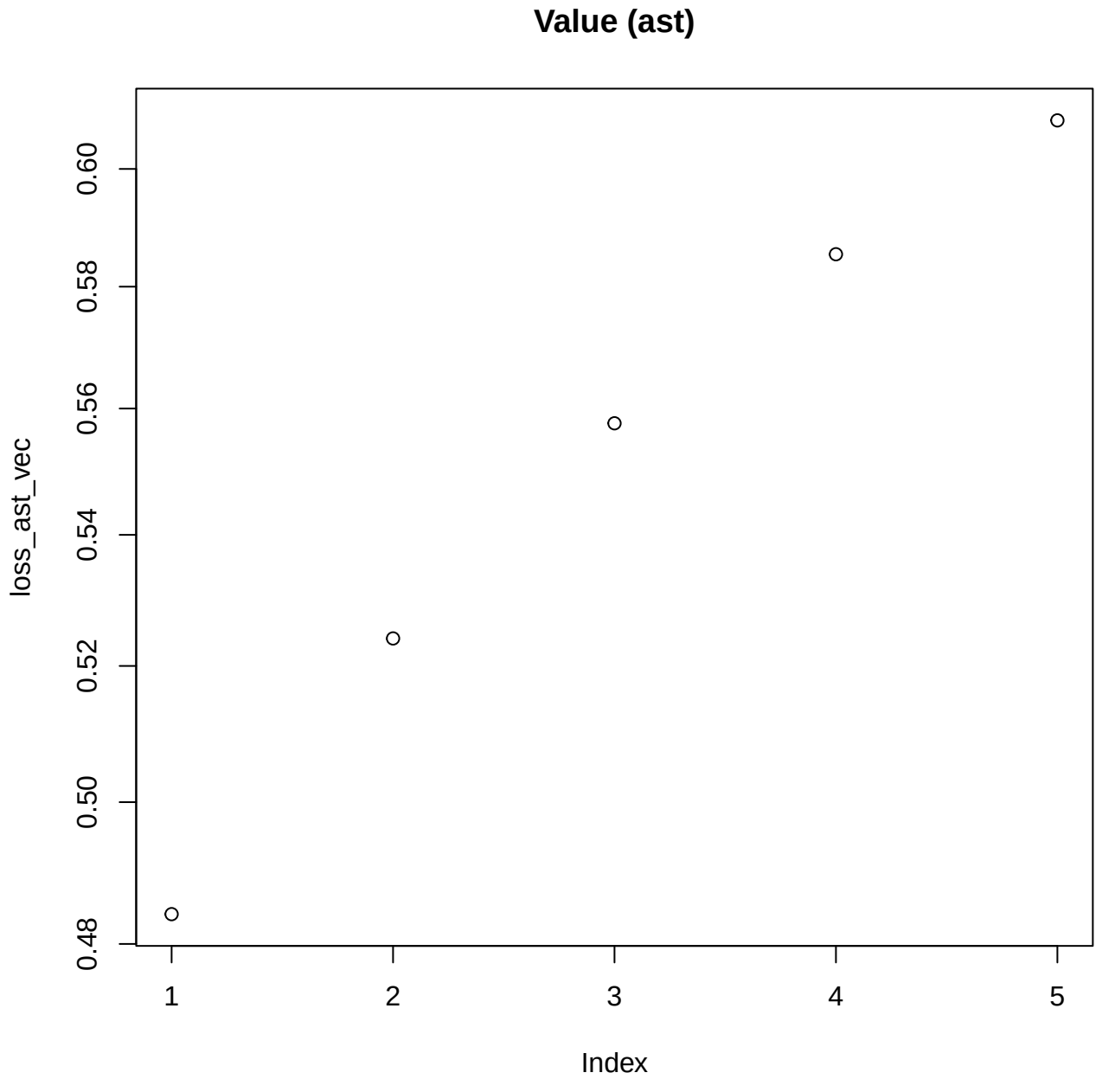
Value (ast)
loss_ast_vec
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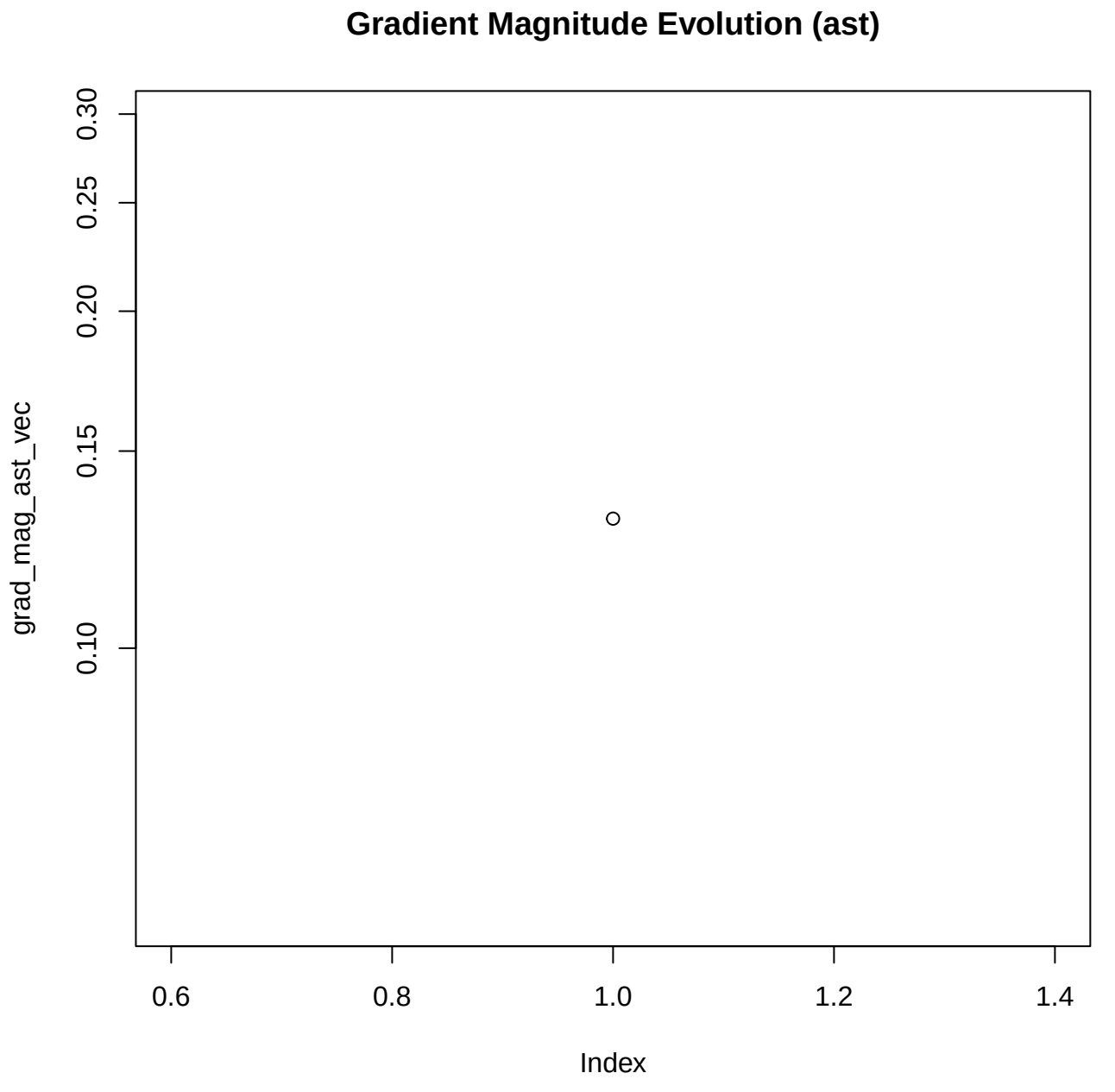
Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.20
0.25
0.30
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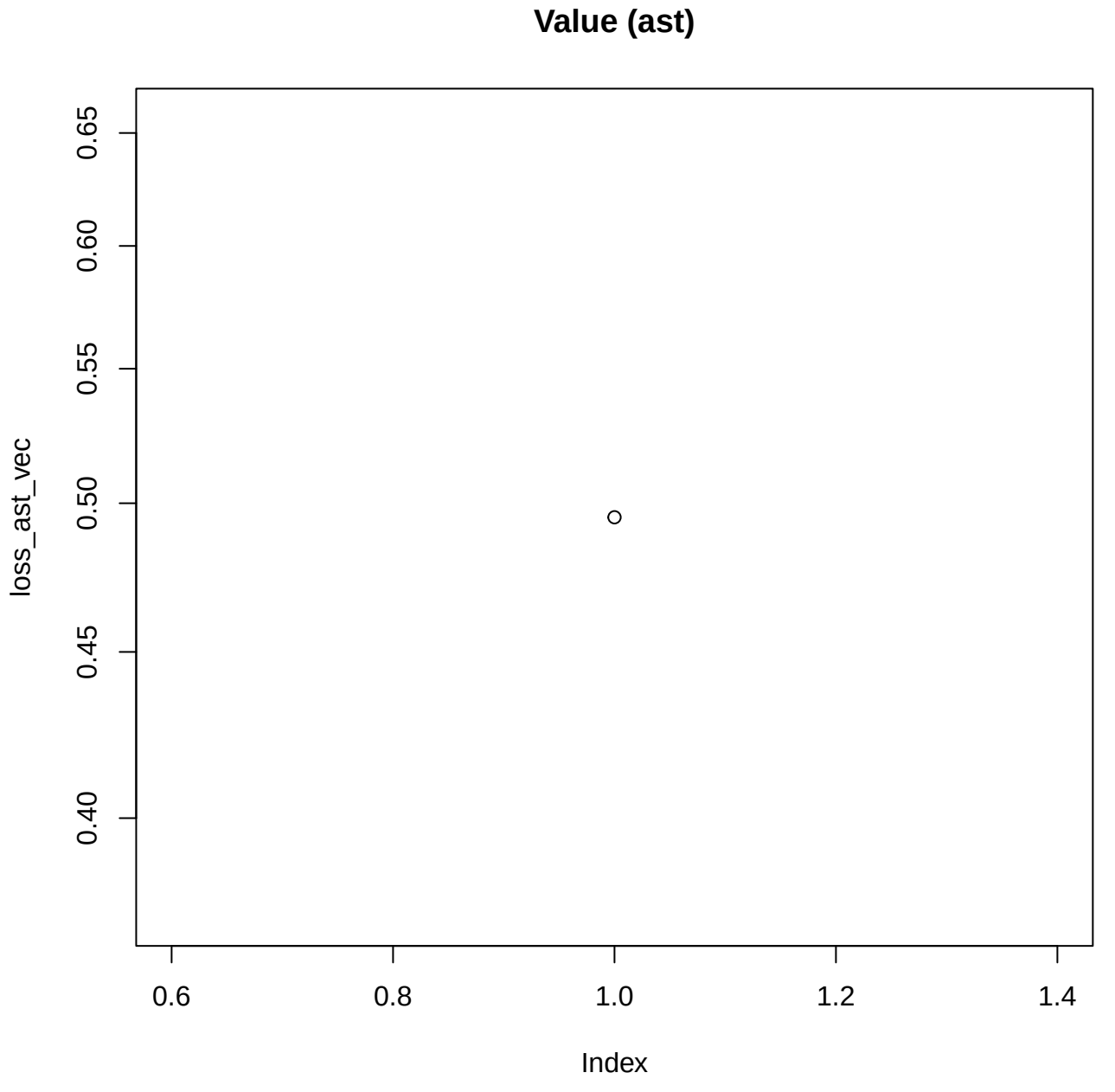
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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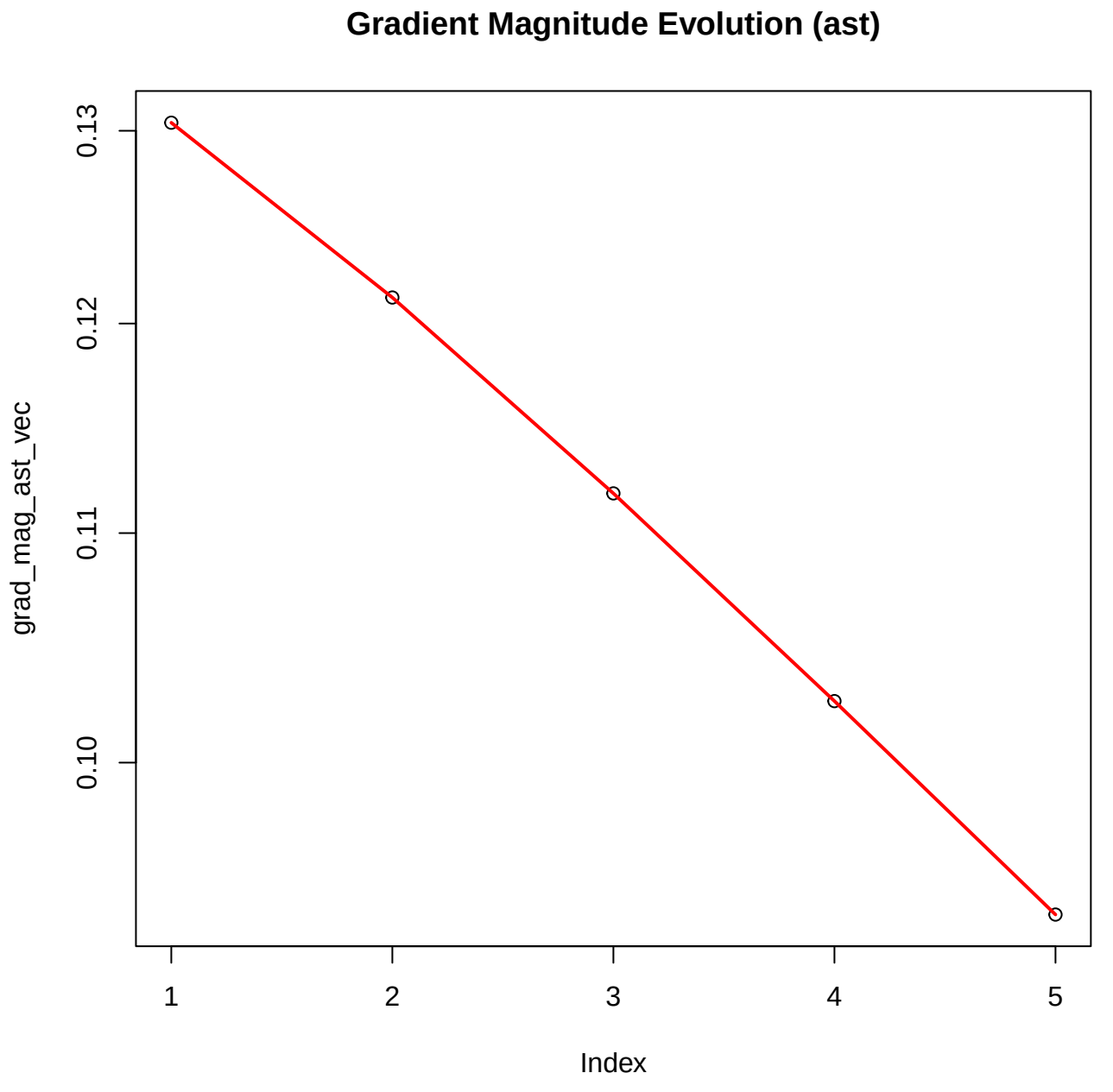
0.50
0.55
0.60
0.65
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.13
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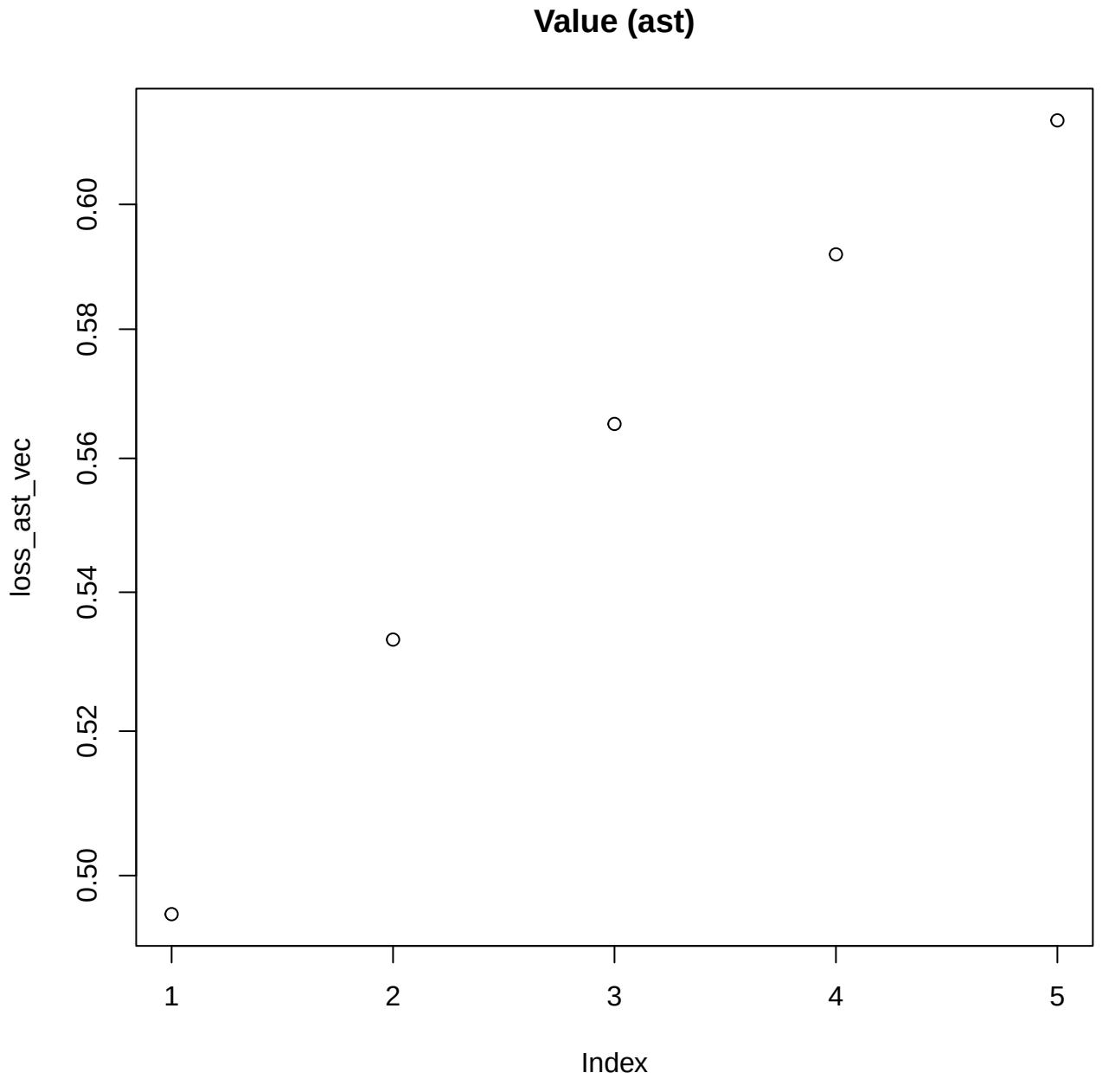
2
3
4
5
Index


Value (ast)
loss_ast_vec
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0.52
0.54
0.56
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0.60
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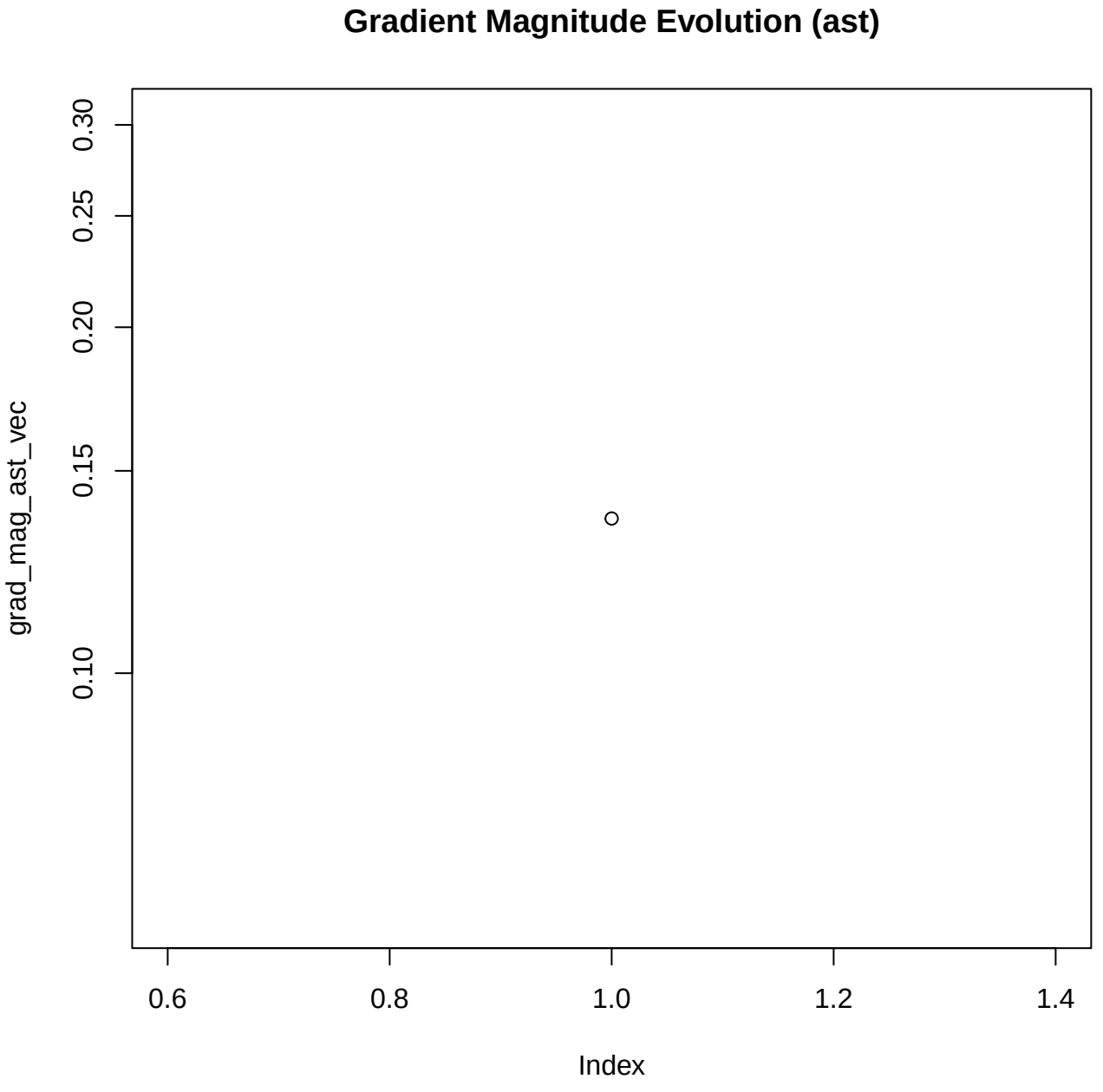
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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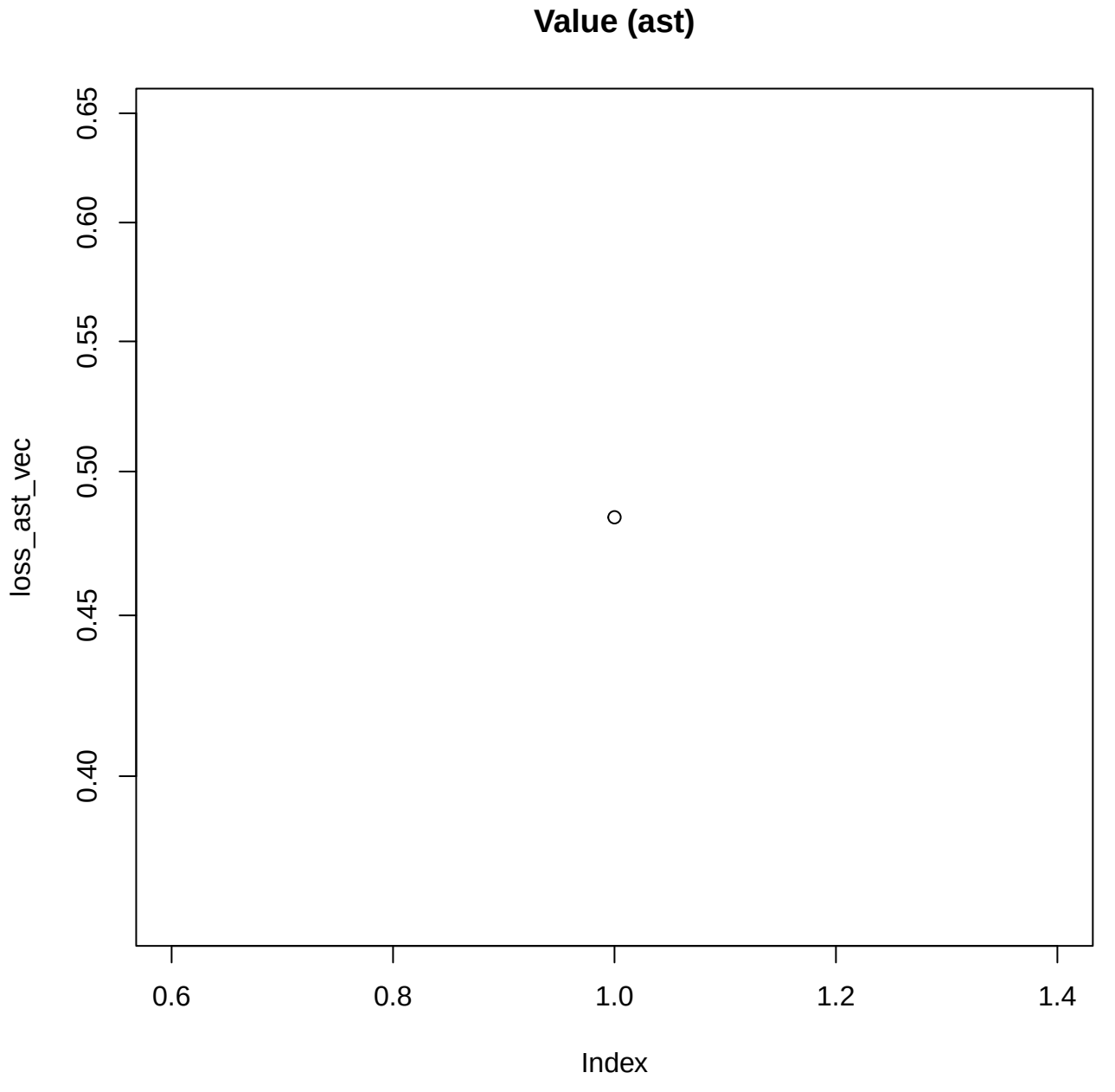
0.20
0.25
0.30
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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0.50
0.45
0.40
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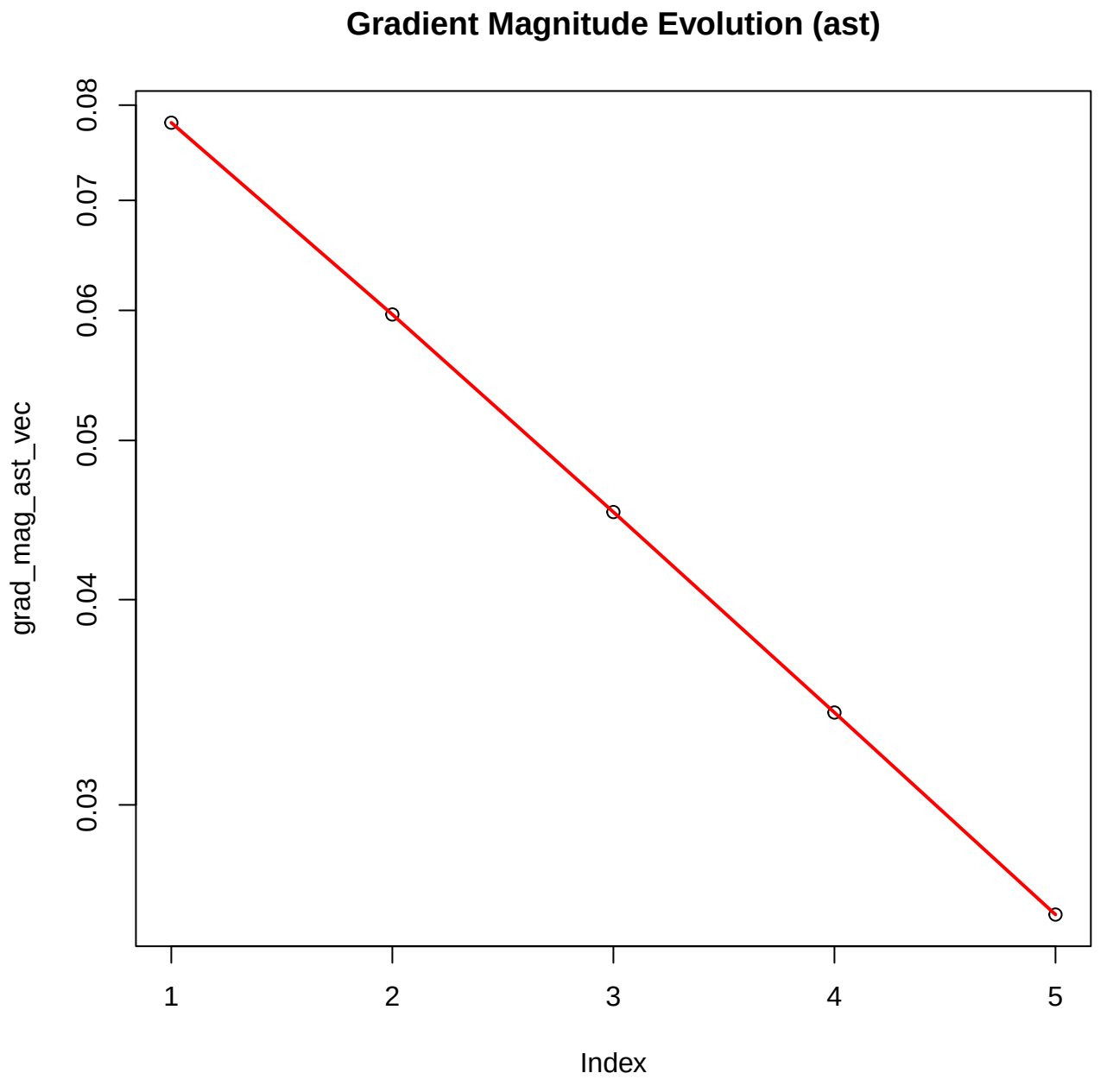
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
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0.03
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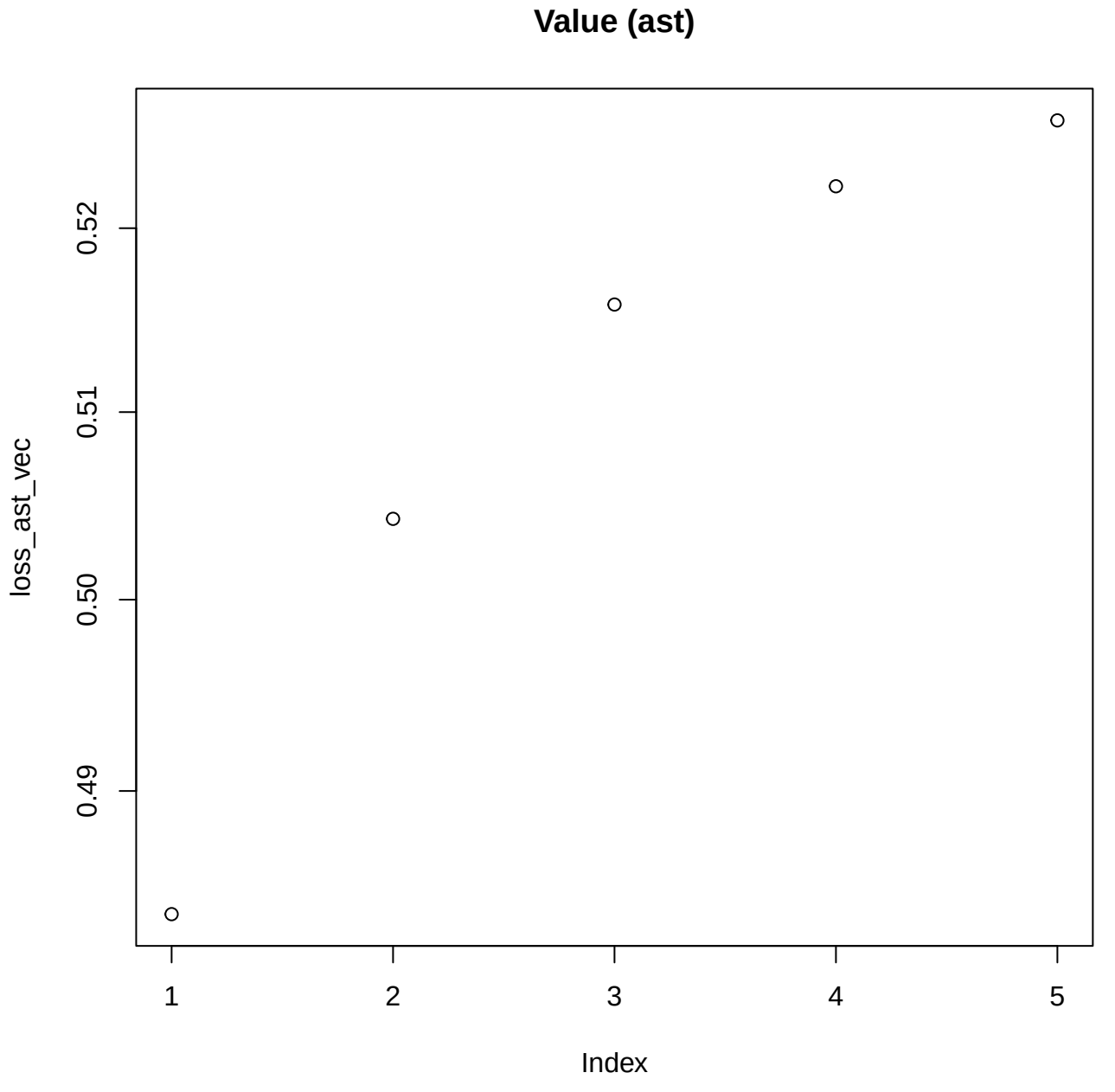
5
Index


Value (ast)
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0.52
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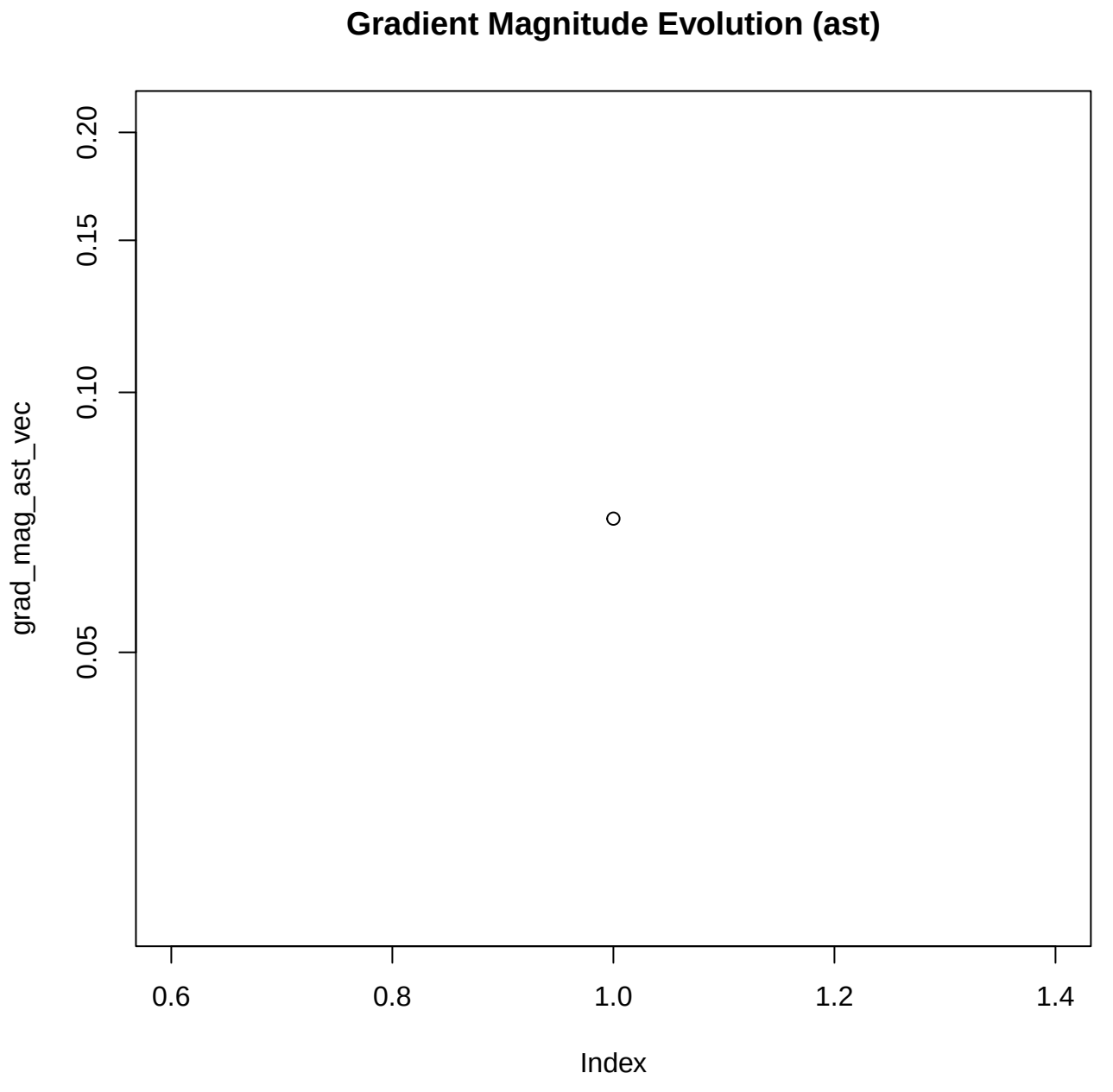
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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0.10
0.15
0.20
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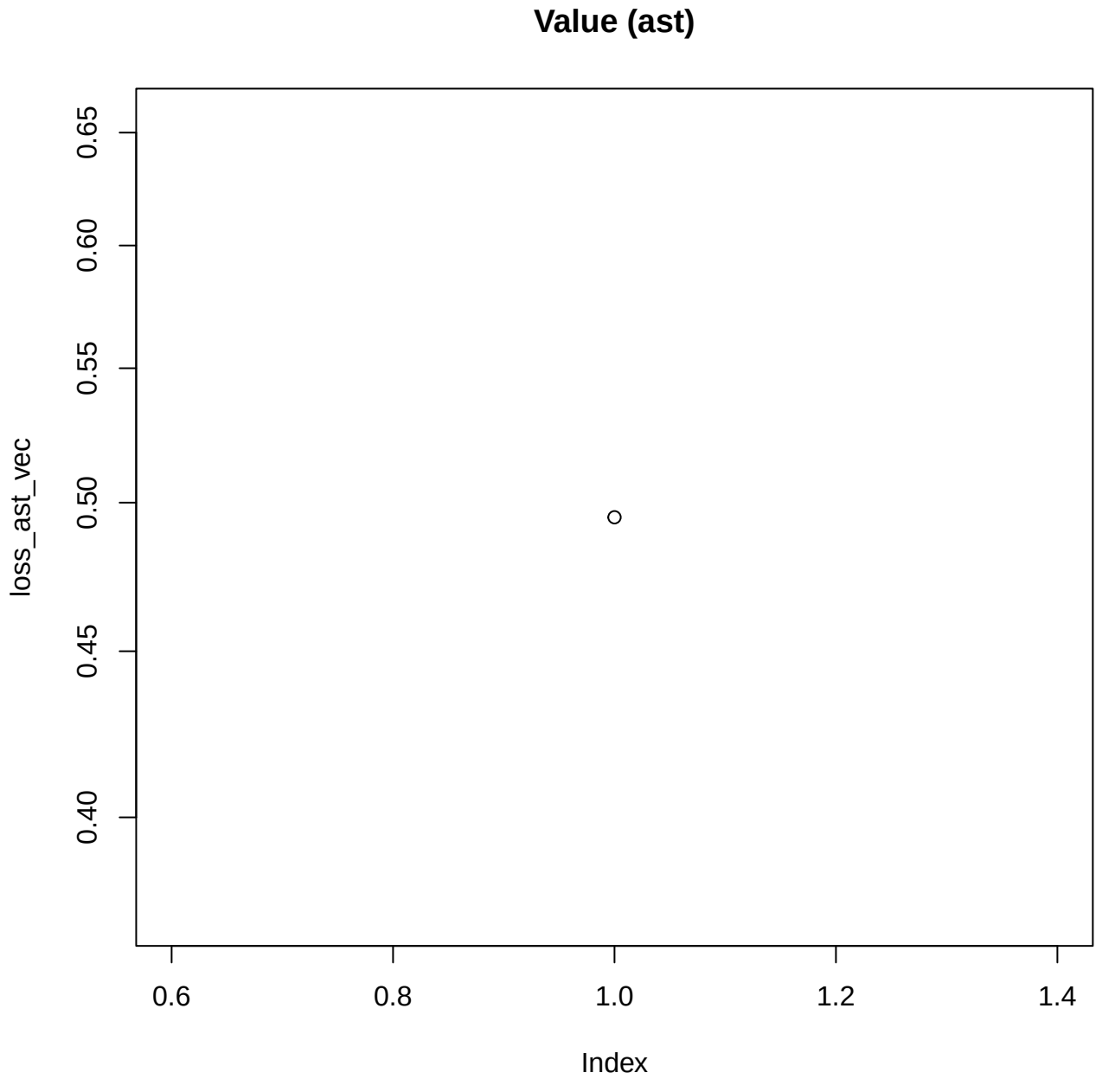
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
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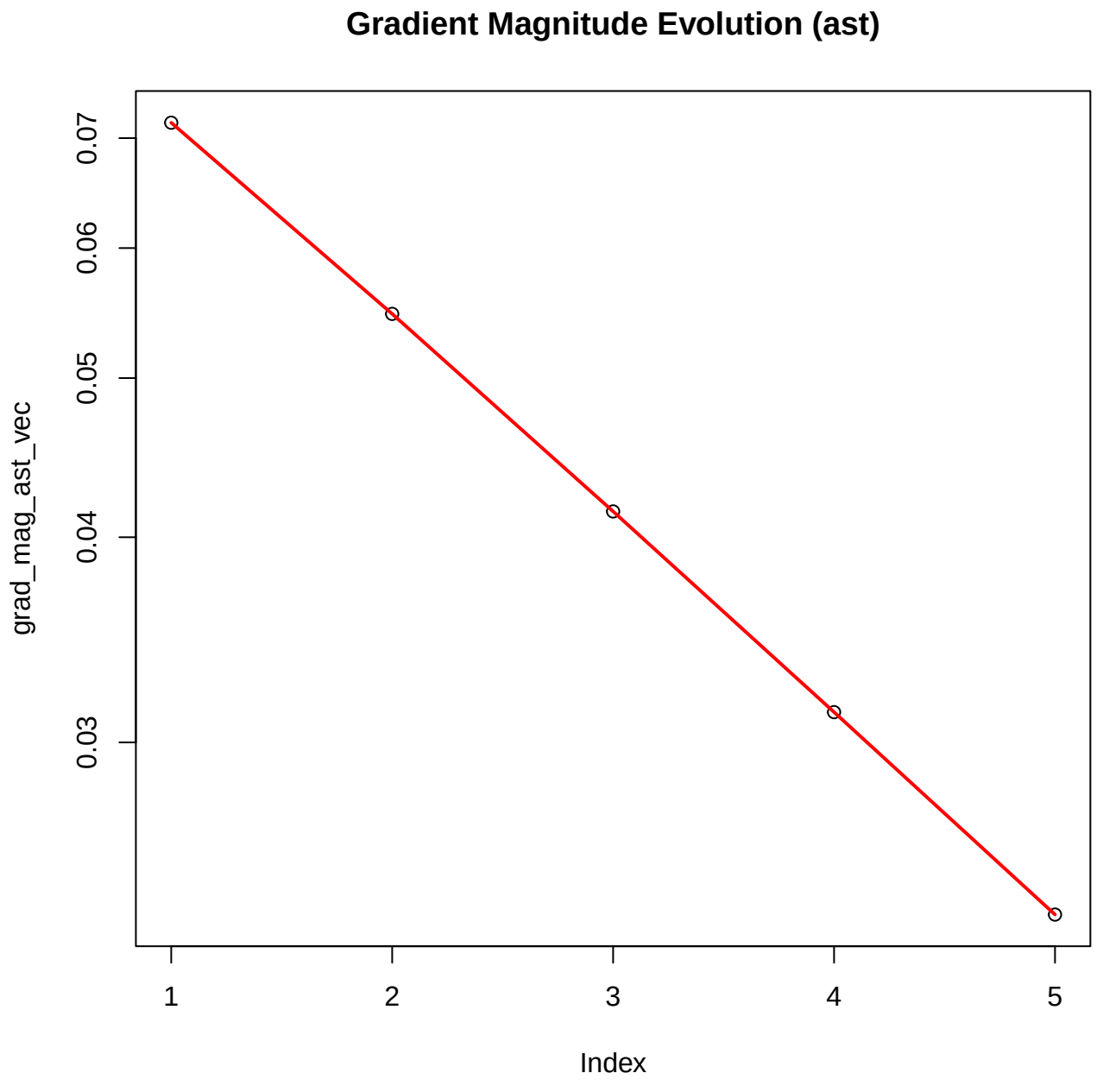
0.55
0.50
0.45
0.40
0.6
0.8
1.0
1.2
1.4
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
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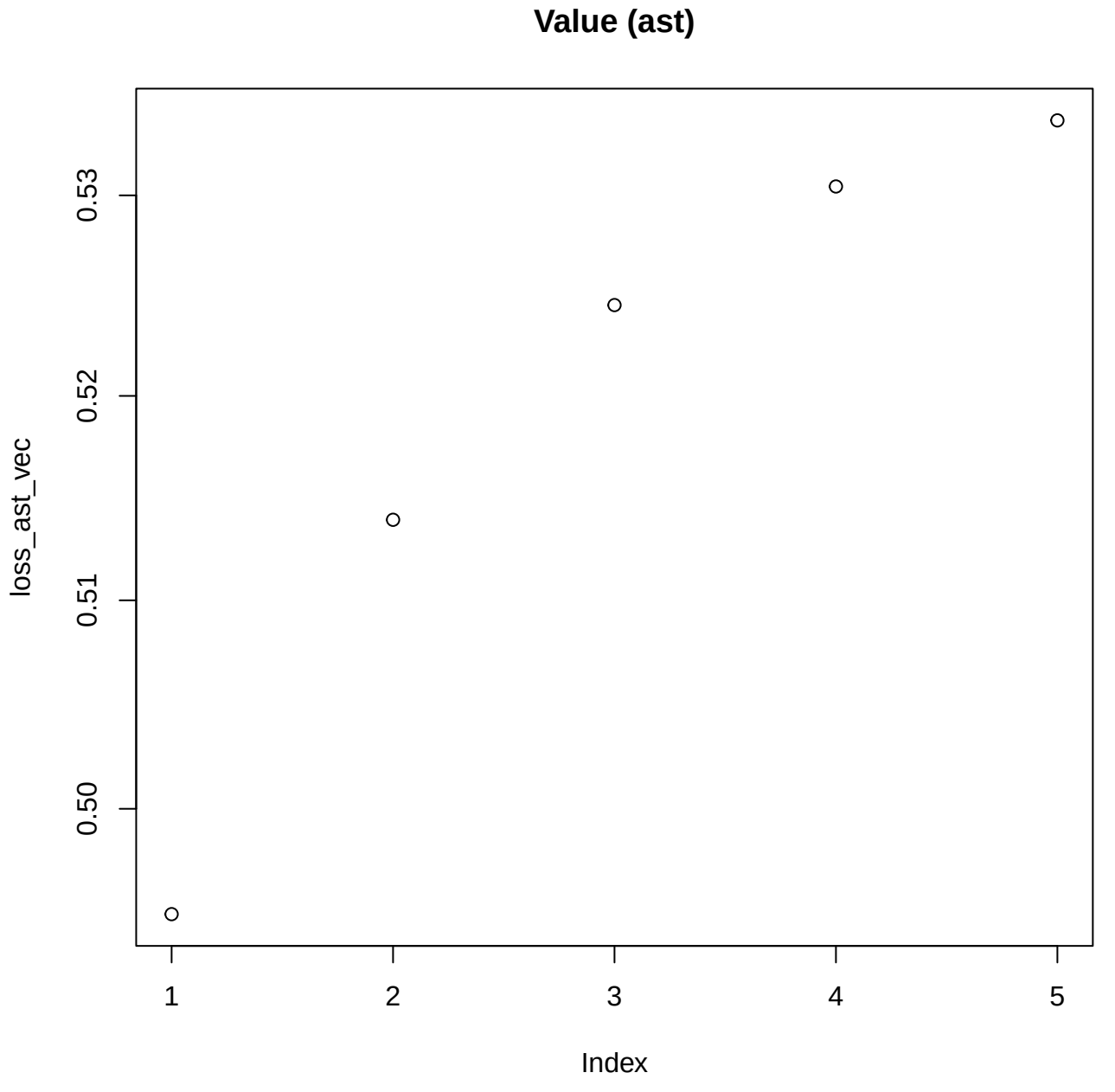
0.04
0.05
0.06
0.07
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.50
0.51
0.52
0.53
1
2
3
4
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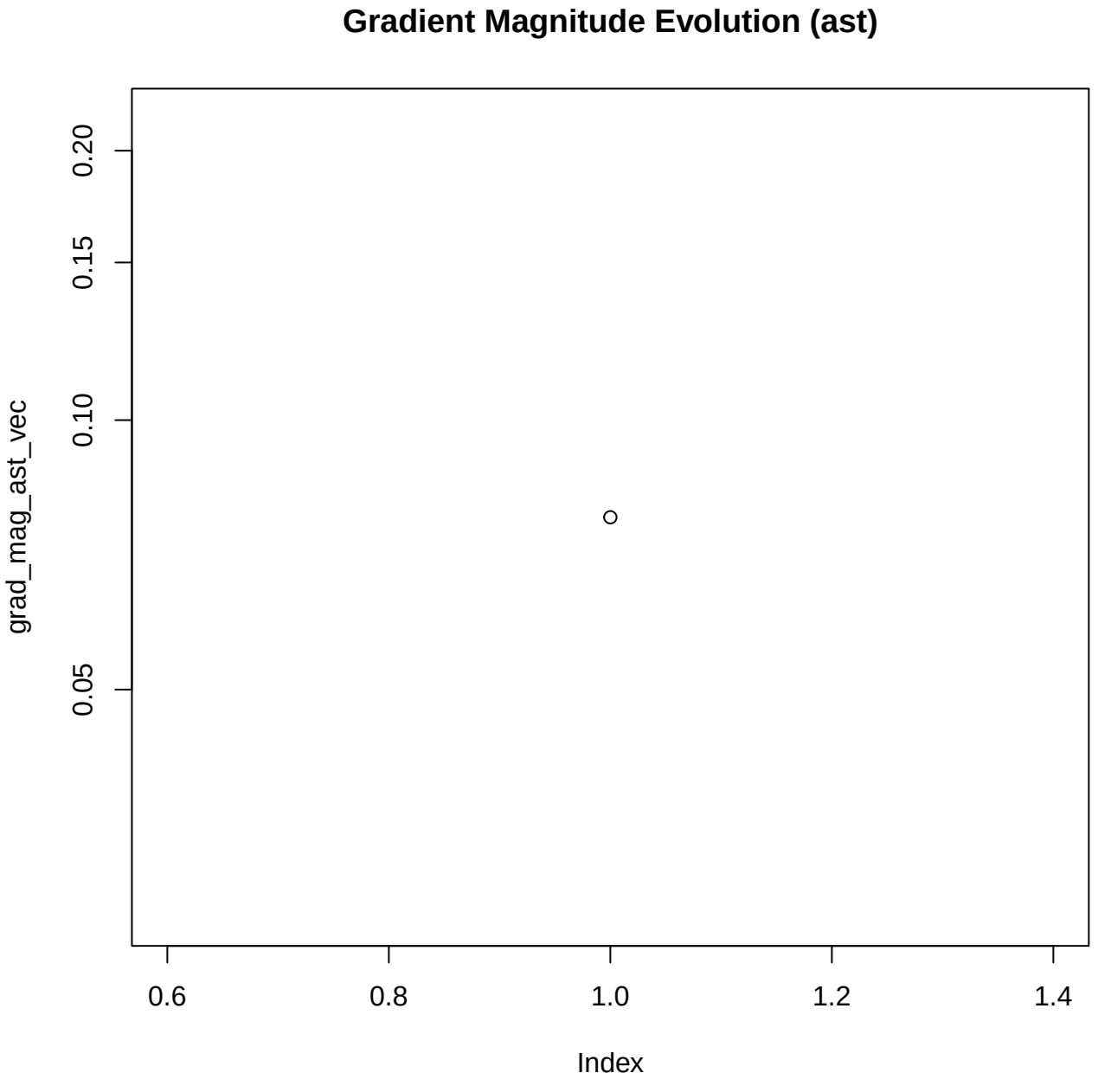
Gradient Magnitude Evolution (ast)
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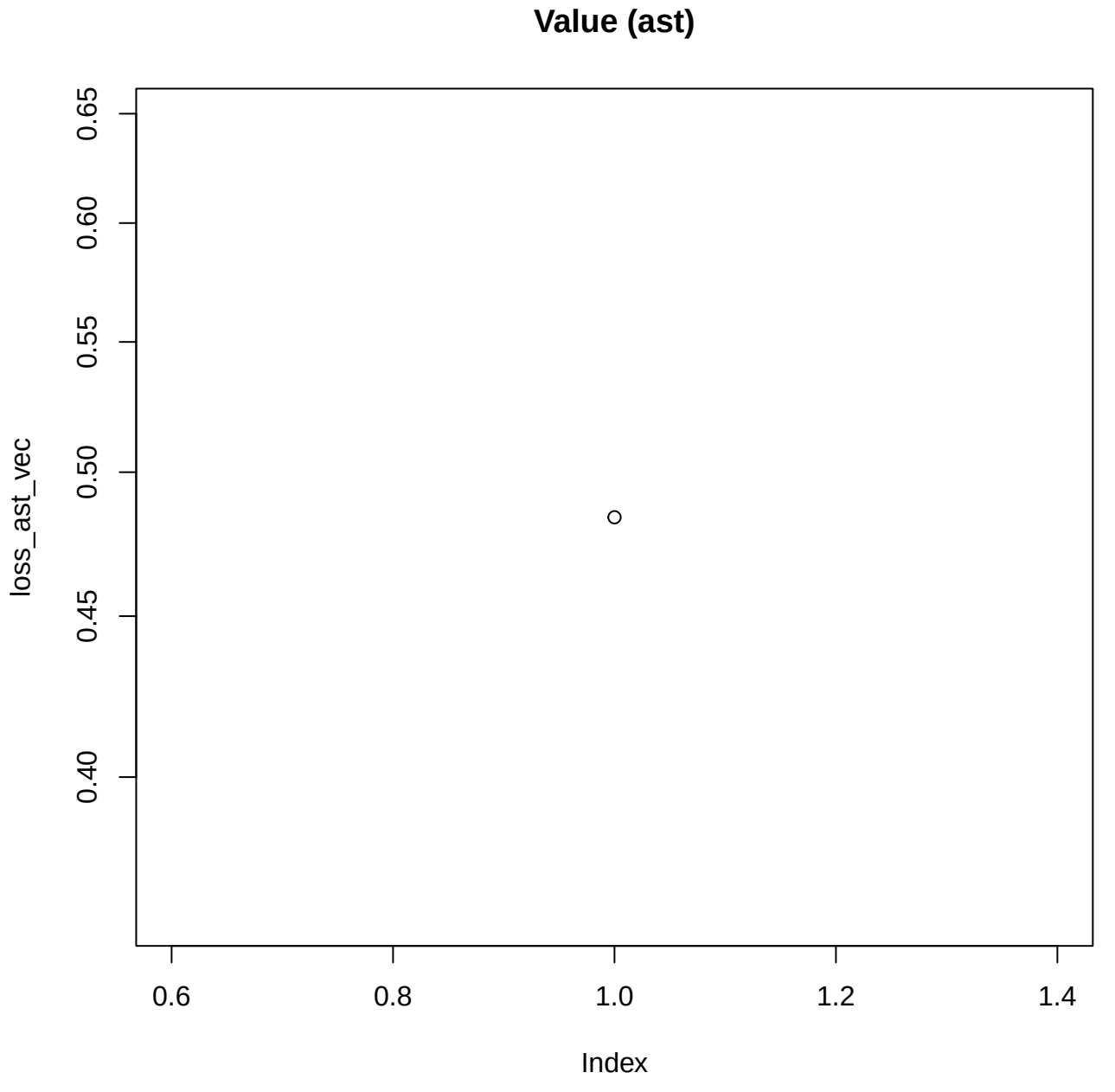
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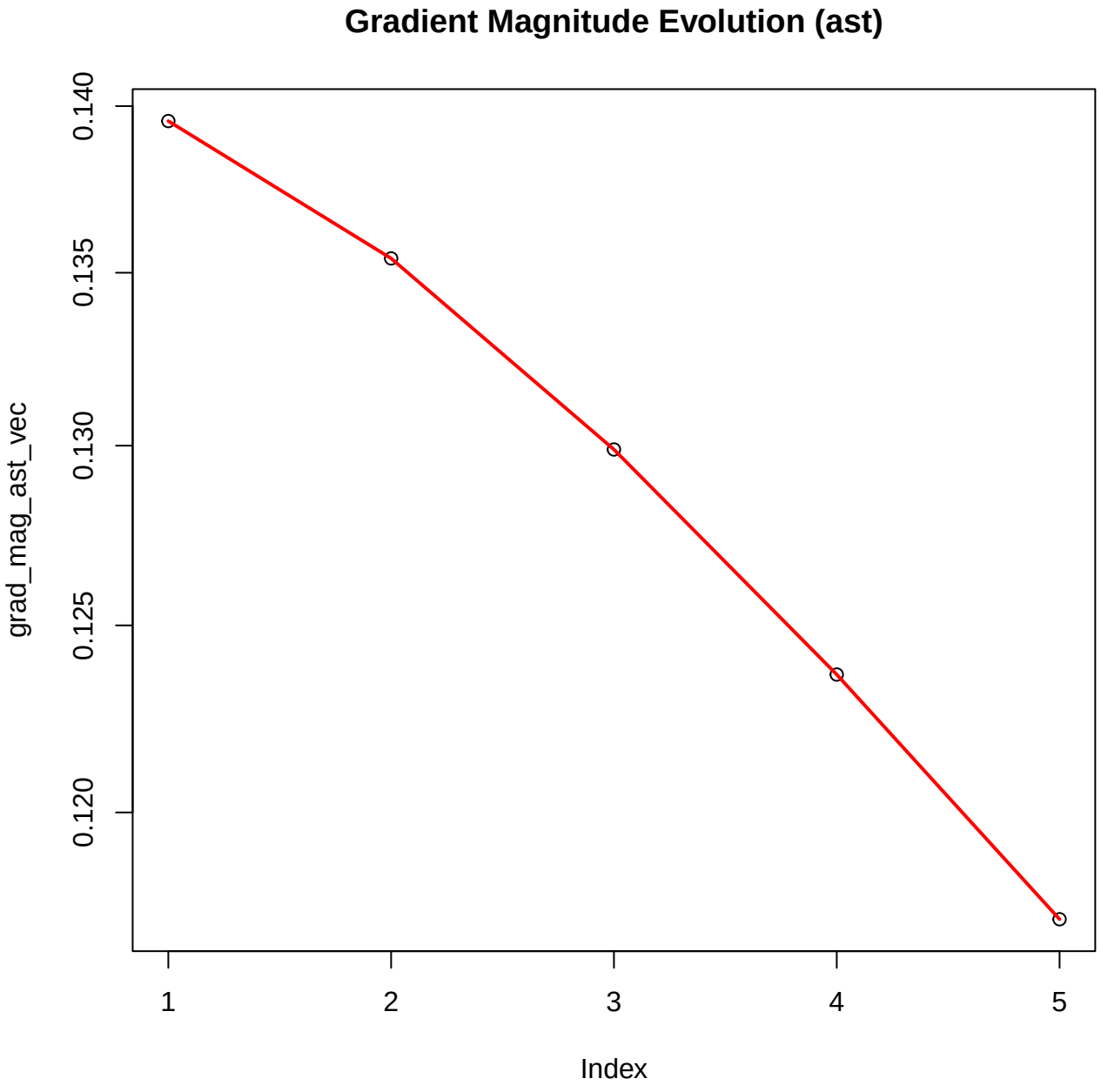
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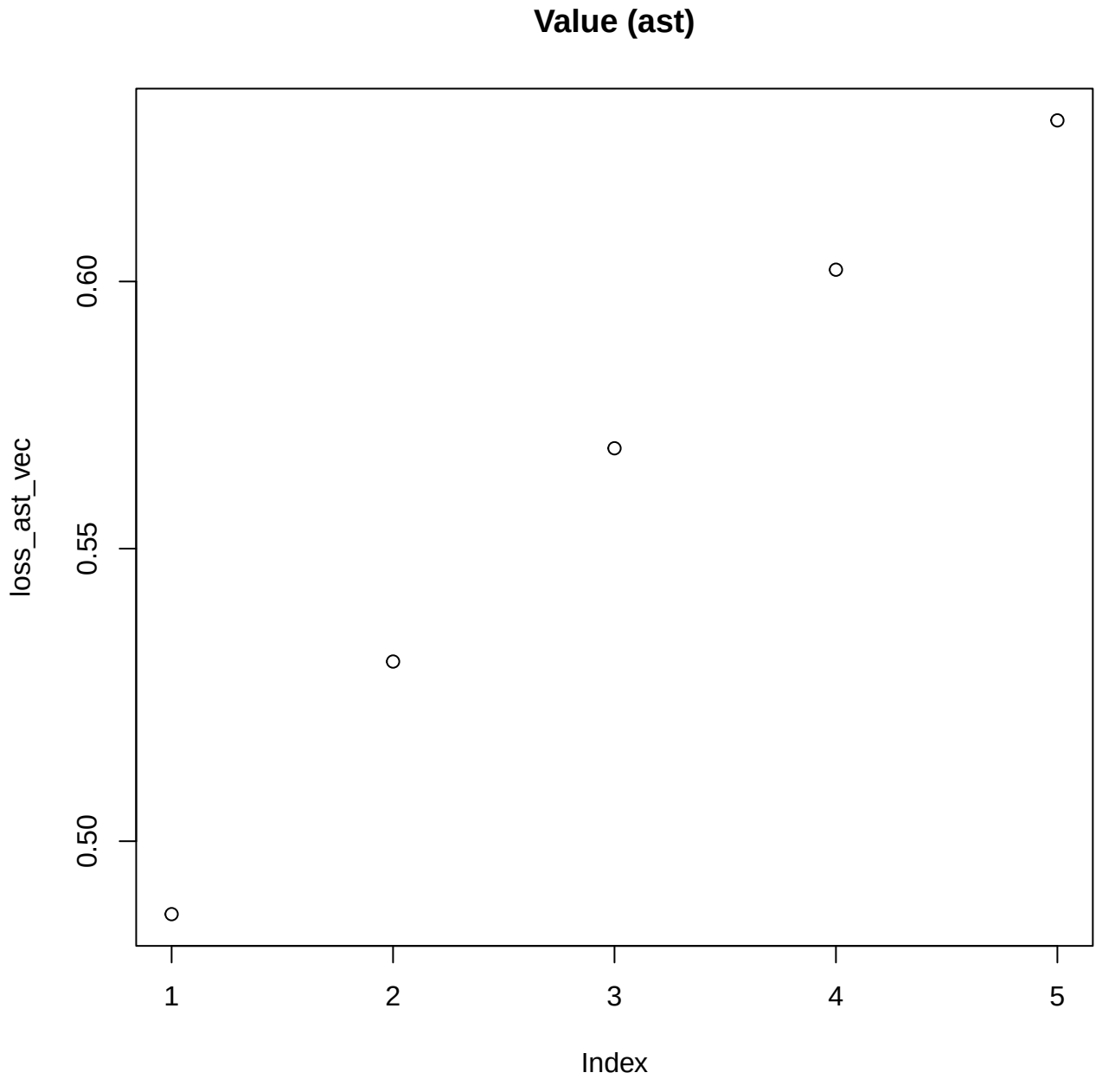
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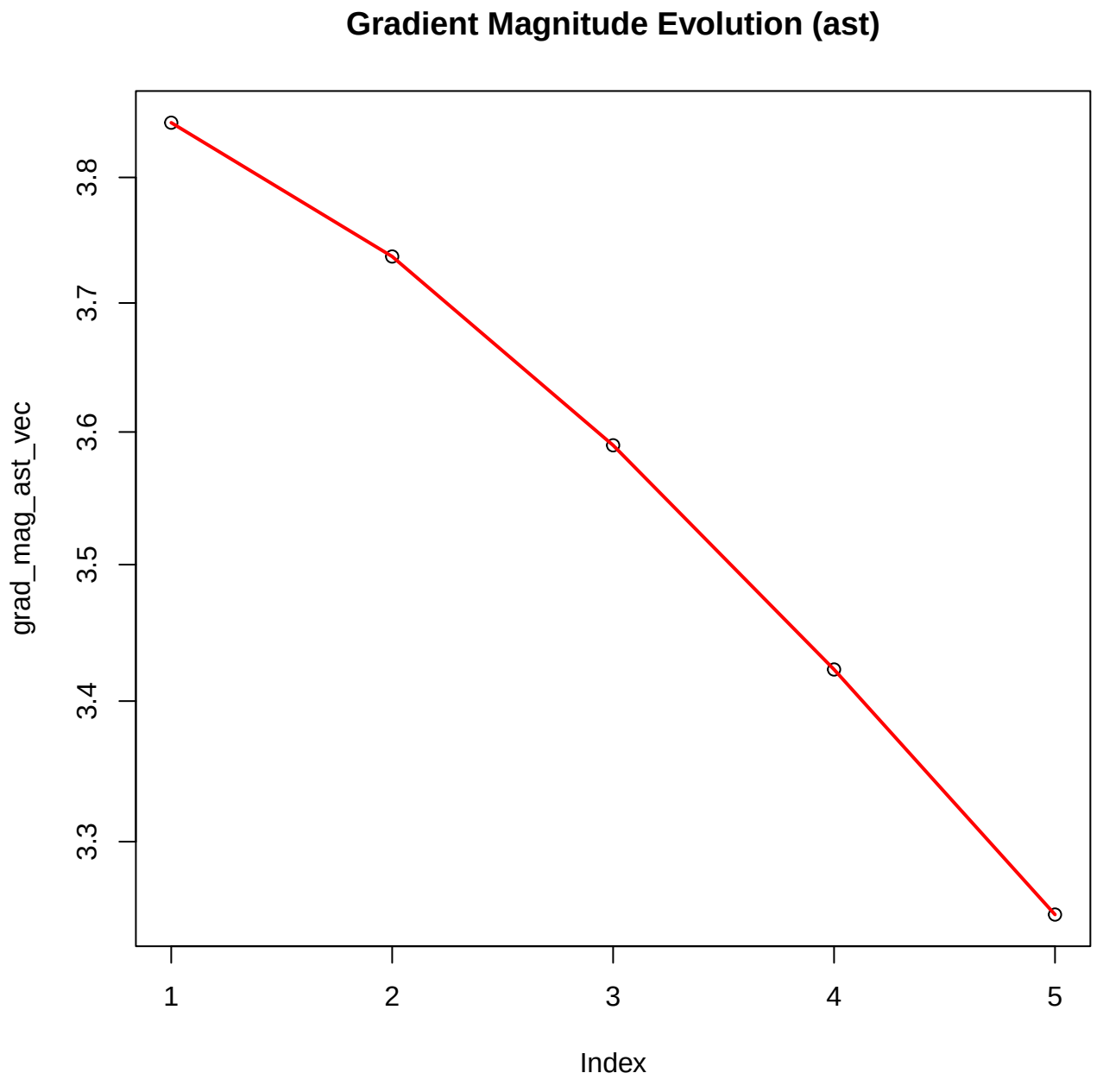
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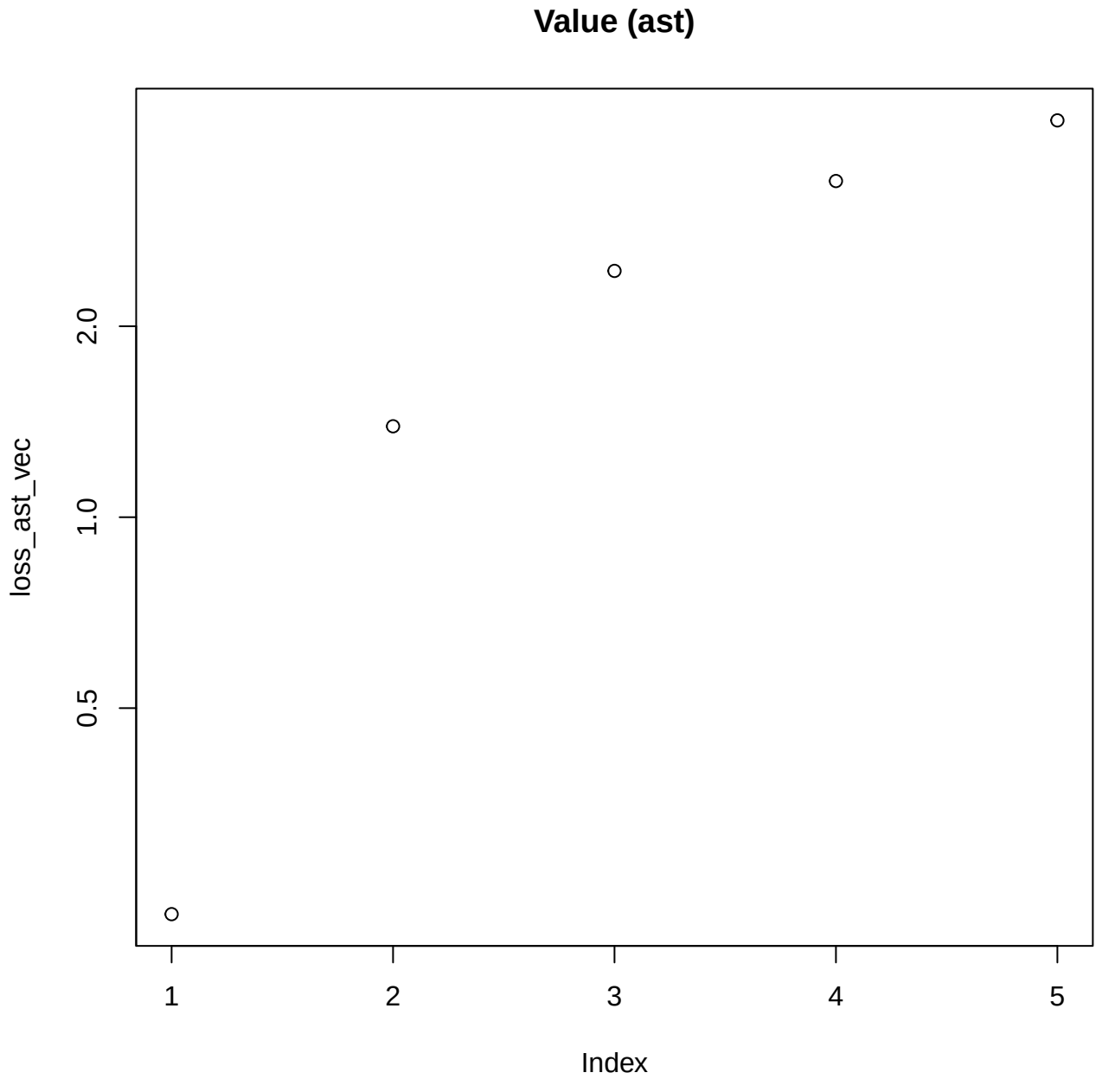
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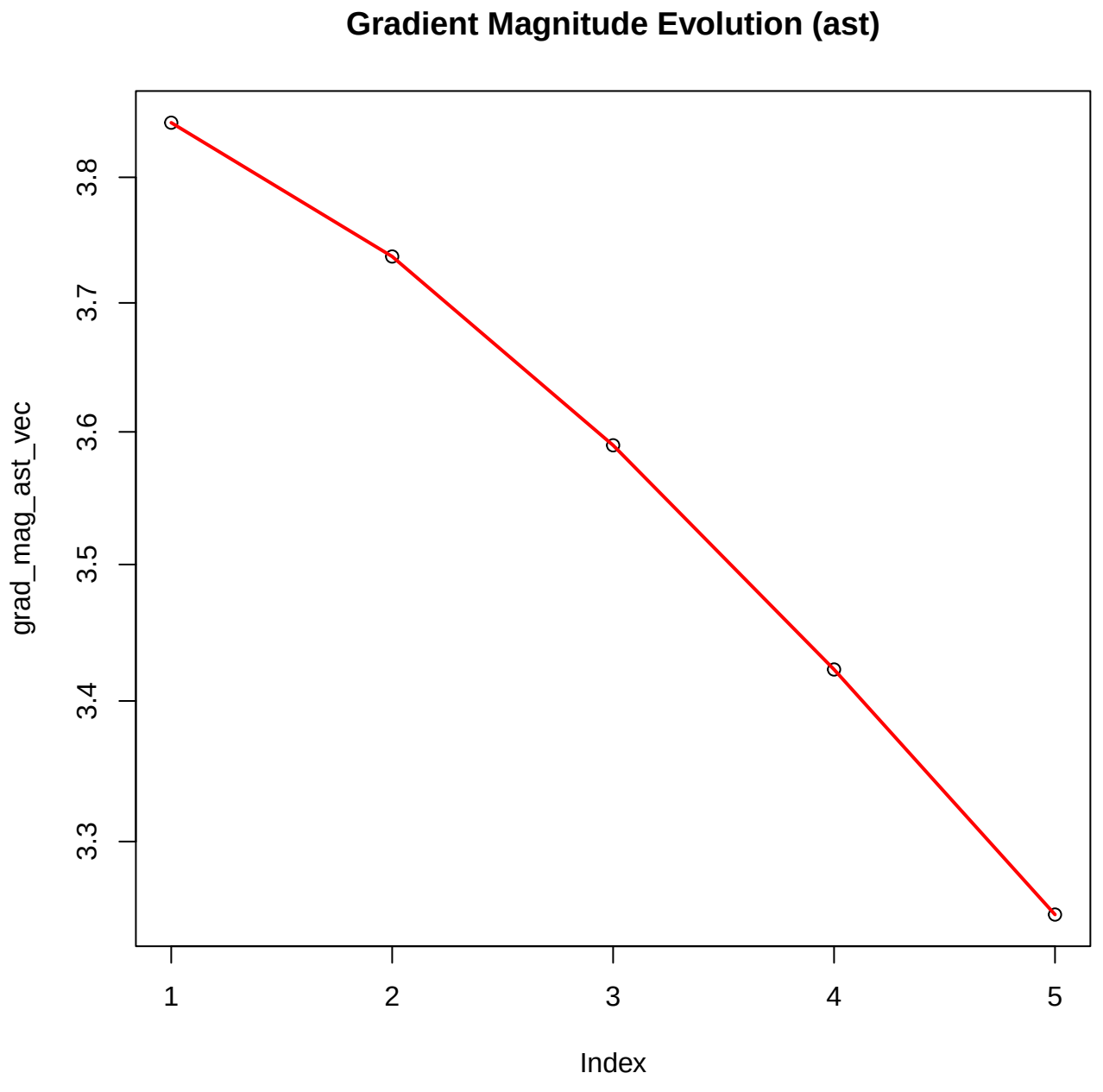
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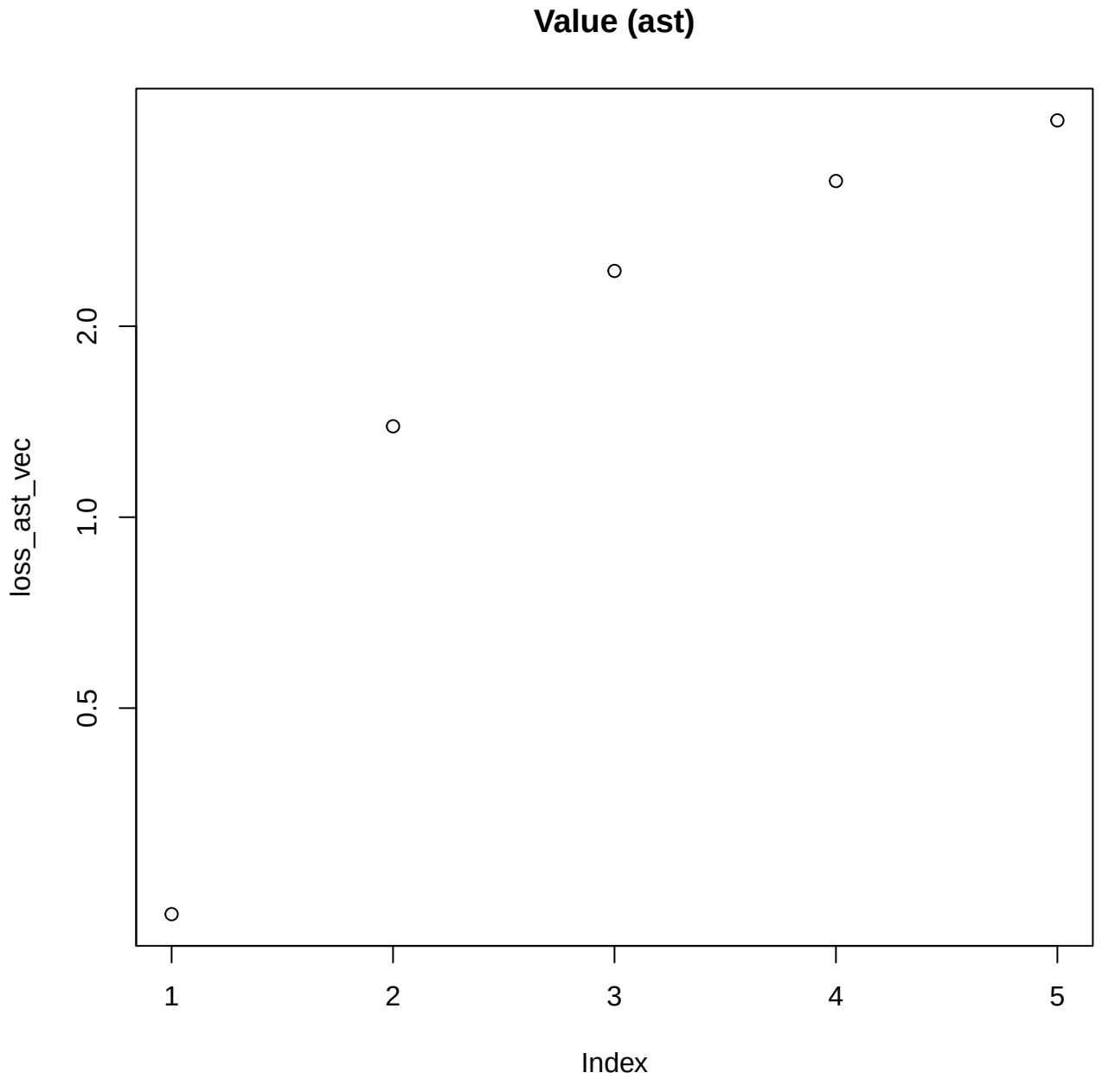
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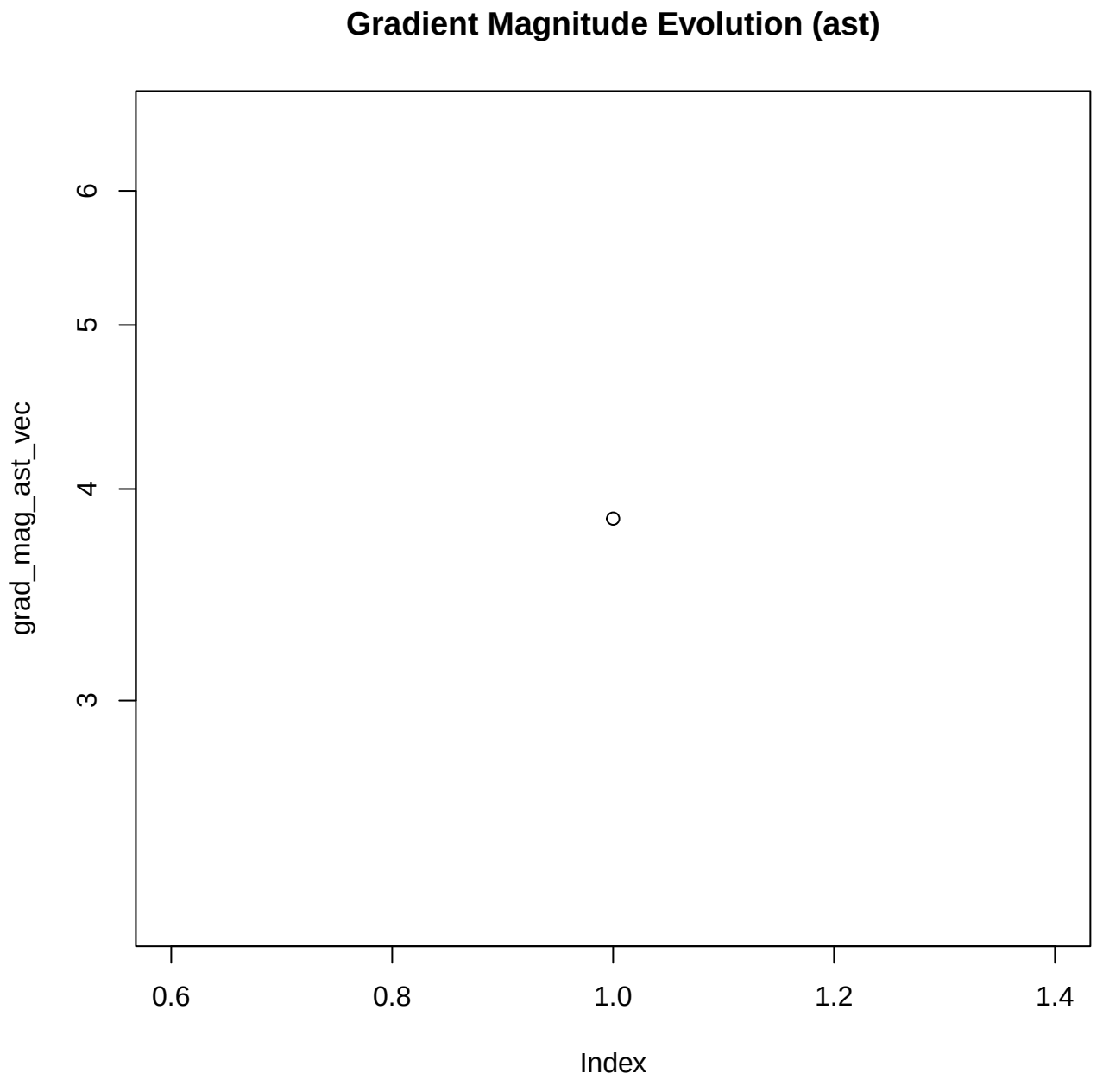
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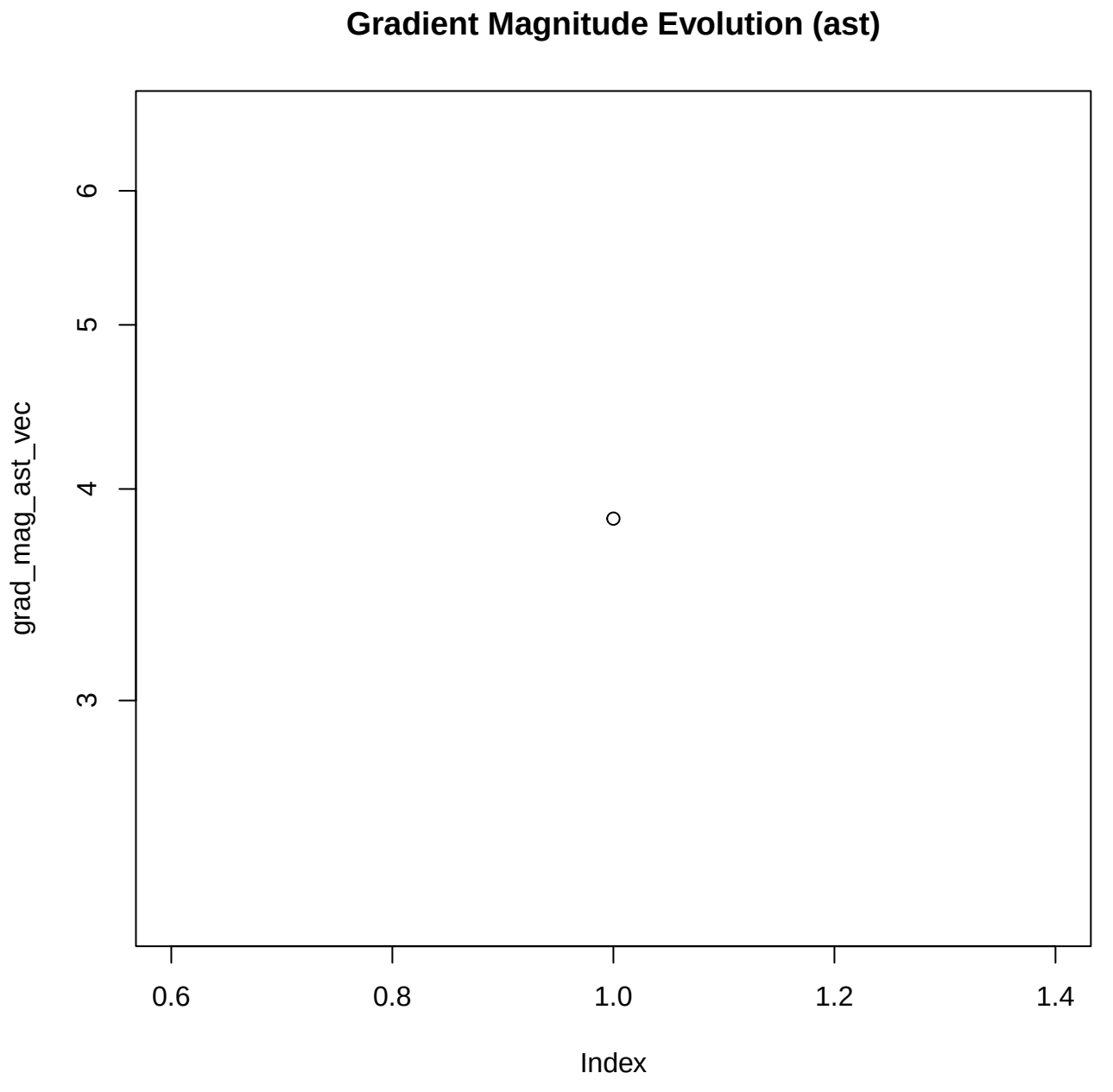
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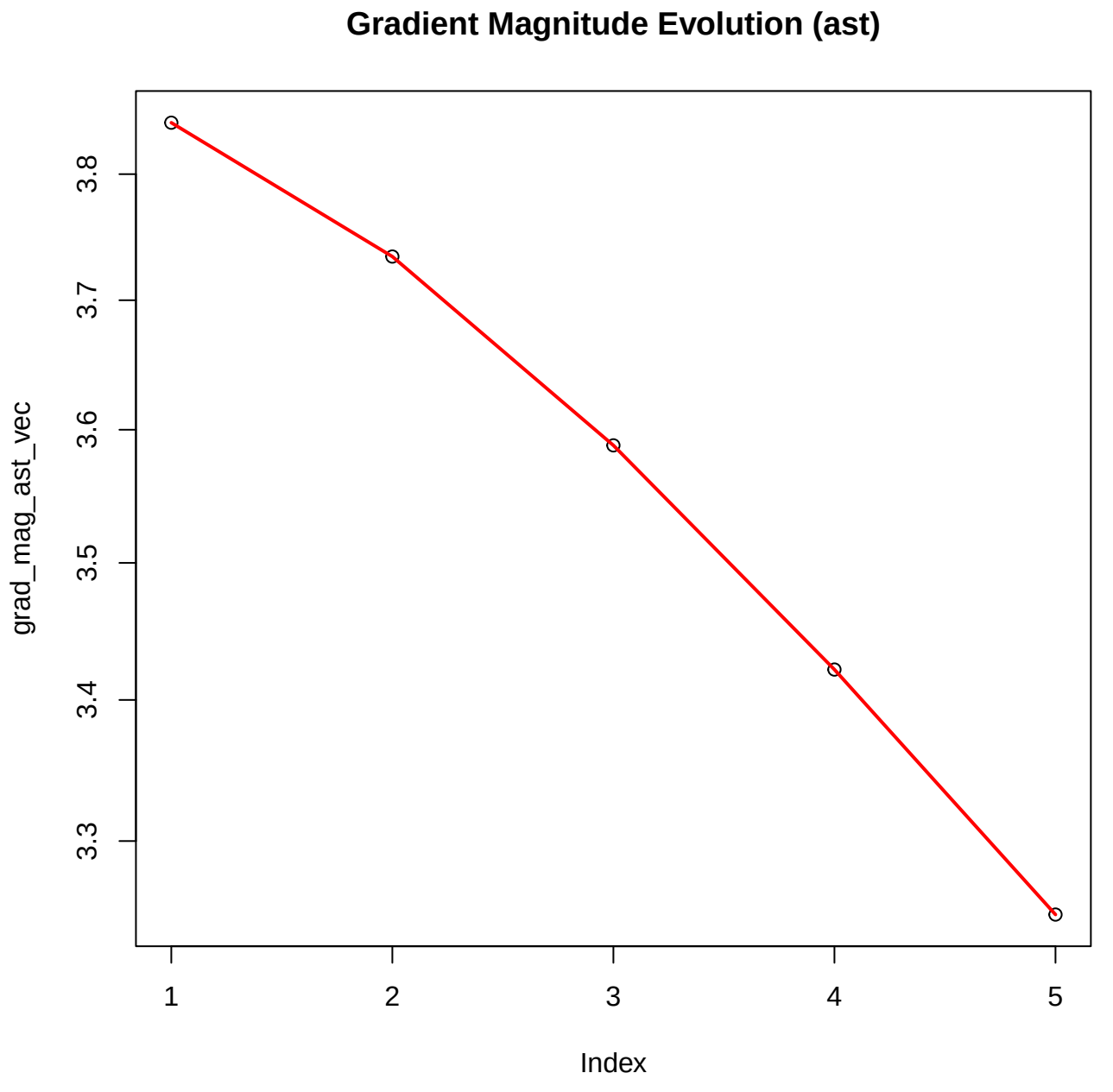
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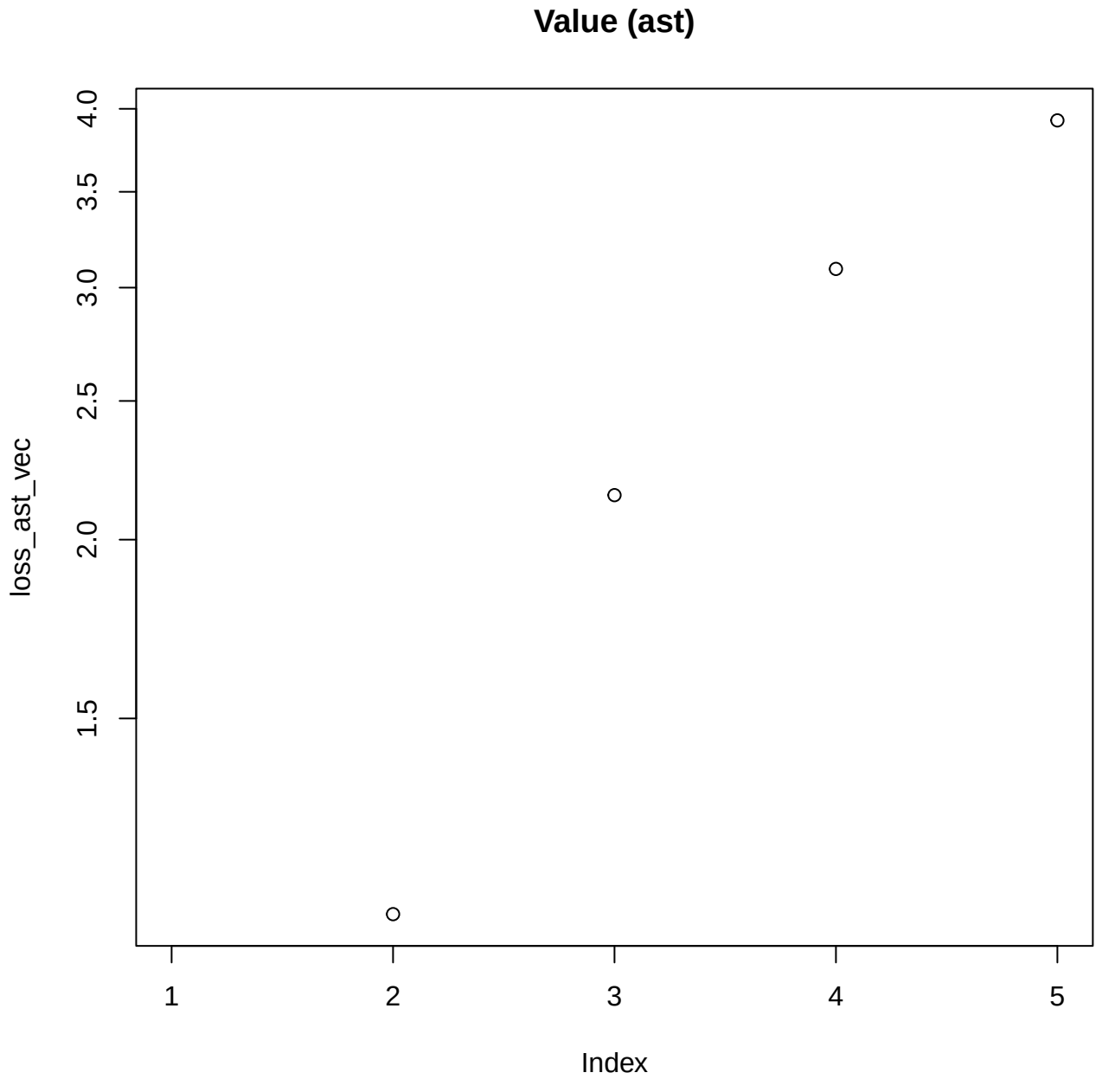
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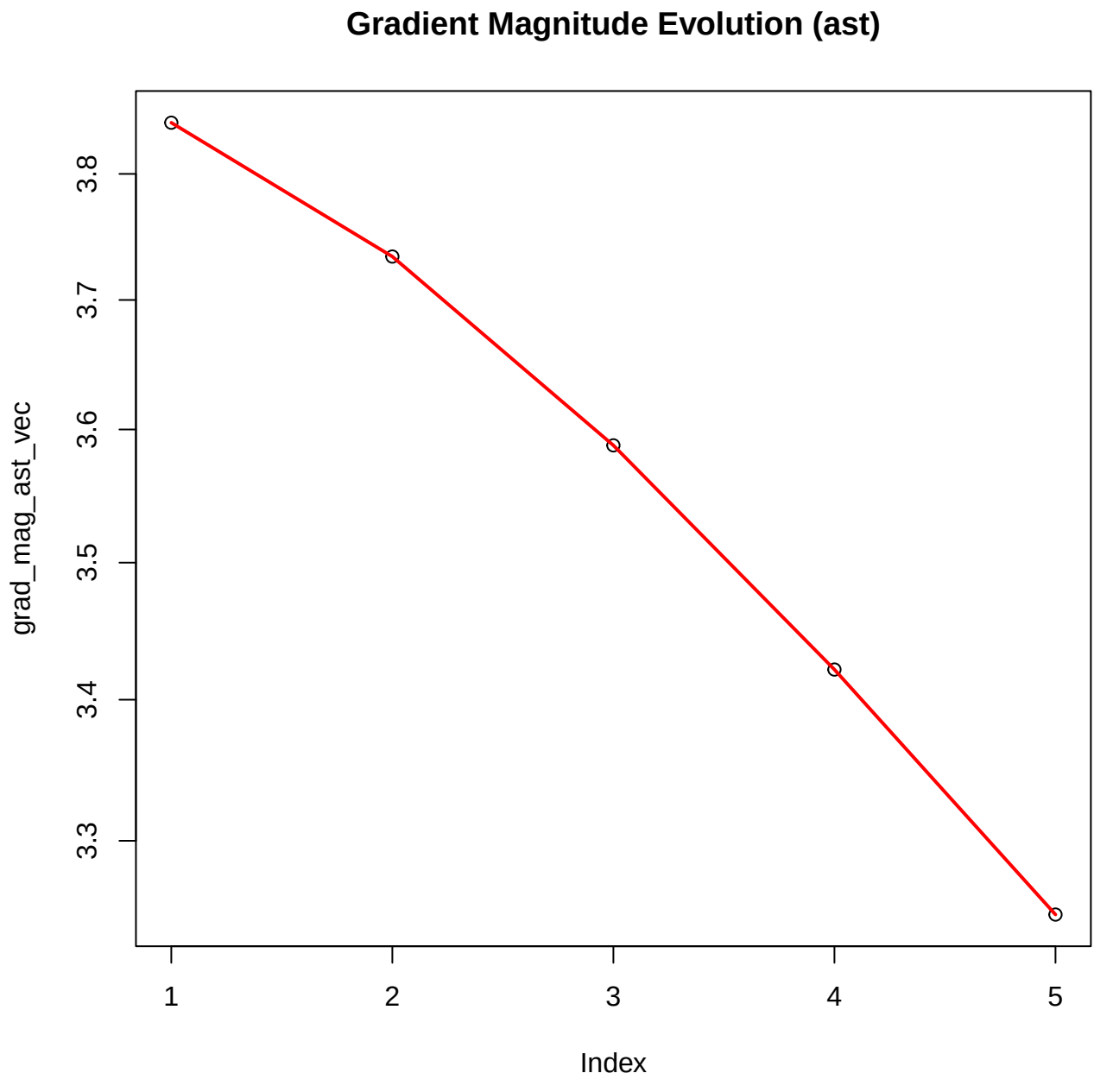
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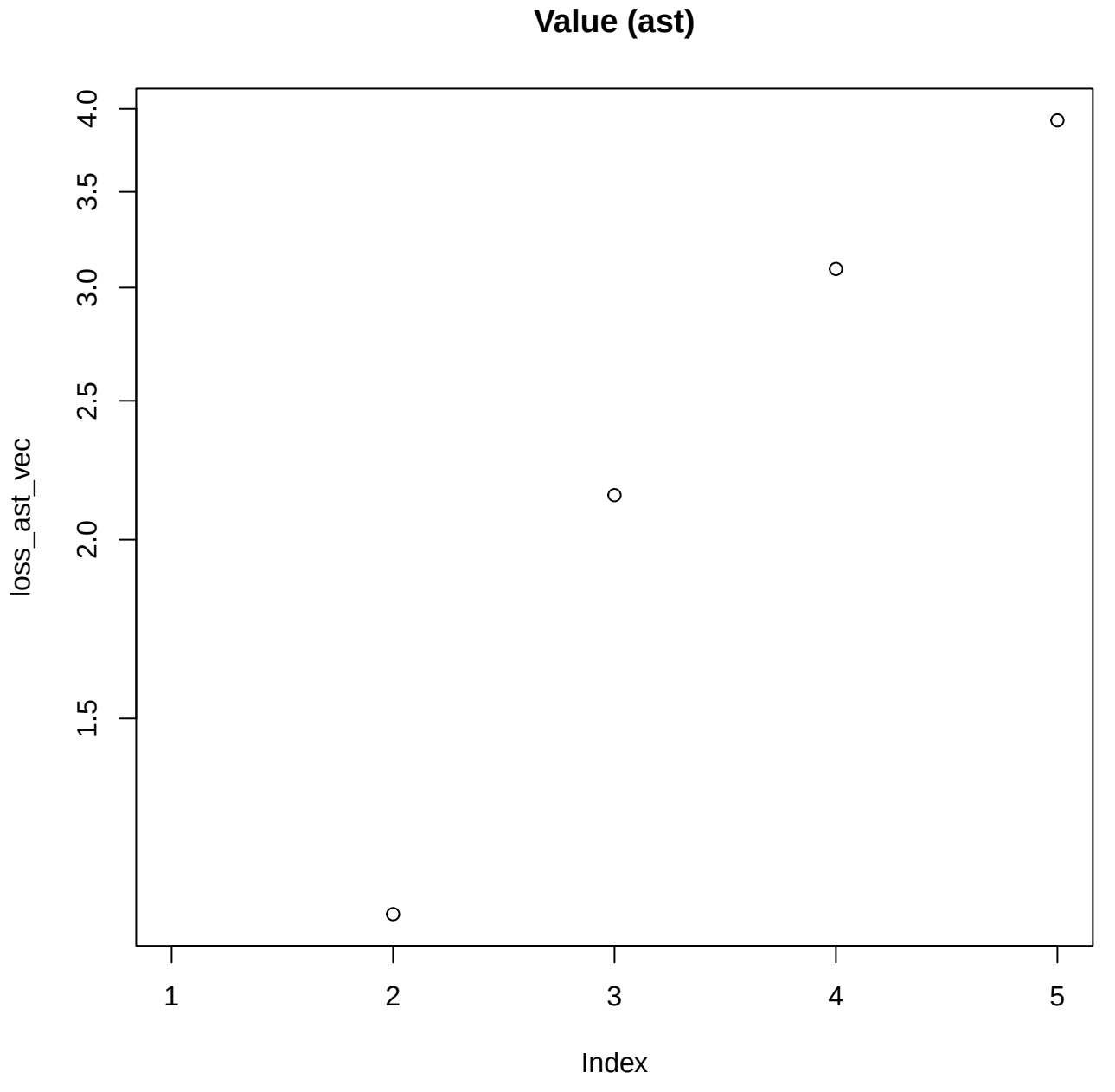
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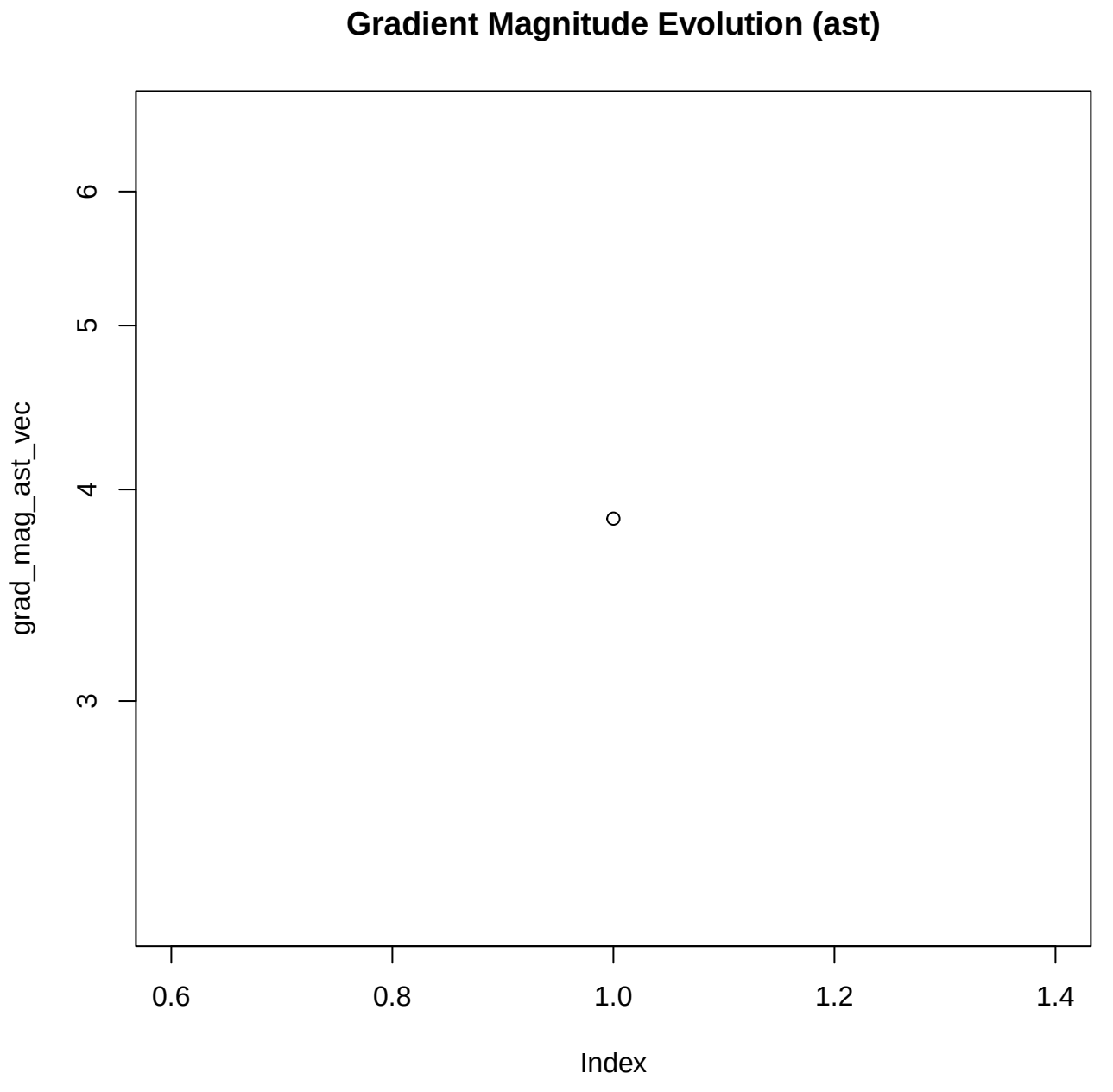
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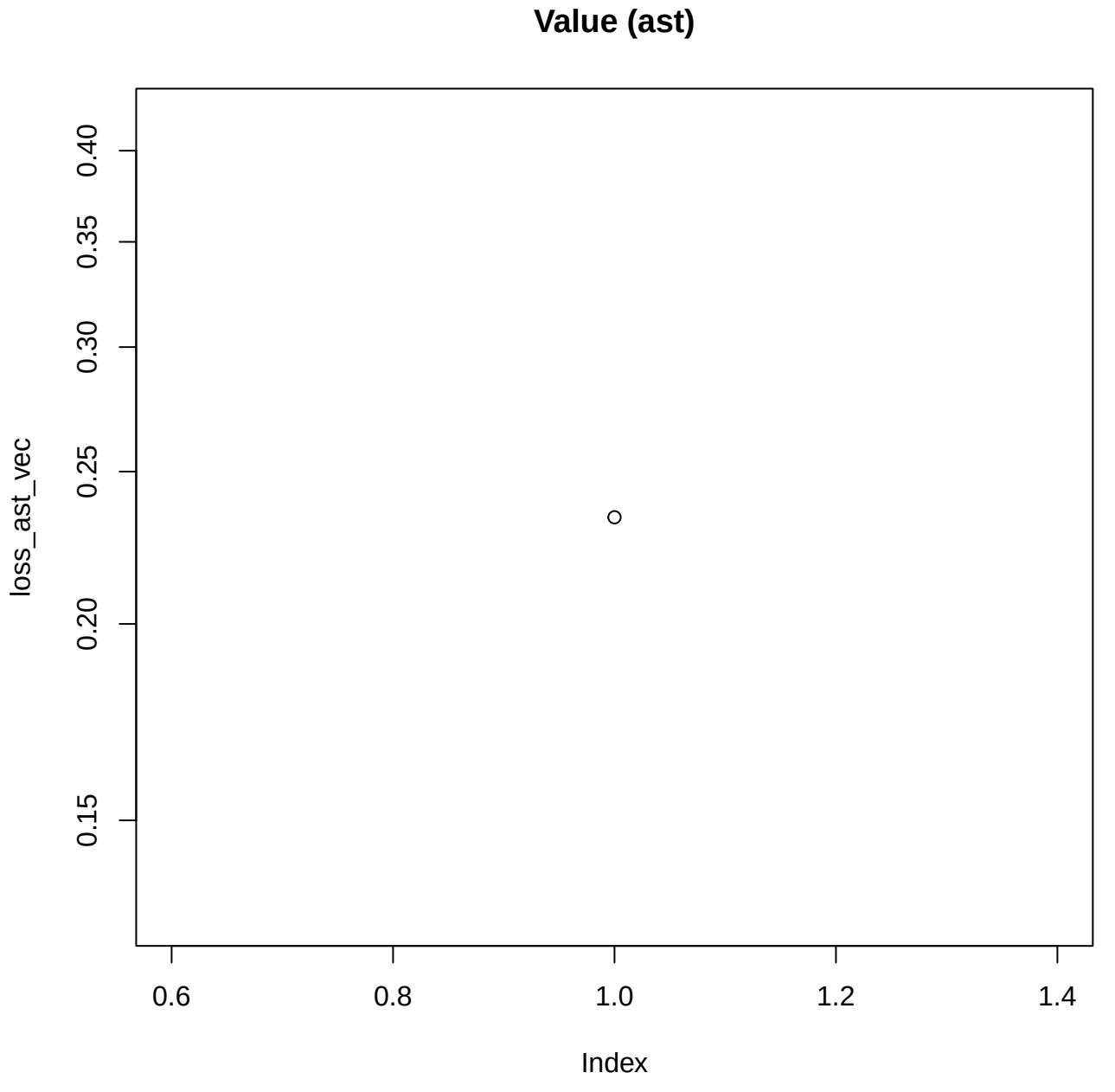
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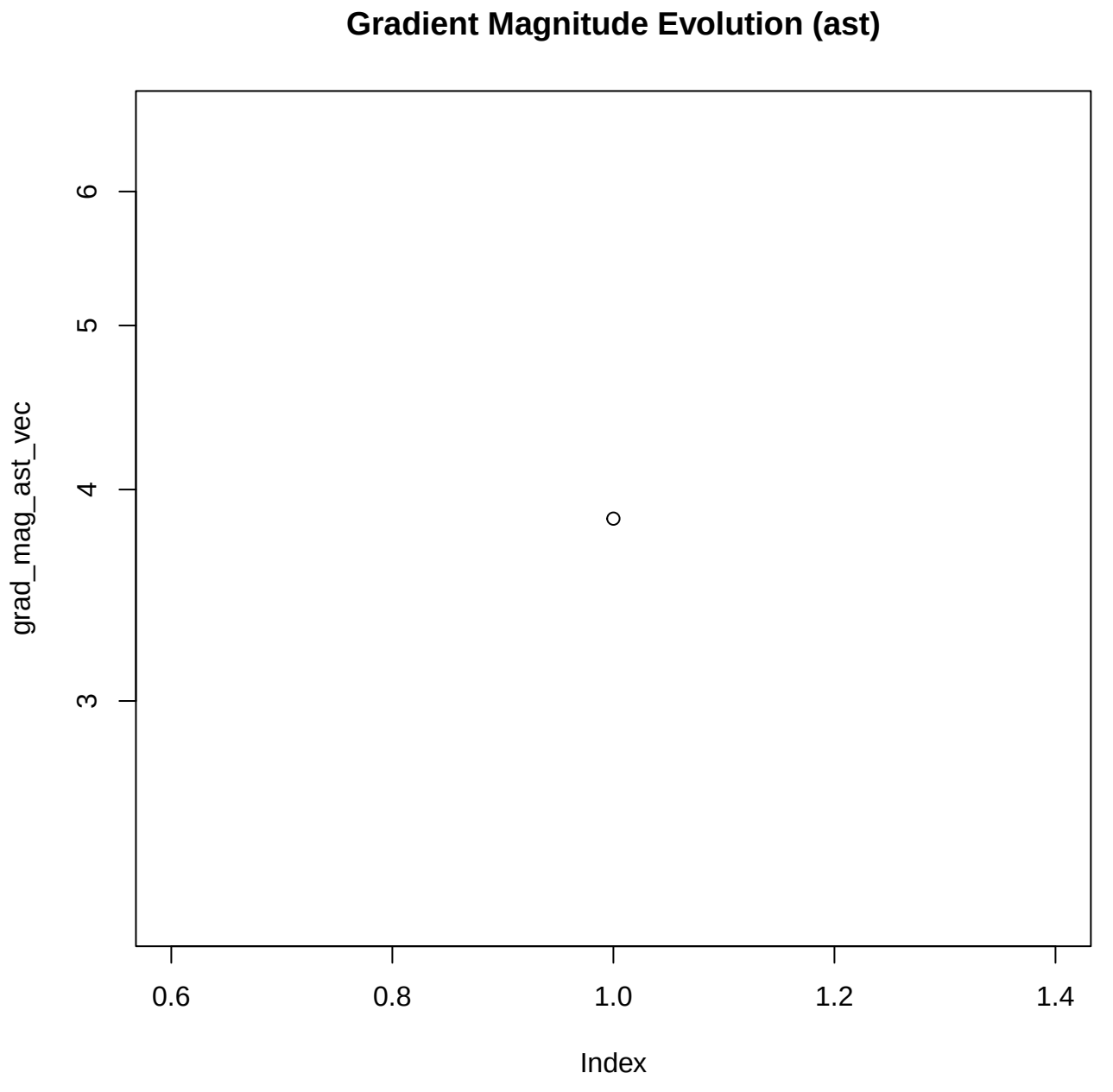
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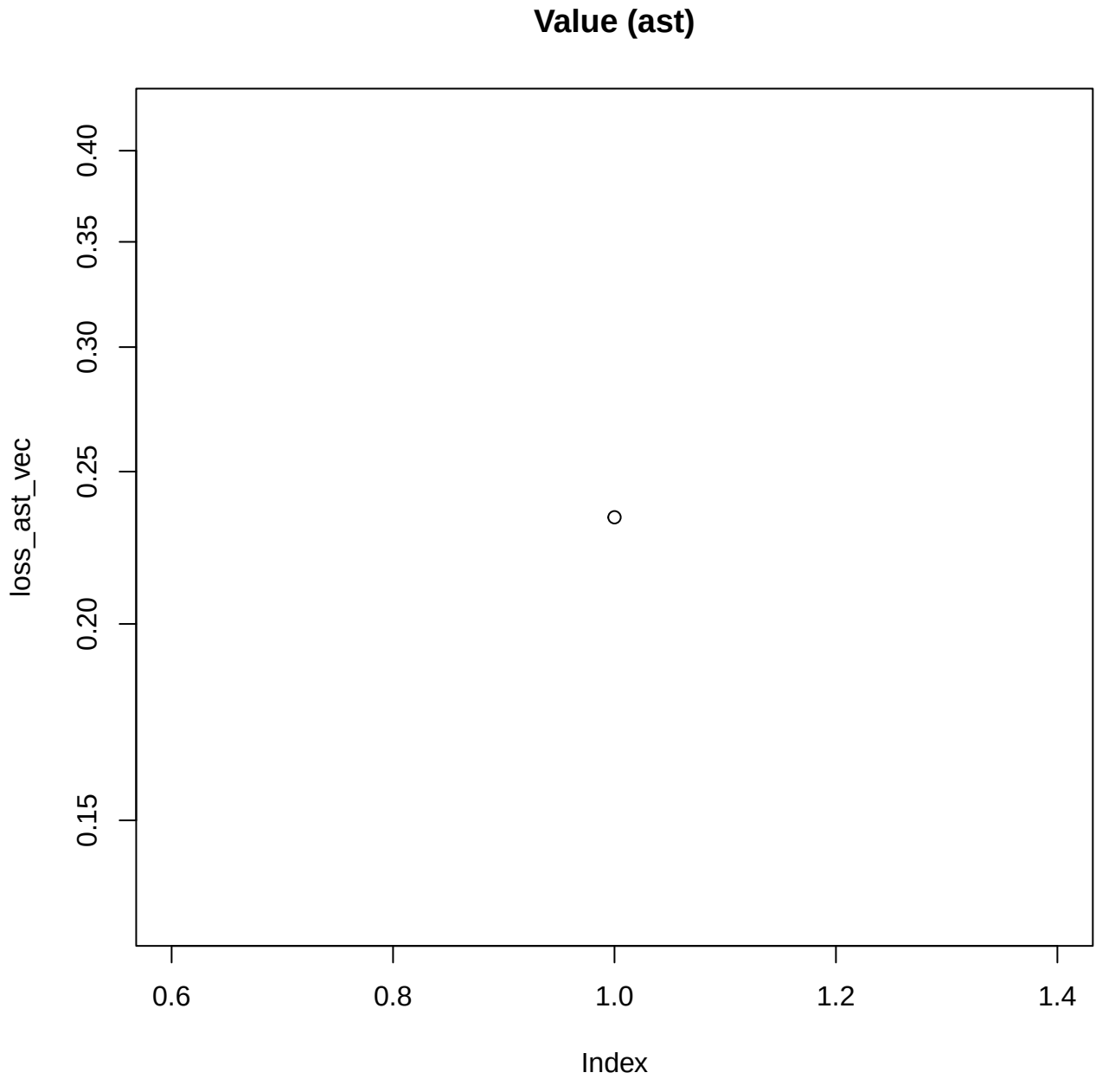
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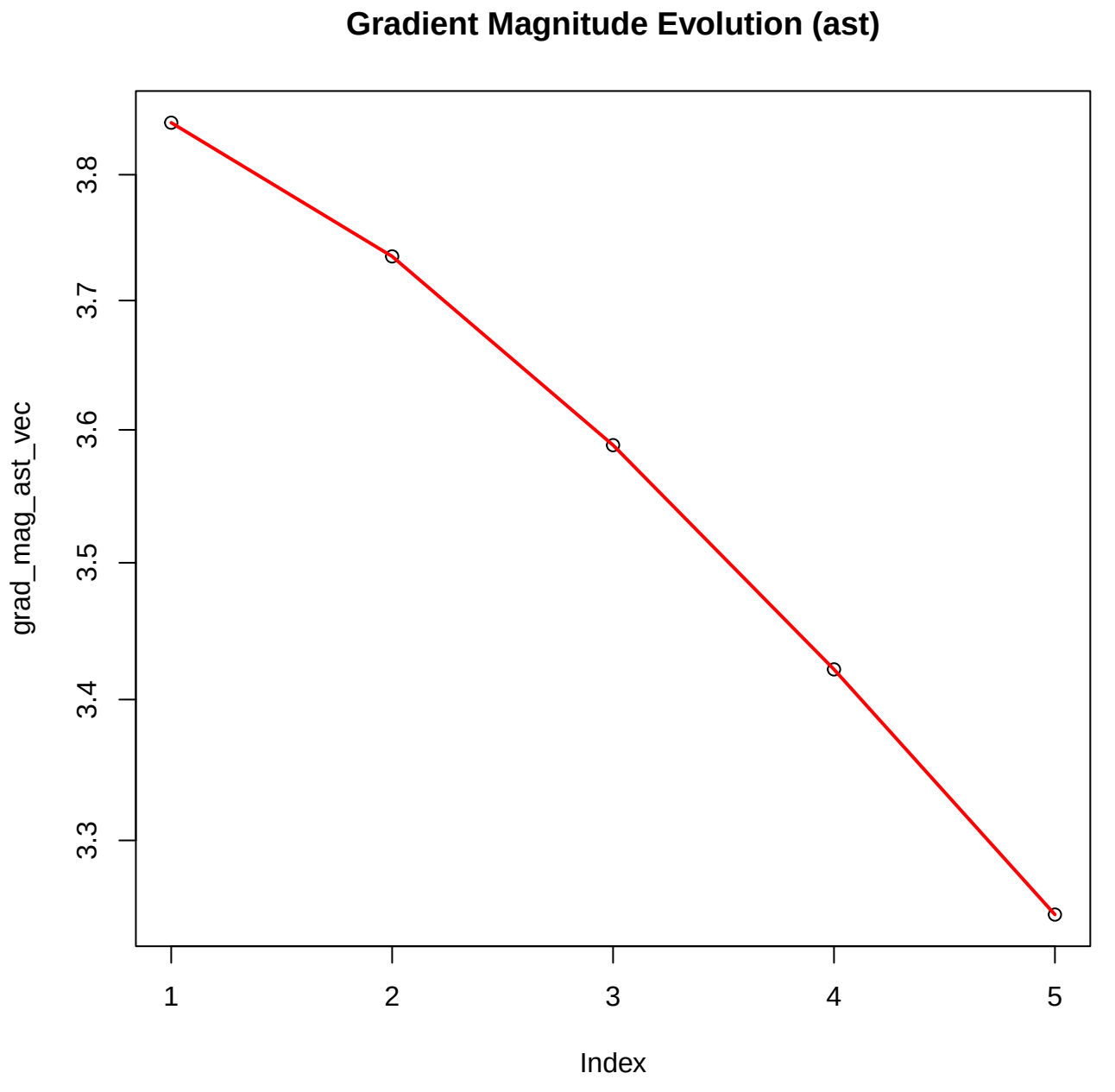
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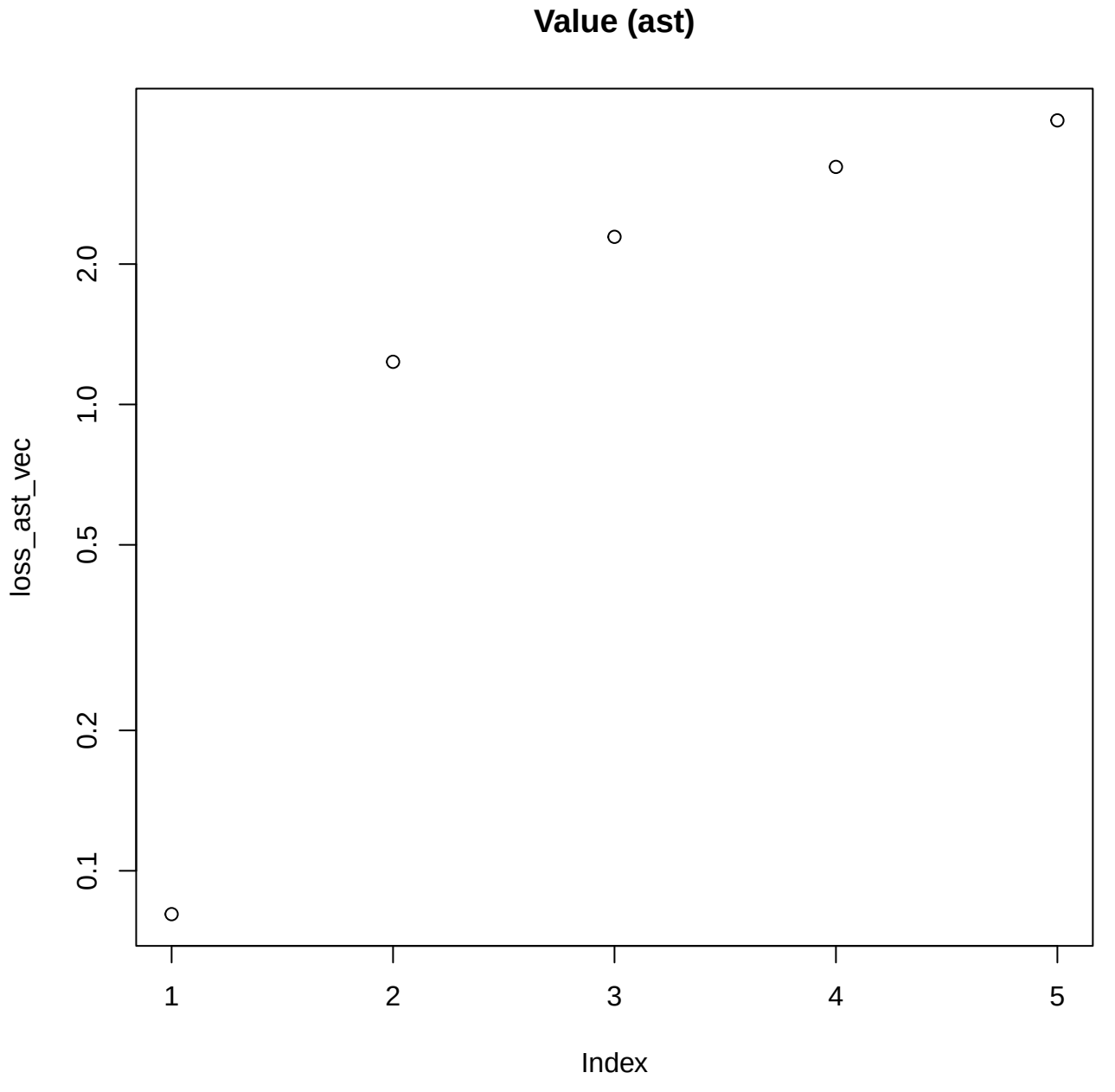
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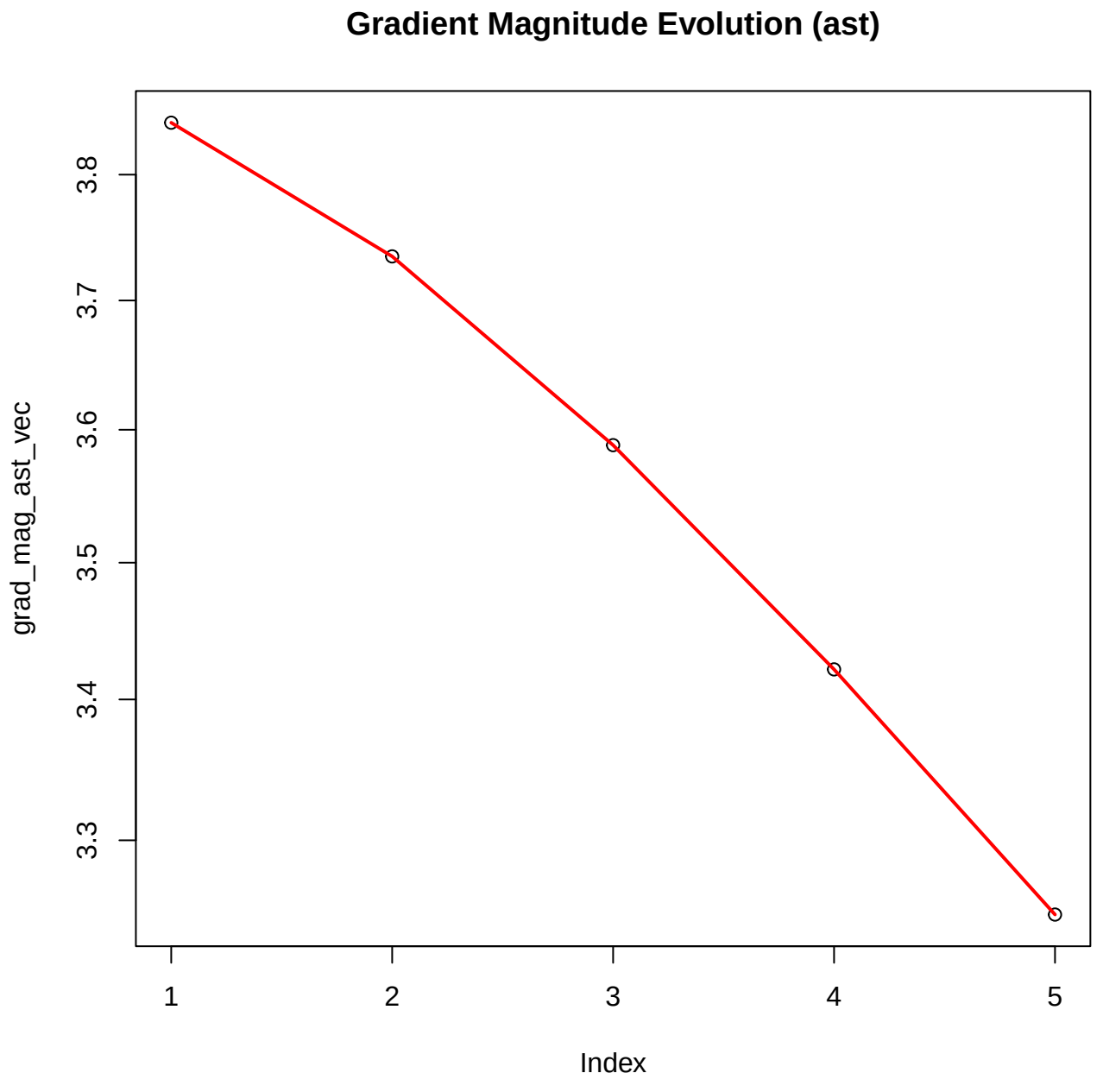
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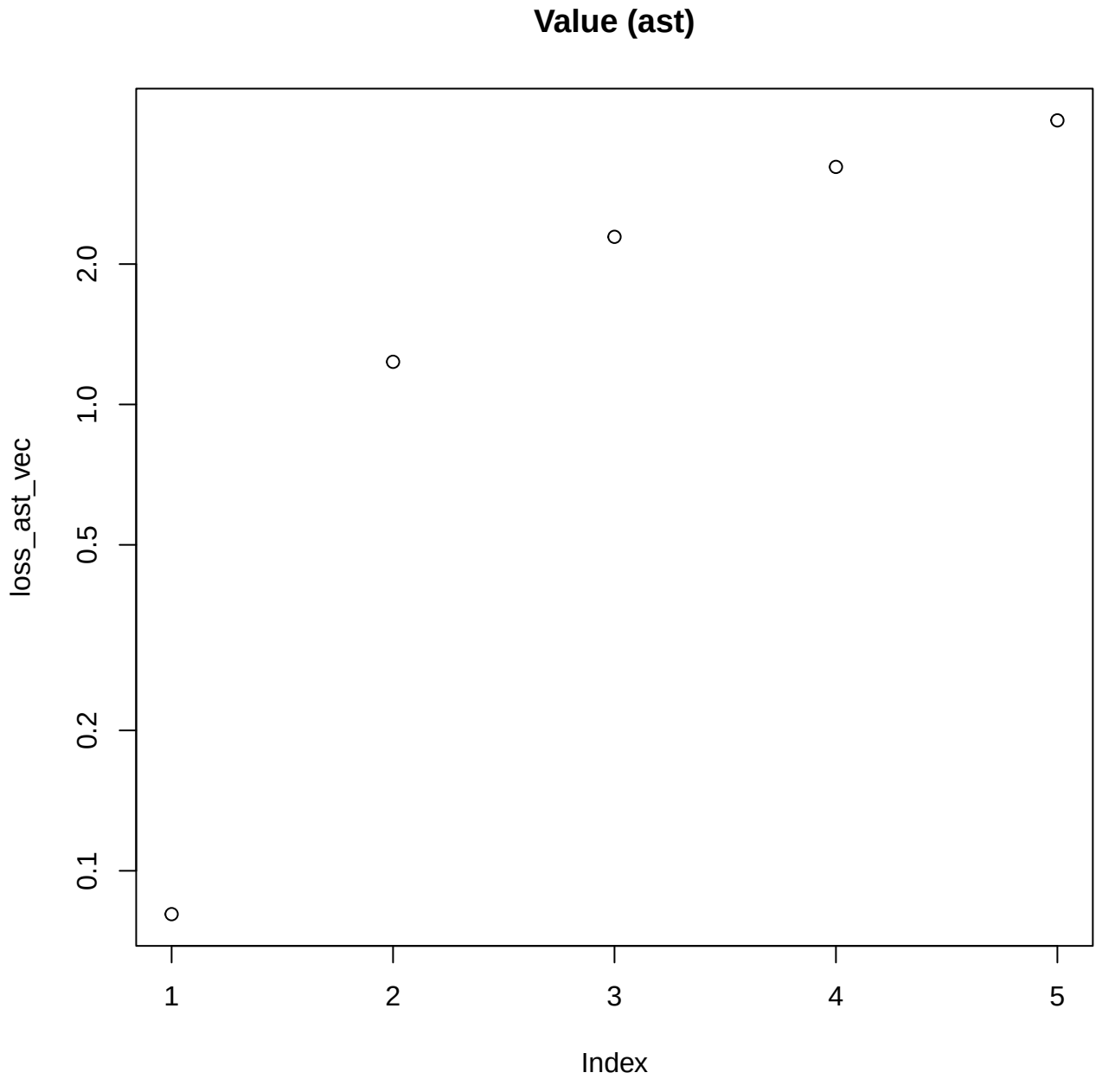
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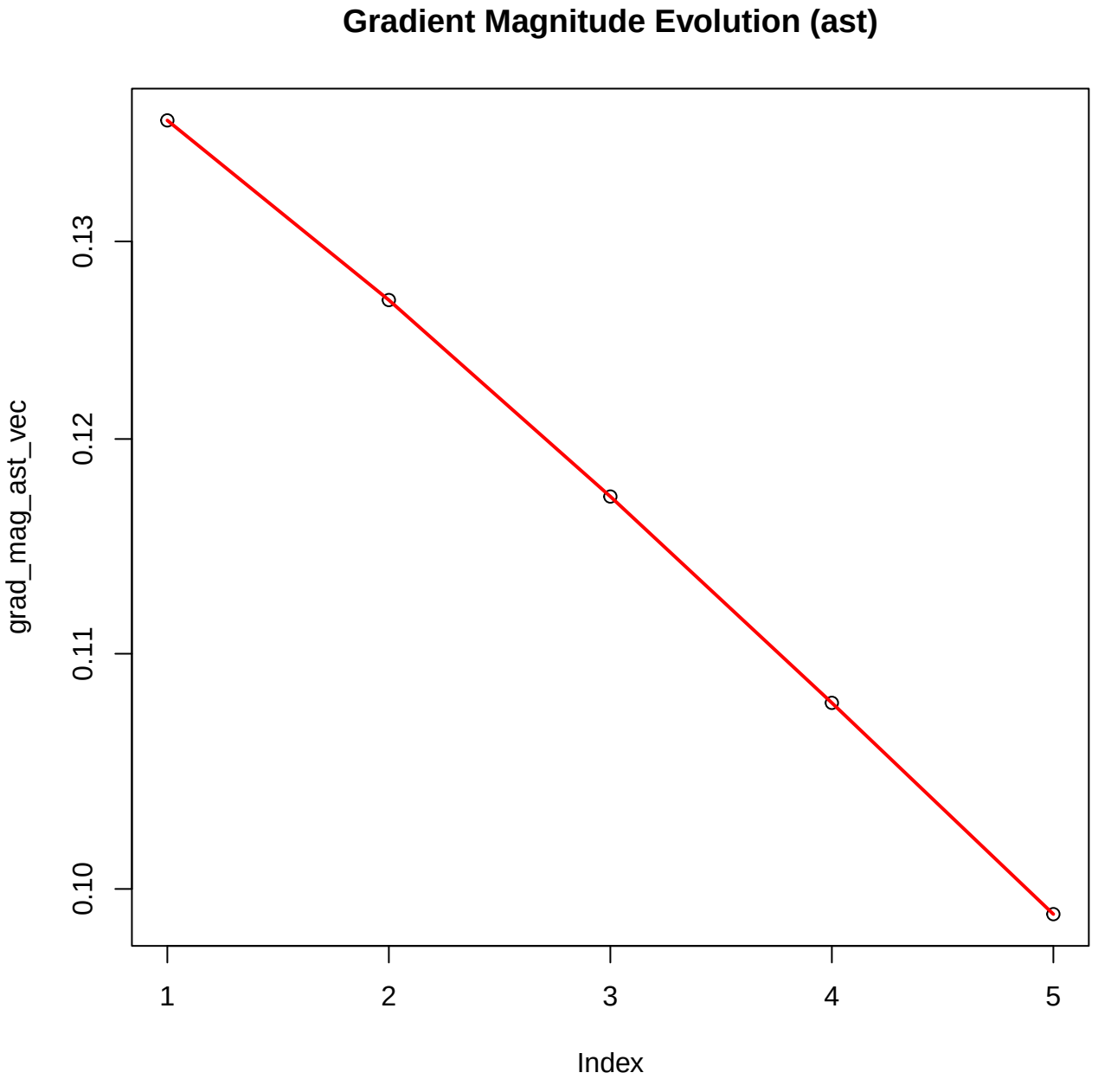
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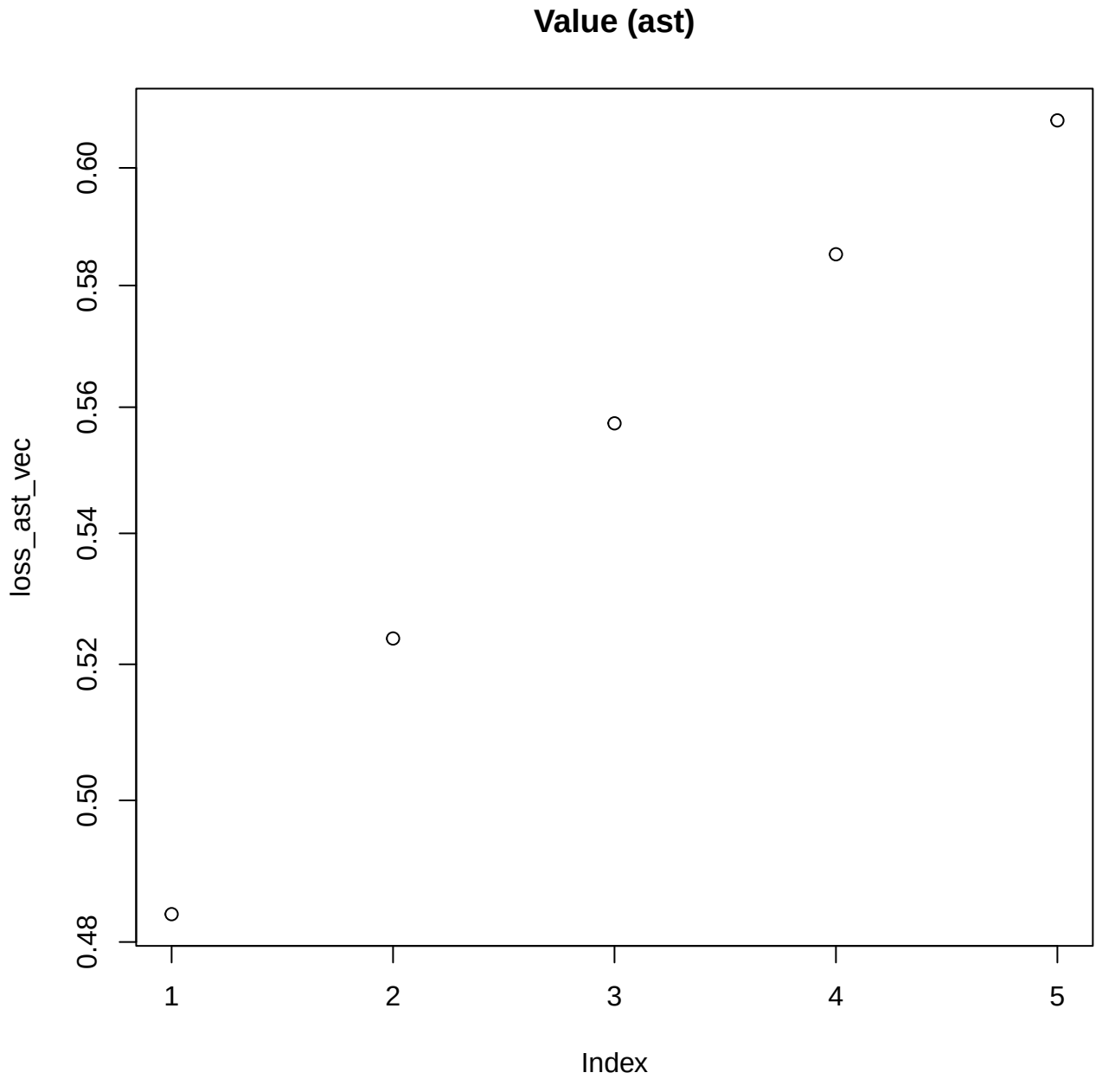
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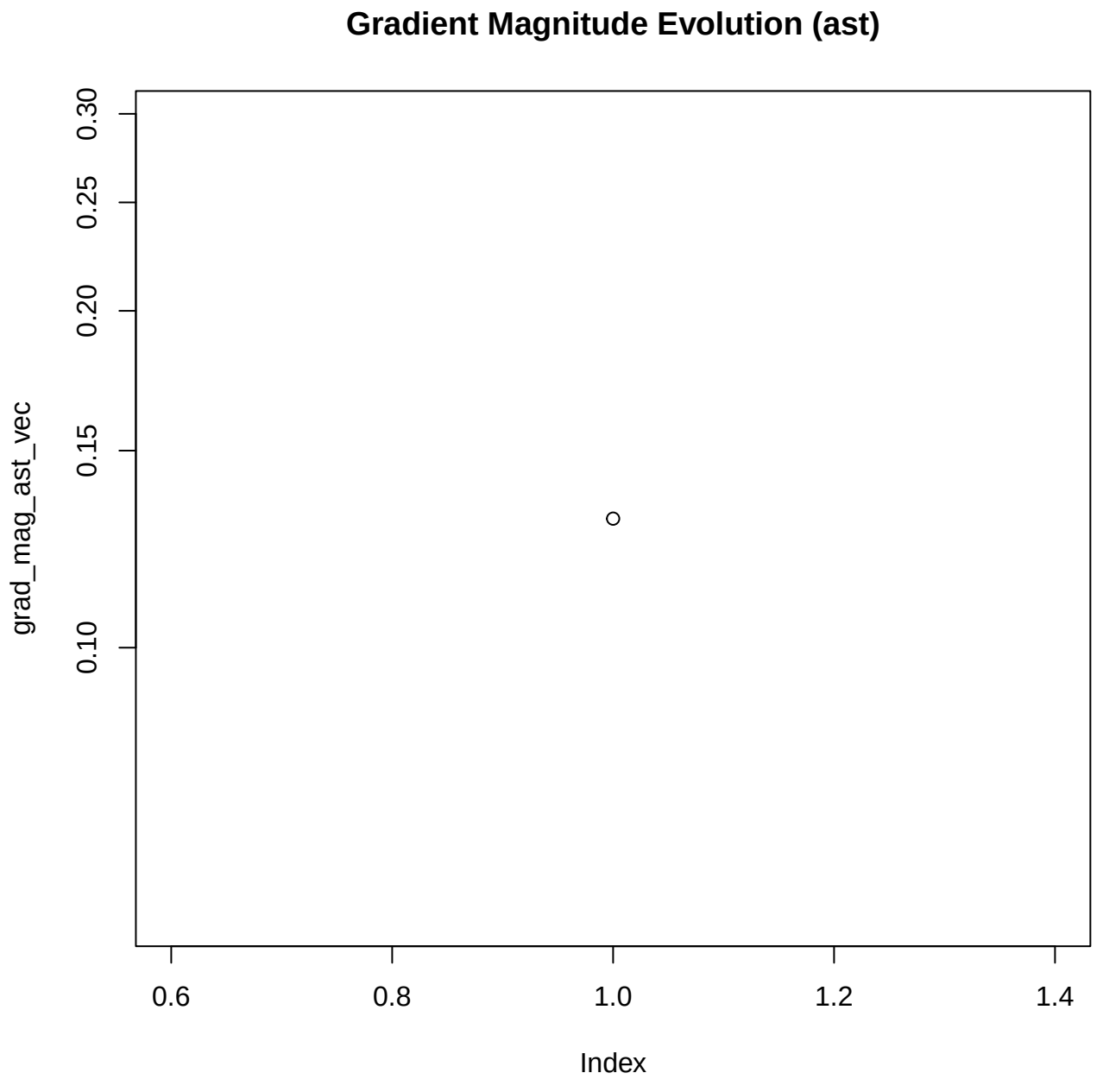
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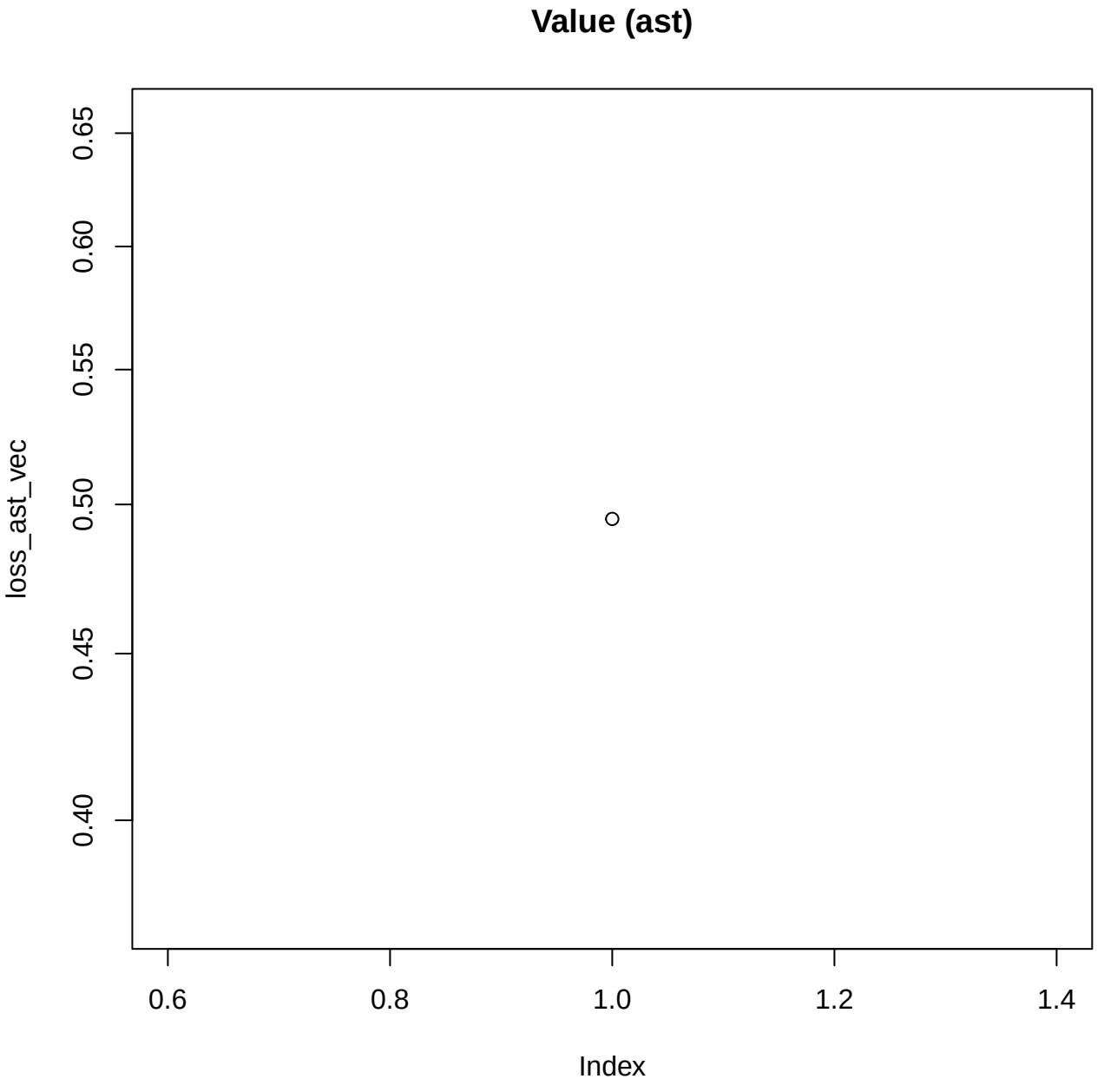
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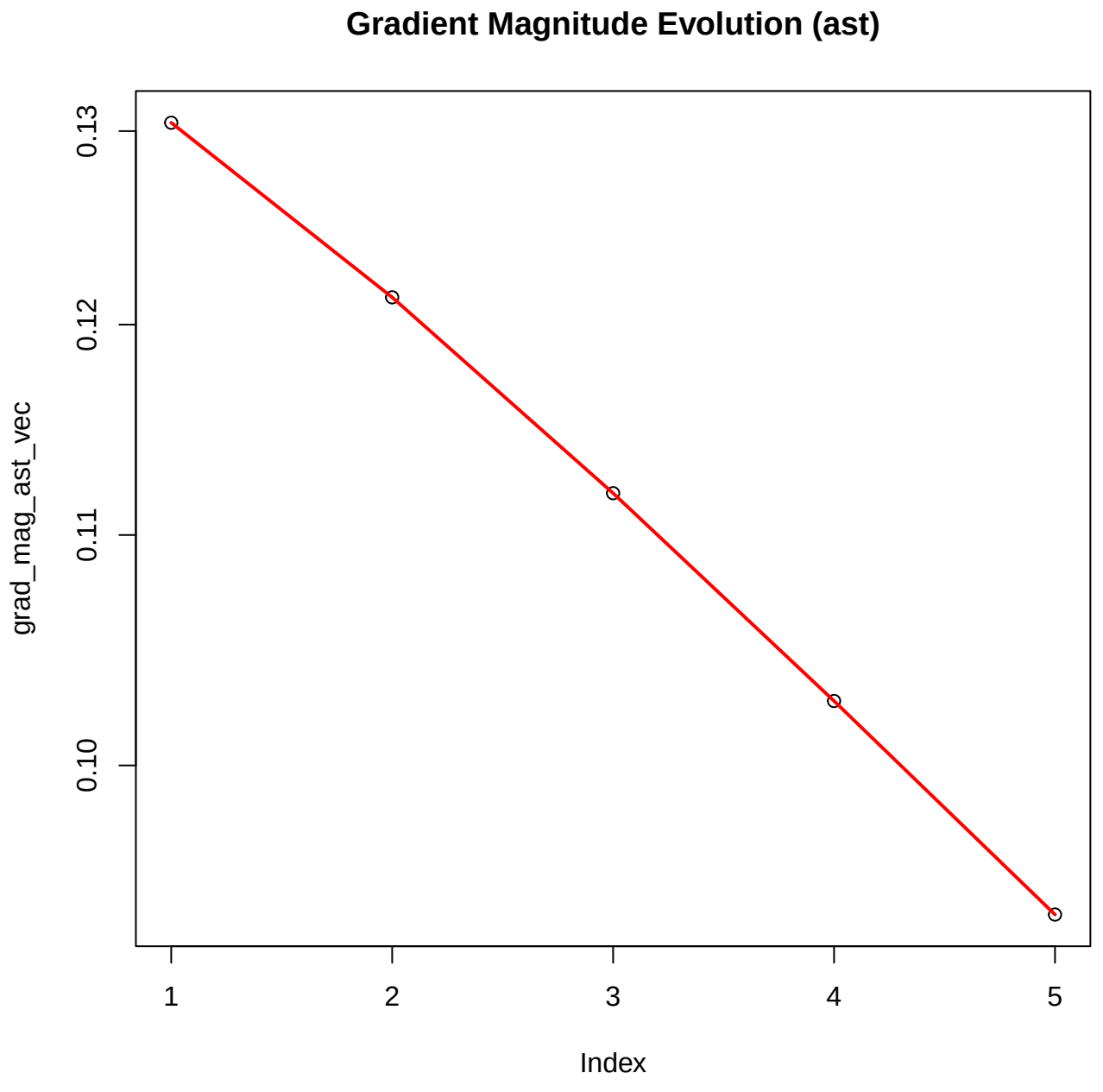
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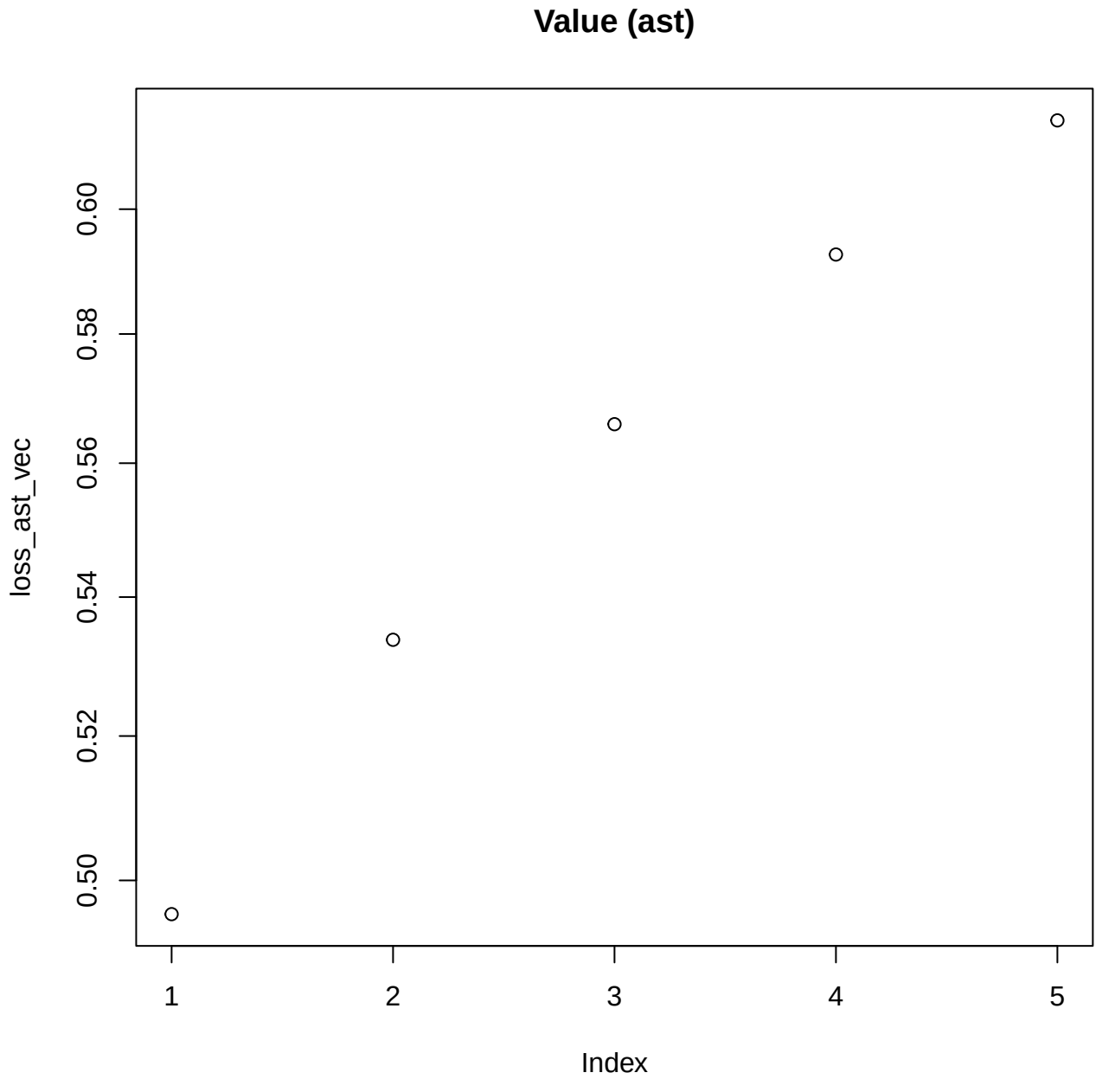
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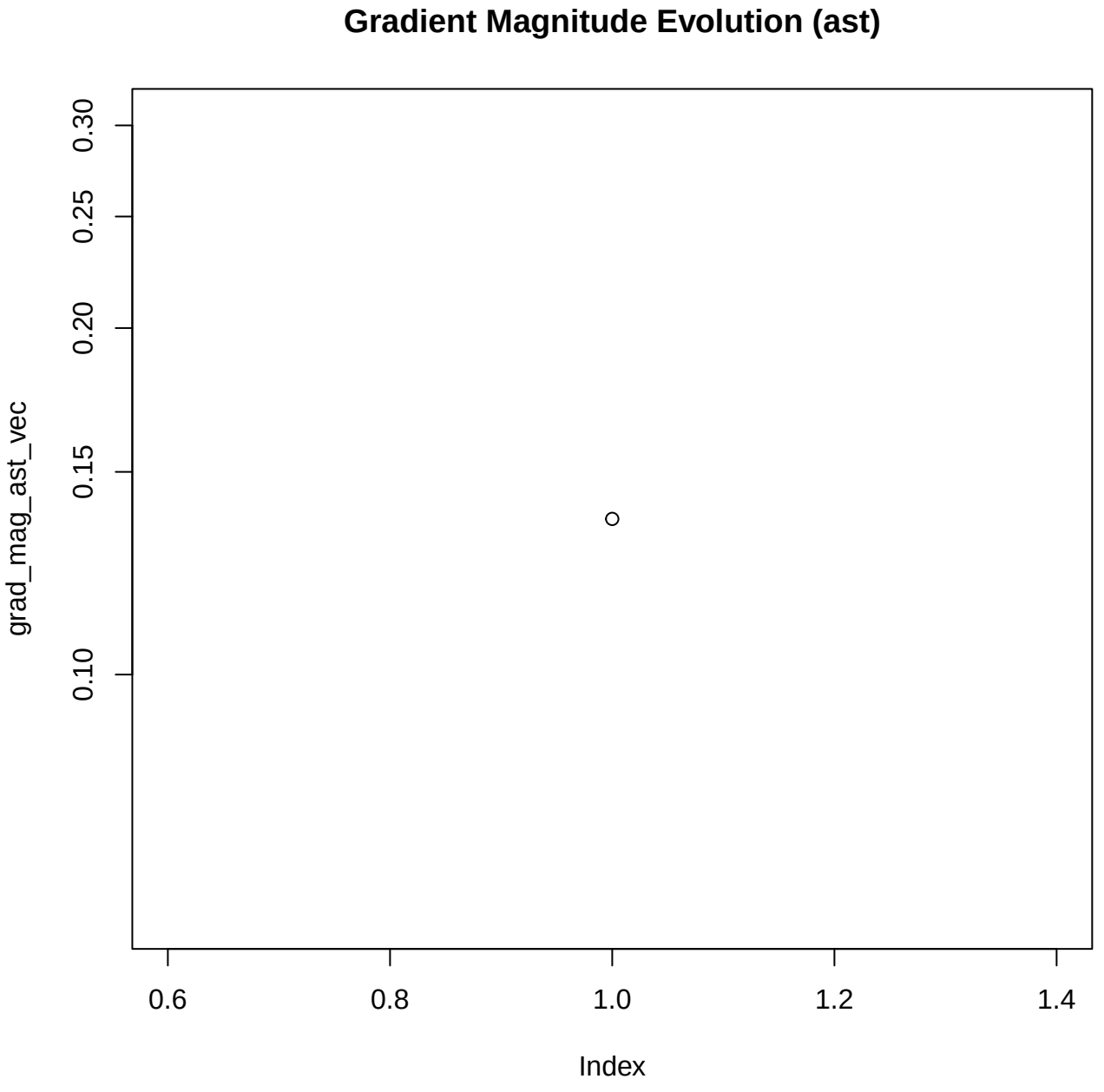
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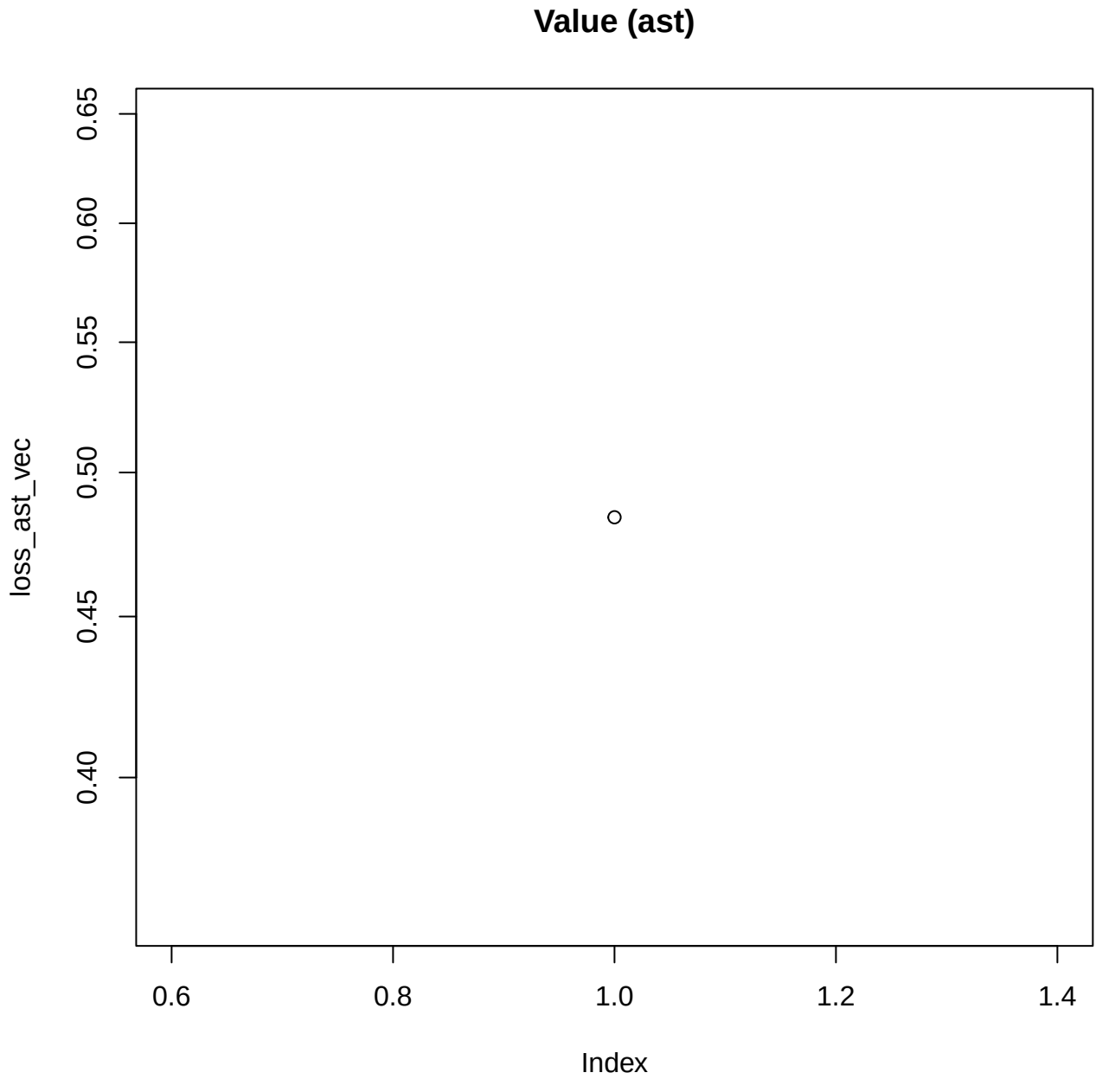
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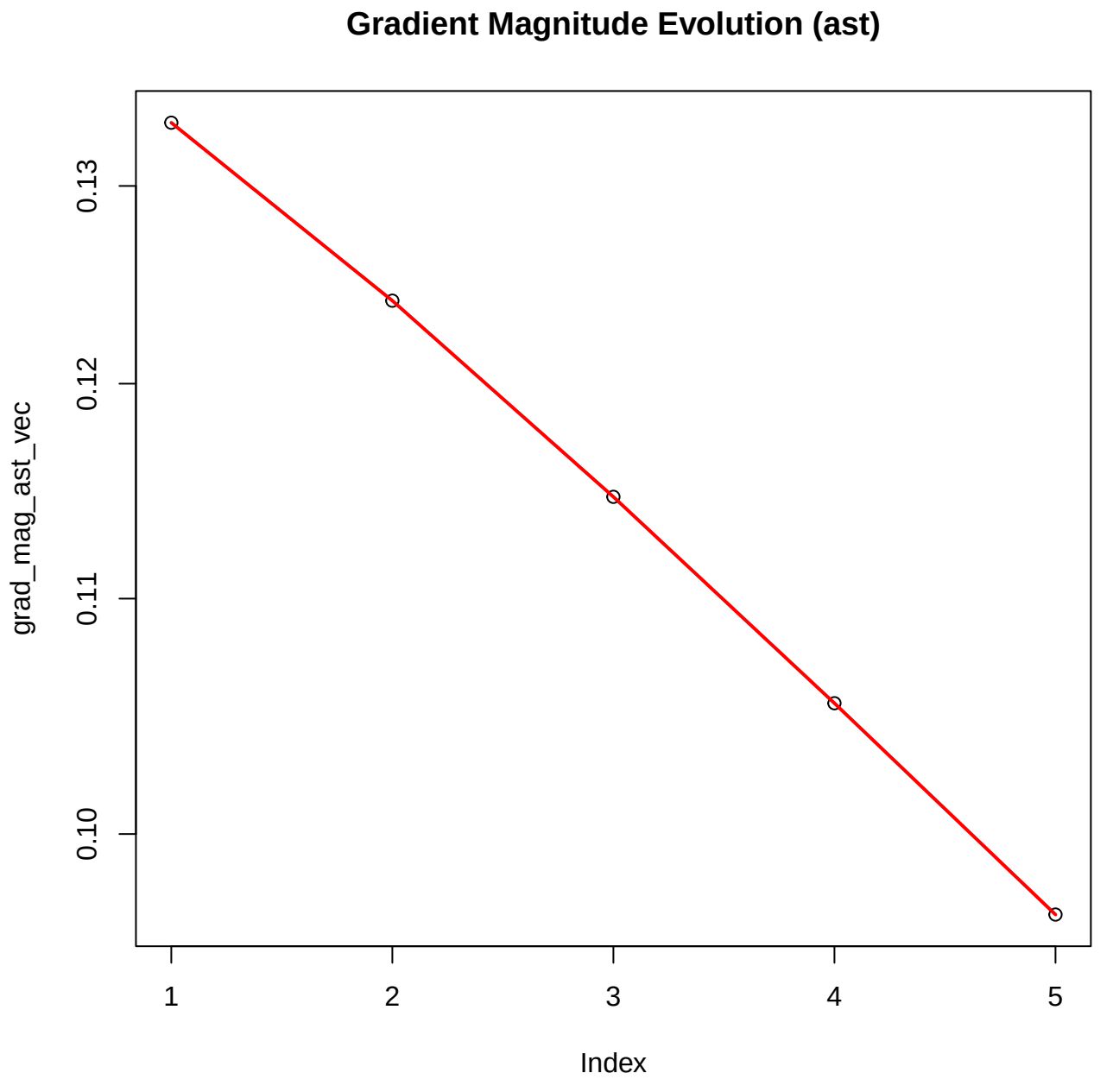
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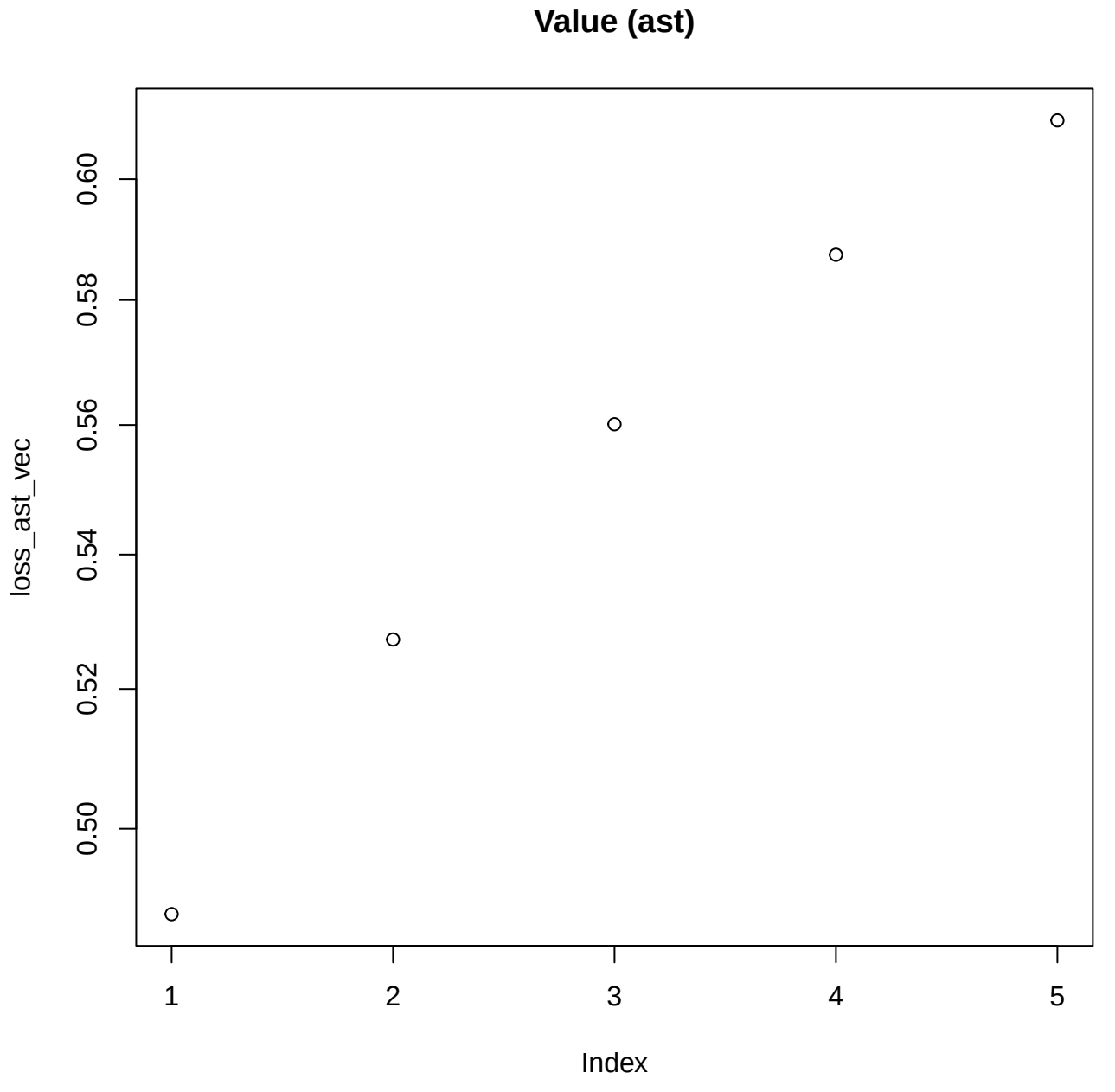
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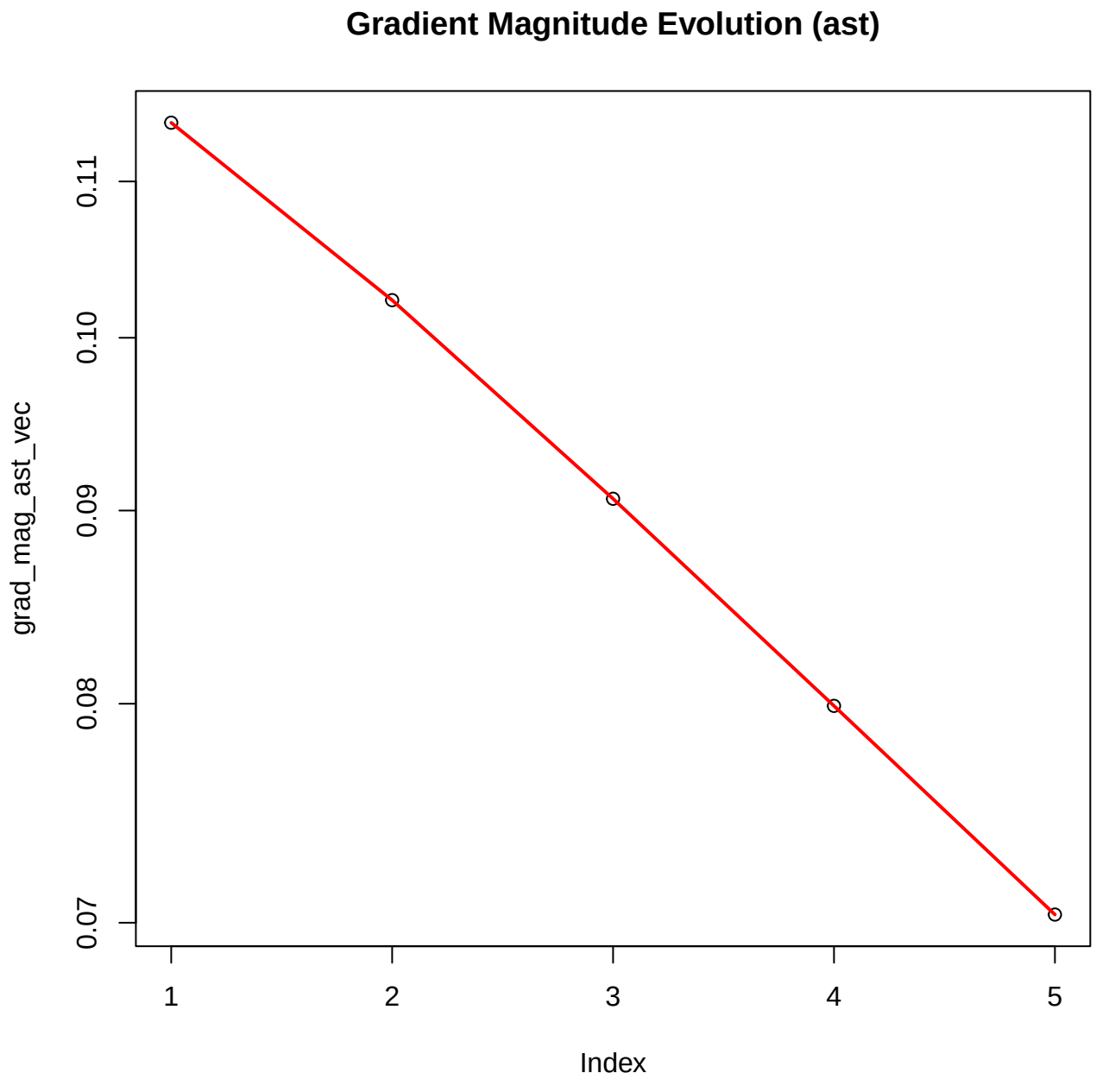
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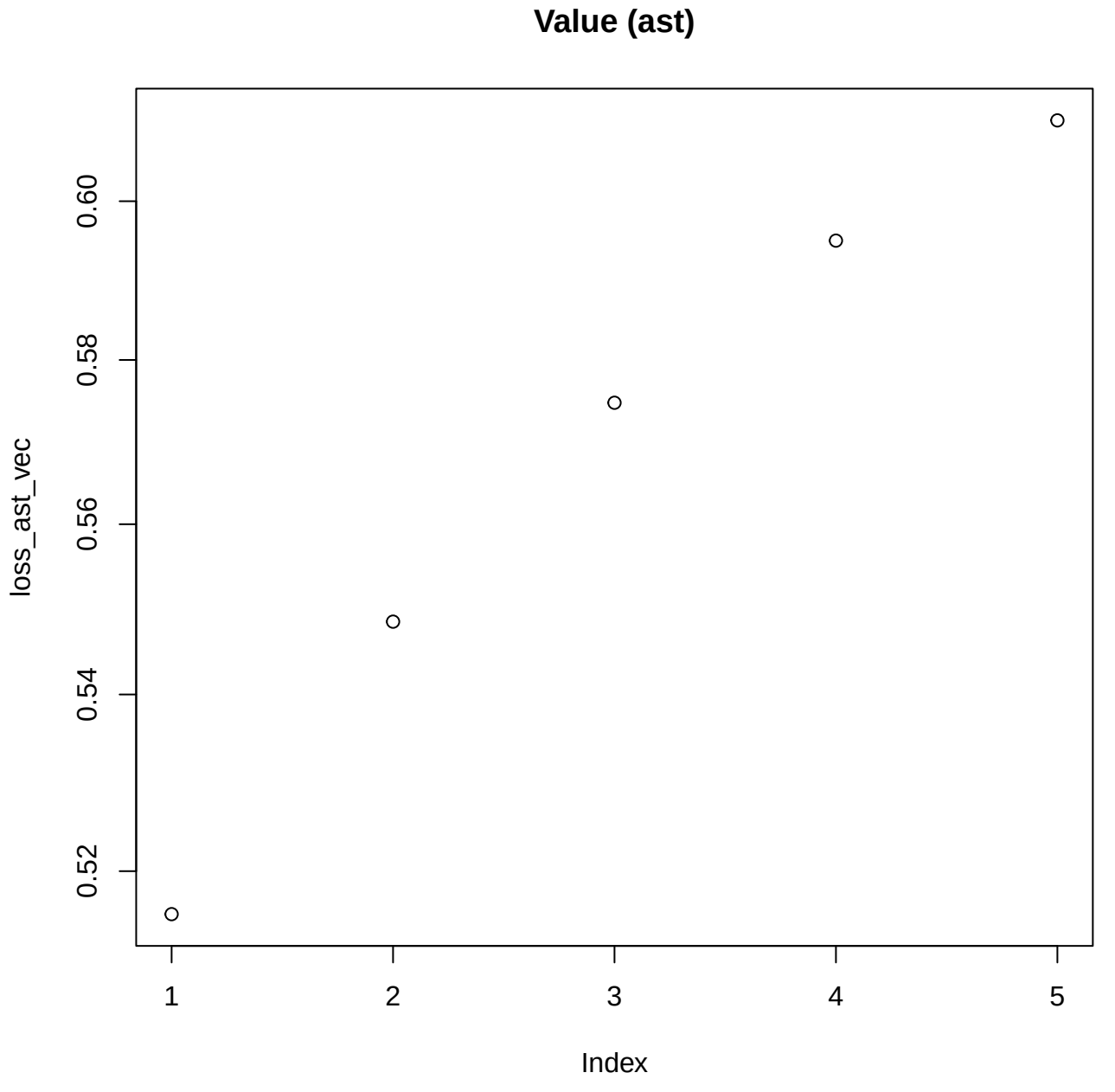
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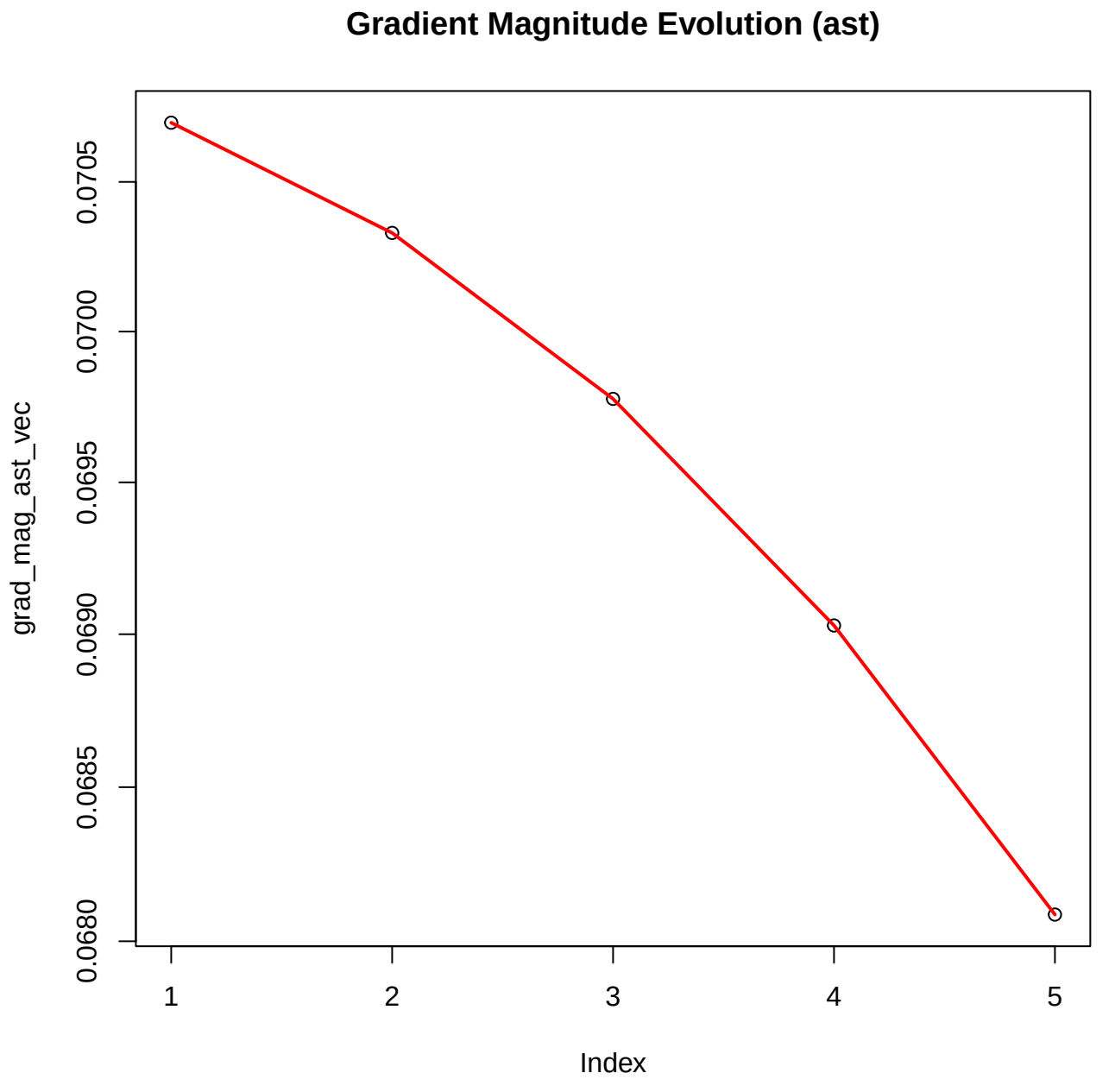
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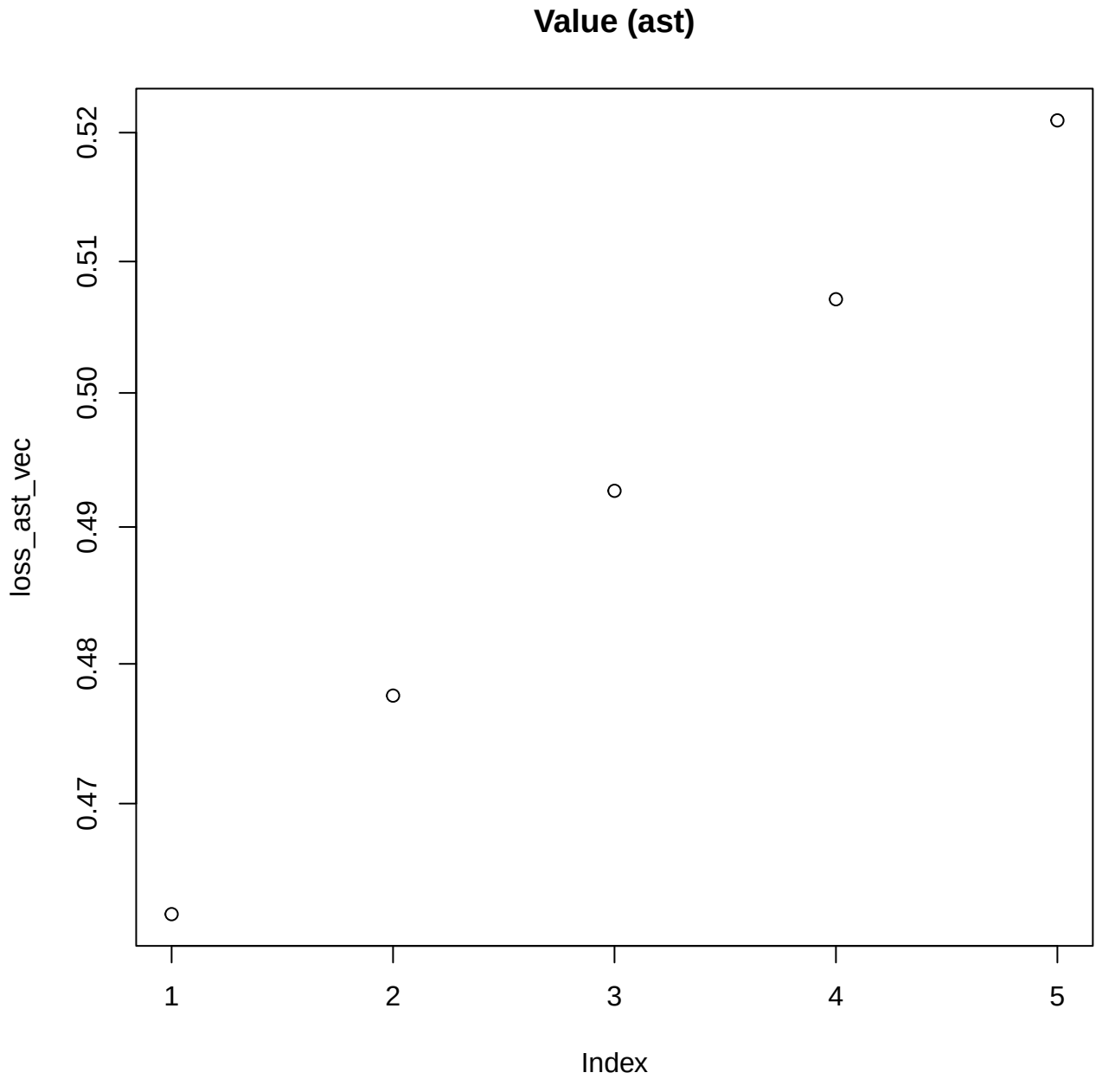
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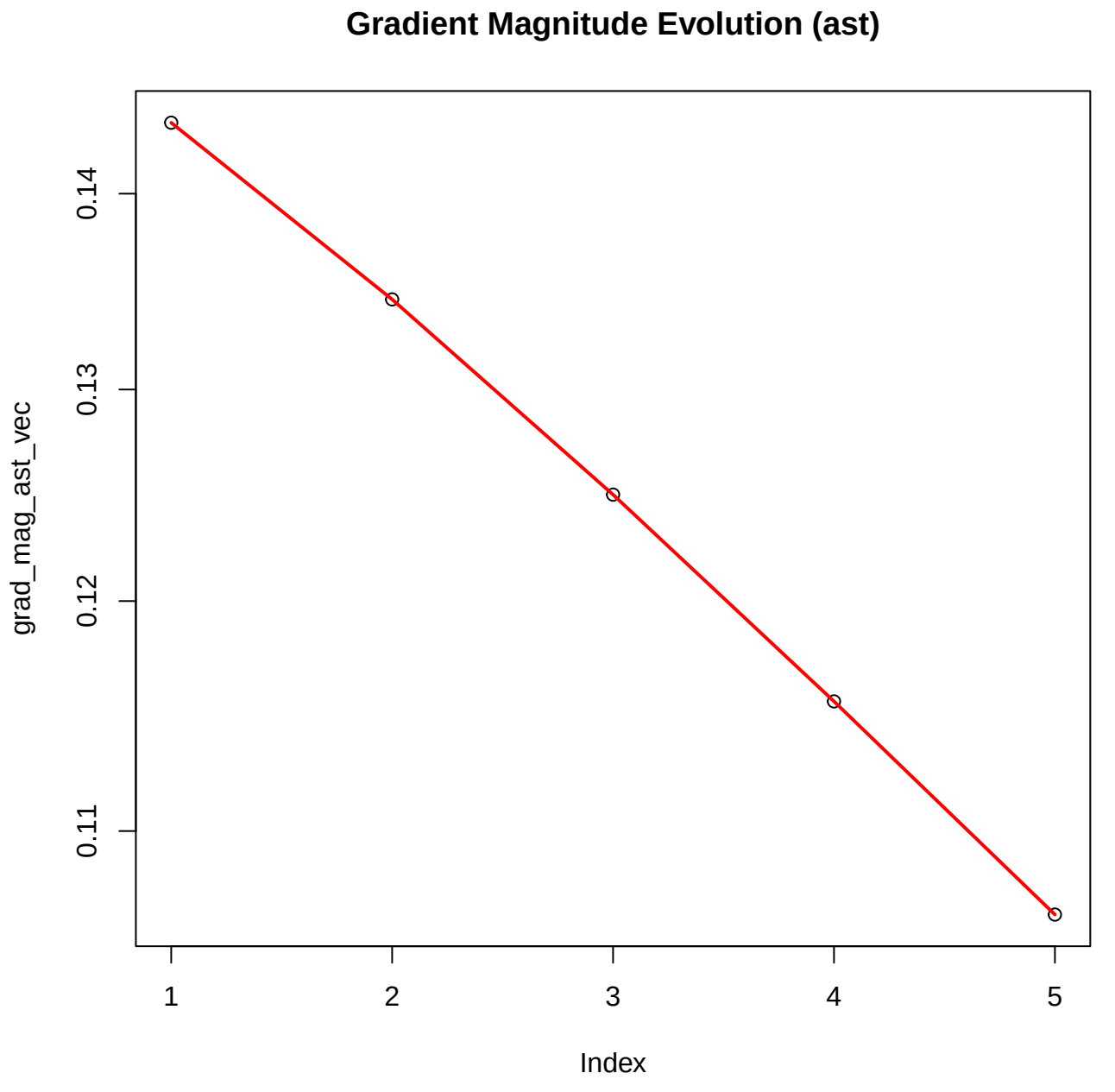
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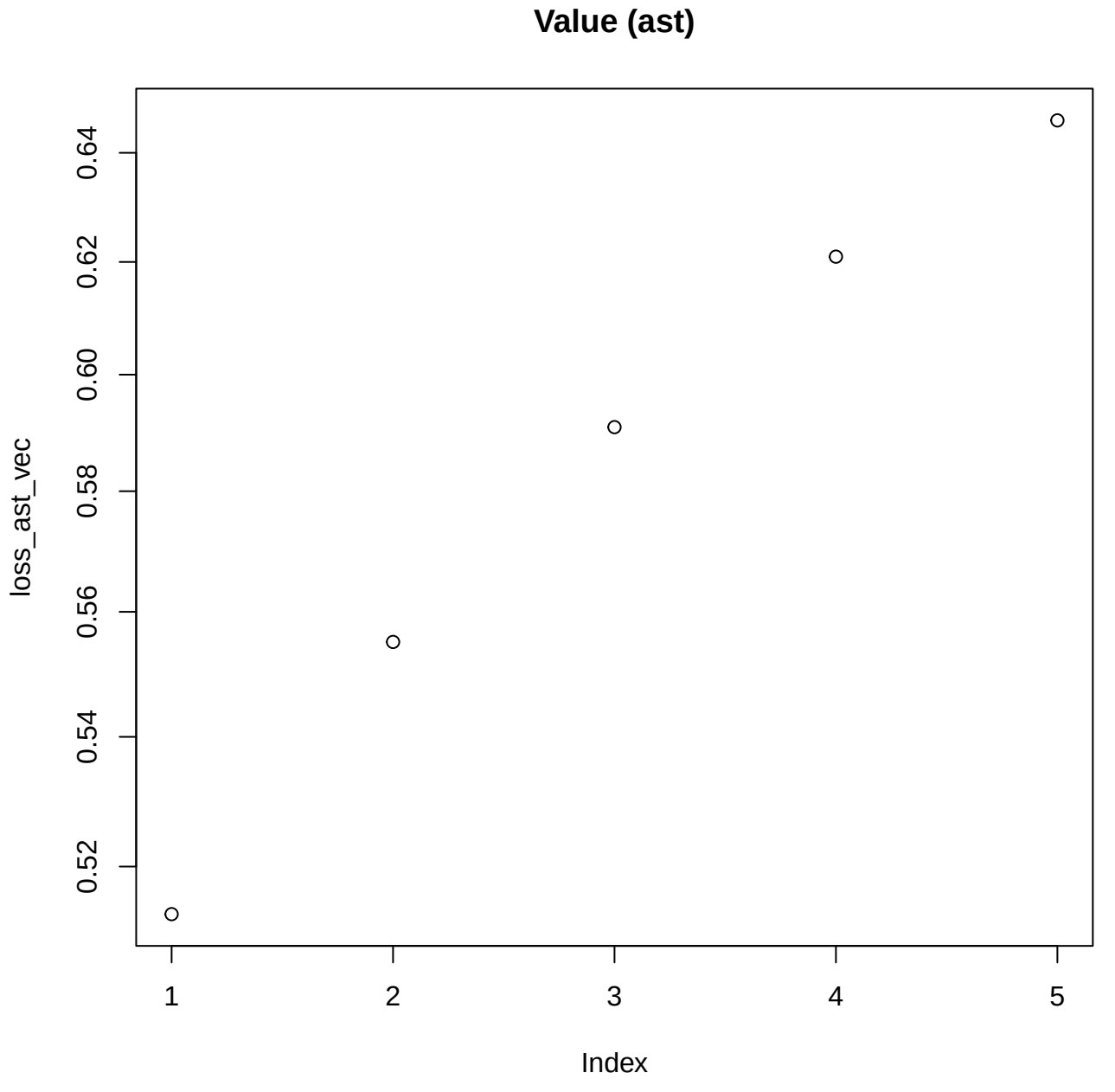
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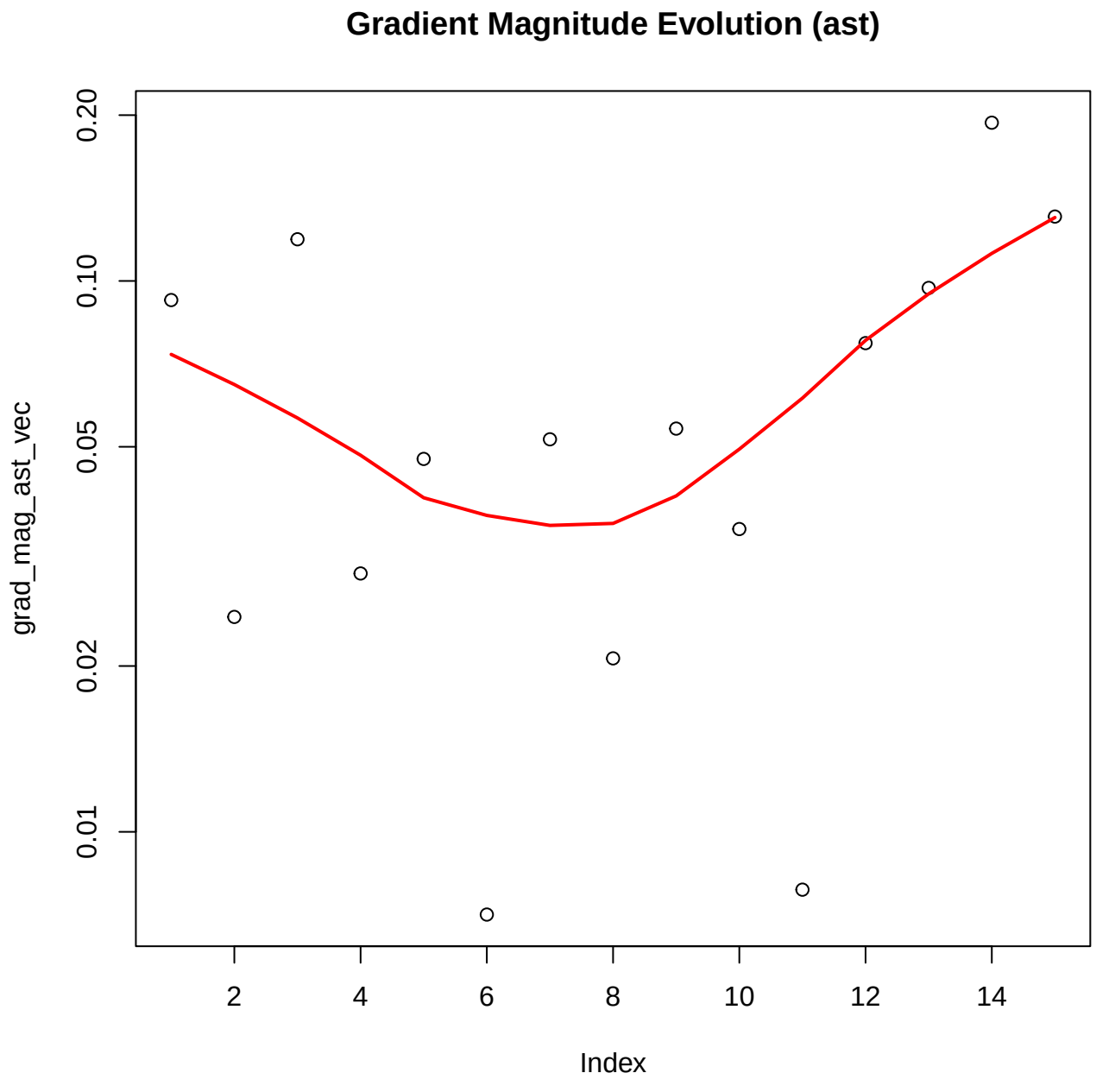
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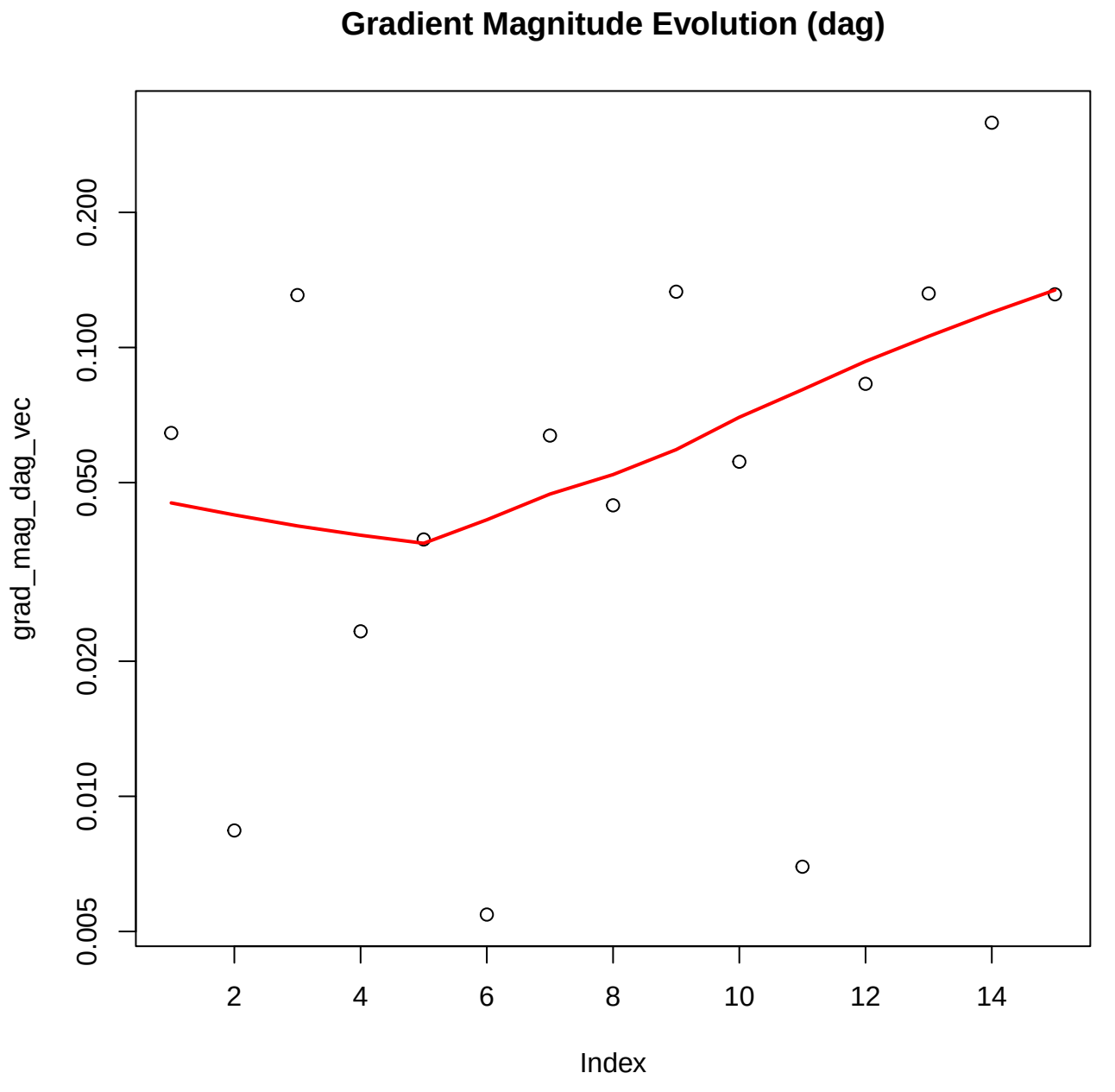
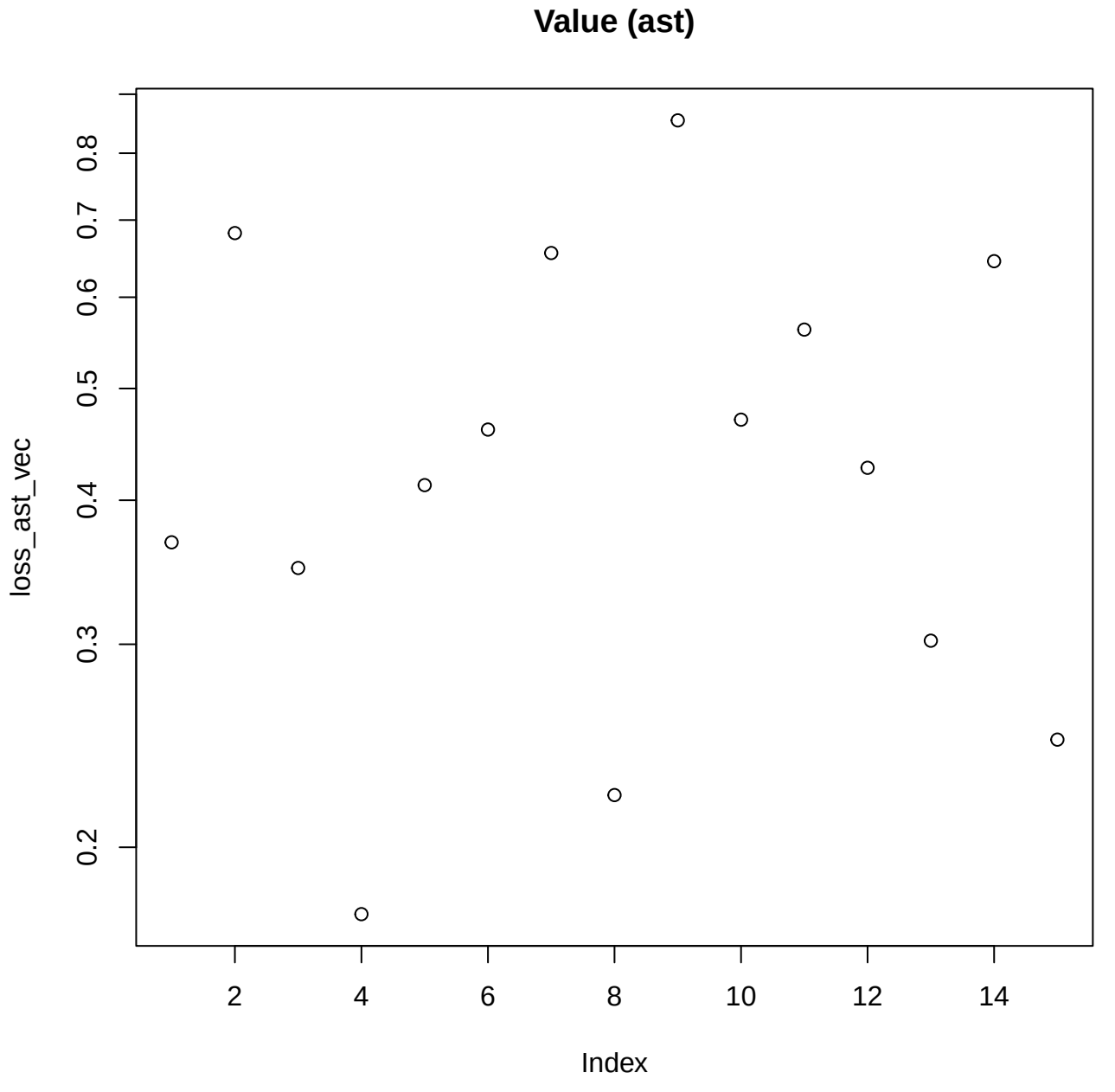
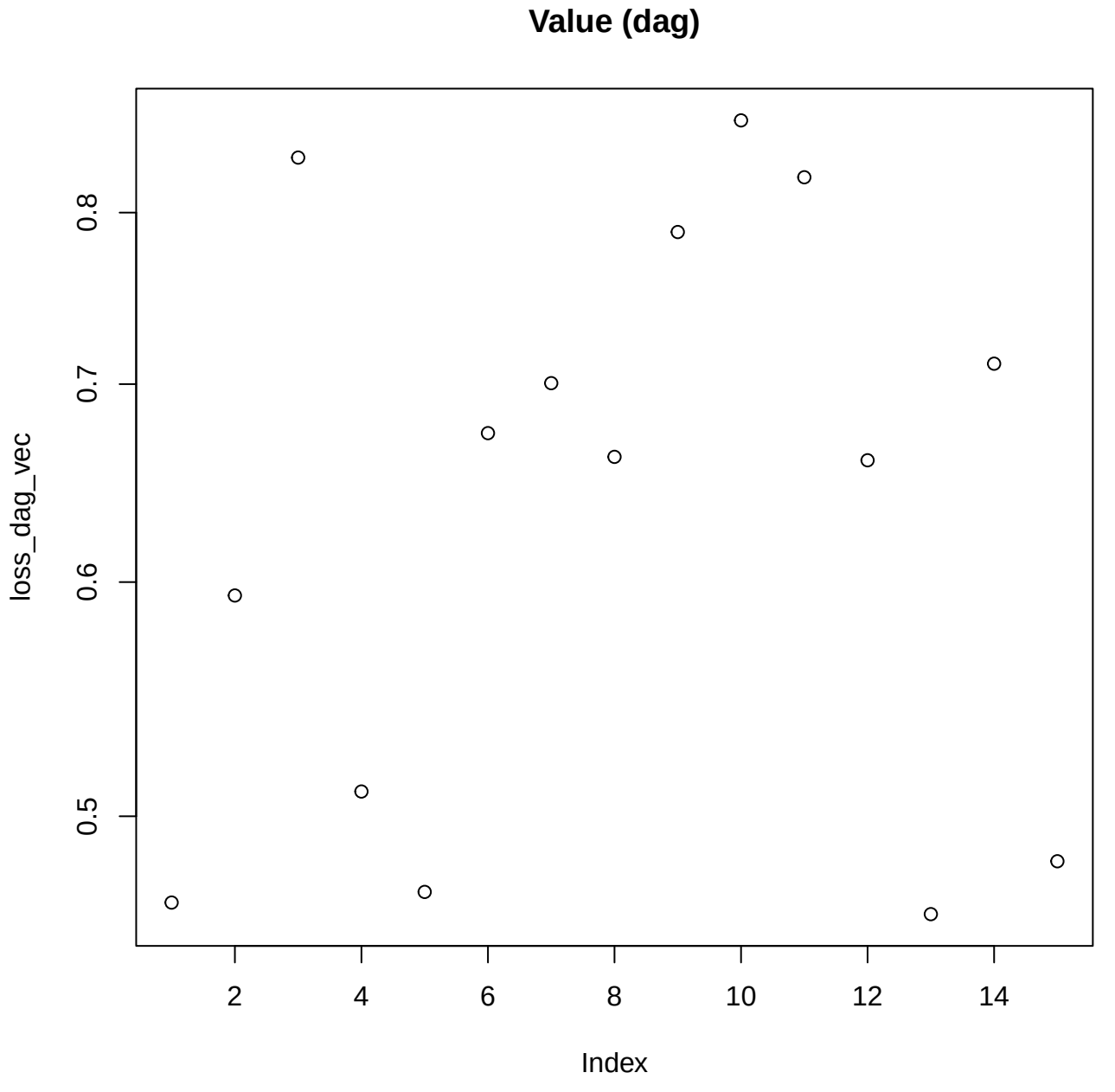
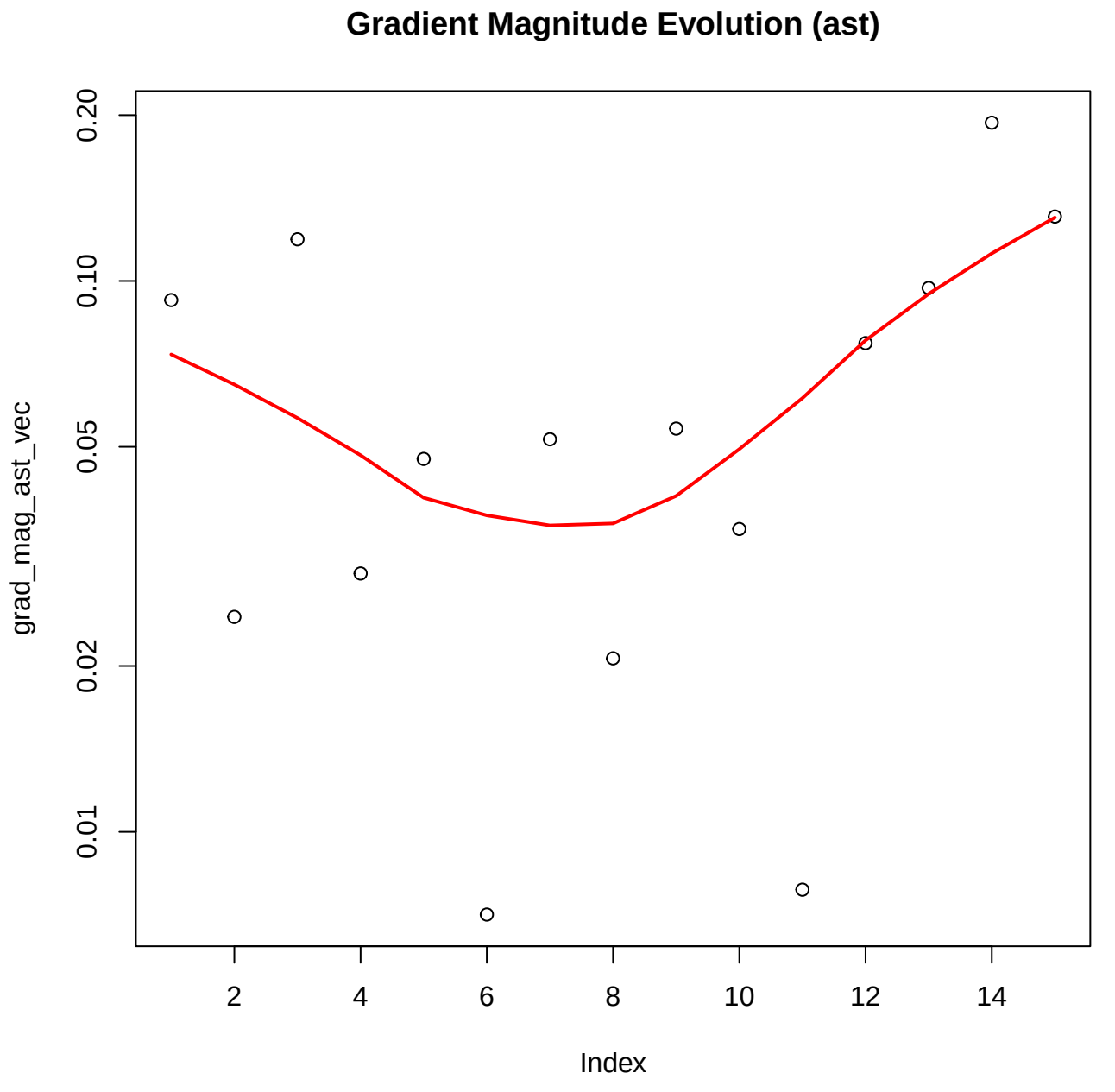
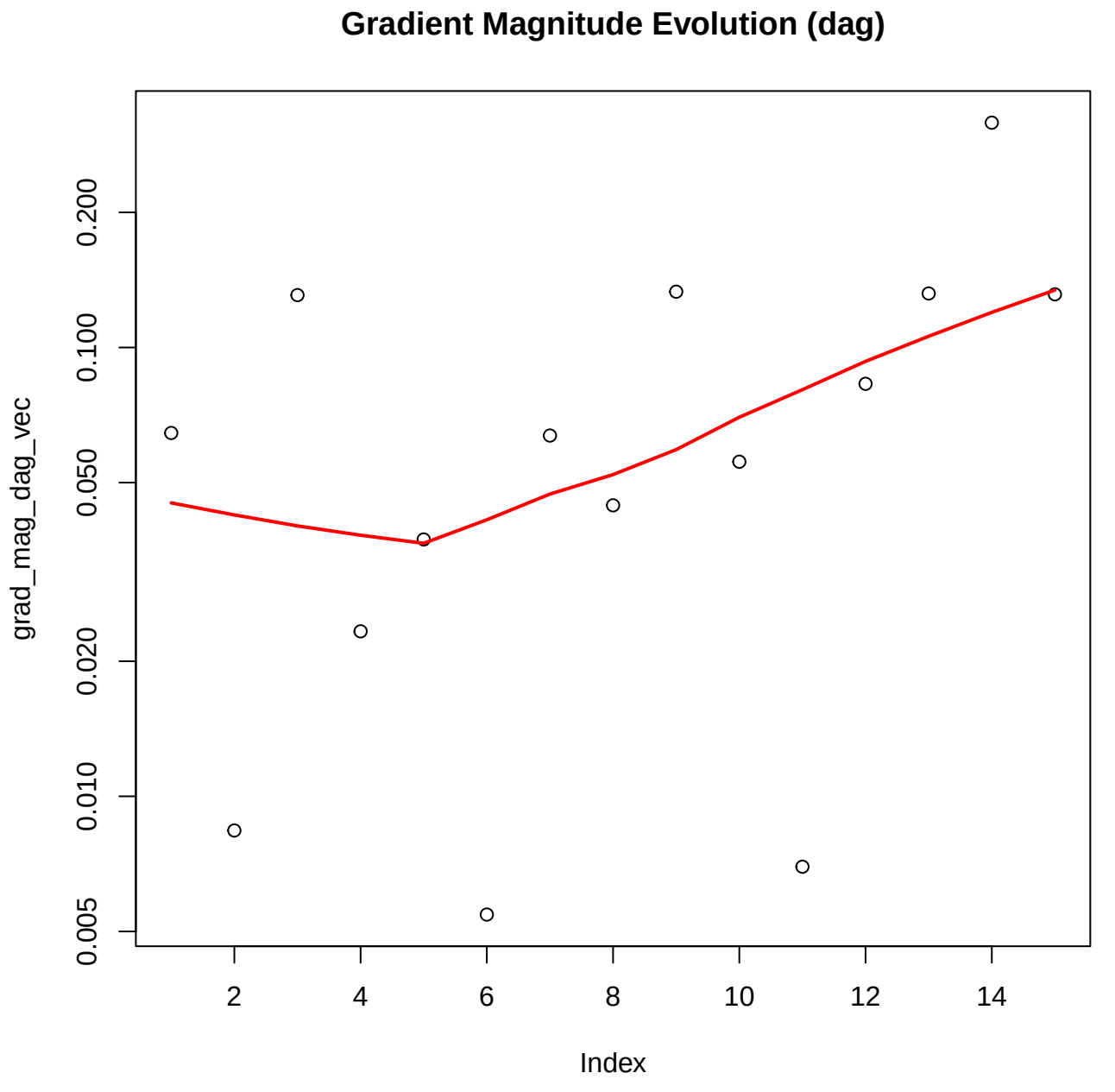

Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
4
6
8
10
12
14
Index


Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index


Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.01
0.02
0.05
0.10
0.20
2
4
6
8
10
12
14
Index


Gradient Magnitude Evolution (dag)
grad_mag_dag_vec
0.005
0.010
0.020
0.050
0.100
0.200
2
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6
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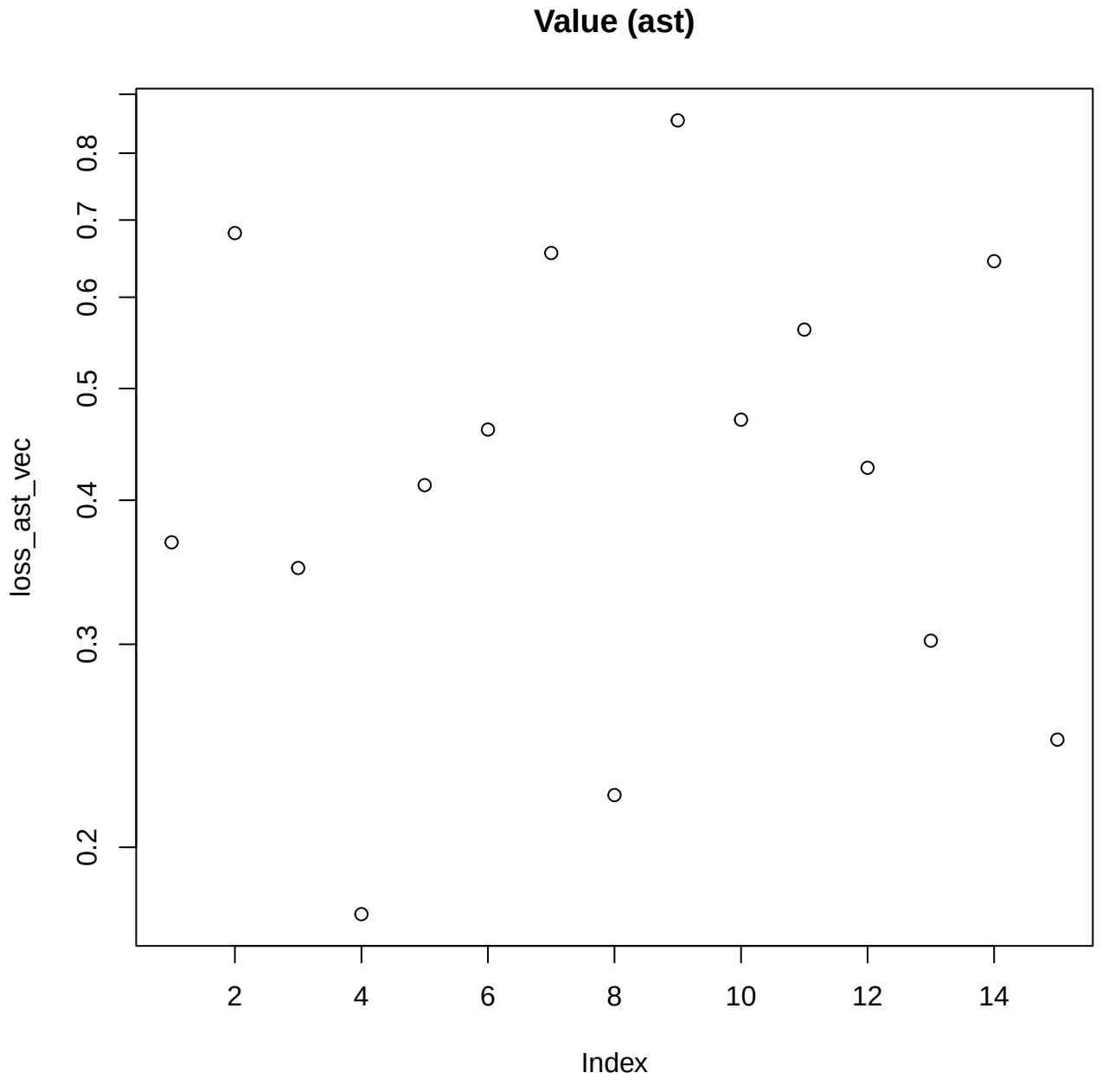
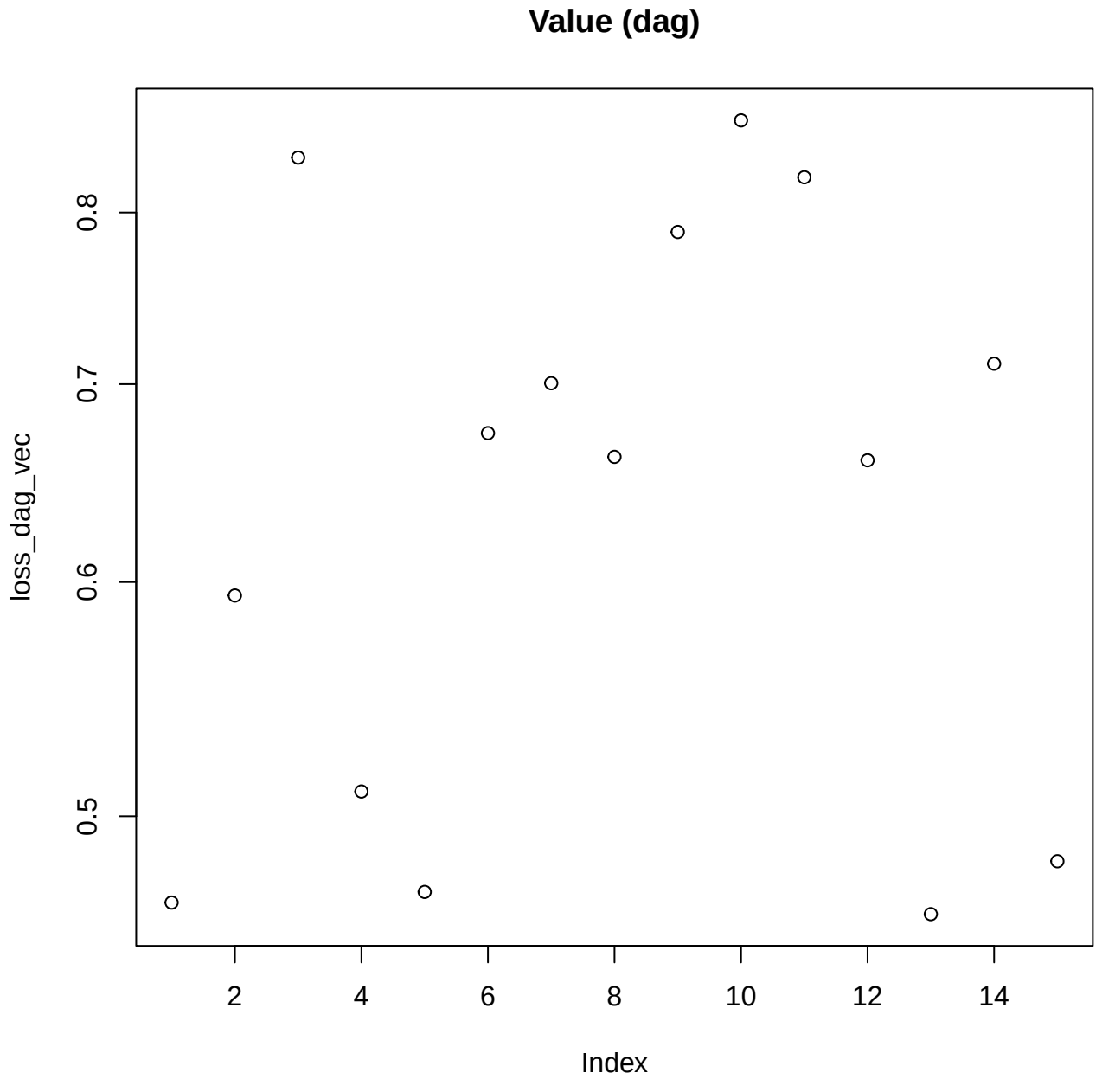
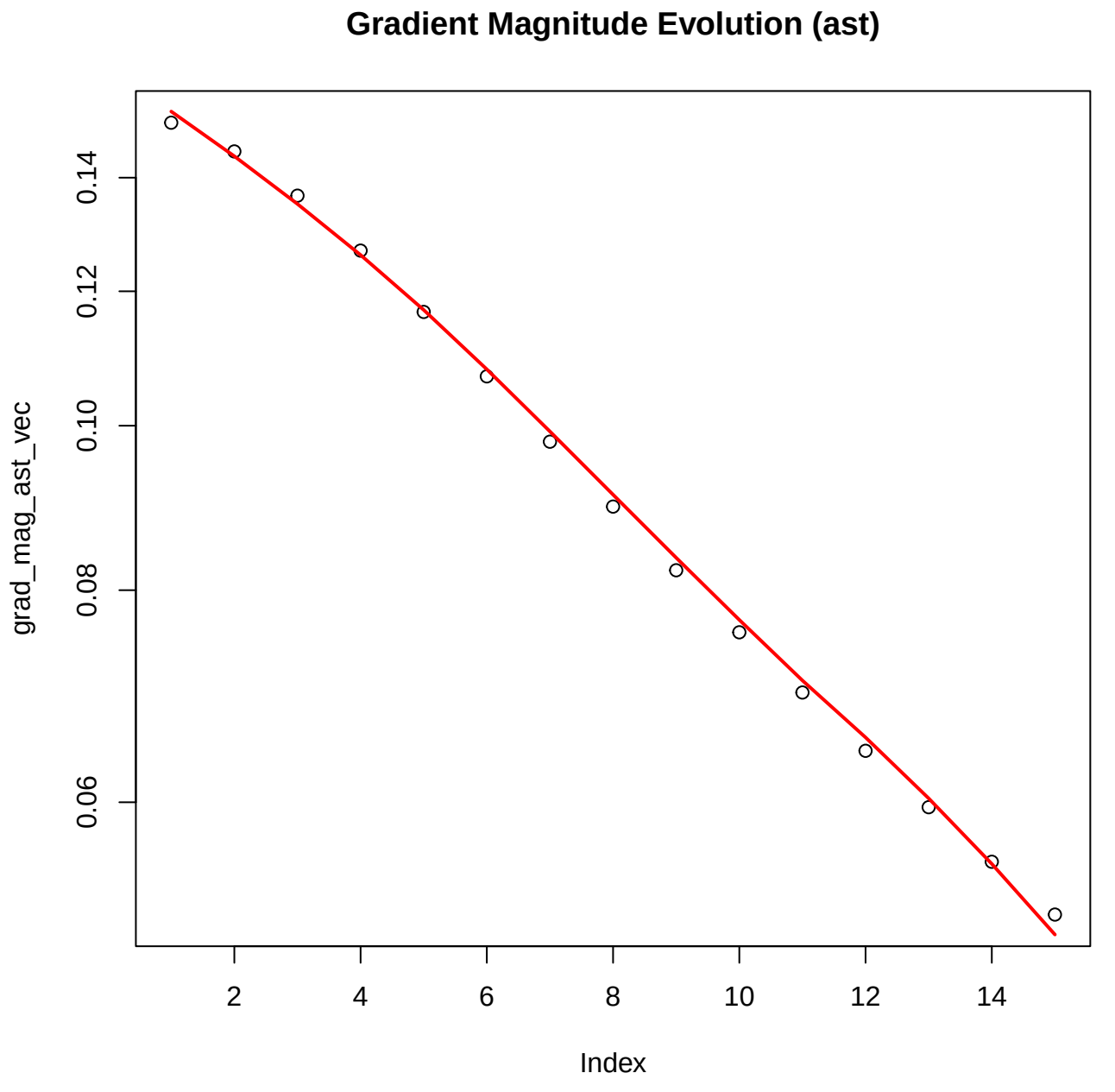
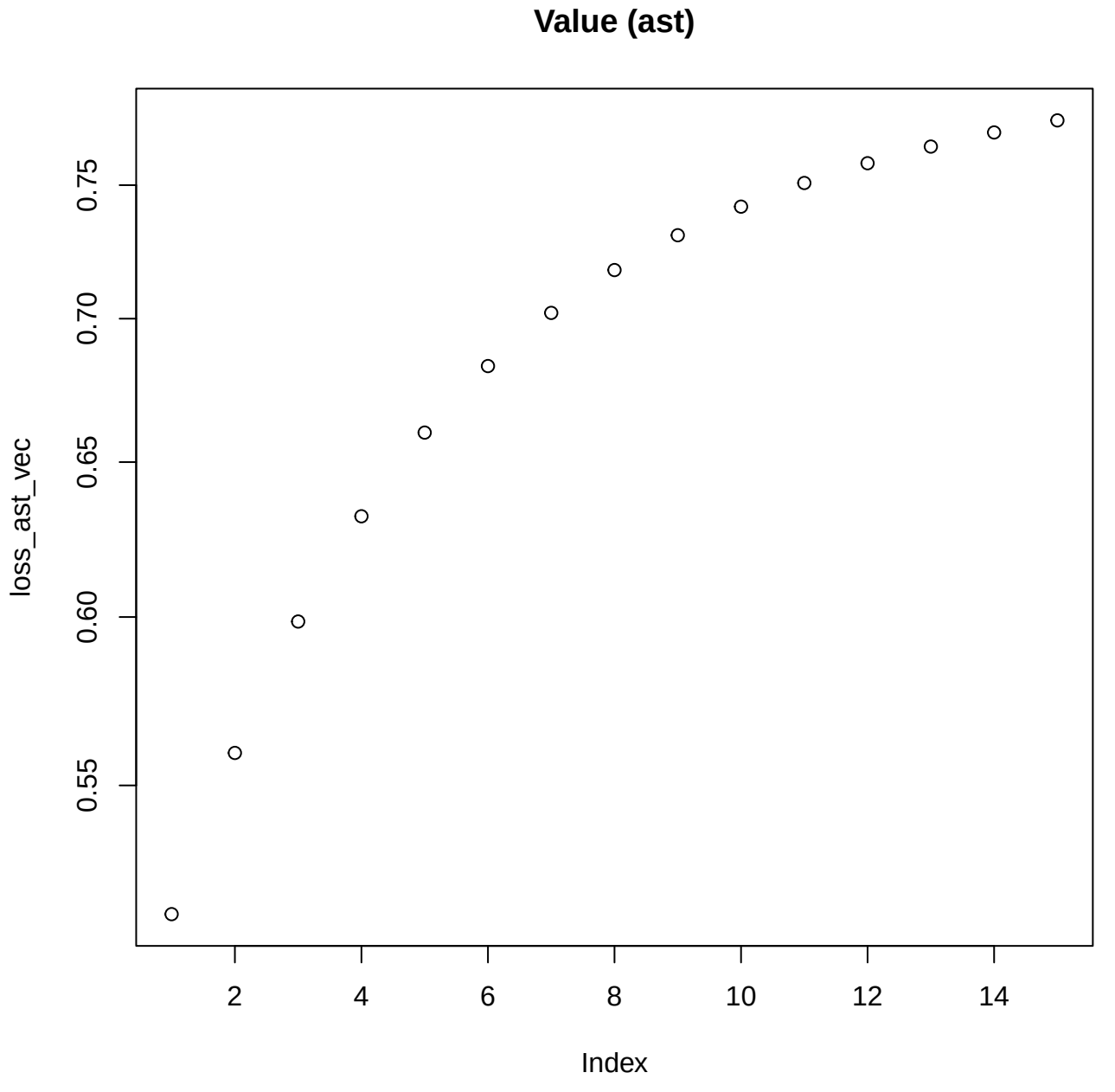
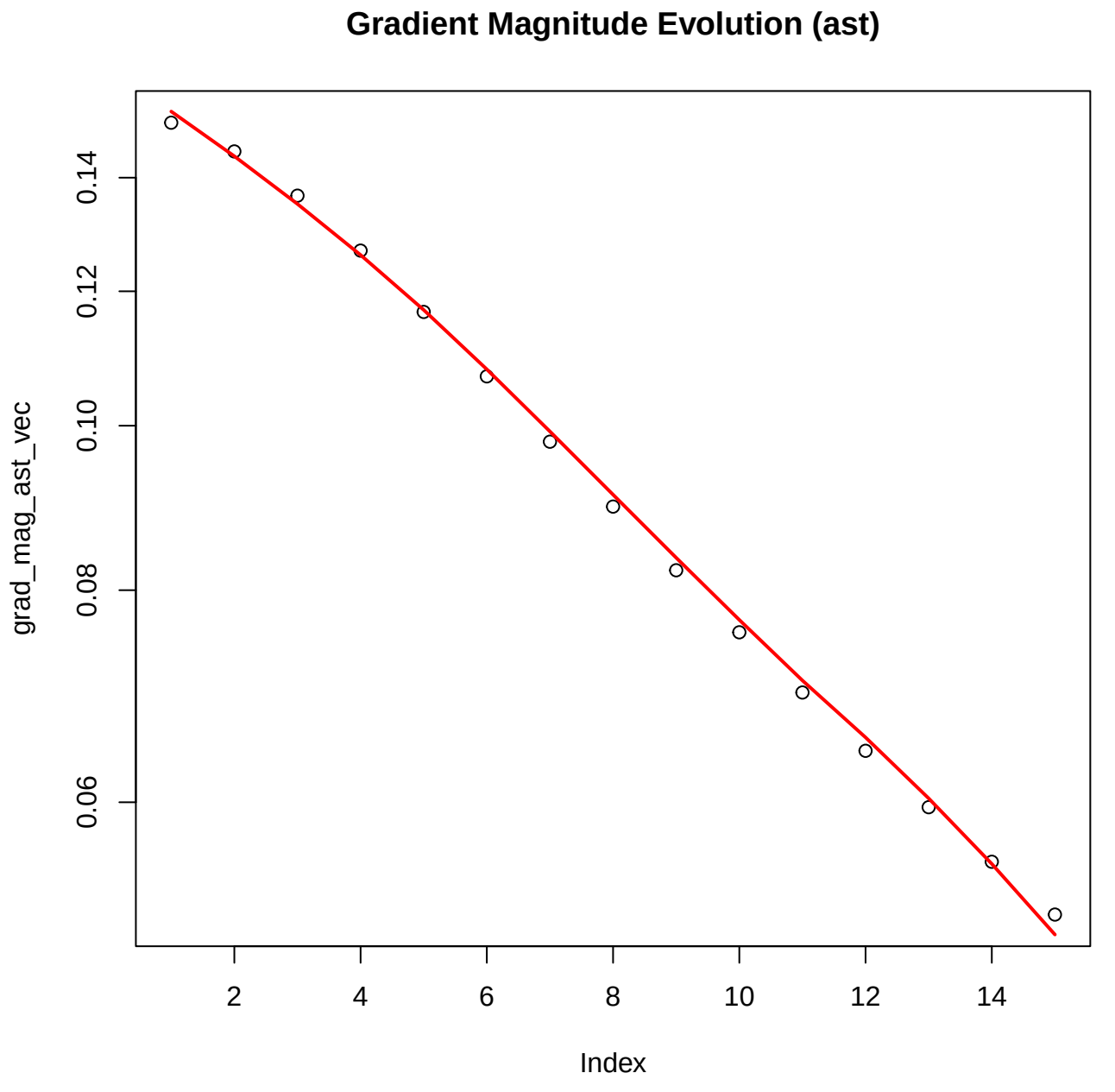
10
12
14
Index

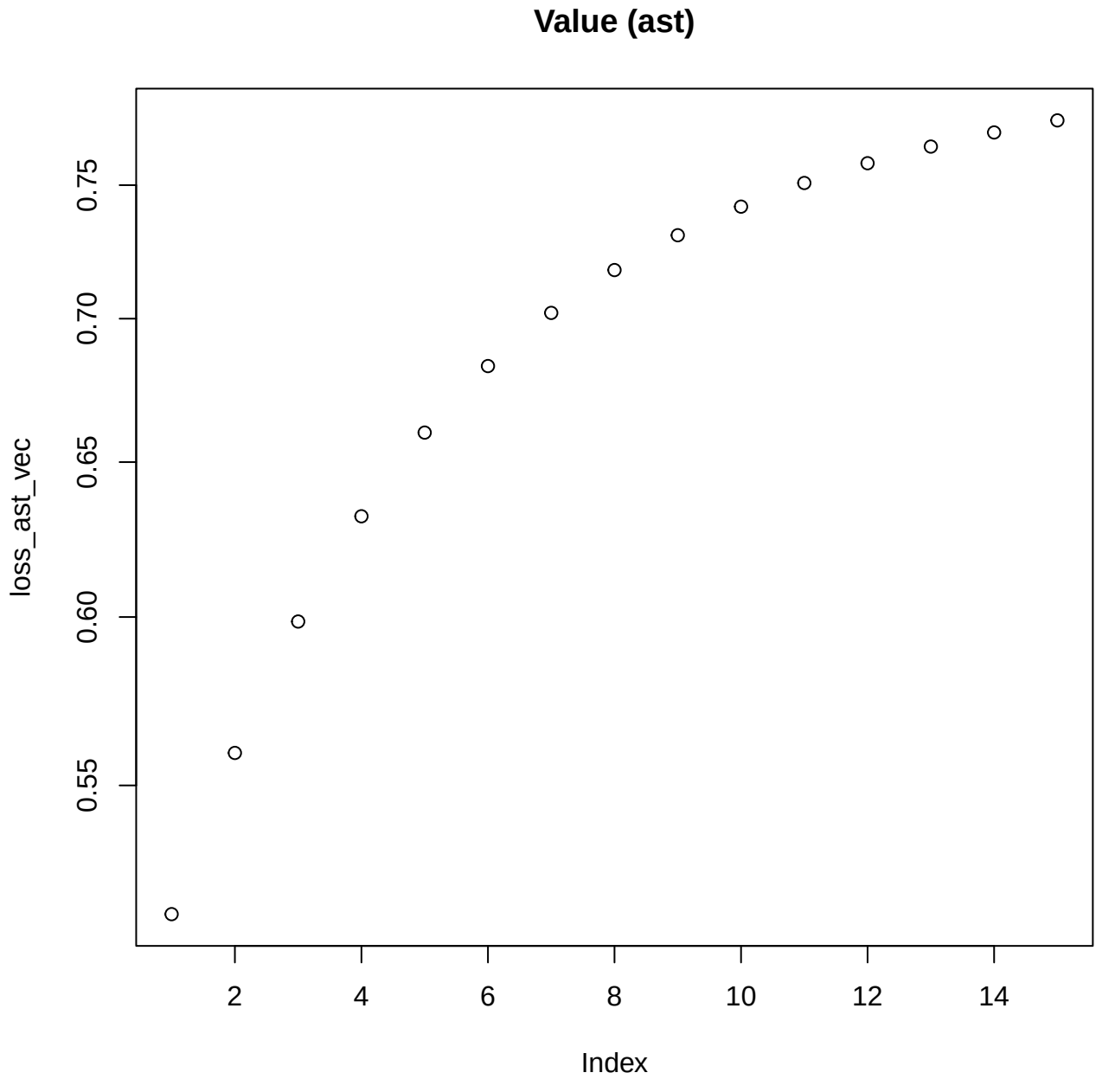

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

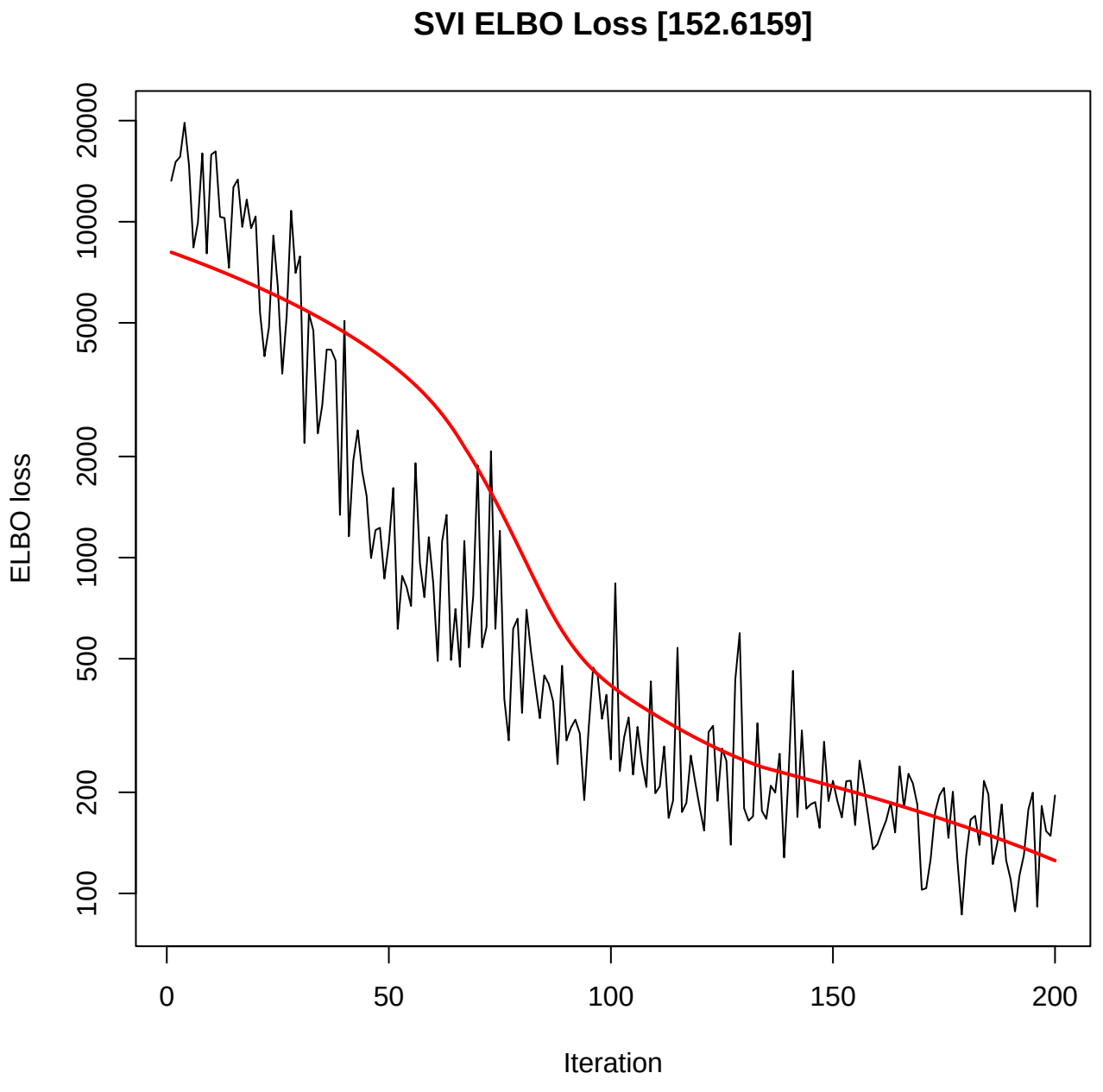

Value (dag)
loss_dag_vec
0.5
0.6
0.7
0.8
2
4
6
8
10
12
14
Index

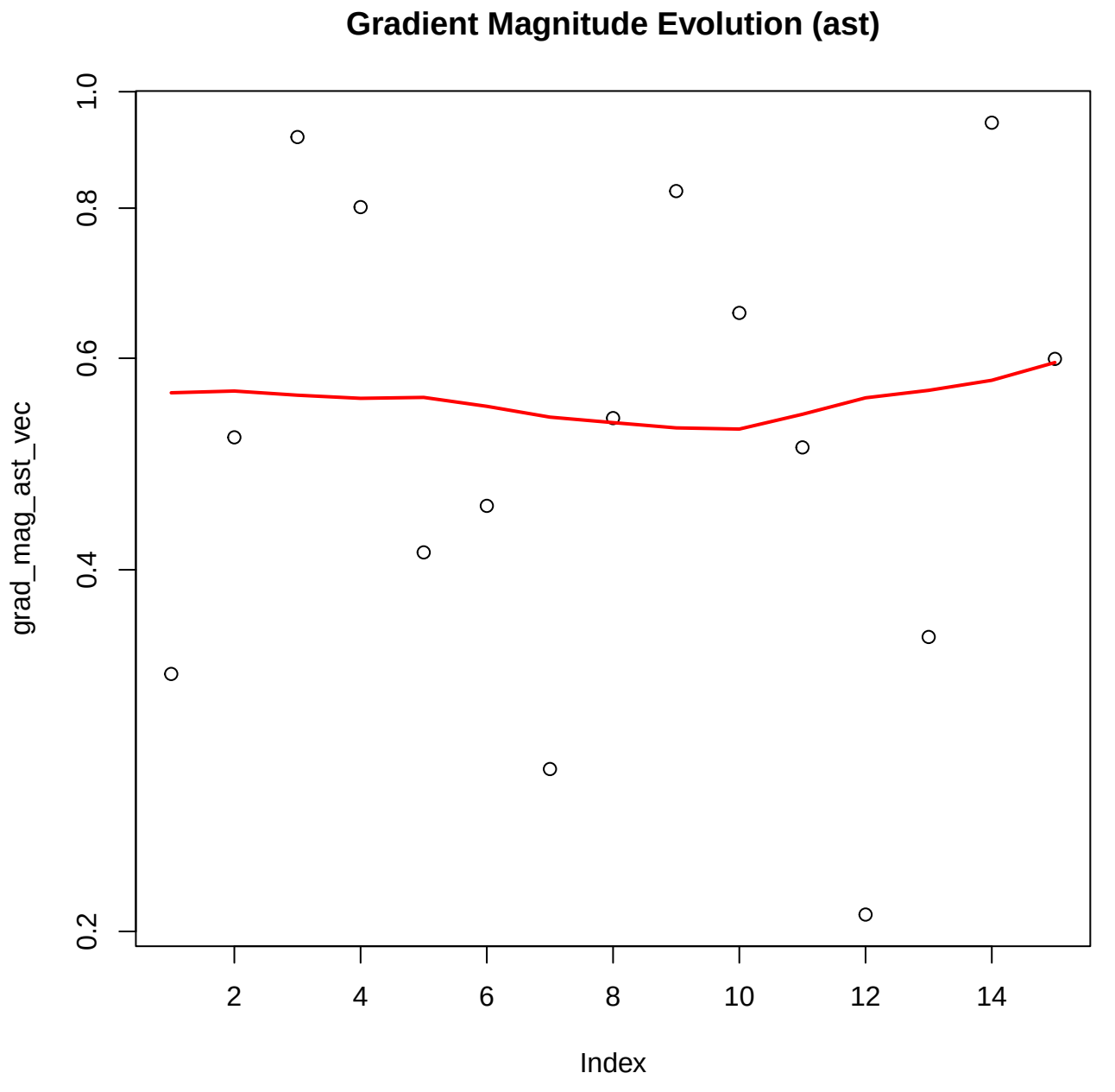

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

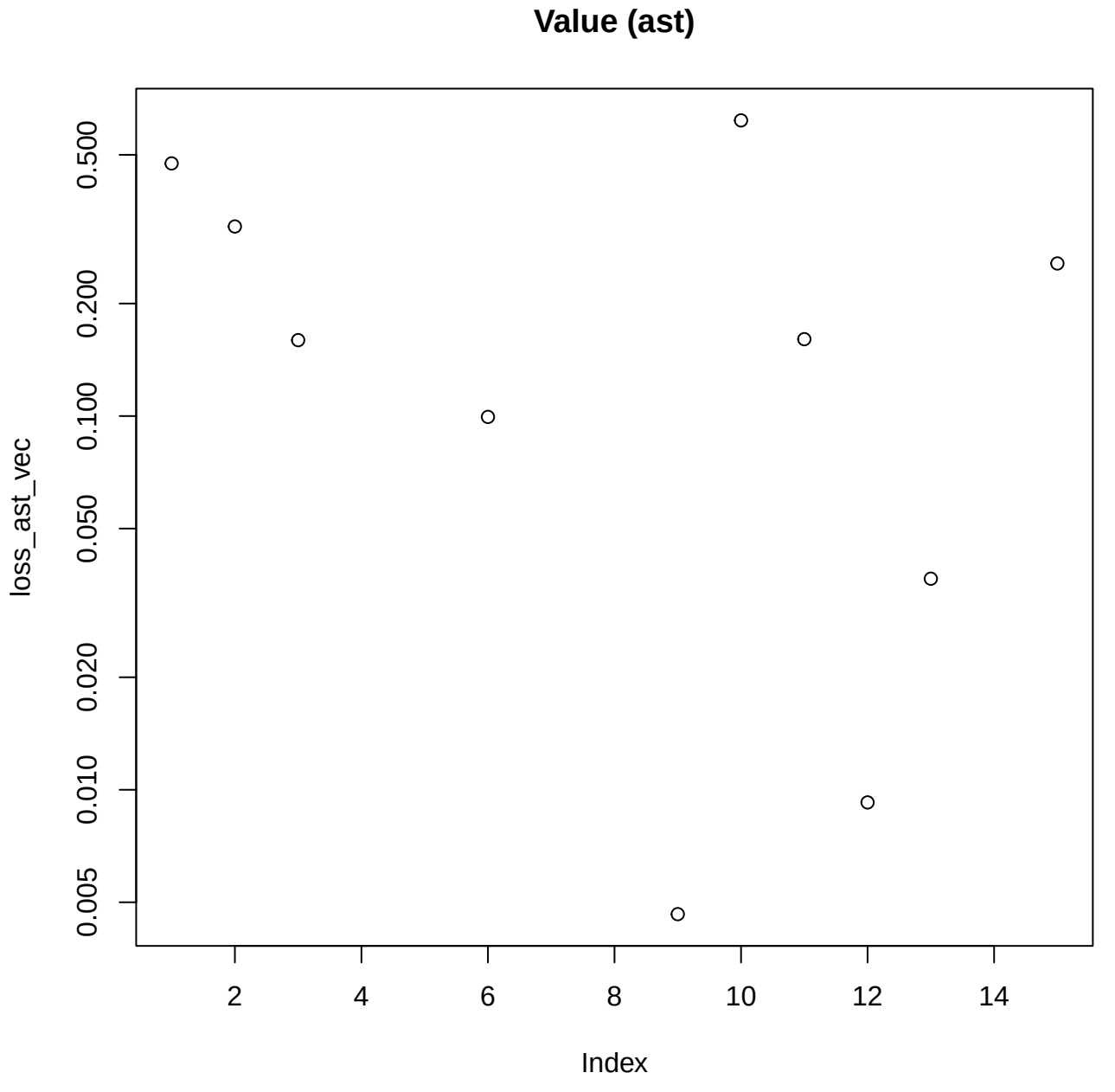

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

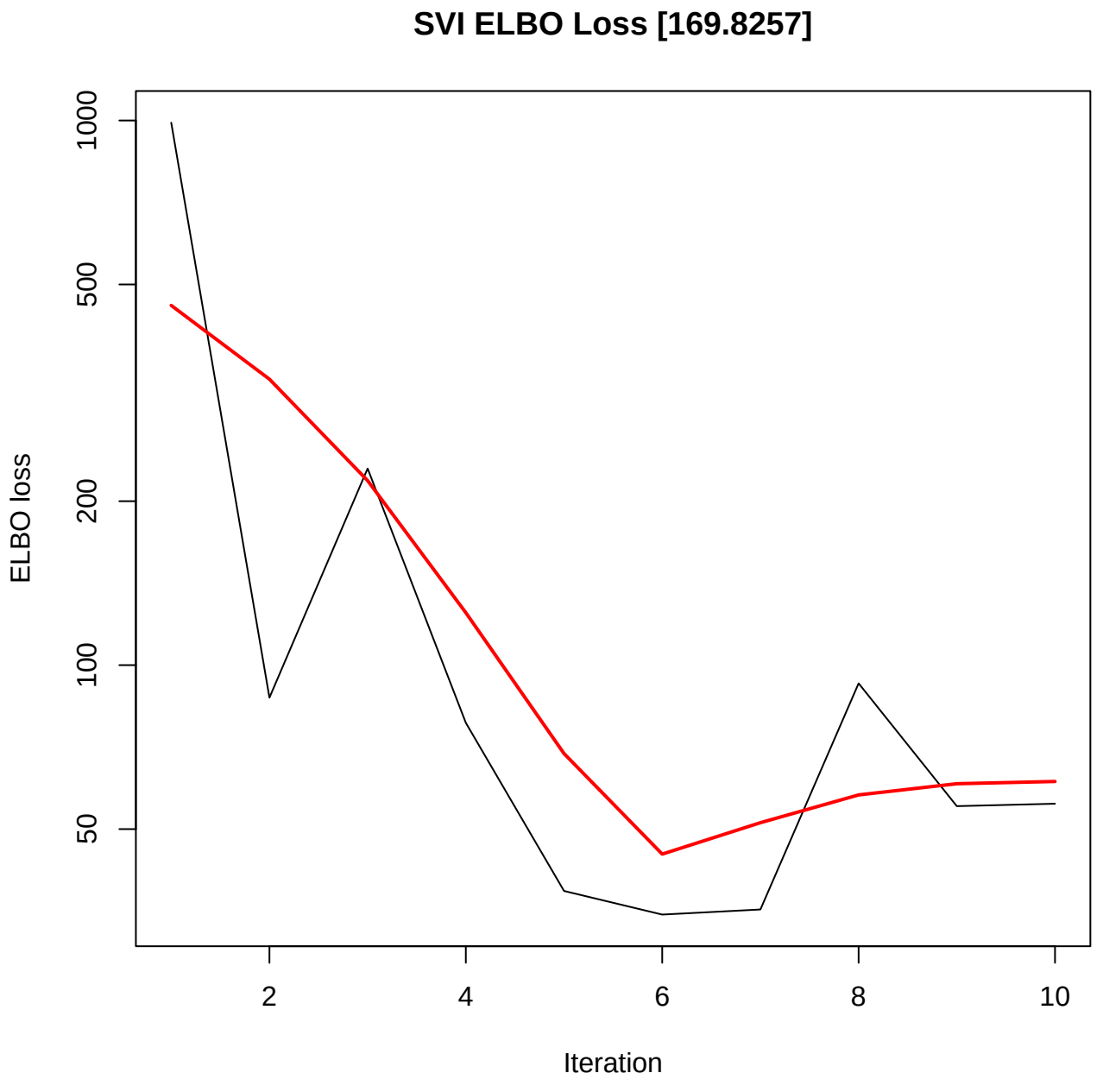

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.06
0.08
0.10
0.12
0.14
2
4
6
8
10
12
14
Index

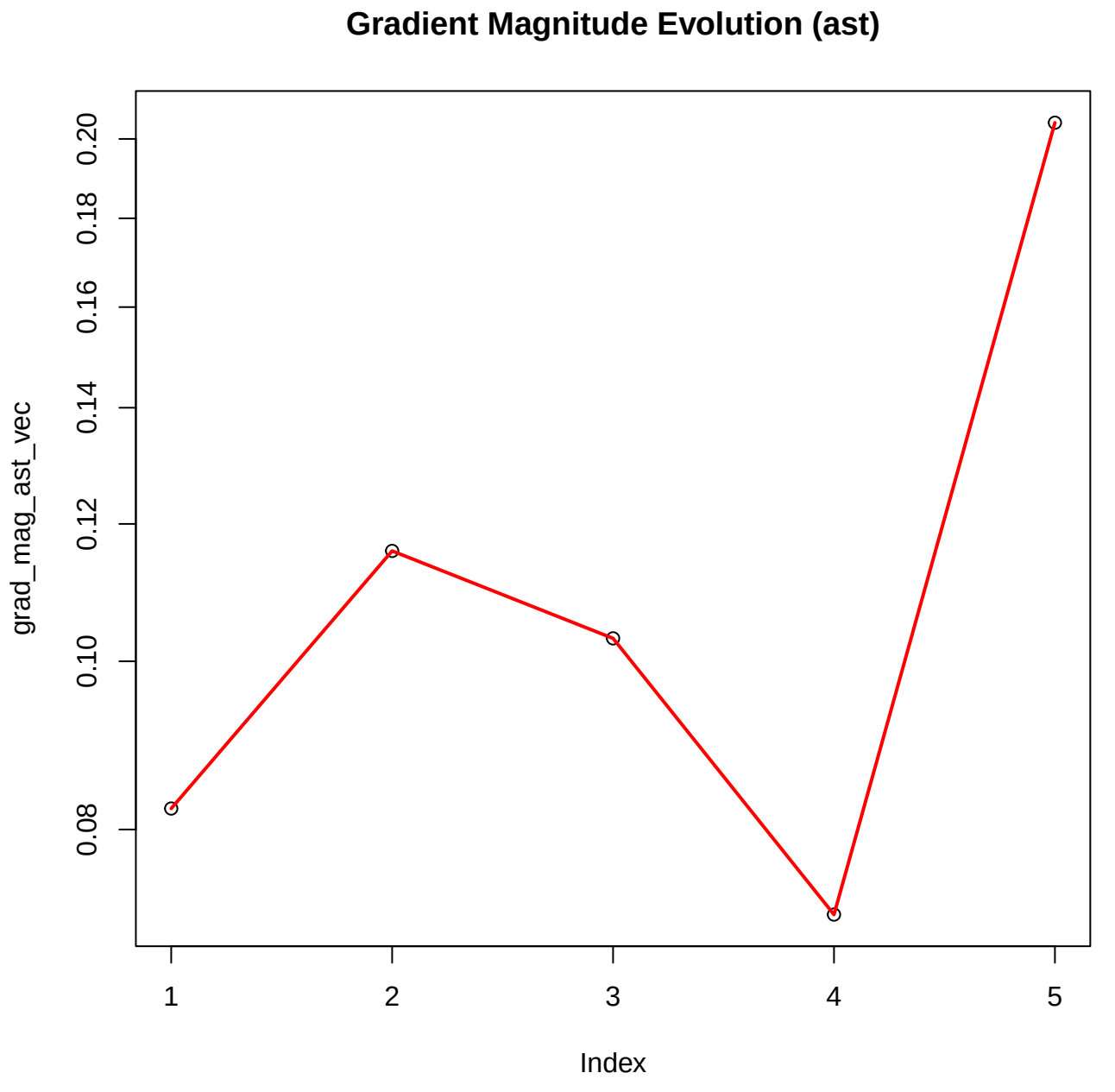

Value (ast)
loss_ast_vec
0.55
0.60
0.65
0.70
0.75
2
4
6
8
10
12
14
Index

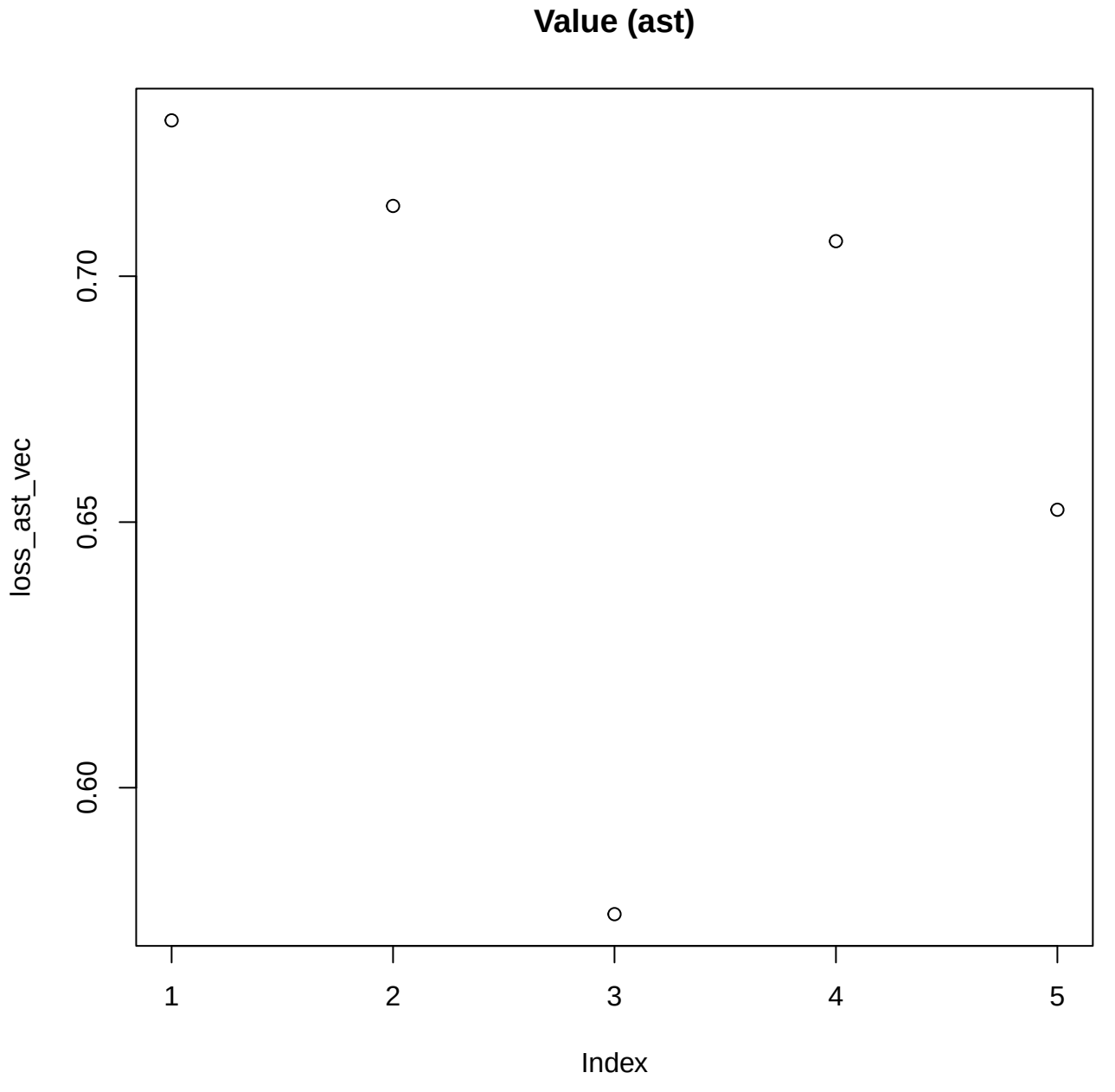

SVI ELBO Loss [152.6159]
ELBO loss
20000
10000
5000
2000
1000
500
200
100
0
50
100
150
200
Iteration

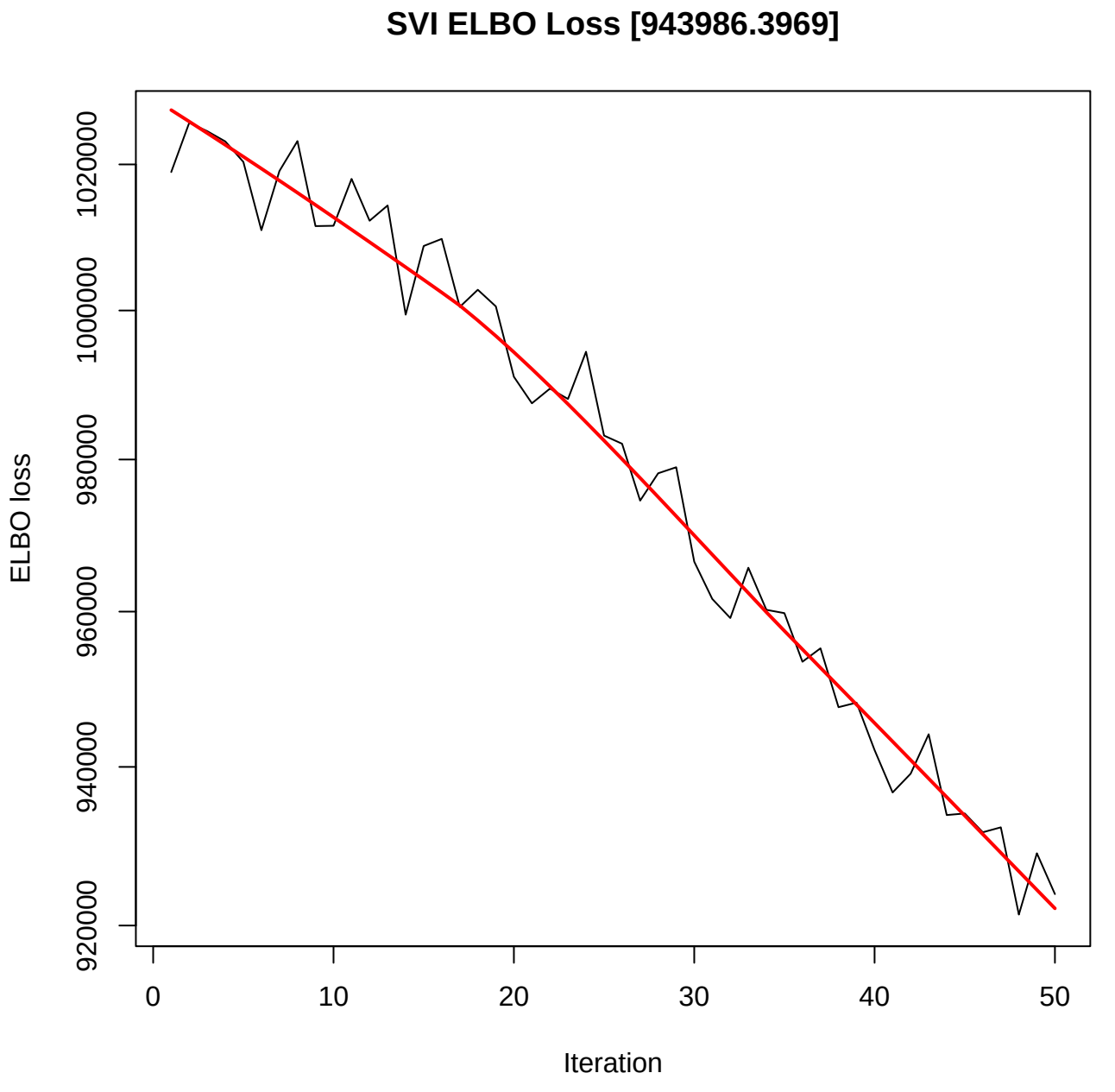

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.2
0.4
0.6
0.8
1.0
2
4
6
8
10
12
14
Index

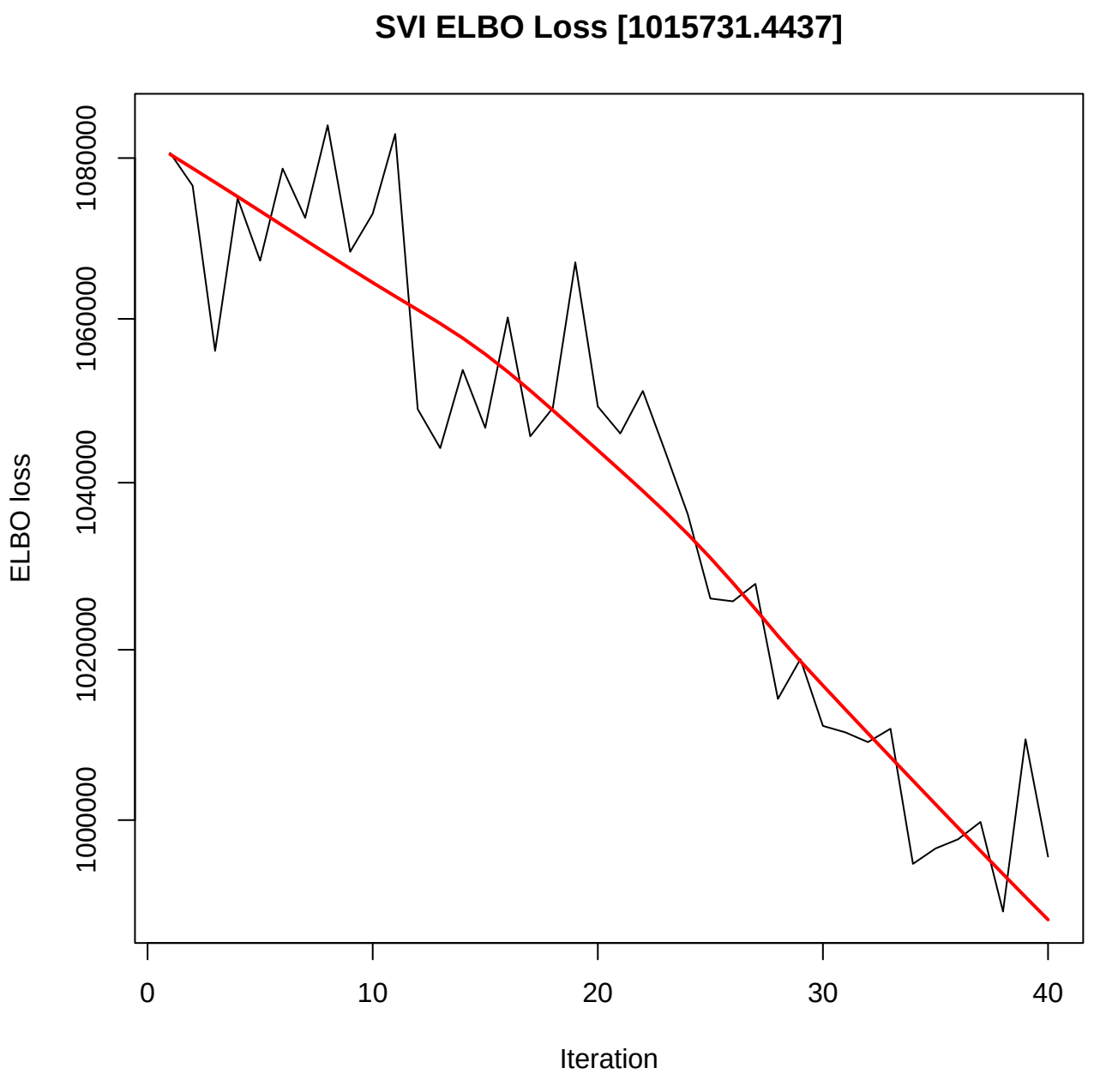

Value (ast)
loss_ast_vec
0.005
0.010
0.020
0.050
0.100
0.200
0.500
2
4
6
8
10
12
14
Index

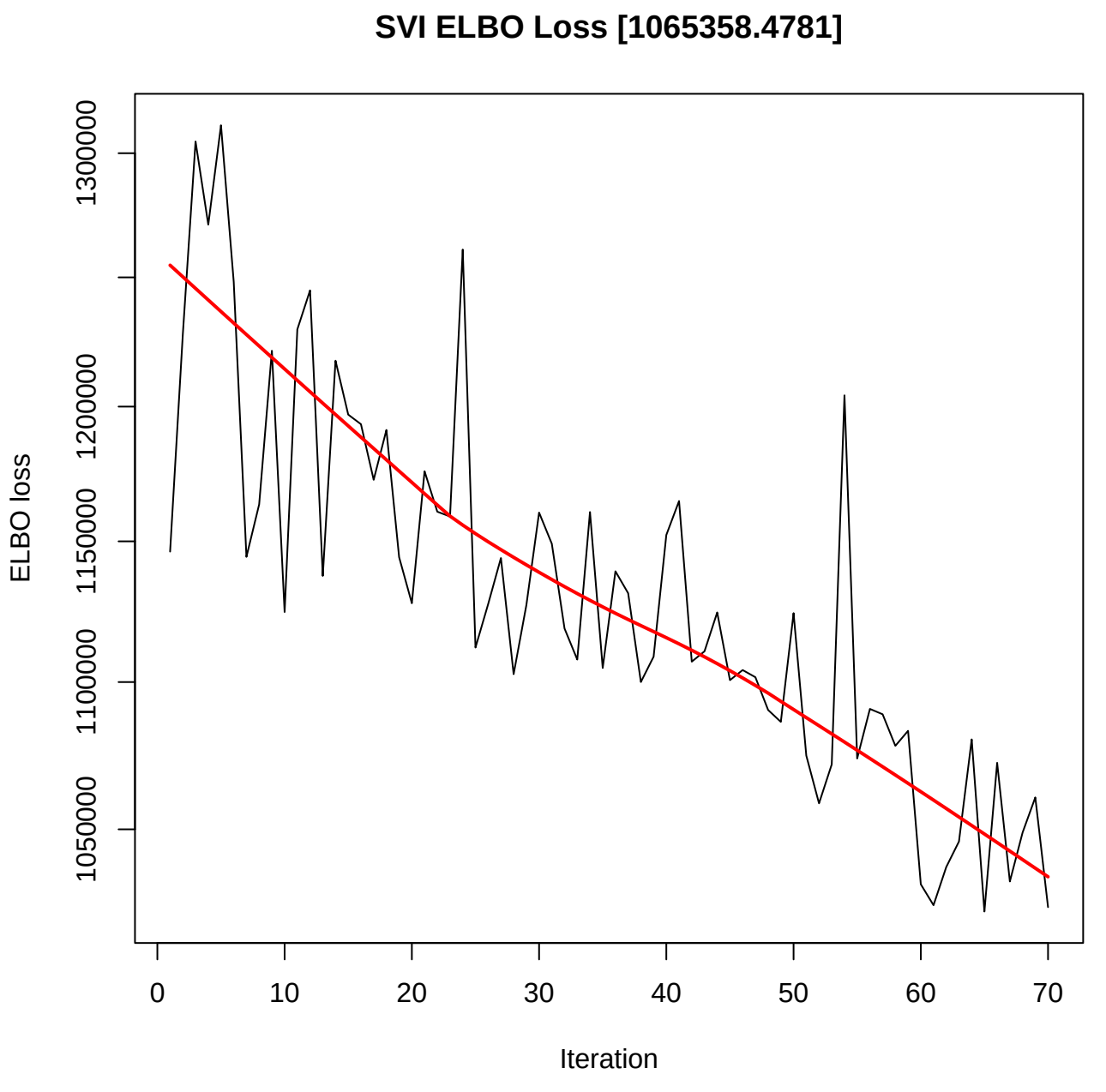

SVI ELBO Loss [169.8257]
ELBO loss
1000
500
200
100
50
2
4
6
8
10
Iteration

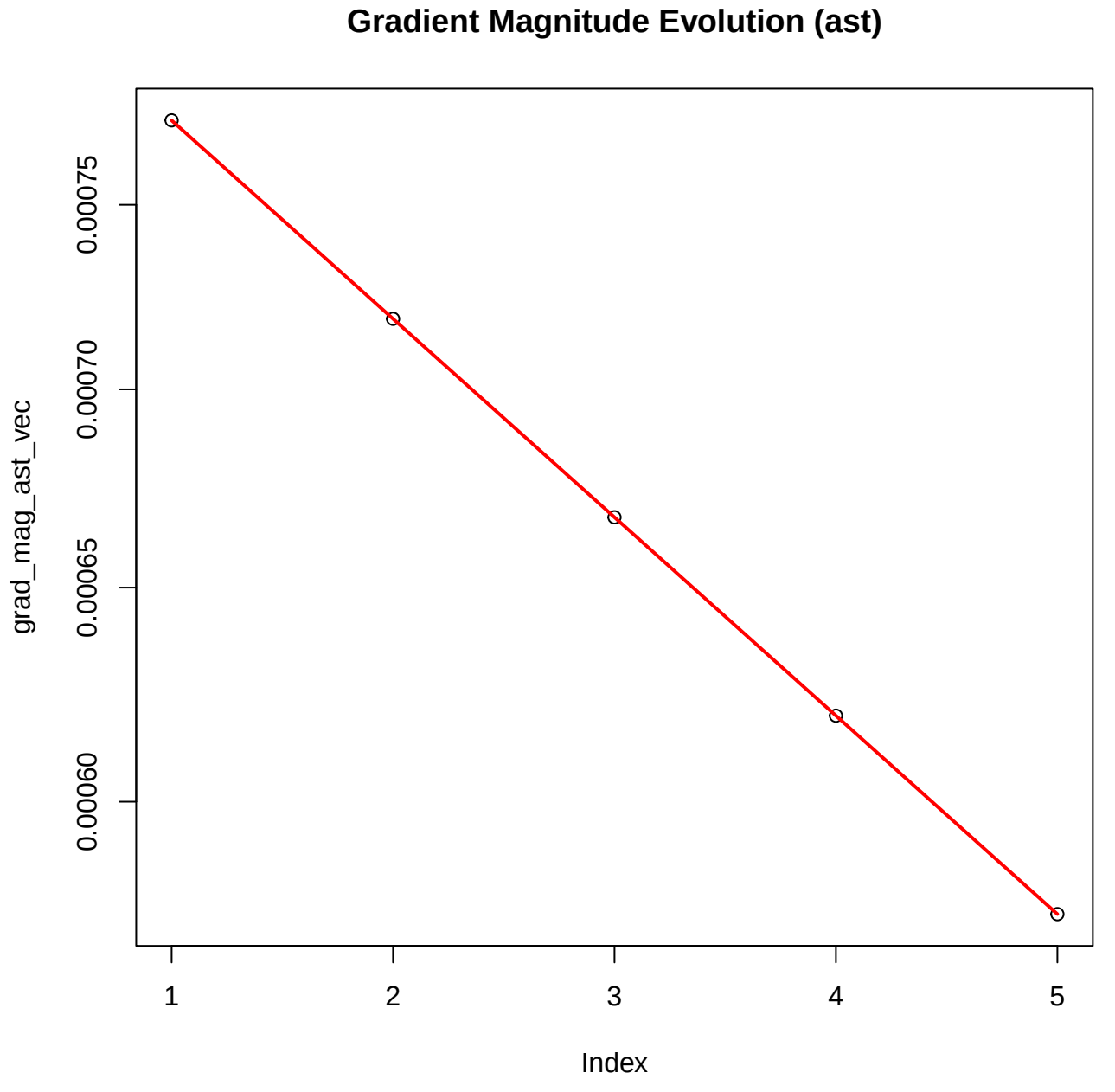

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.10
0.12
0.14
0.16
0.18
0.20
1
2
3
4
5
Index

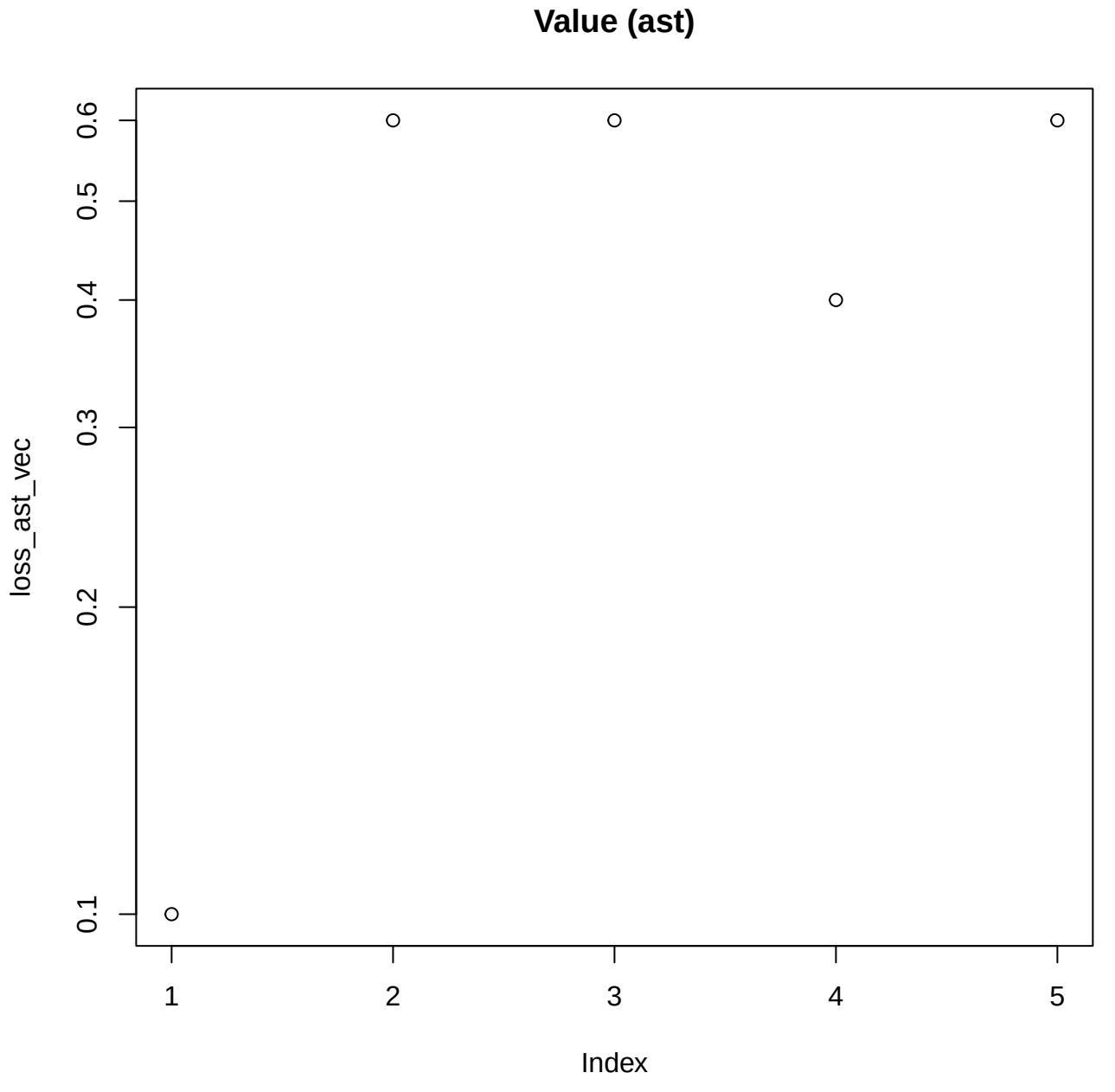

Value (ast)
loss_ast_vec
0.60
0.65
0.70
1
2
3
4
5
Index

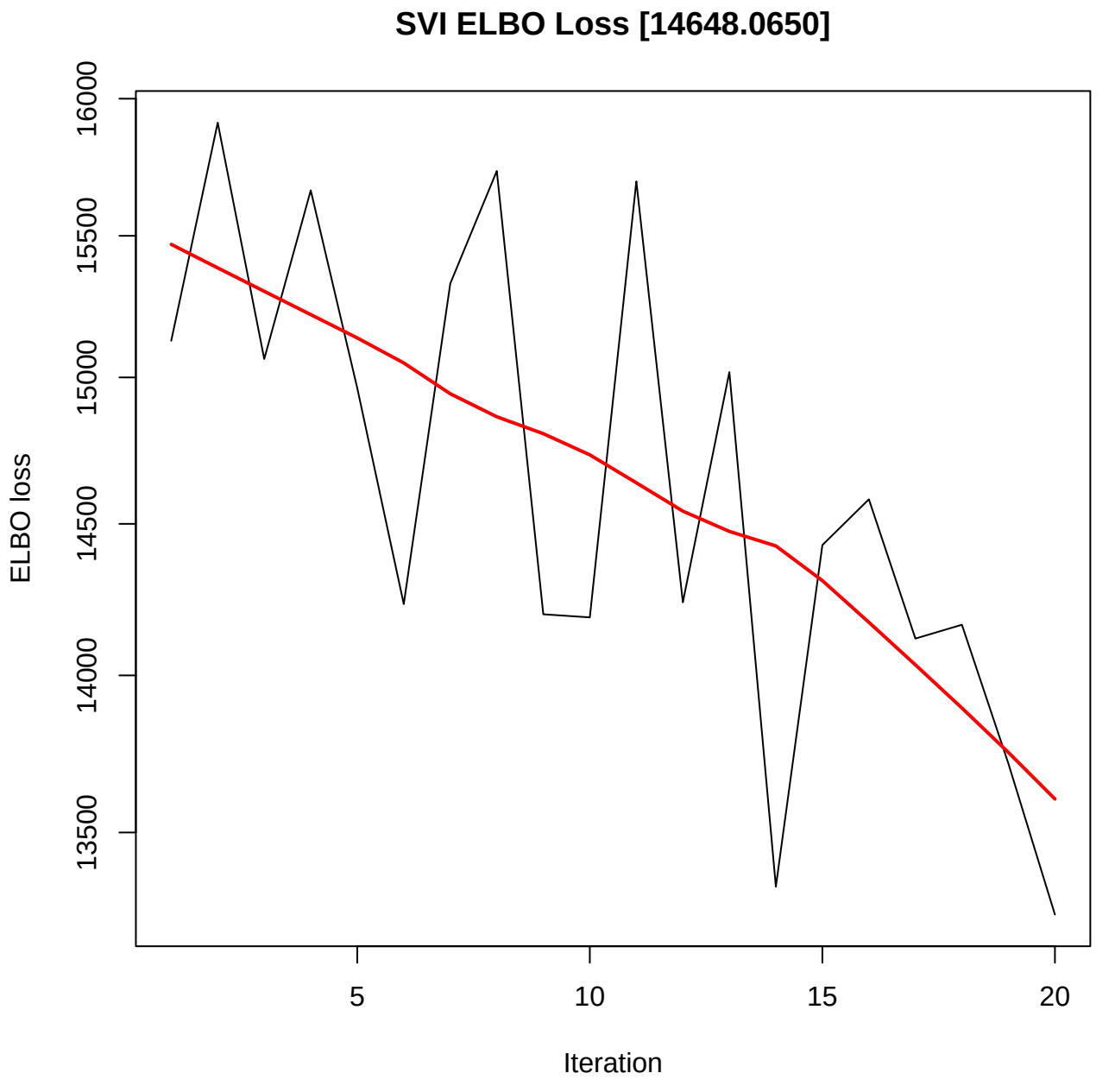

SVI ELBO Loss [943986.3969]
ELBO loss
920000
940000
960000
980000
1000000
1020000
0
10
20
30
40
50
Iteration

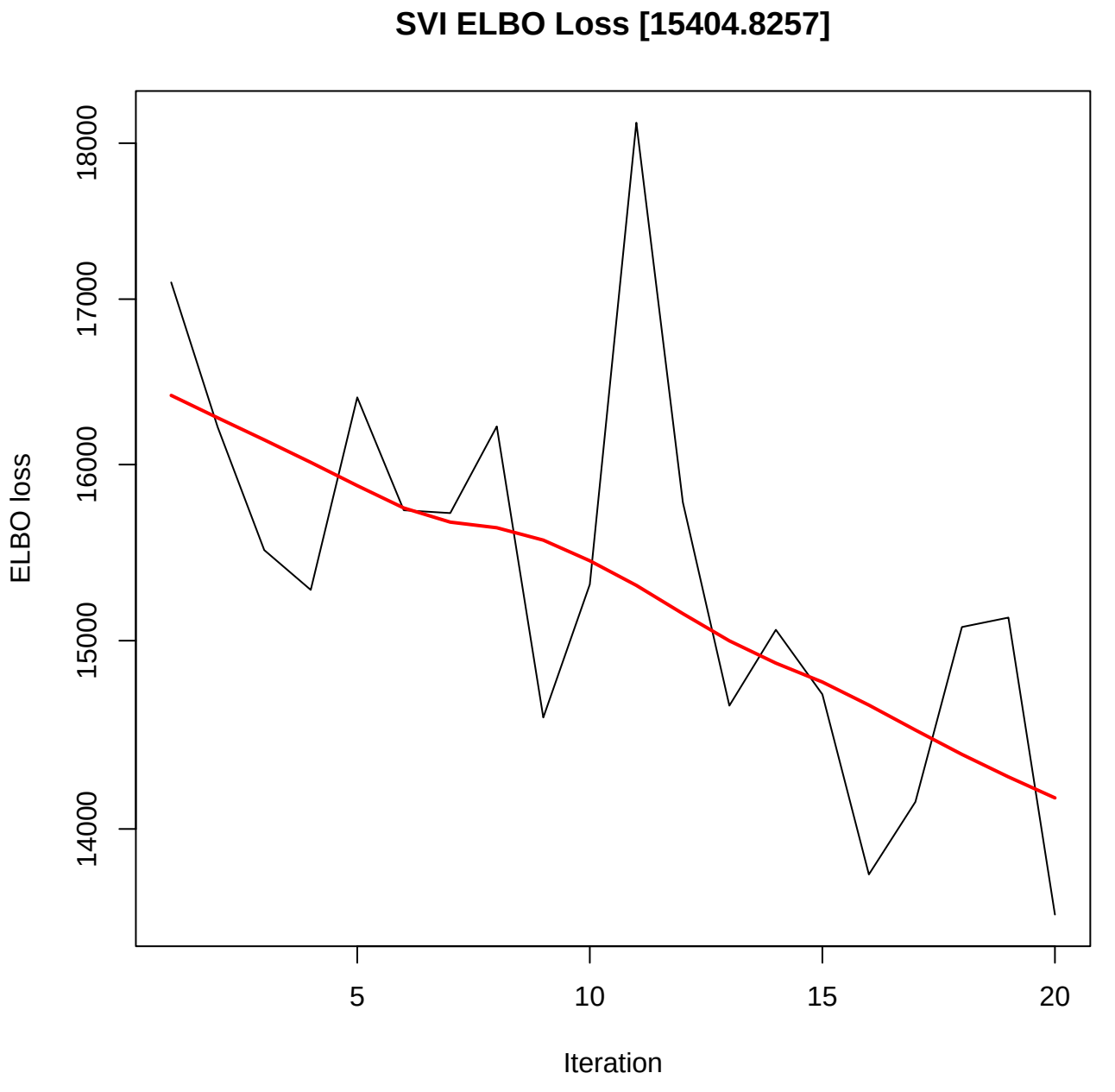

SVI ELBO Loss [1015731.4437]
ELBO loss
1000000
1020000
1040000
1060000
1080000
0
10
20
30
40
Iteration

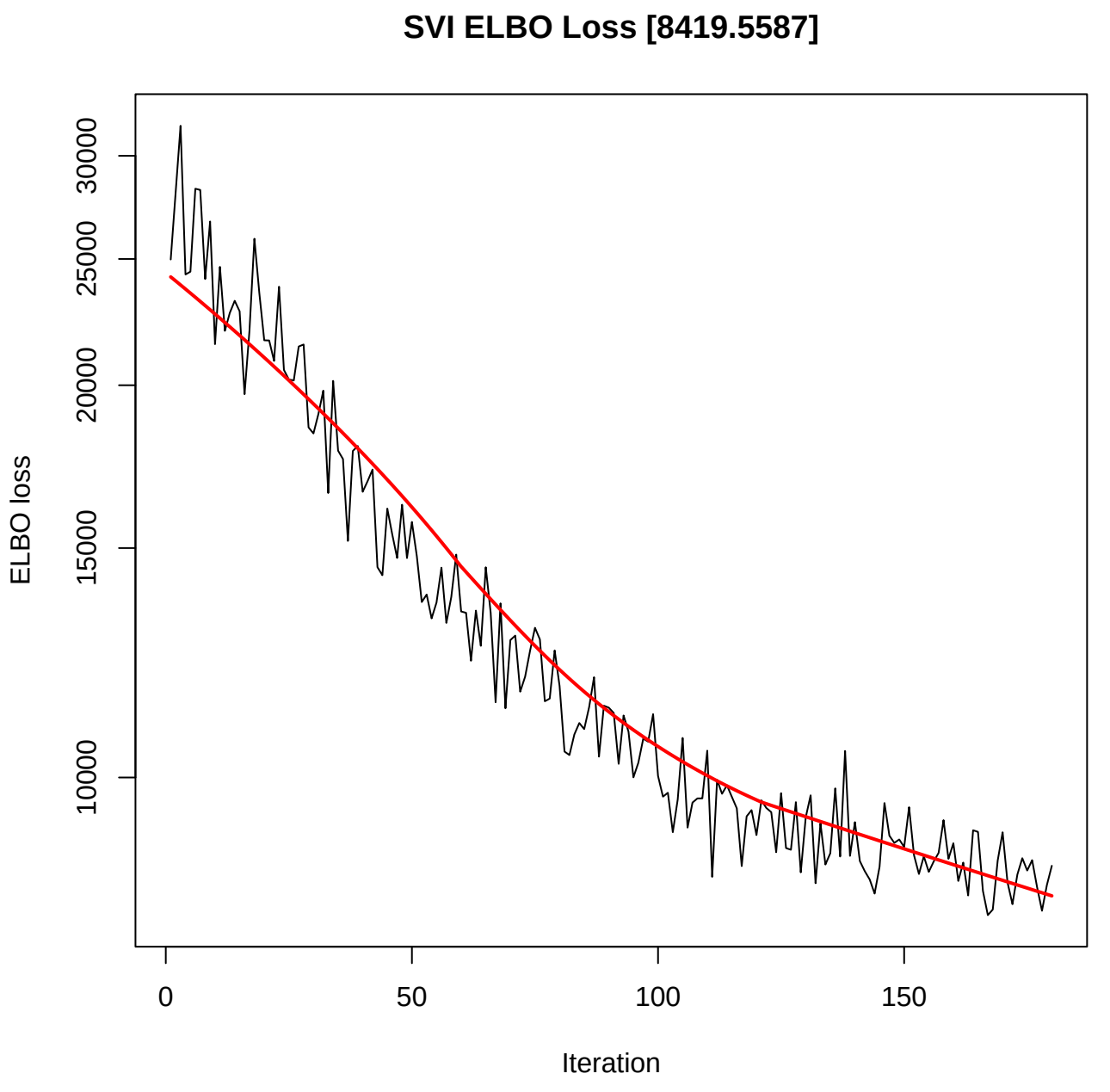

SVI ELBO Loss [1065358.4781]
ELBO loss
1300000
1200000
1150000
1100000
1050000
0
10
20
30
40
50
60
70
Iteration

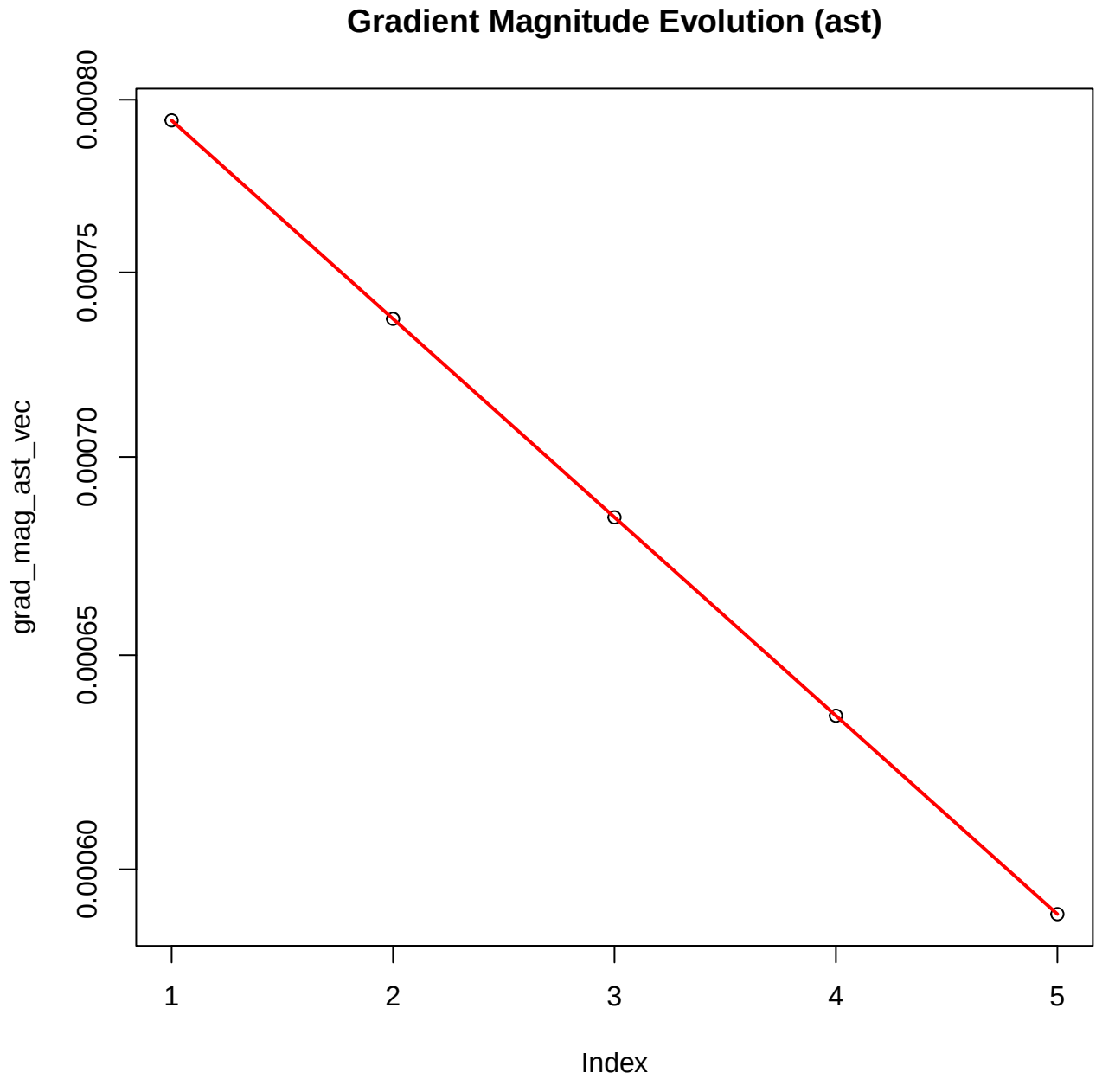

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

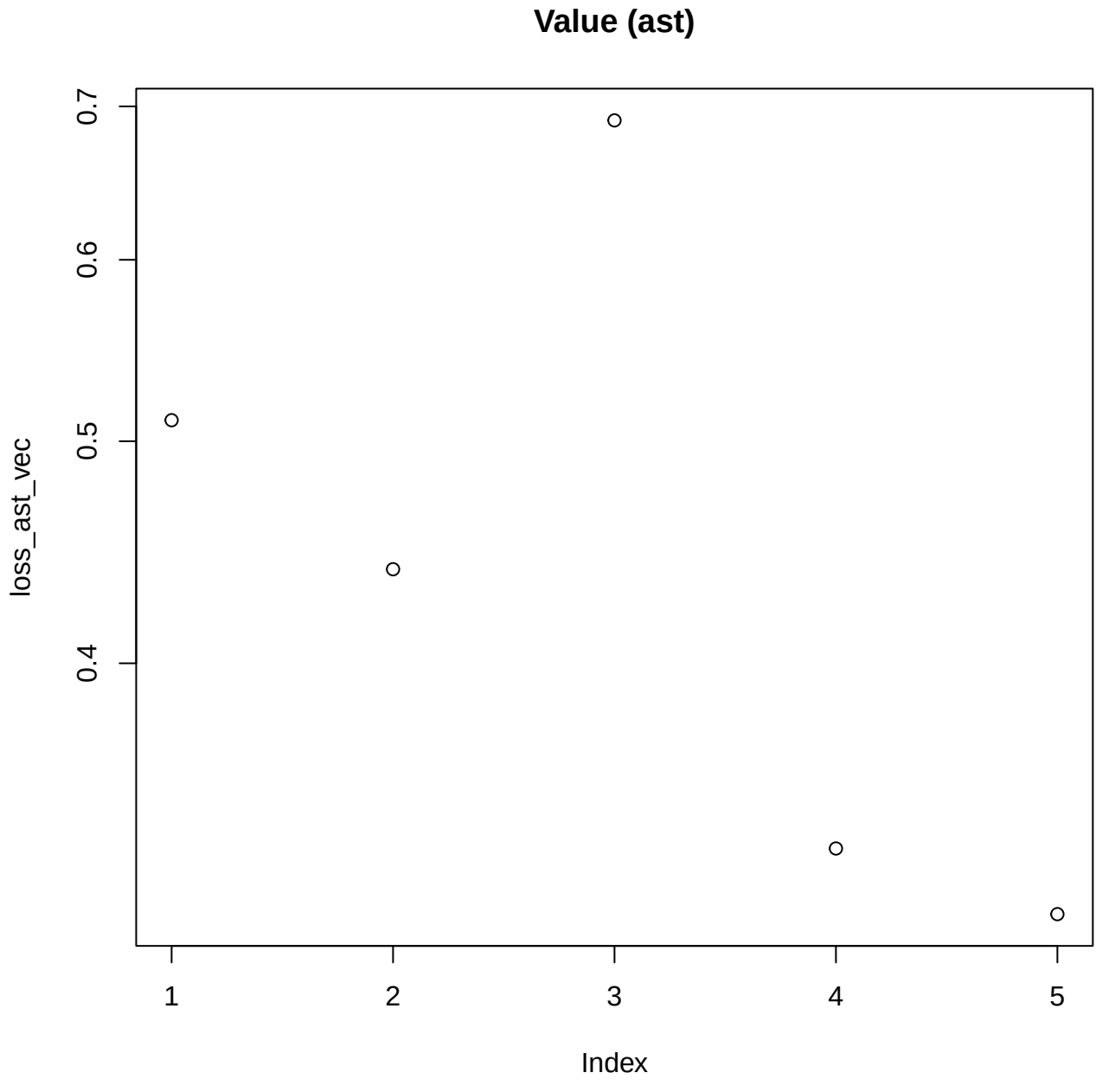

Value (ast)
loss_ast_vec
0.1
0.2
0.3
0.4
0.5
0.6
1
2
3
4
5
Index

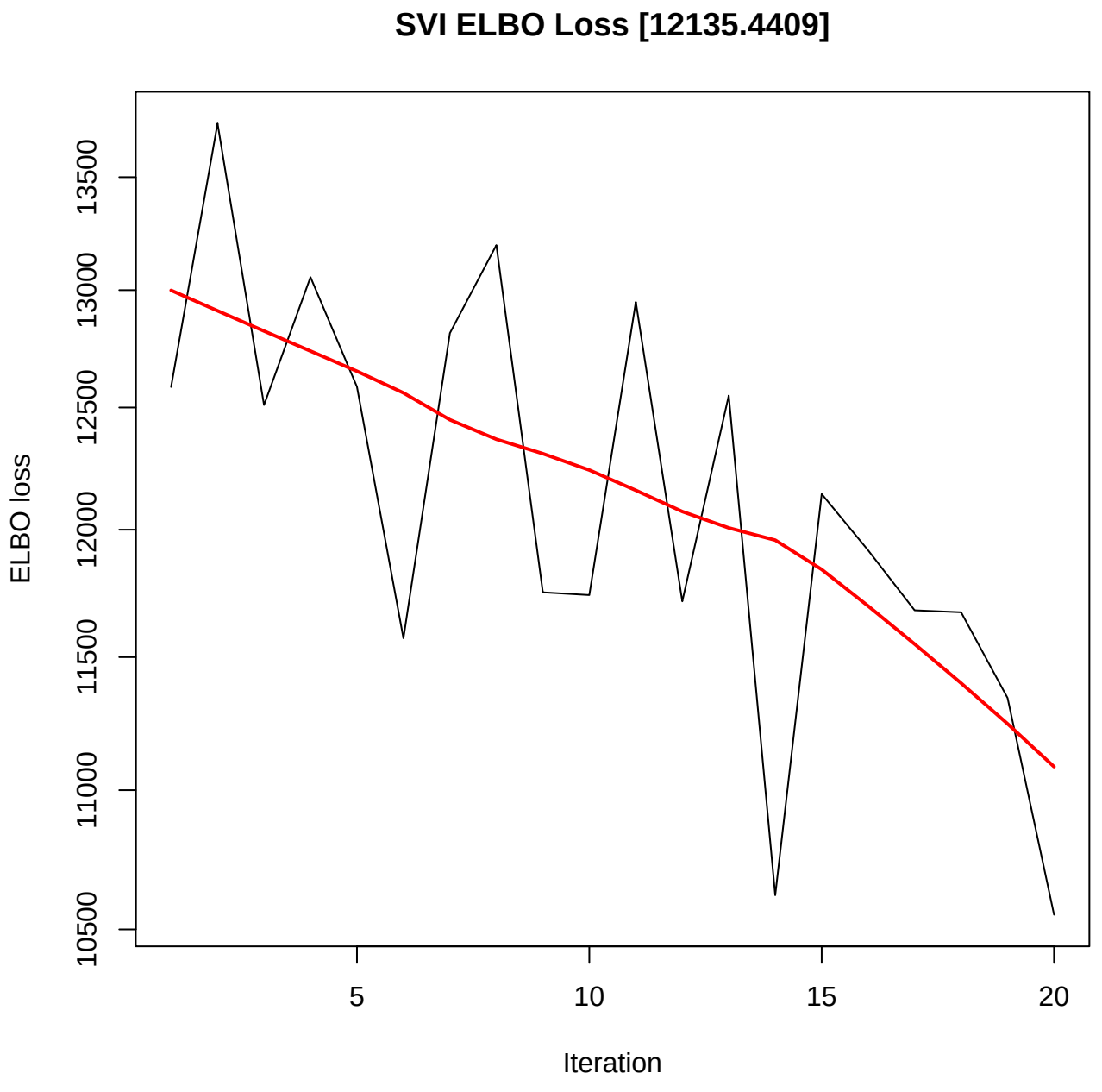

SVI ELBO Loss [14648.0650]
ELBO loss
16000
15500
15000
14500
14000
13500
5
10
15
20
Iteration

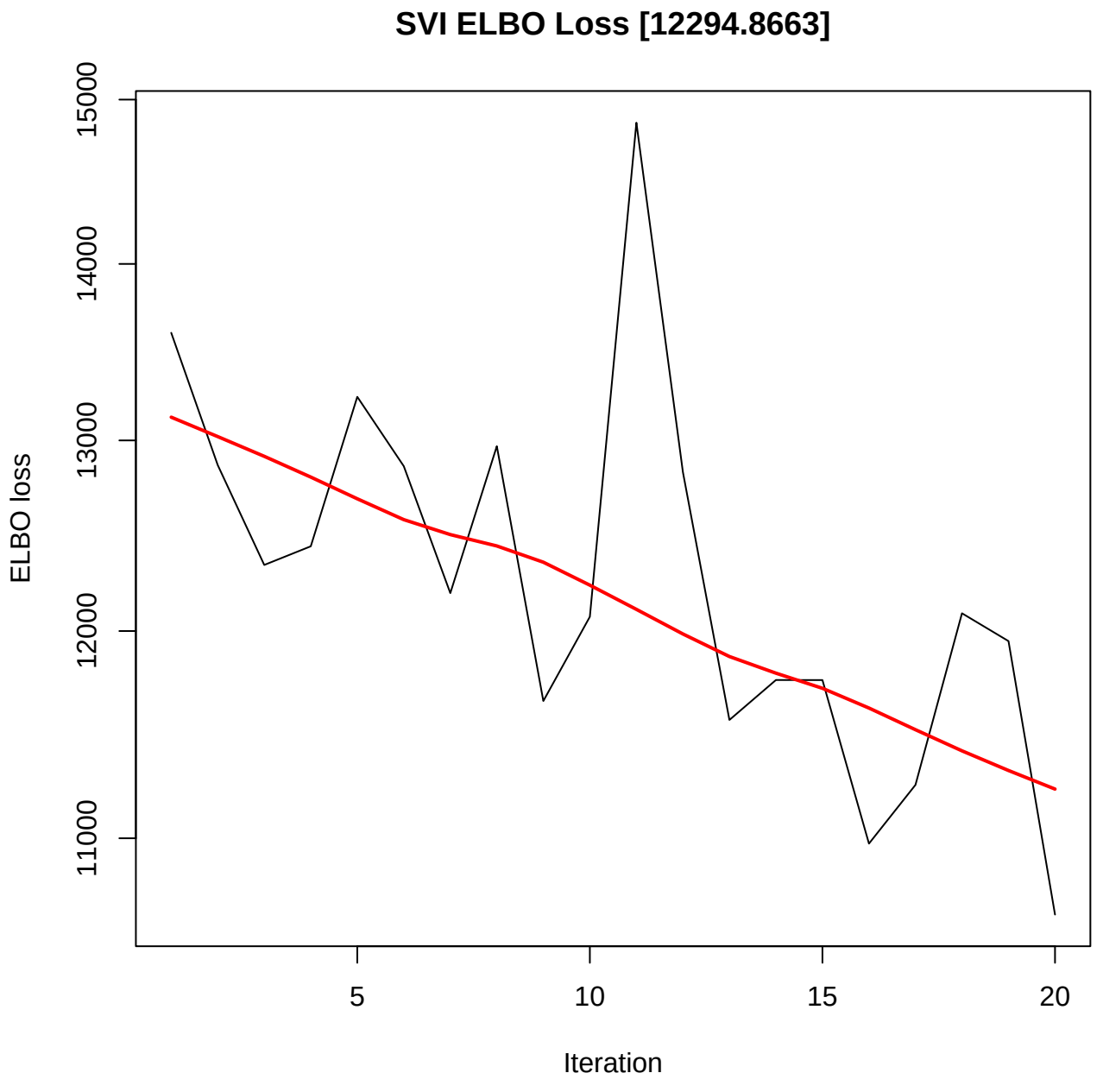

SVI ELBO Loss [15404.8257]
ELBO loss
18000
17000
16000
15000
14000
5
10
15
20
Iteration

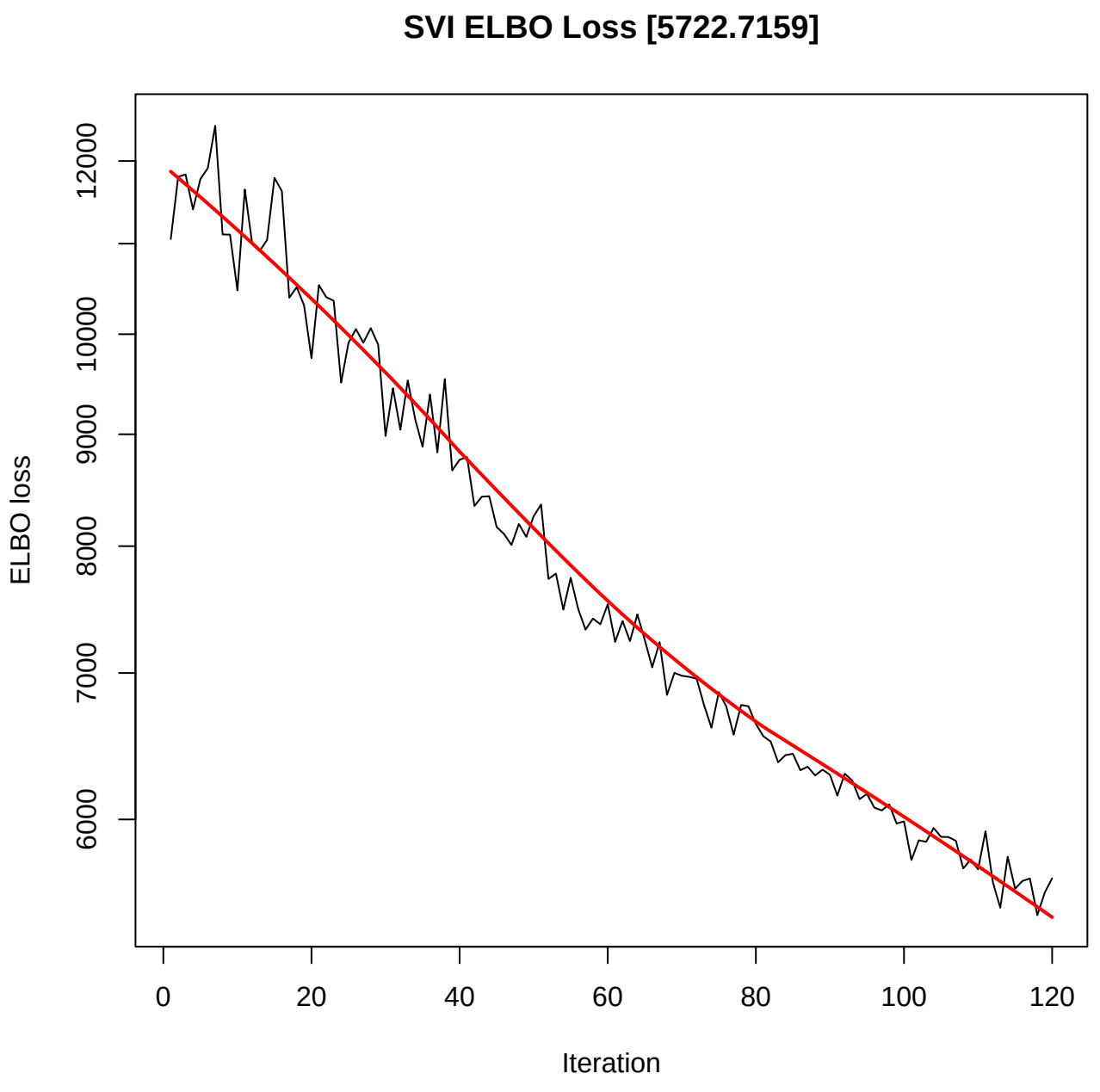

SVI ELBO Loss [8419.5587]
ELBO loss
10000
15000
20000
25000
30000
0
50
100
150
Iteration

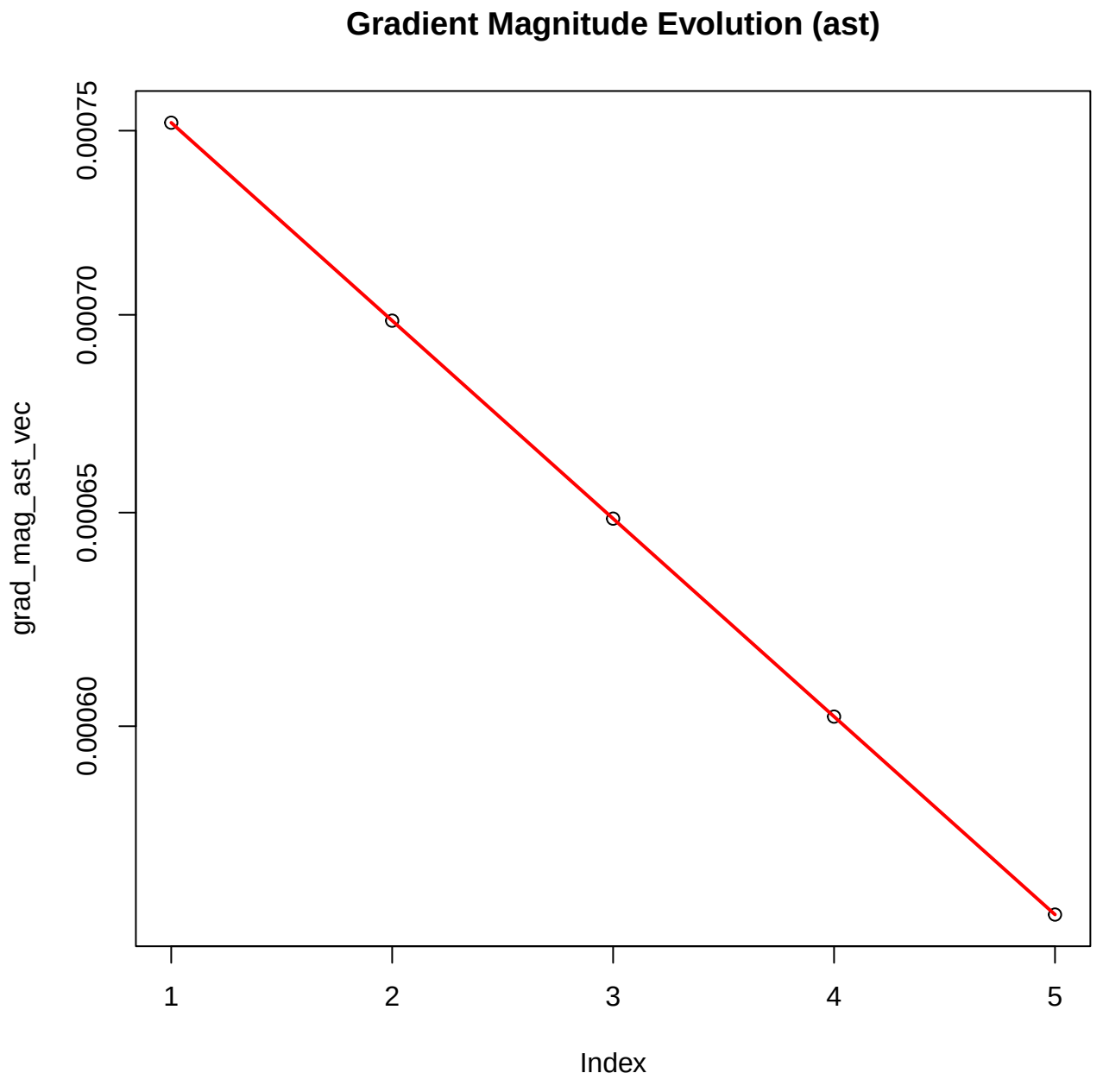

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

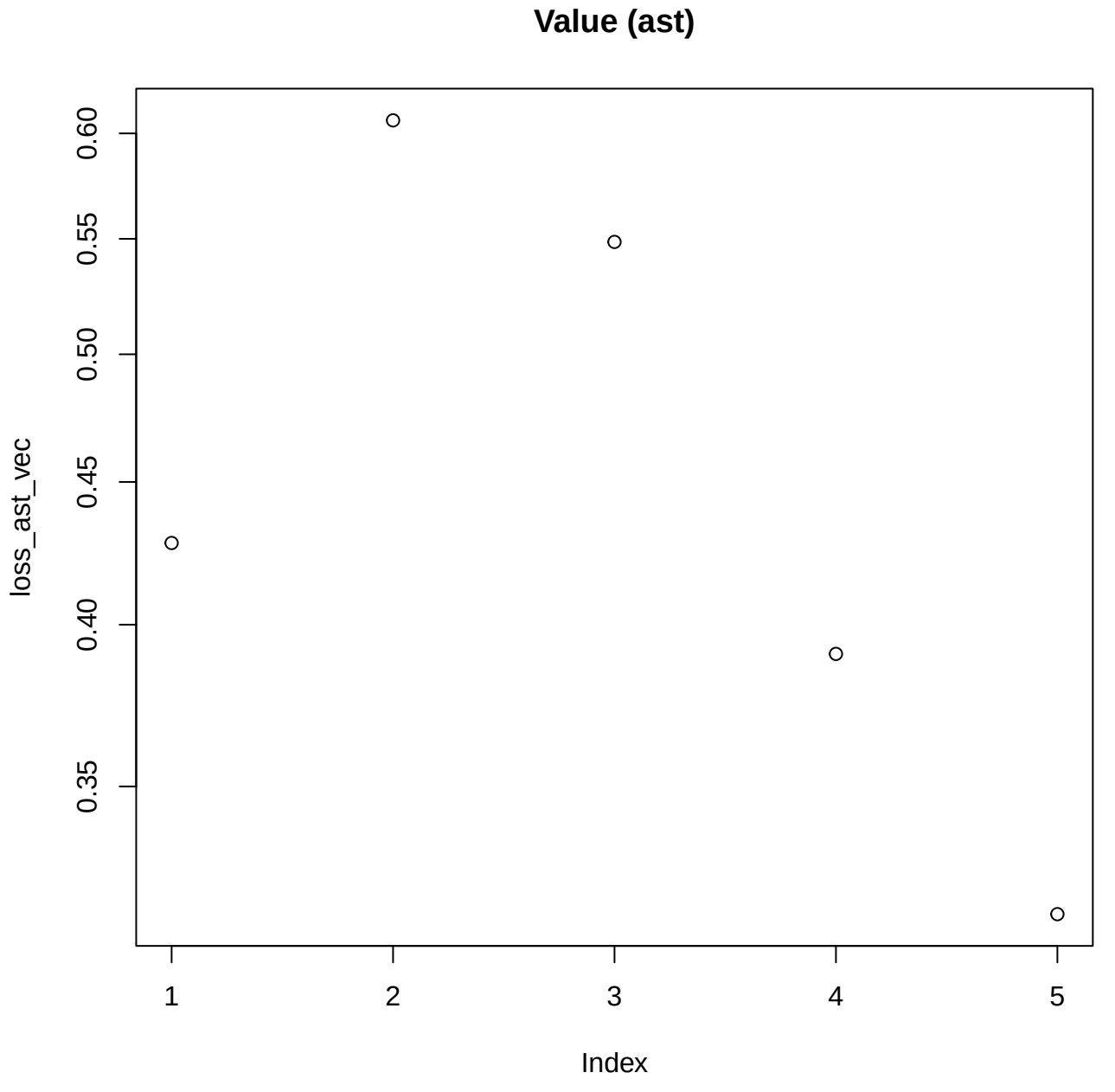

Value (ast)
loss_ast_vec
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

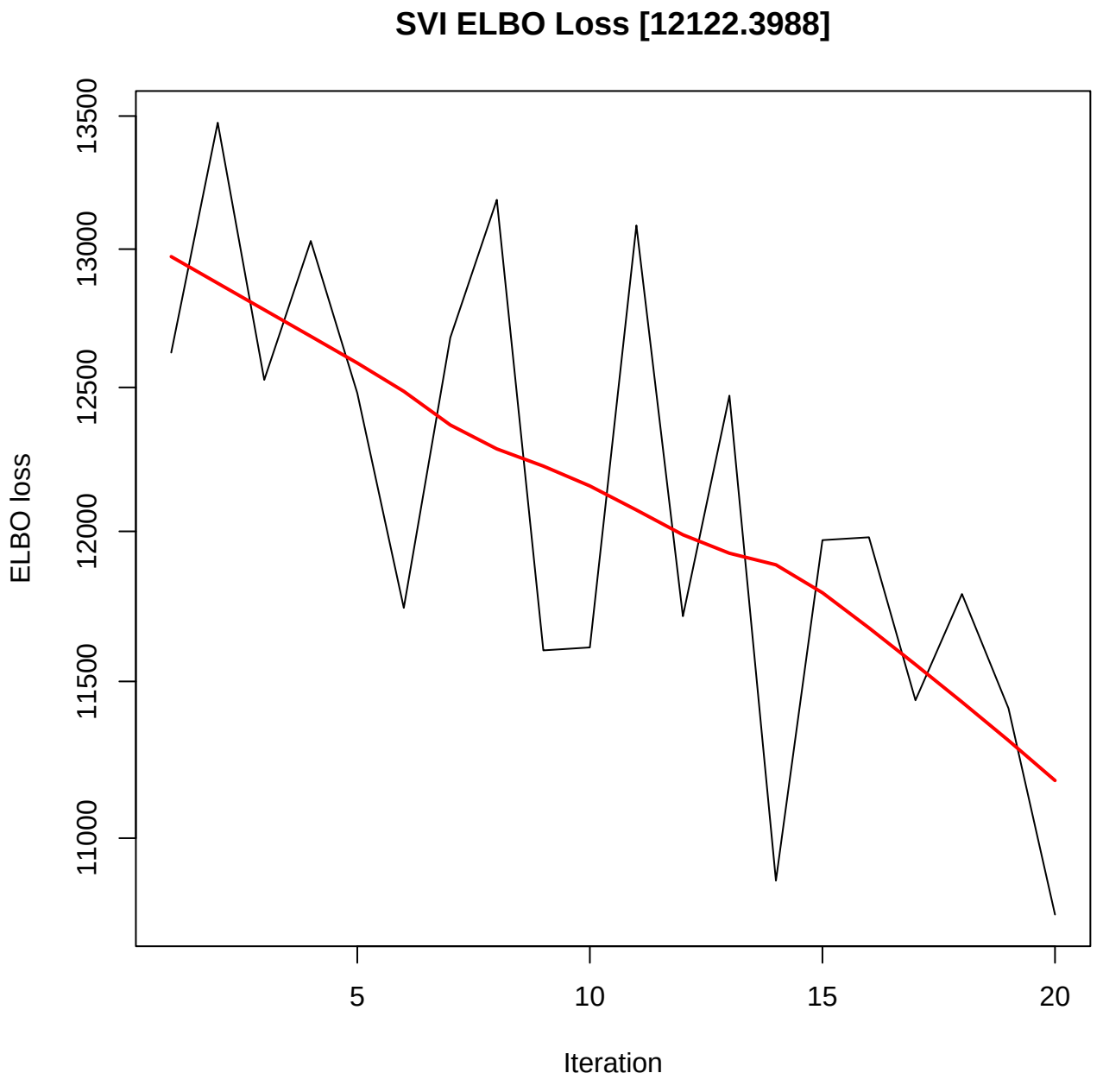

SVI ELBO Loss [12135.4409]
ELBO loss
10500
11000
11500
12000
12500
13000
13500
5
10
15
20
Iteration

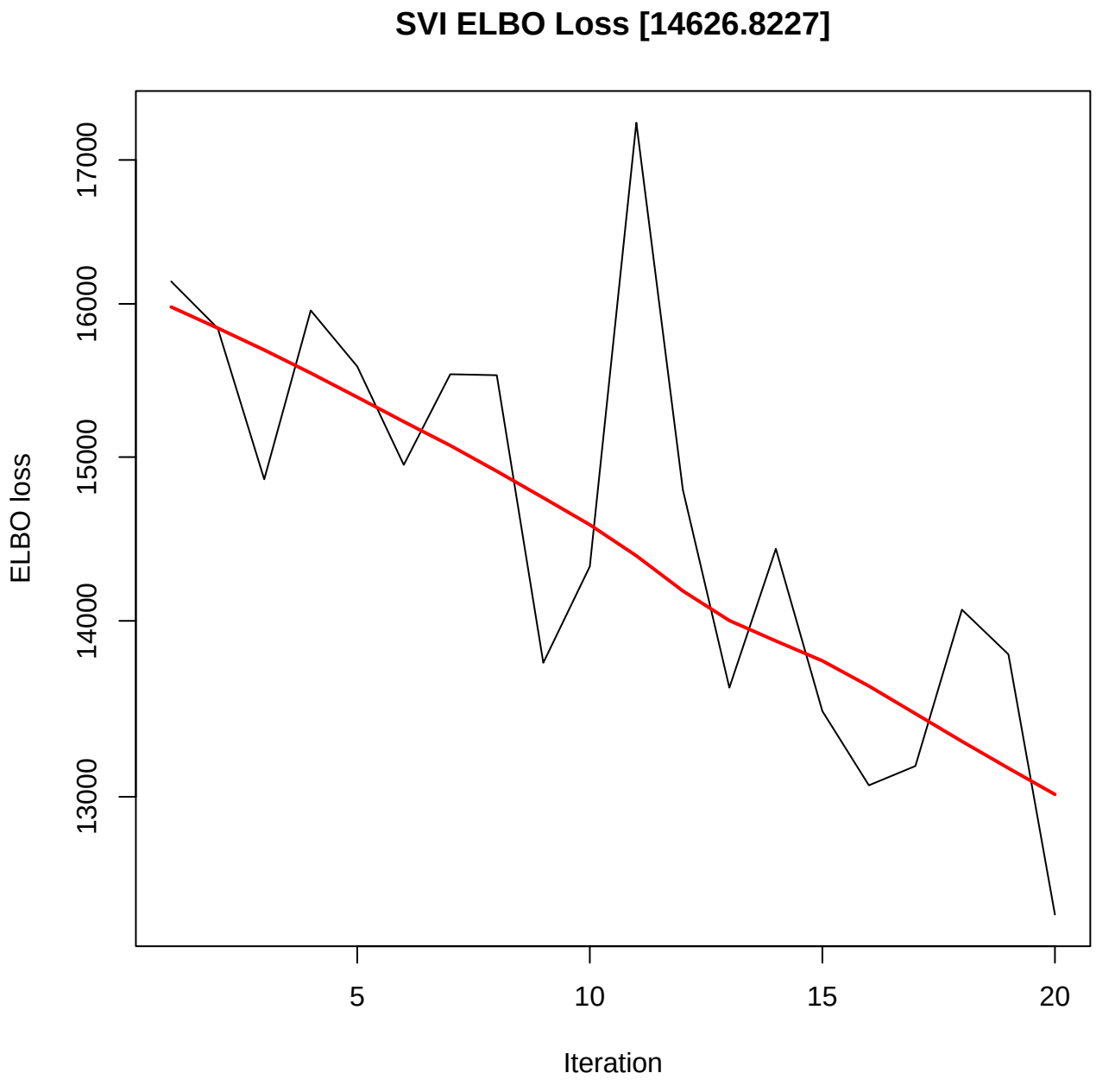

SVI ELBO Loss [12294.8663]
ELBO loss
15000
14000
13000
12000
11000
5
10
15
20
Iteration

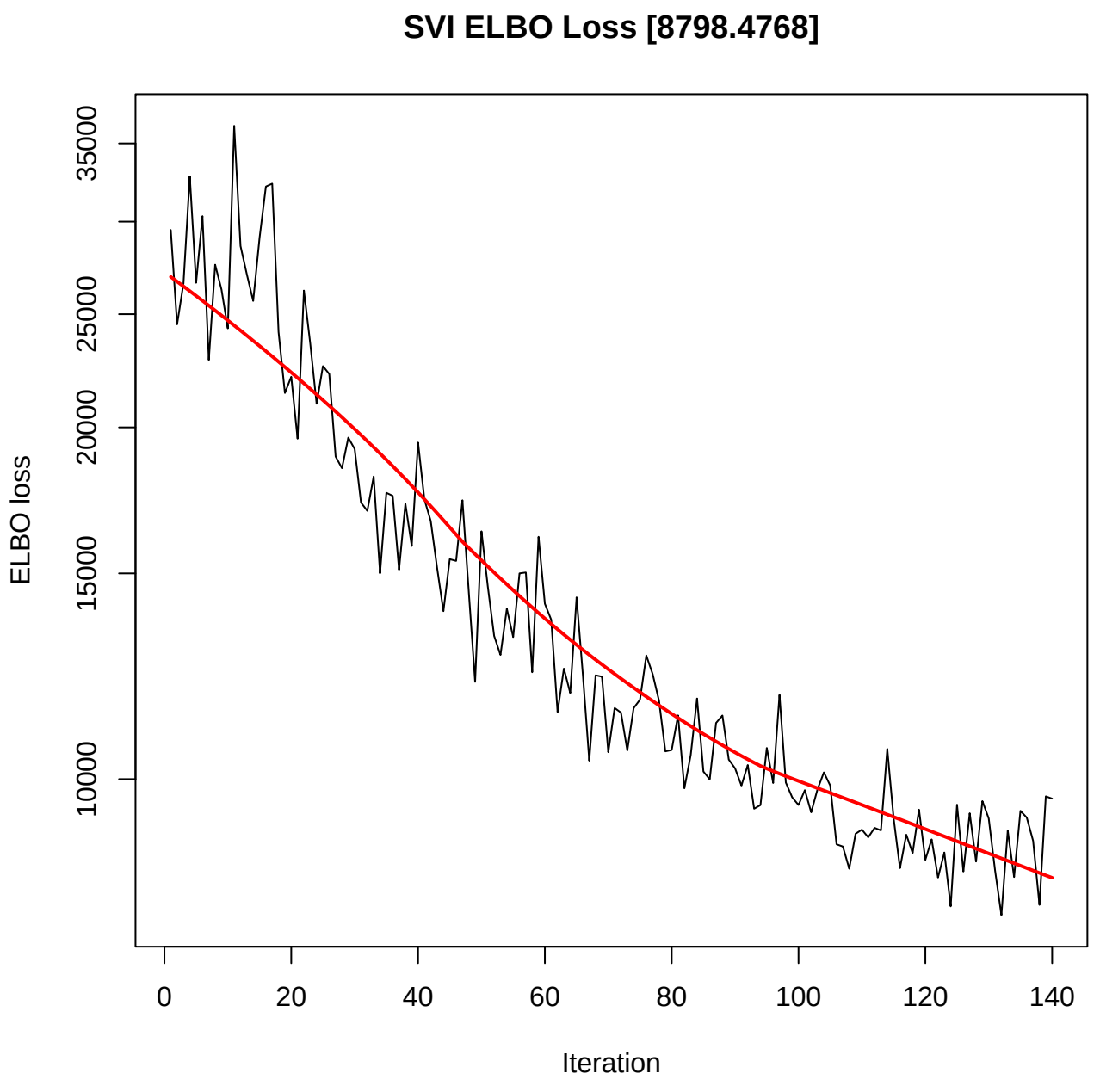

SVI ELBO Loss [5722.7159]
ELBO loss
12000
10000
9000
8000
7000
6000
0
20
40
60
80
100
120
Iteration

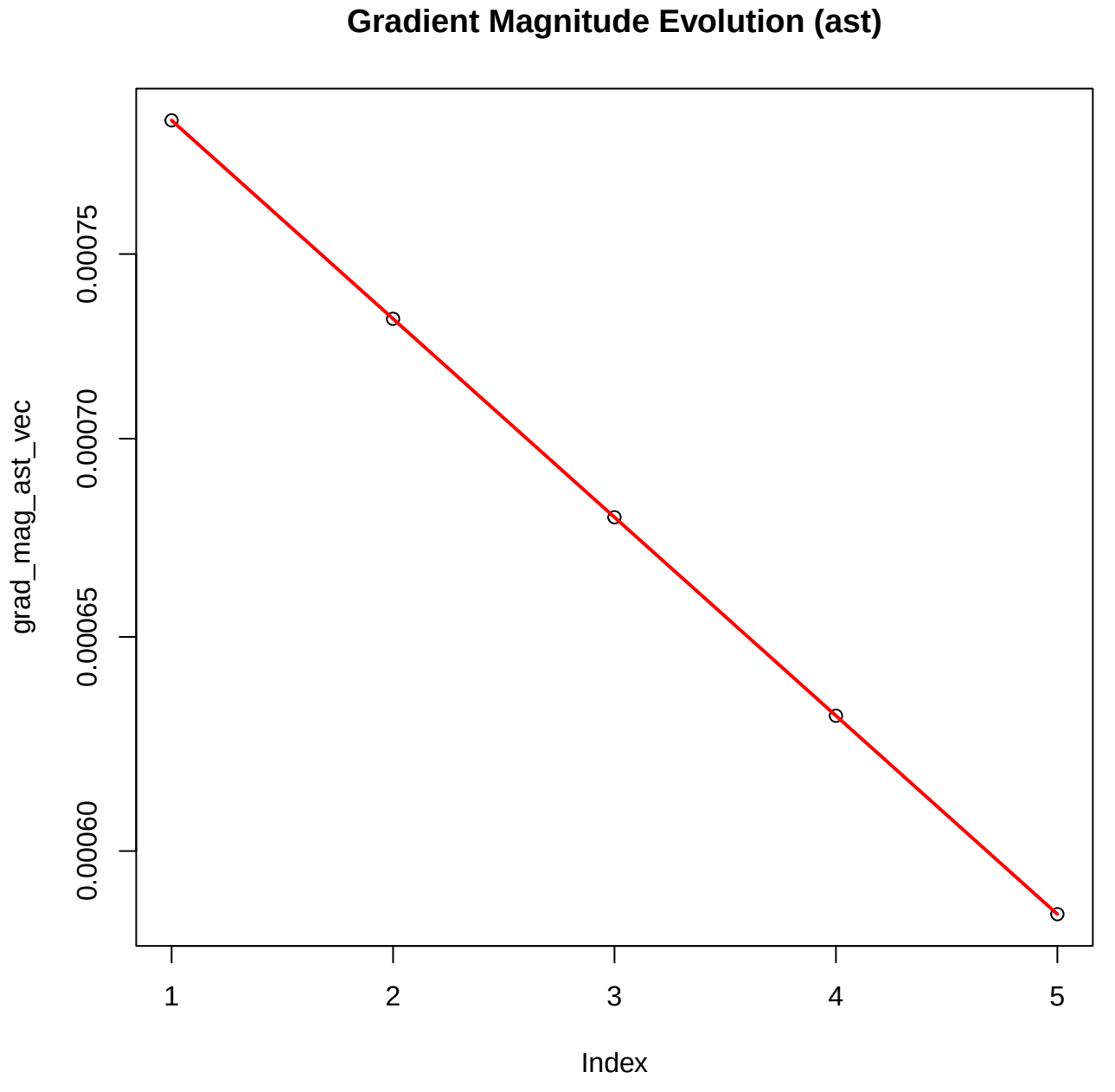

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

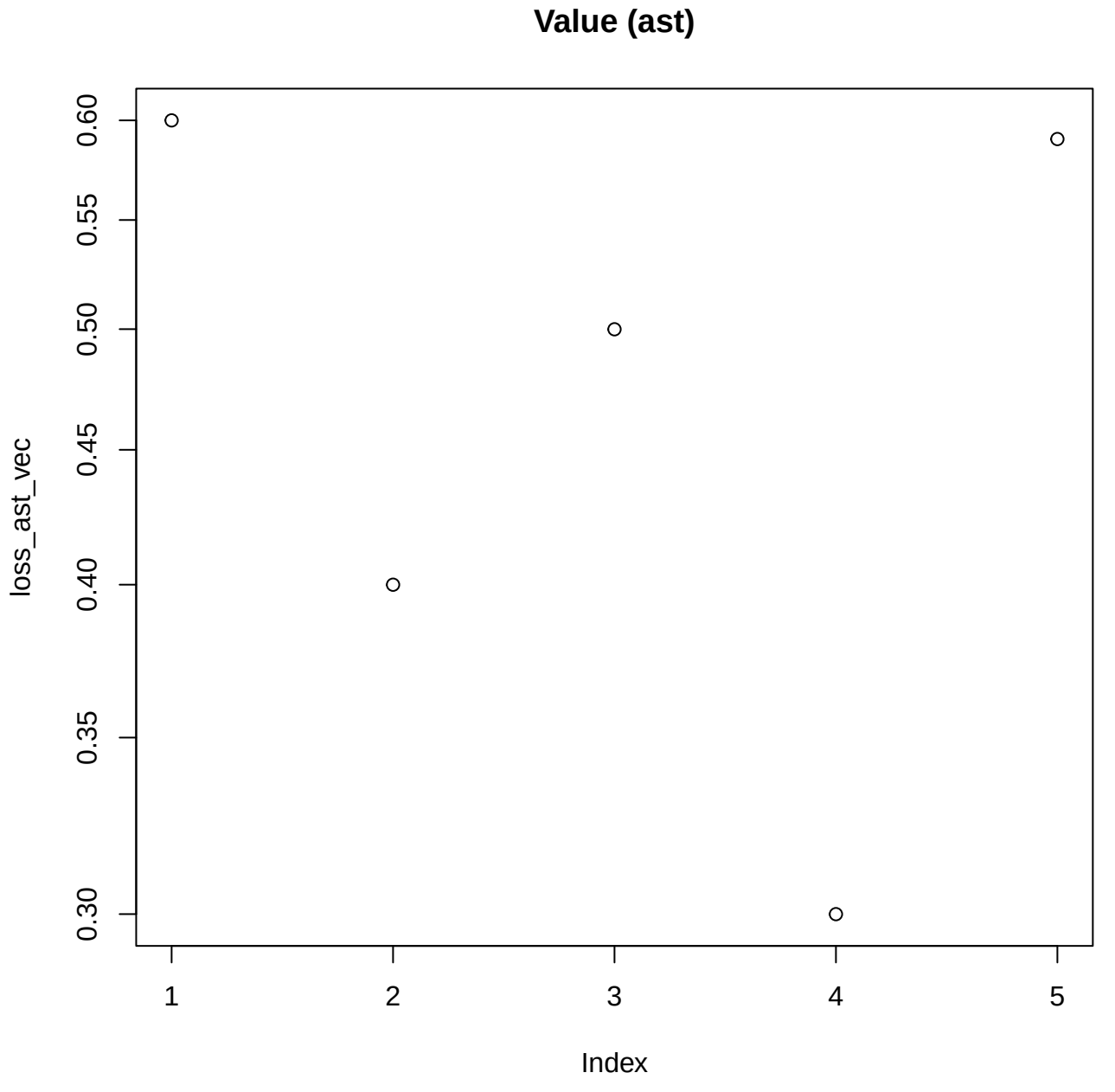

Value (ast)
loss_ast_vec
0.35
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

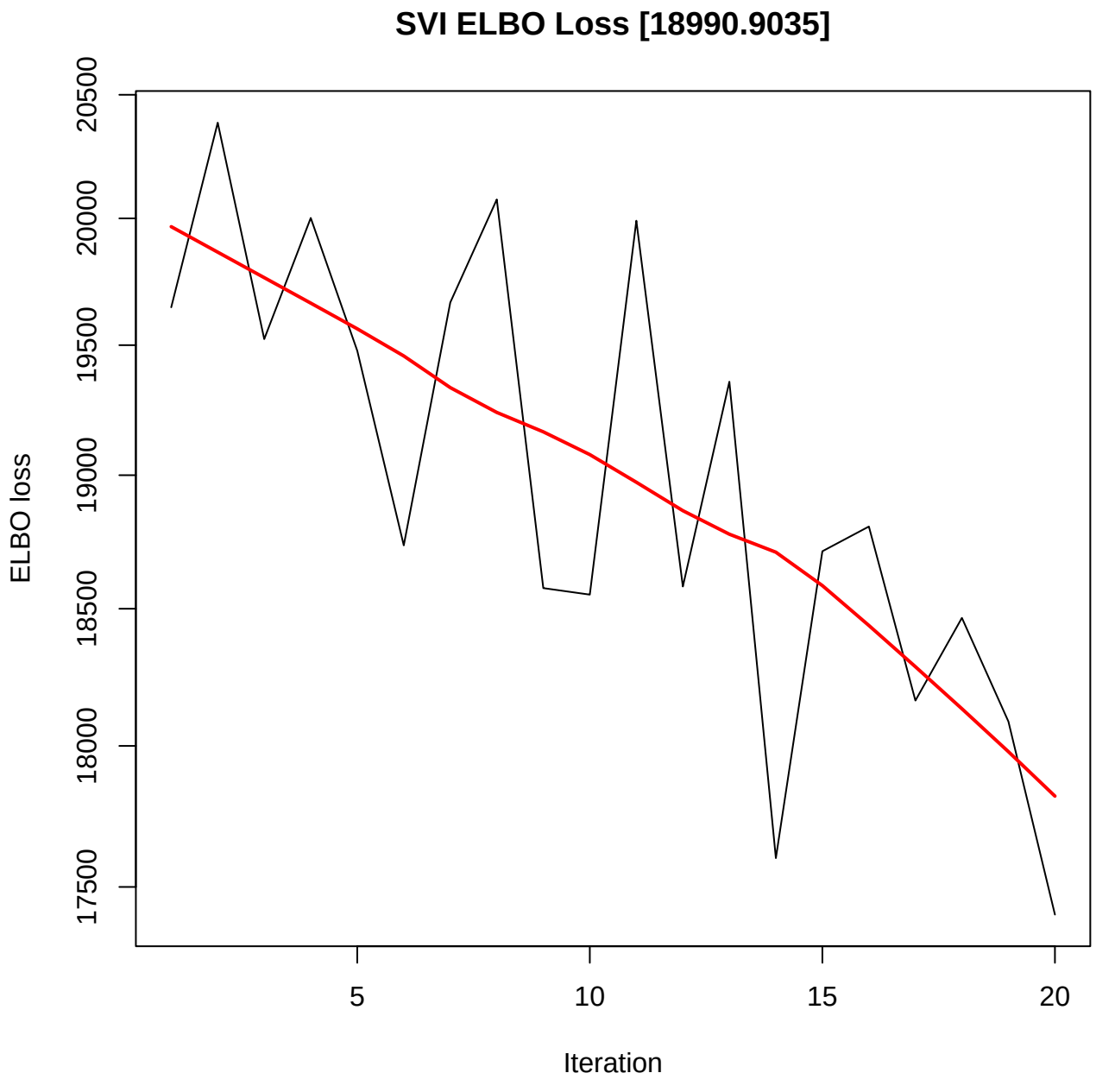

SVI ELBO Loss [12122.3988]
ELBO loss
13500
13000
12500
12000
11500
11000
5
10
15
20
Iteration

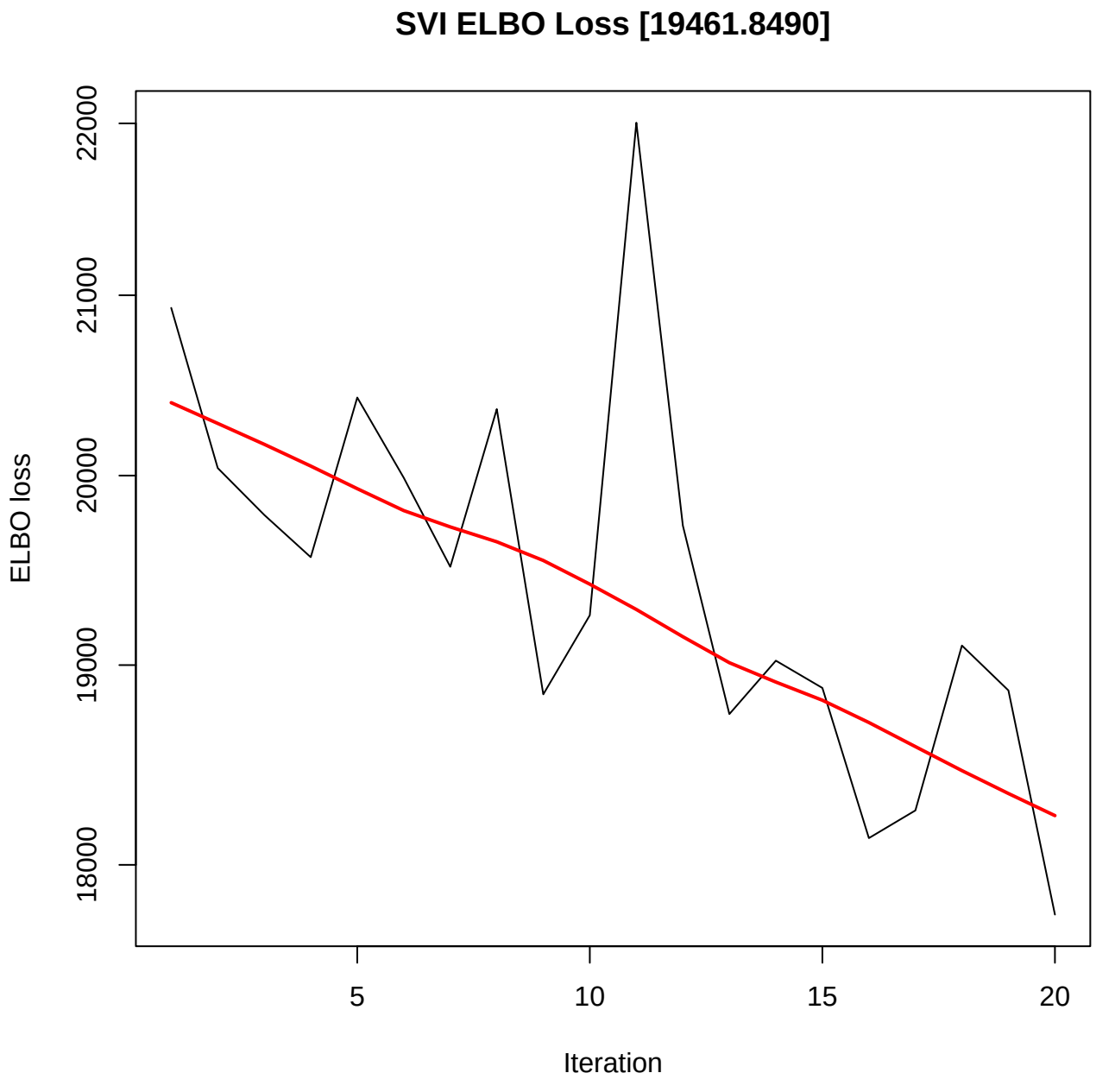

SVI ELBO Loss [14626.8227]
ELBO loss
17000
16000
15000
14000
13000
5
10
15
20
Iteration

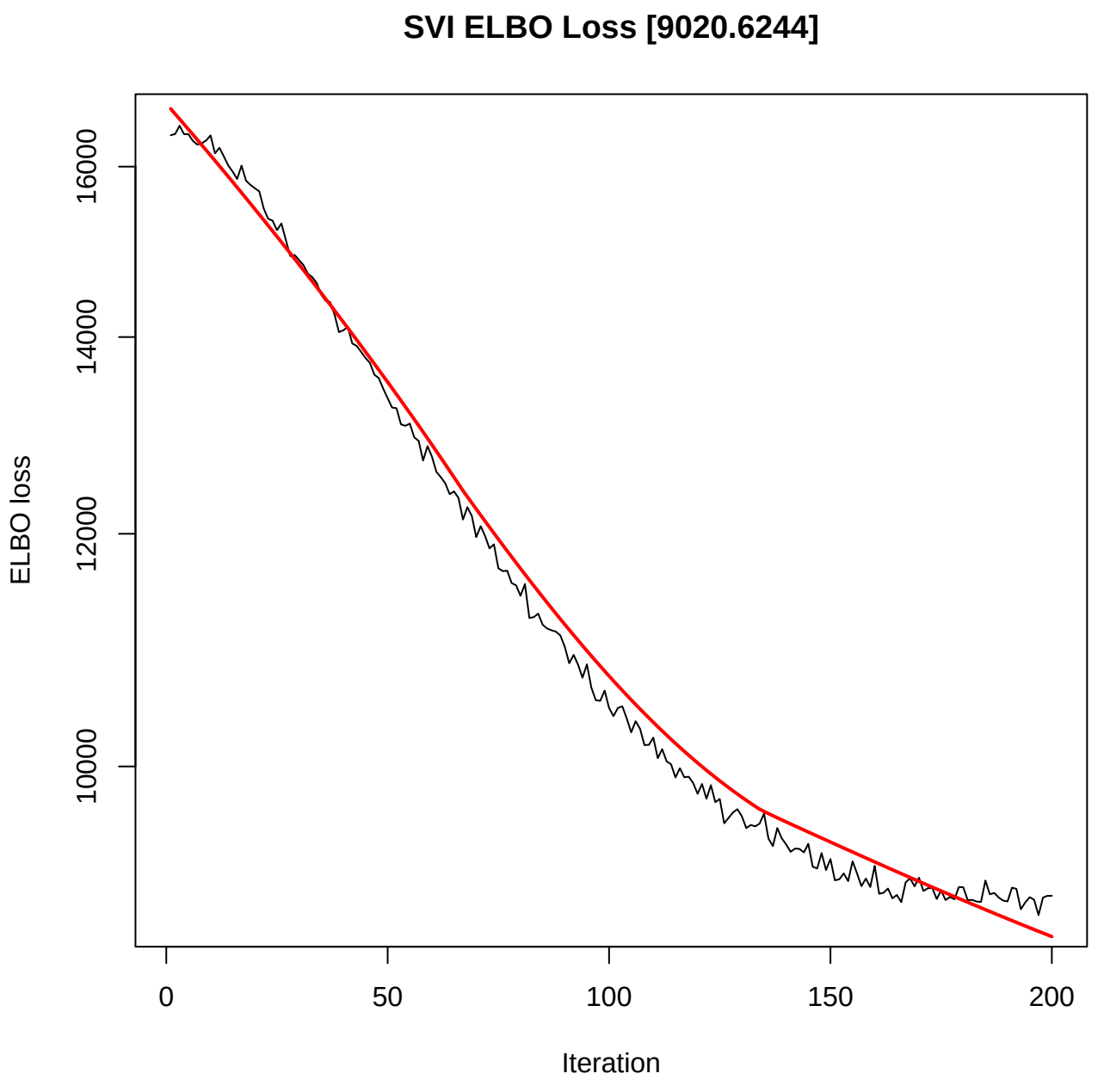

SVI ELBO Loss [8798.4768]
ELBO loss
35000
25000
20000
15000
10000
0
20
40
60
80
100
120
140
Iteration

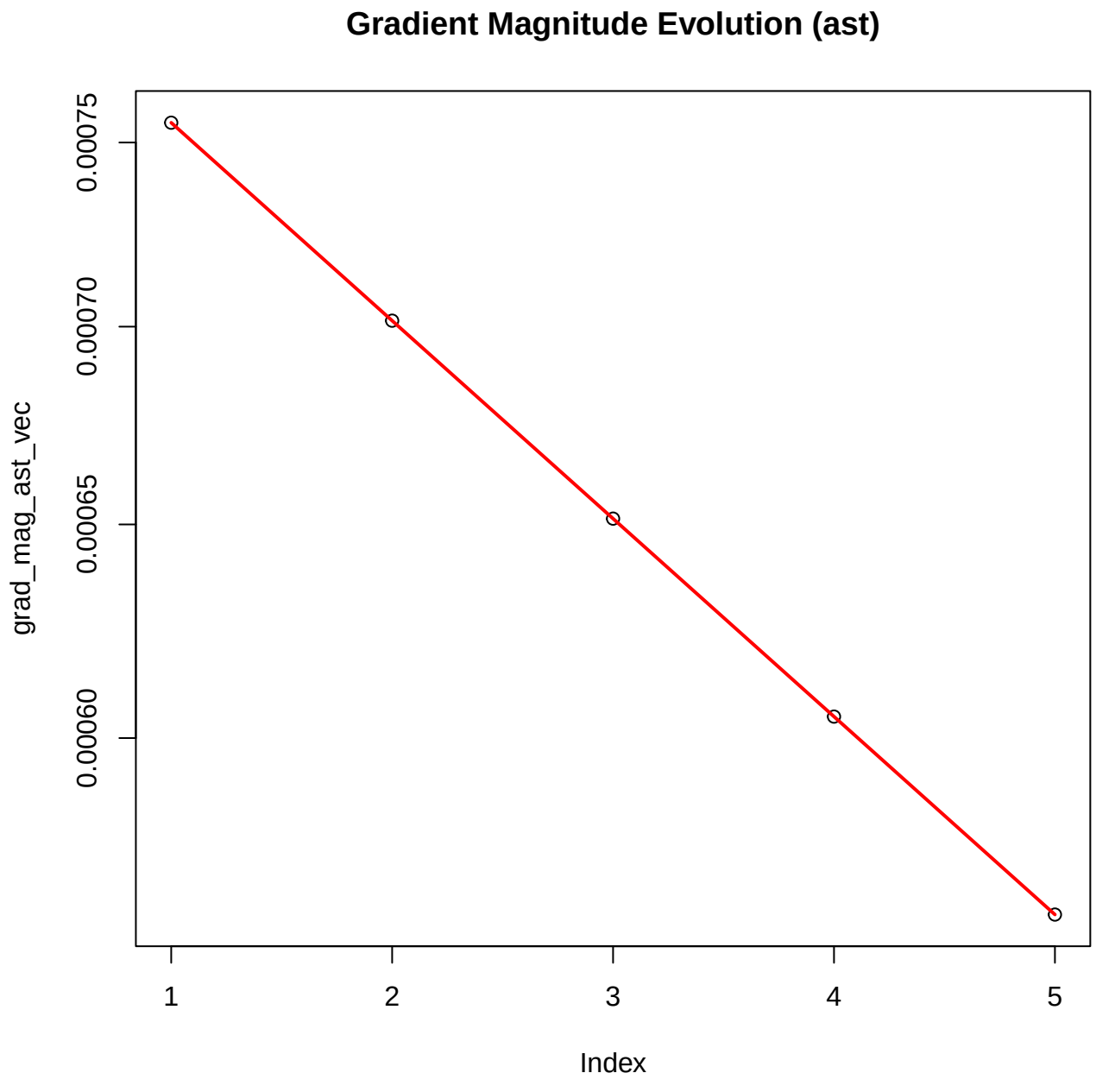

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

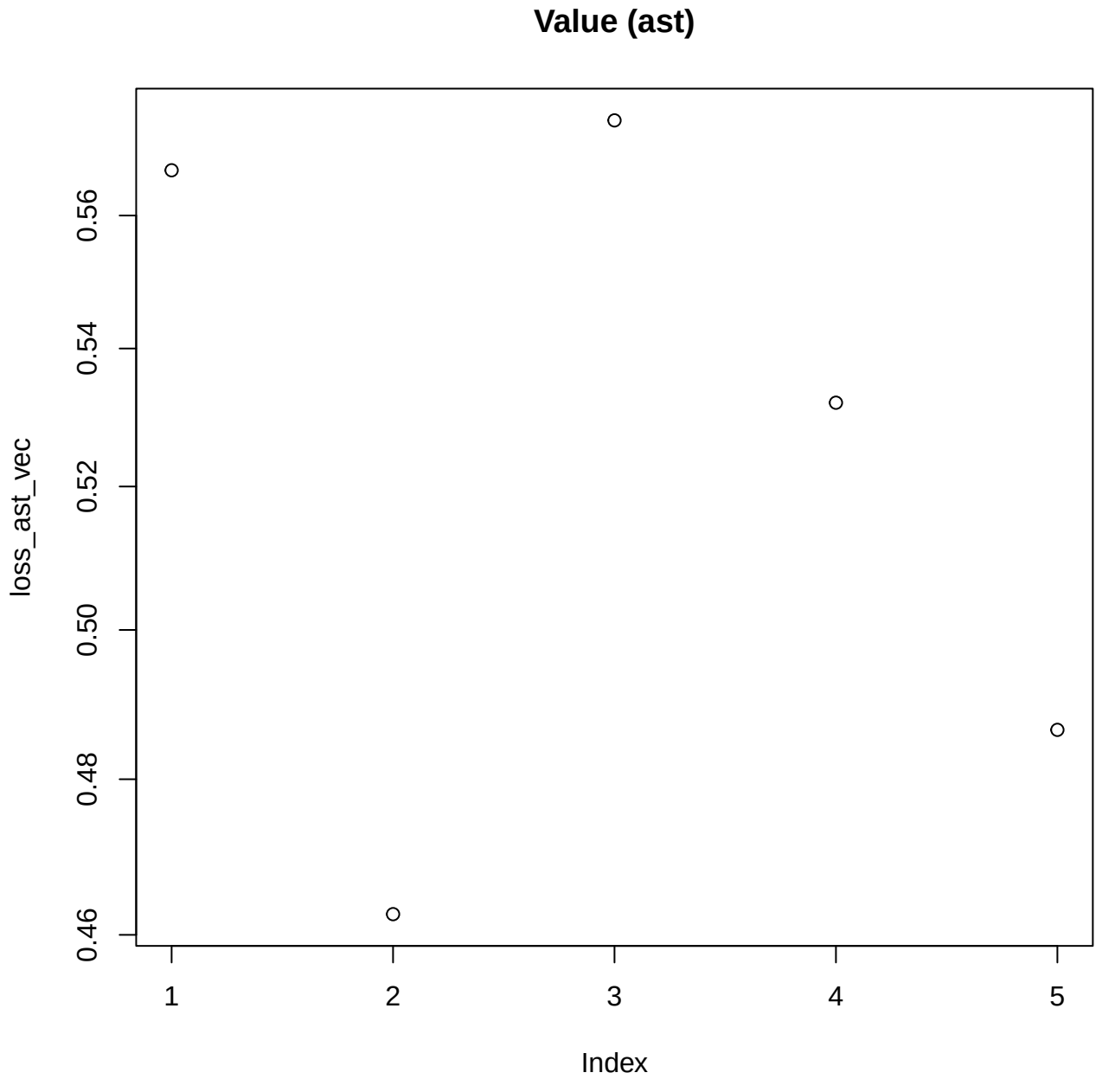

Value (ast)
loss_ast_vec
0.30
0.35
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

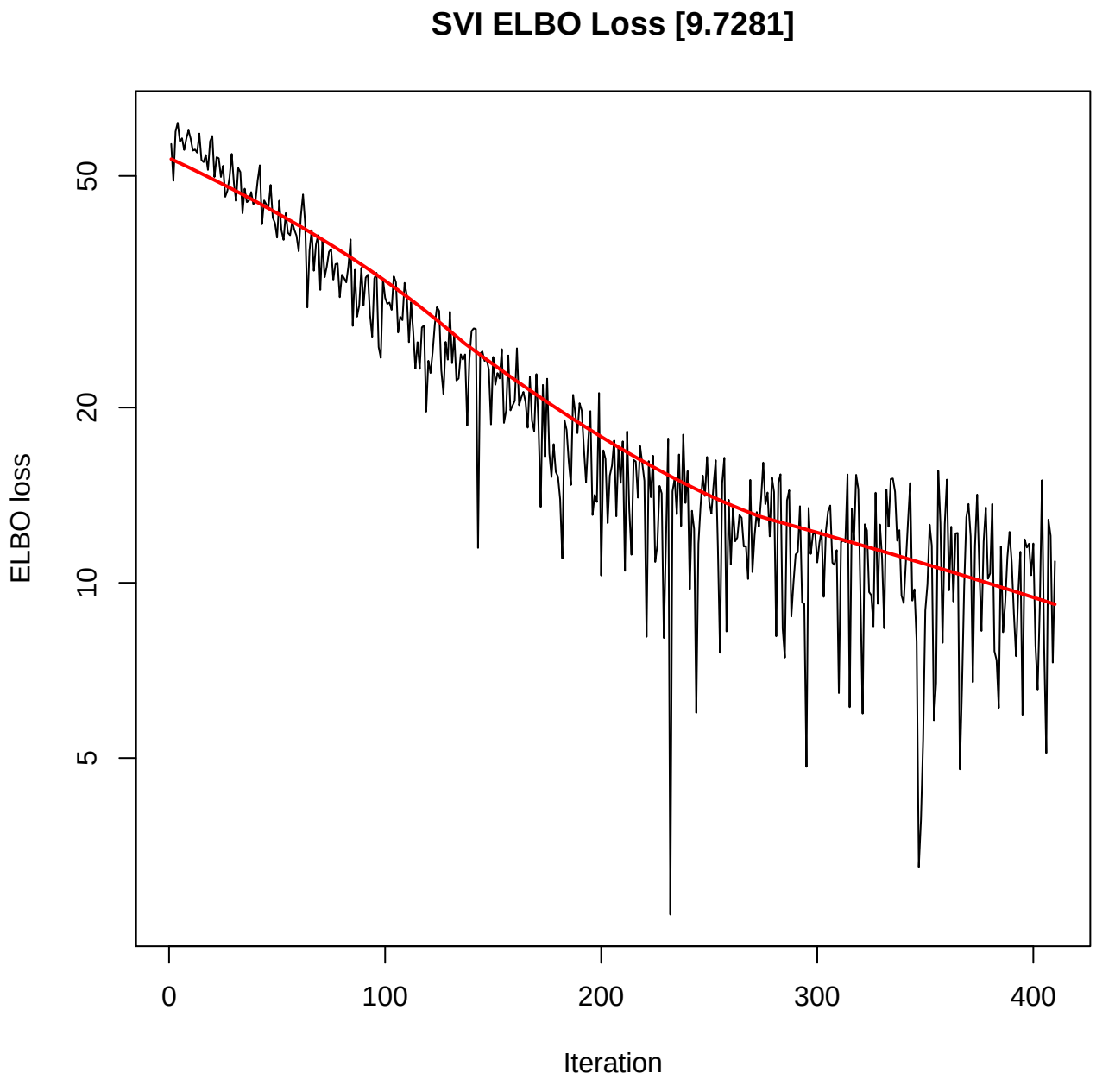

SVI ELBO Loss [18990.9035]
ELBO loss
20500
20000
19500
19000
18500
18000
17500
5
10
15
20
Iteration

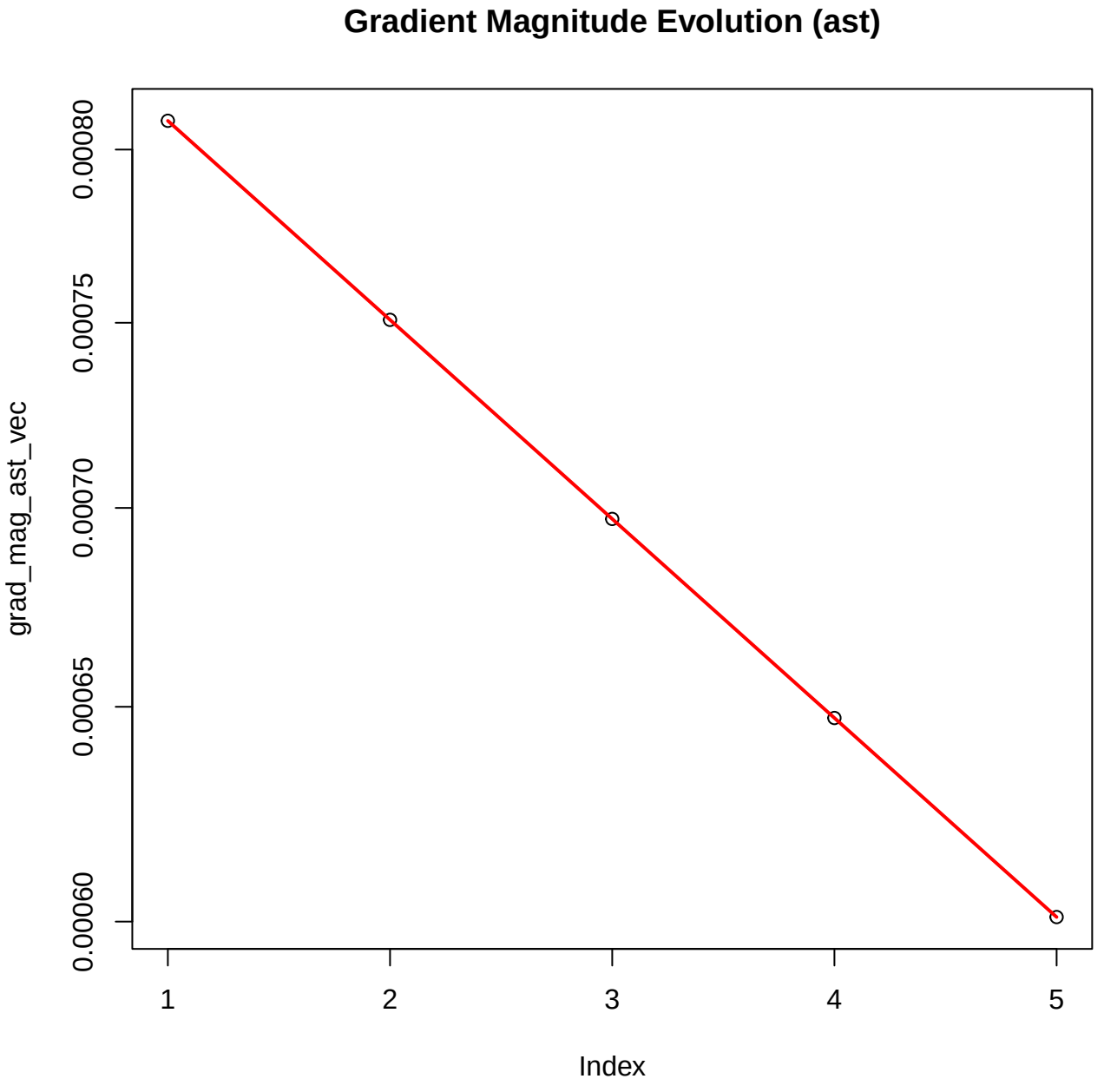

SVI ELBO Loss [19461.8490]
ELBO loss
22000
21000
20000
19000
18000
5
10
15
20
Iteration

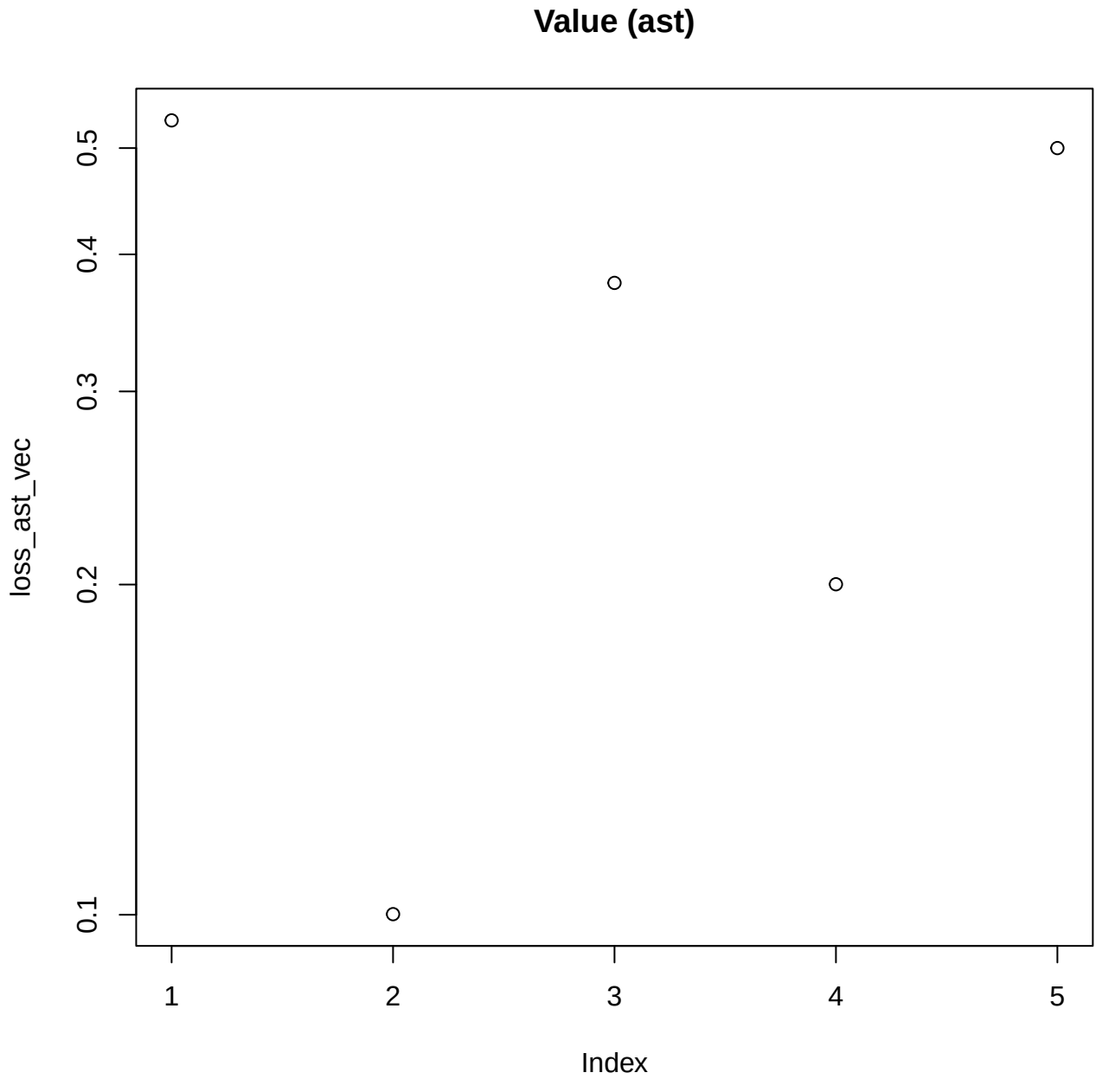

SVI ELBO Loss [9020.6244]
ELBO loss
16000
14000
12000
10000
0
50
100
150
200
Iteration

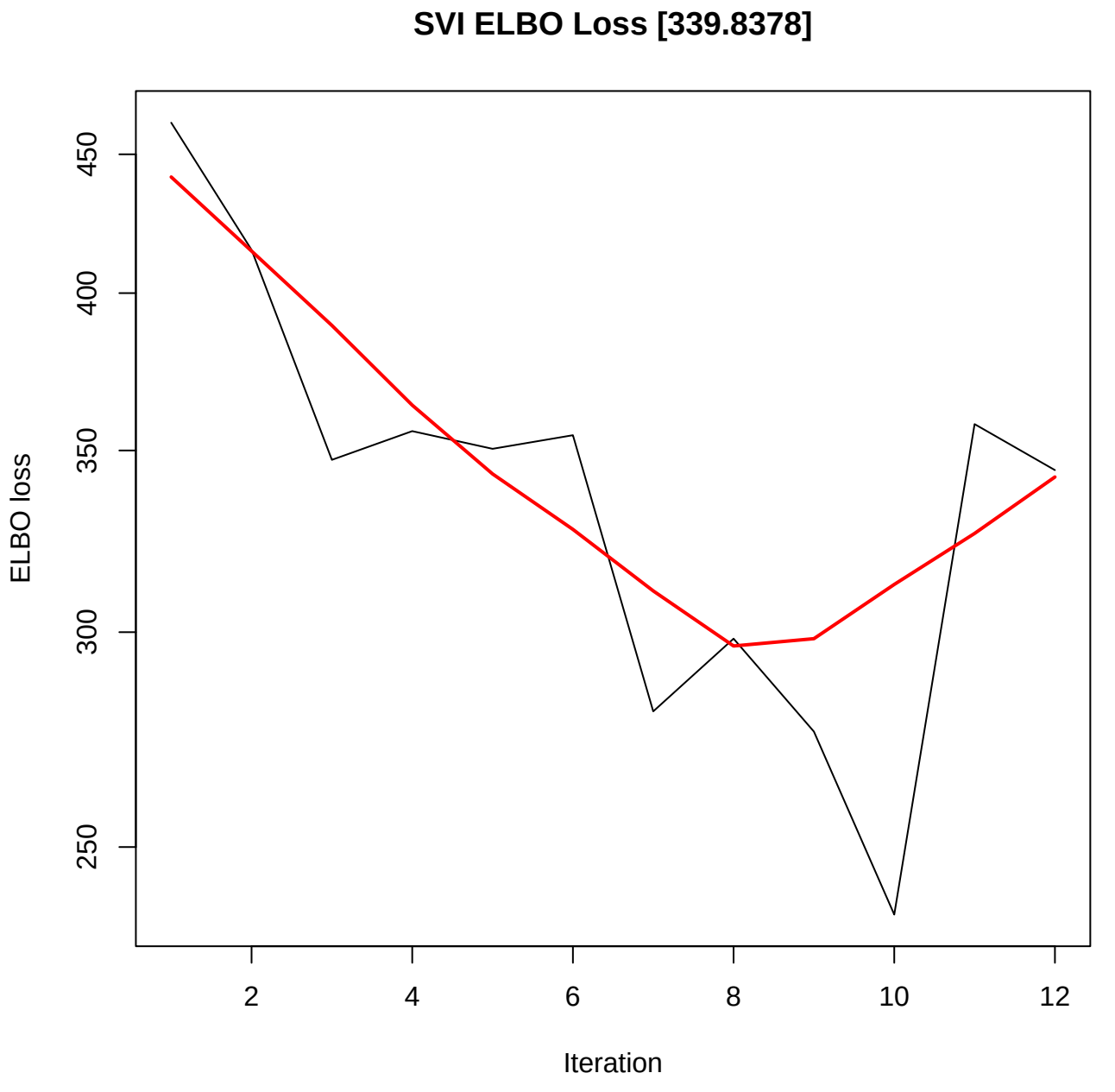

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

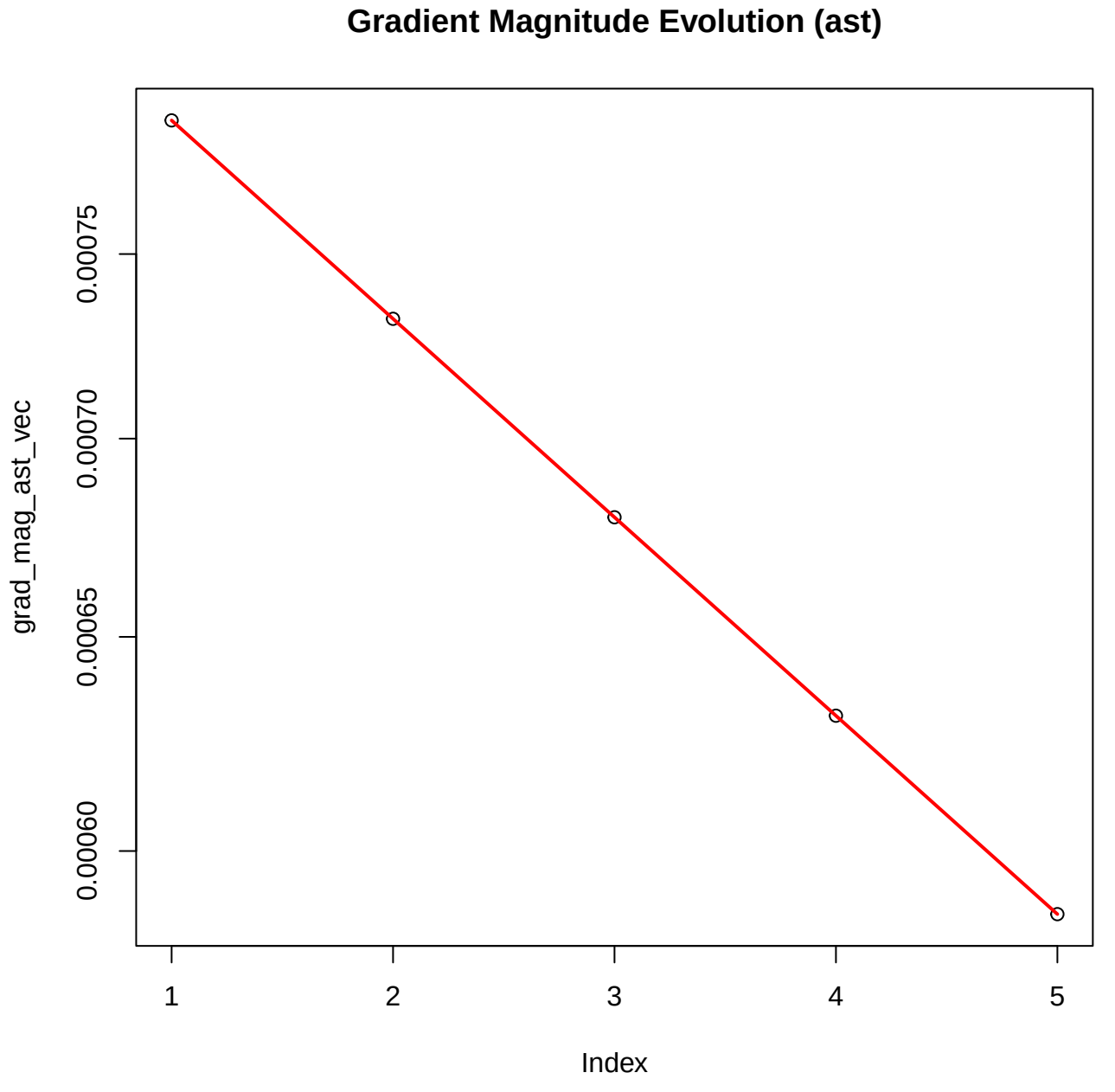

Value (ast)
loss_ast_vec
0.46
0.48
0.50
0.52
0.54
0.56
1
2
3
4
5
Index

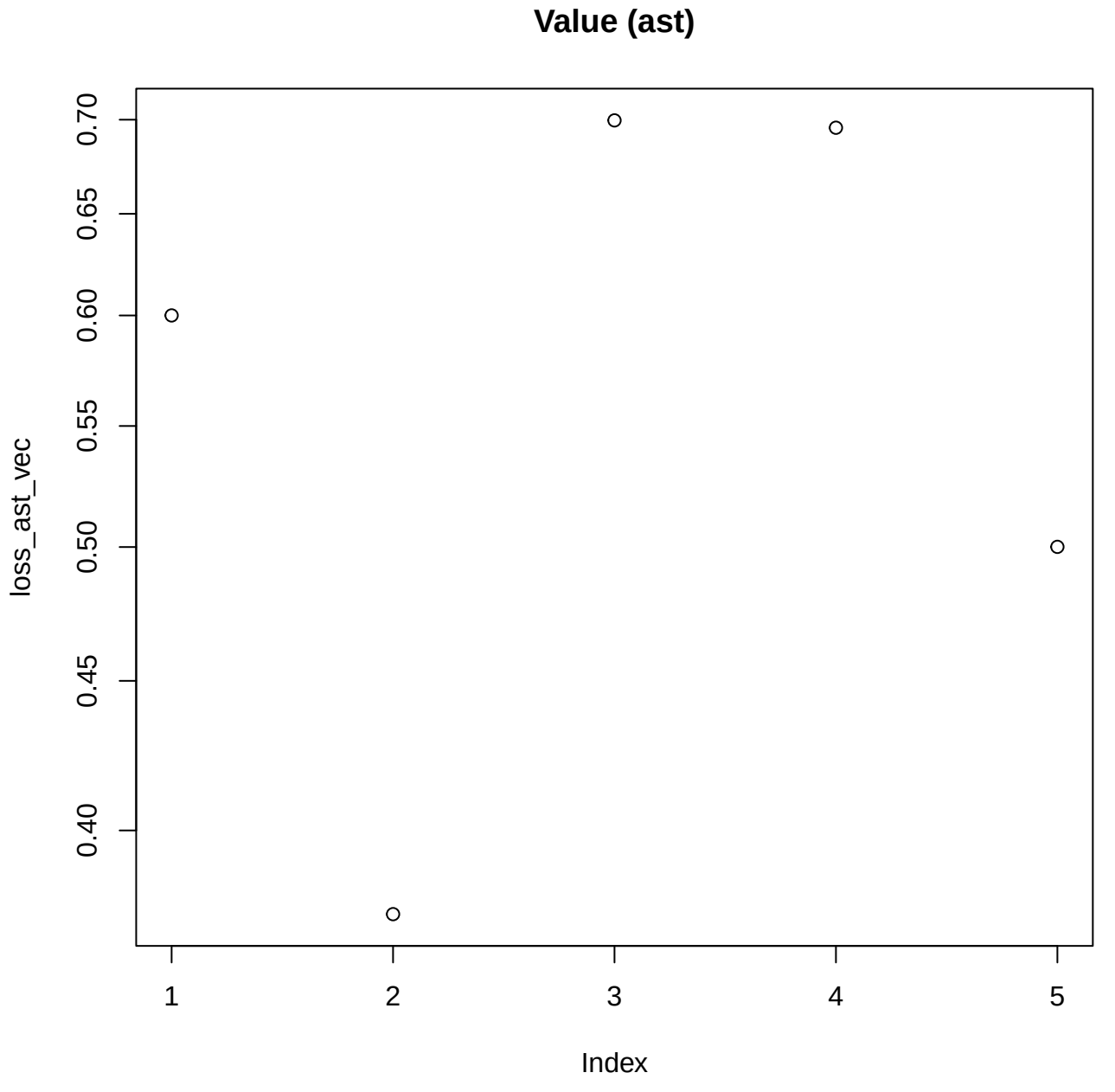

SVI ELBO Loss [9.7281]
ELBO loss
50
20
10
5
0
100
200
300
400
Iteration

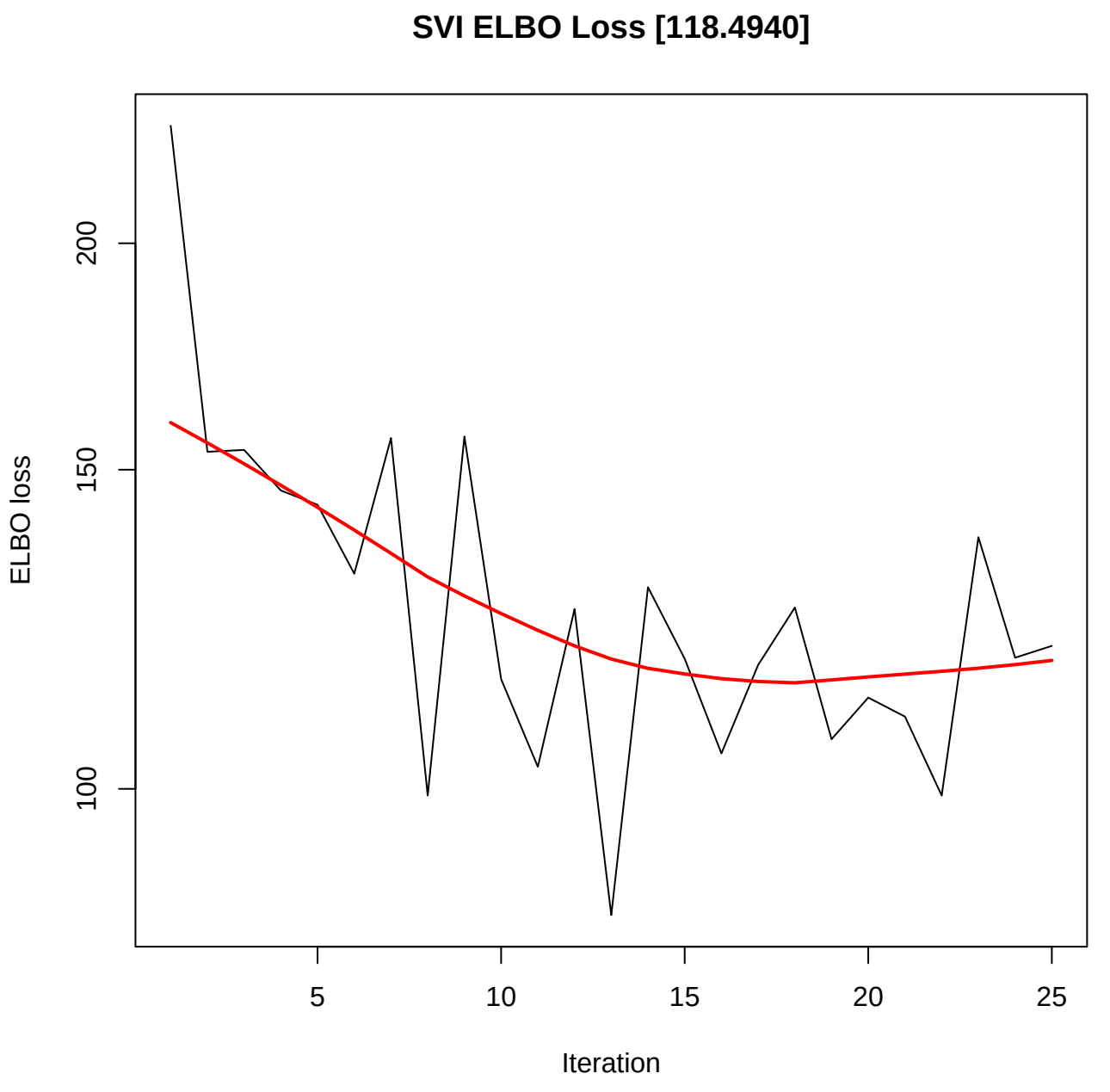

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

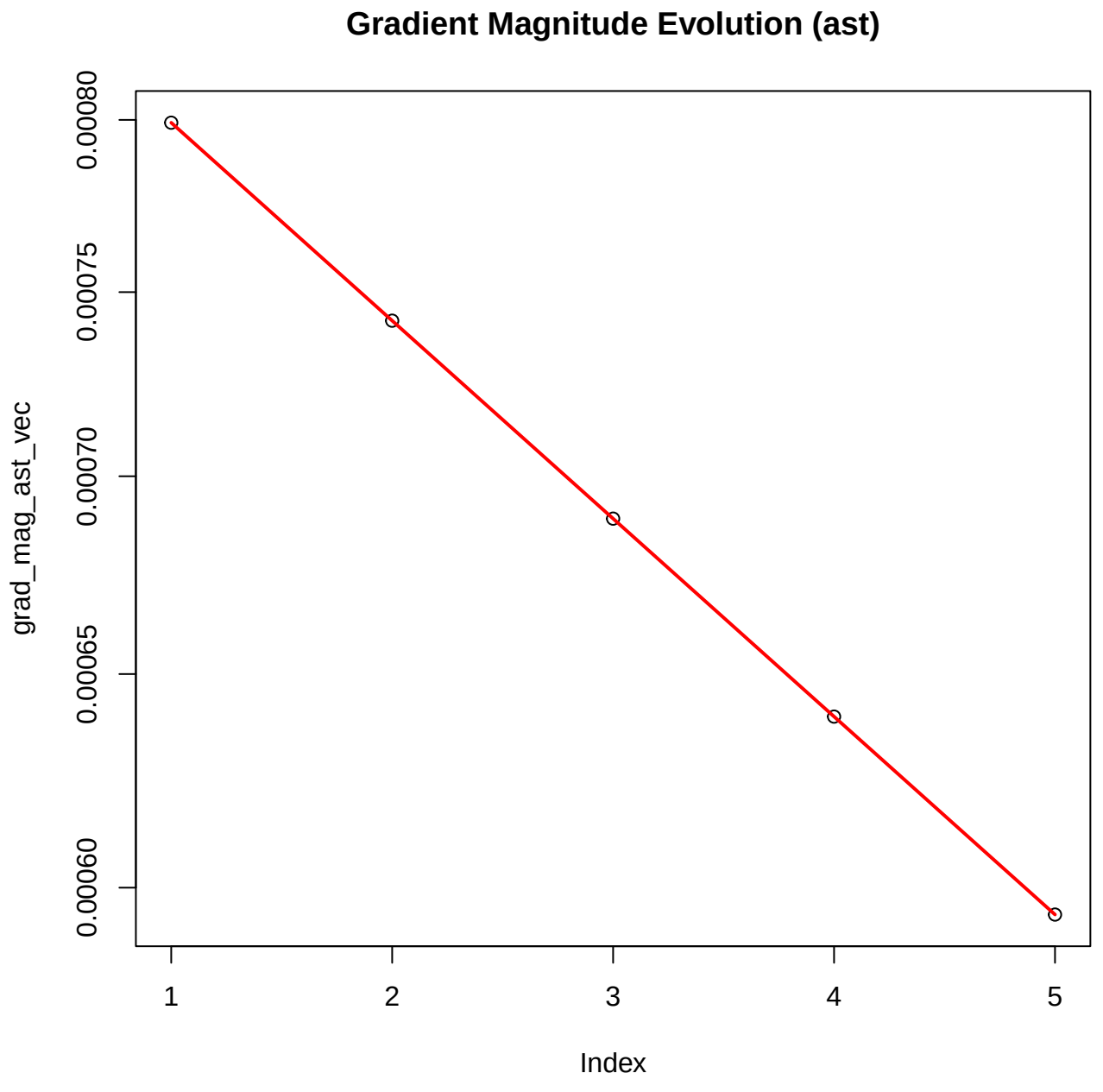

Value (ast)
loss_ast_vec
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
Index

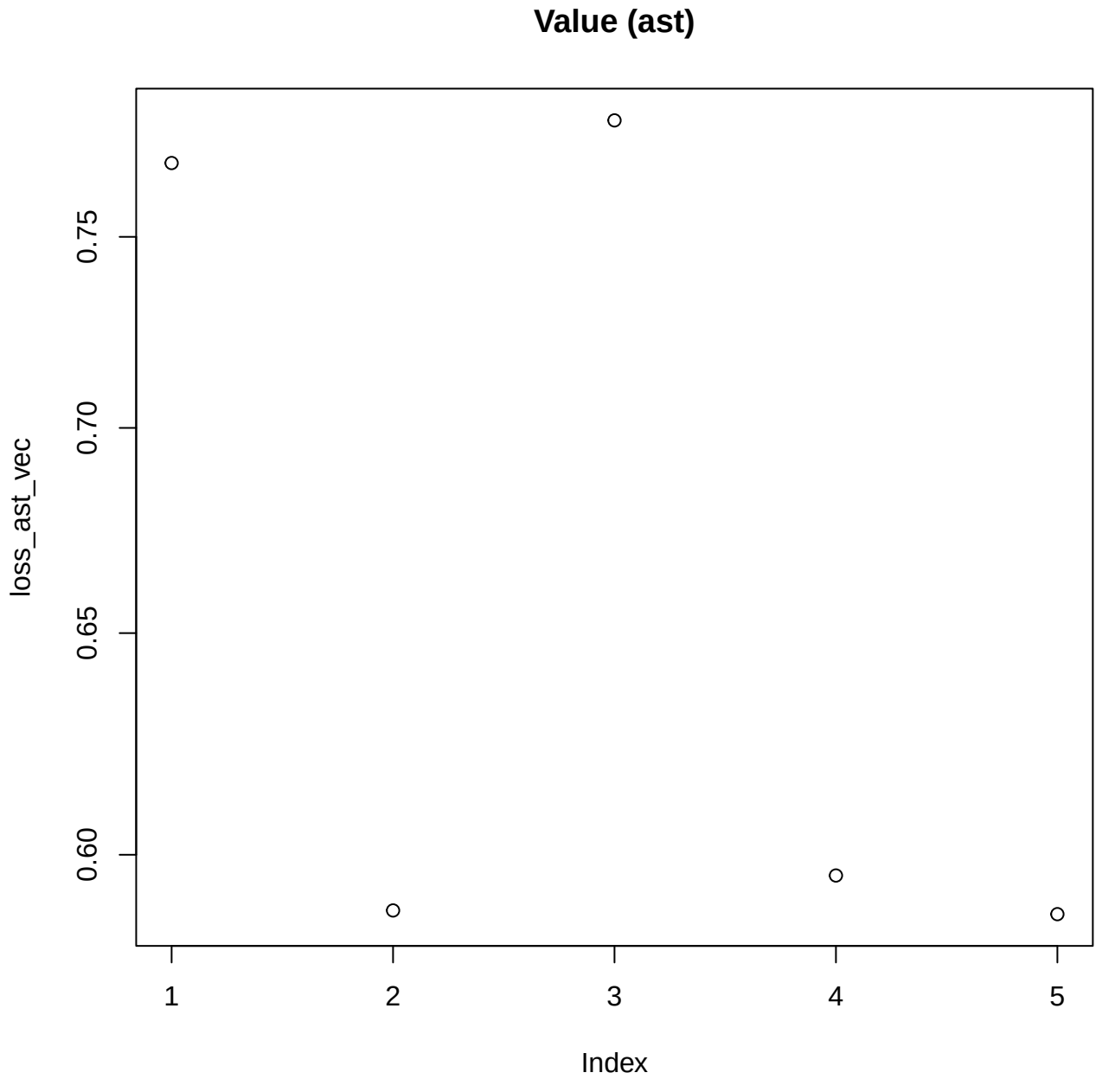

SVI ELBO Loss [339.8378]
ELBO loss
450
400
350
300
250
2
4
6
8
10
12
Iteration

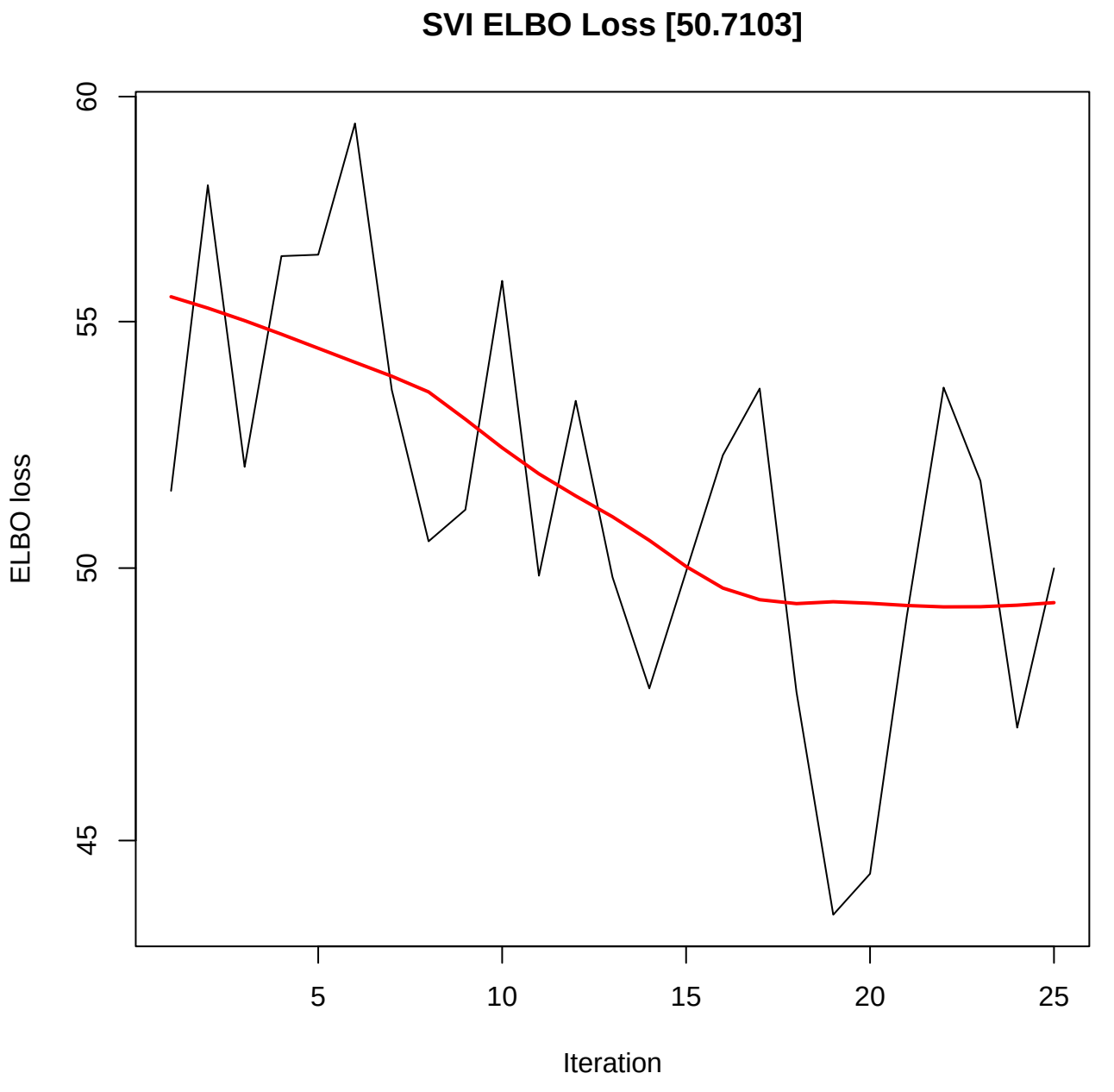

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

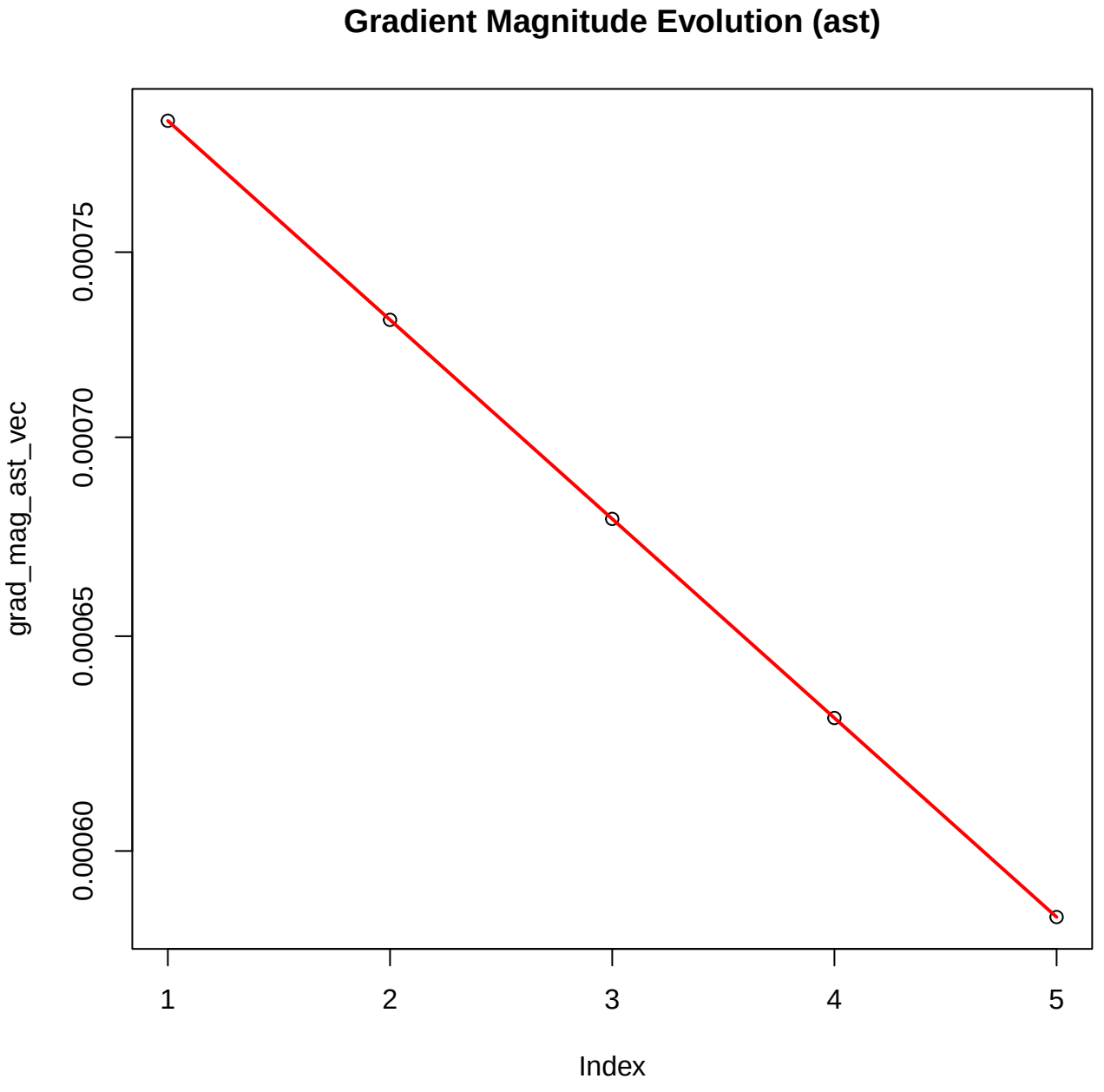

Value (ast)
loss_ast_vec
0.40
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

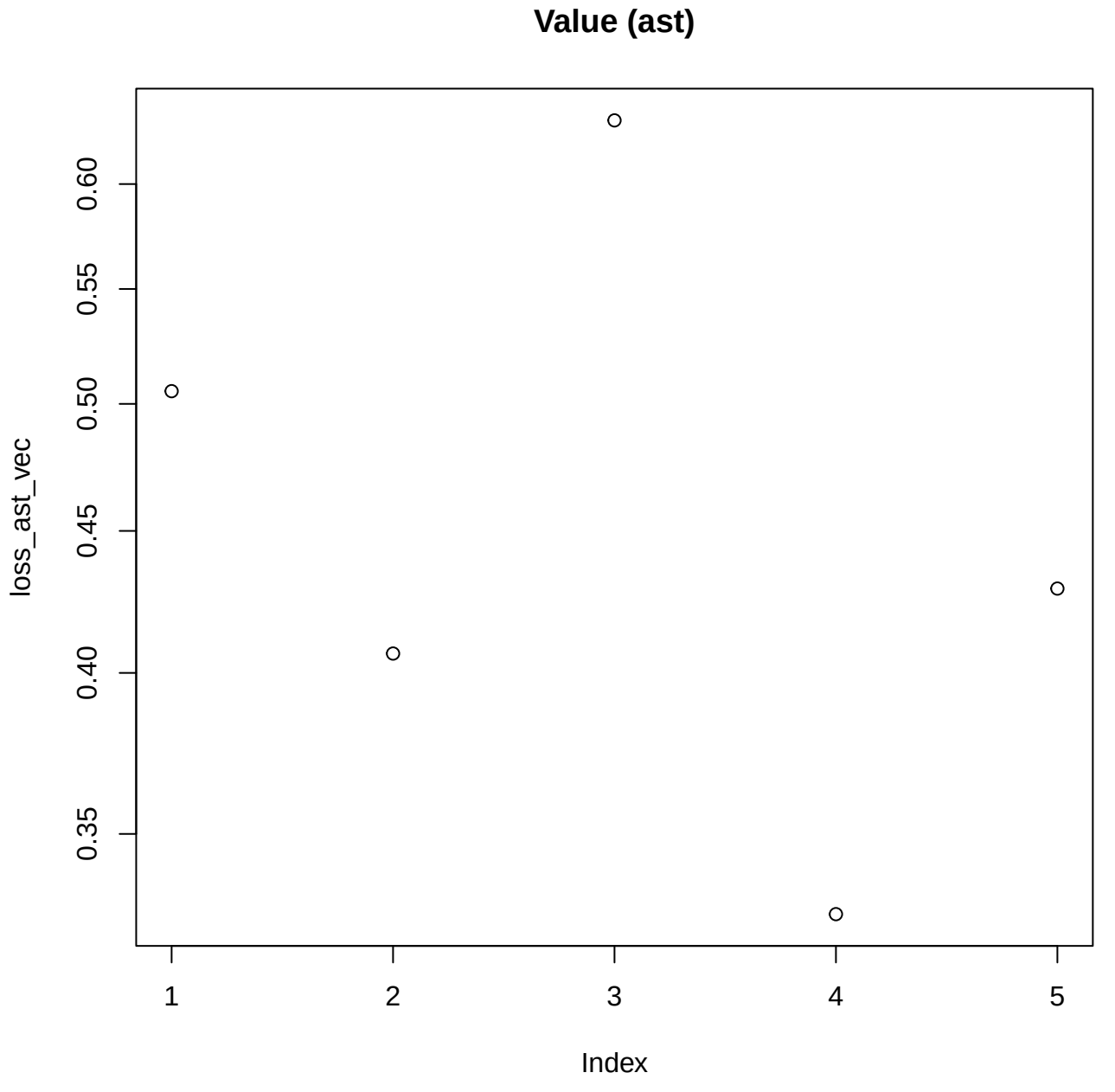

SVI ELBO Loss [118.4940]
ELBO loss
100
150
200
5
10
15
20
25
Iteration

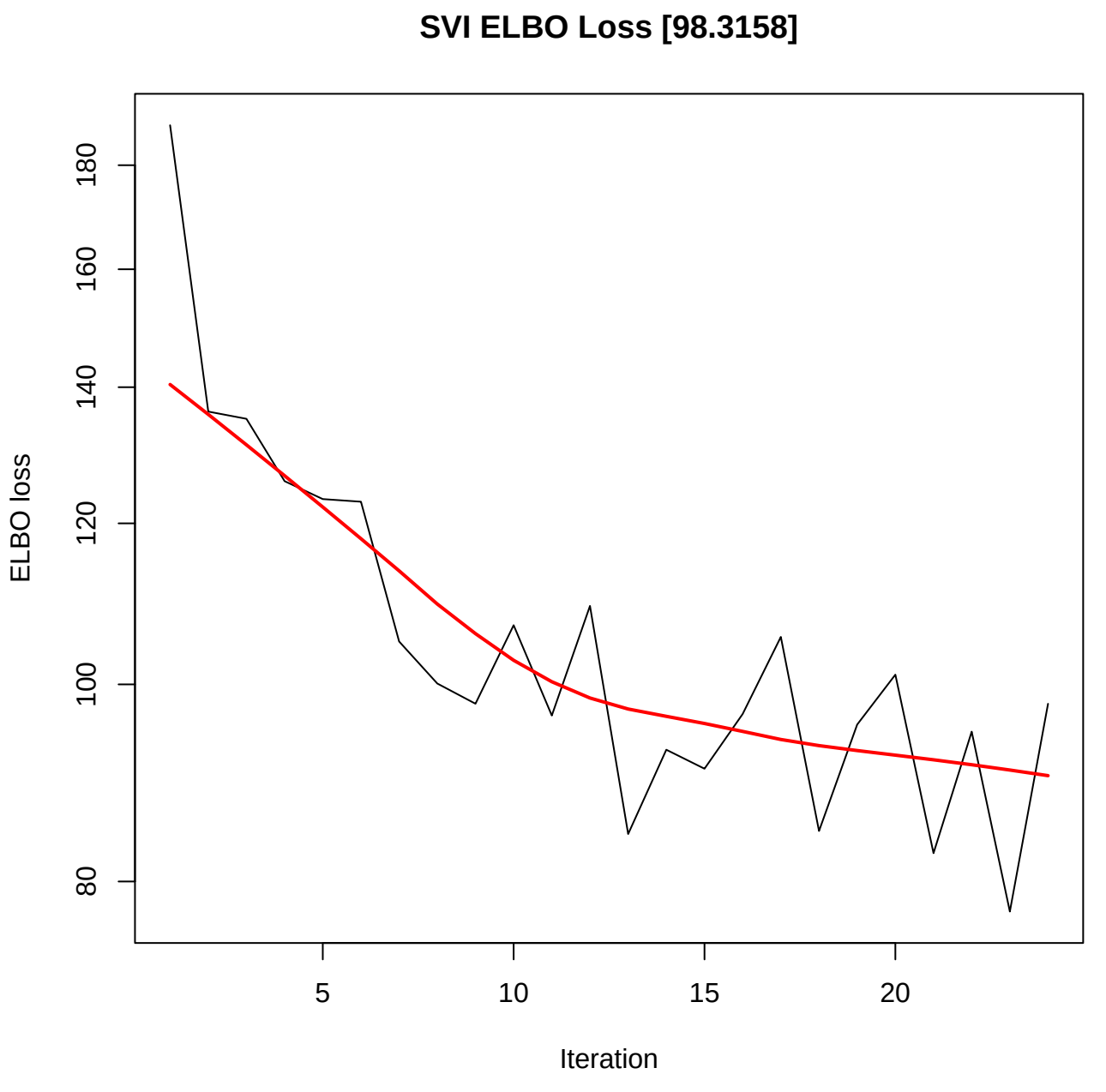

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

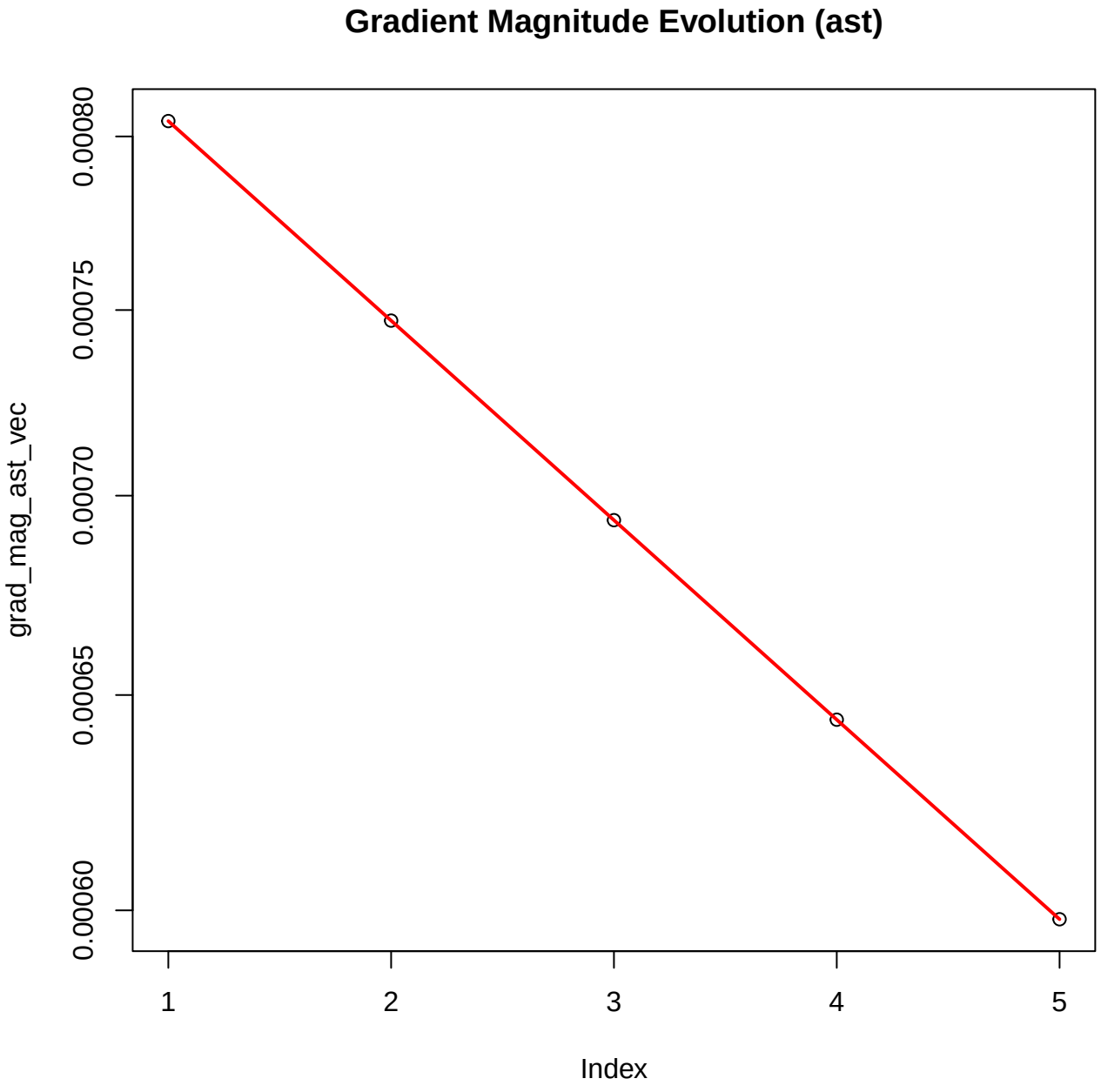

Value (ast)
loss_ast_vec
0.60
0.65
0.70
0.75
1
2
3
4
5
Index

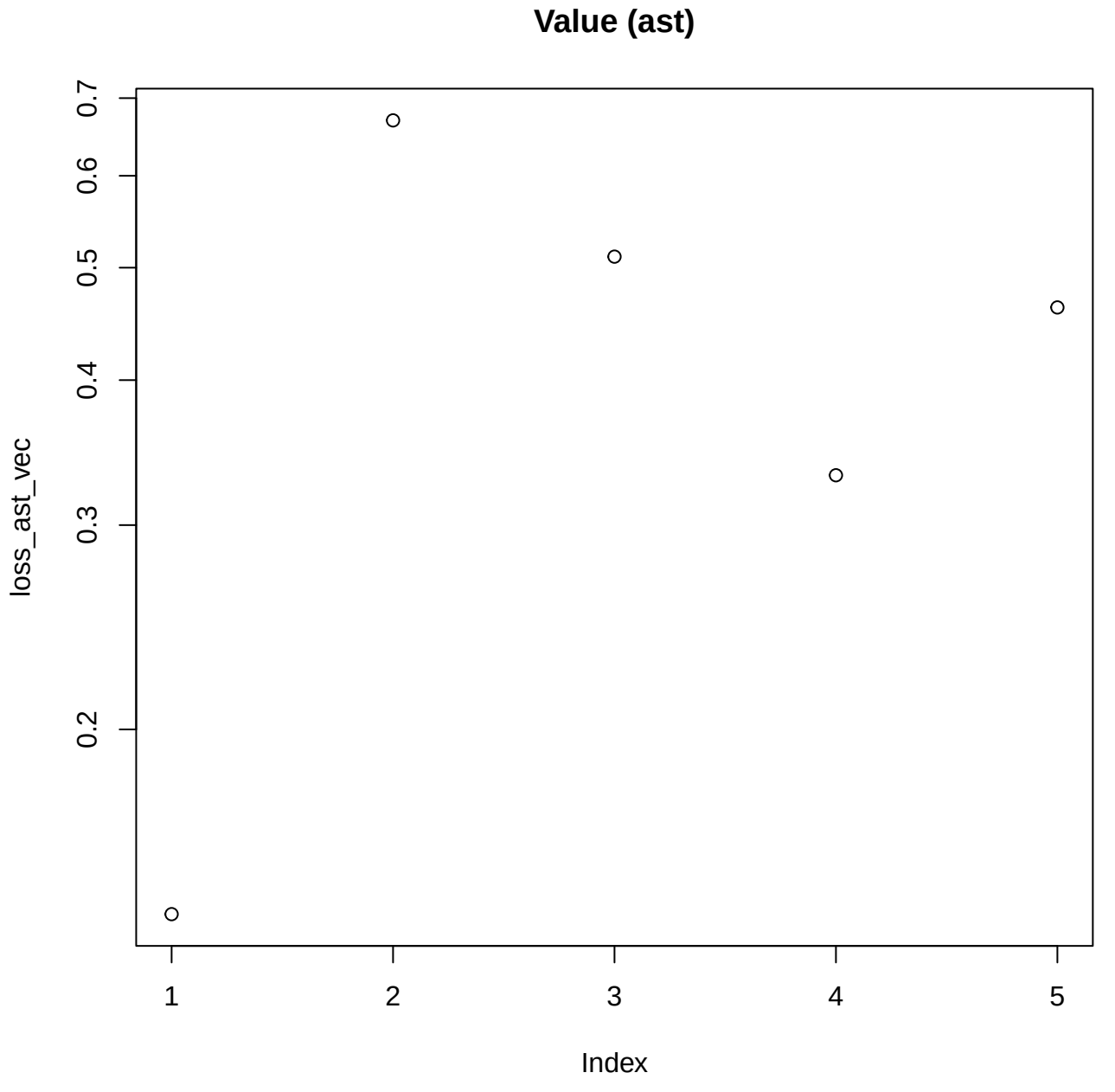

SVI ELBO Loss [50.7103]
ELBO loss
60
55
50
45
5
10
15
20
25
Iteration

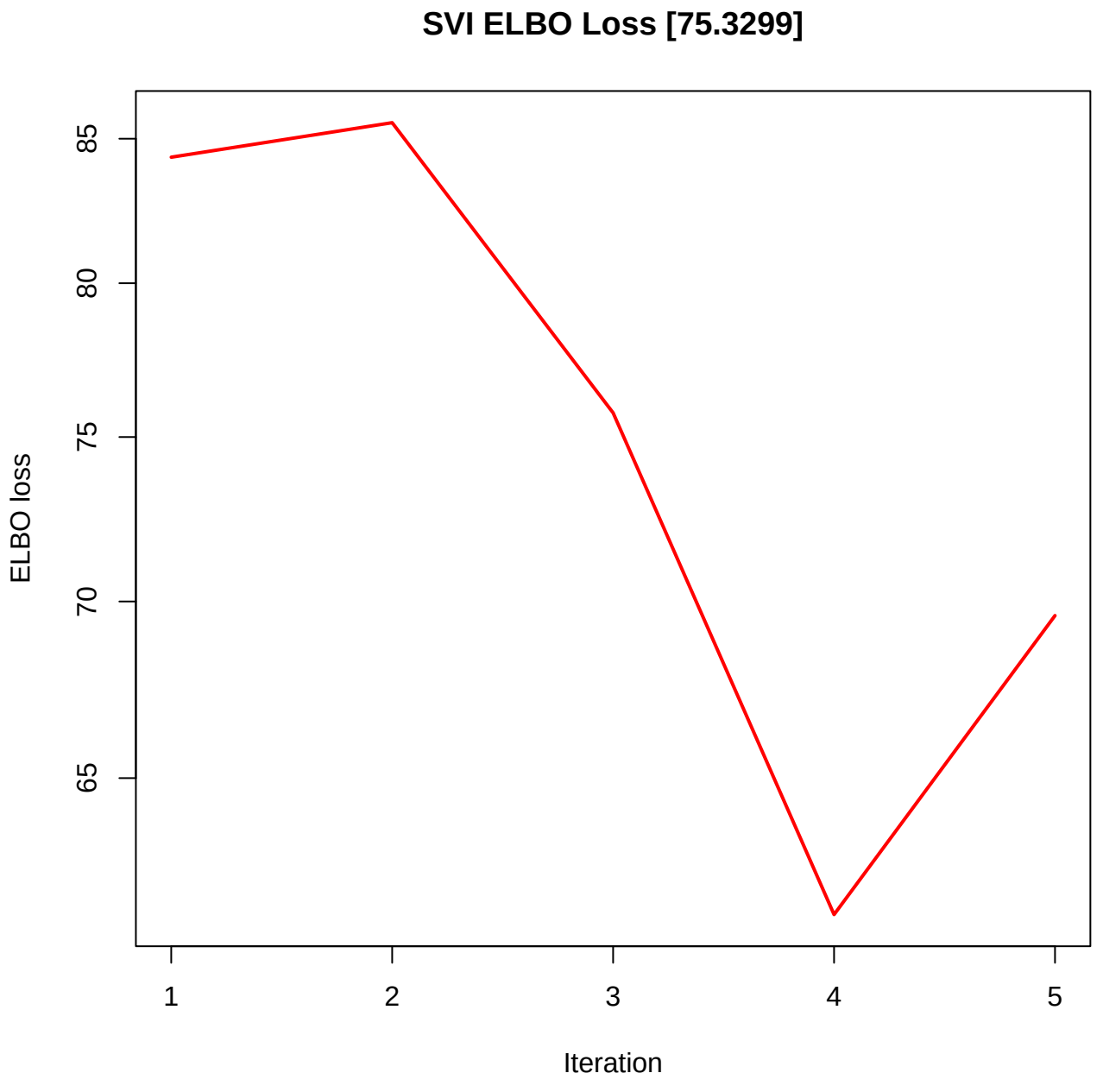

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

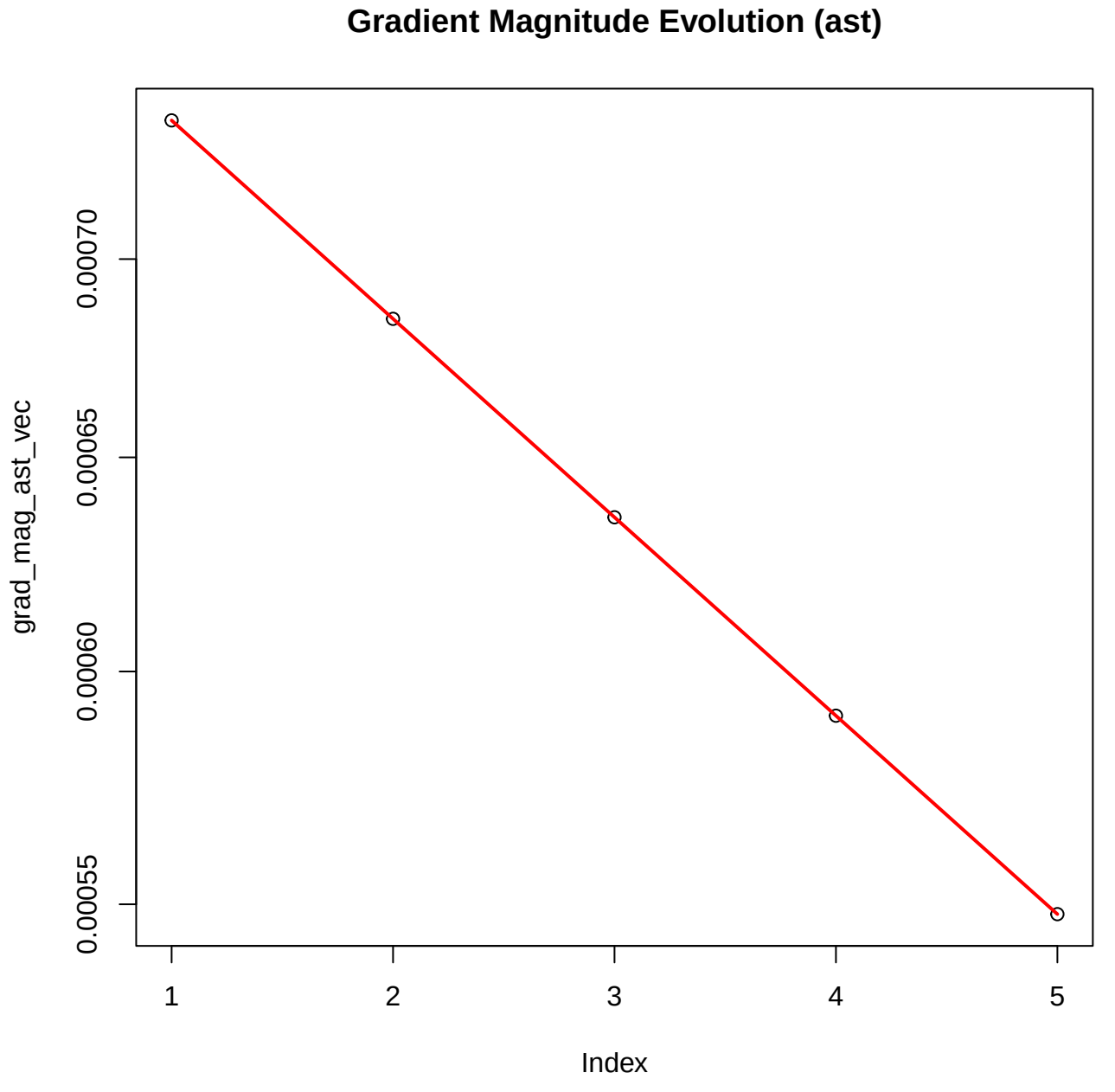

Value (ast)
loss_ast_vec
0.35
0.40
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

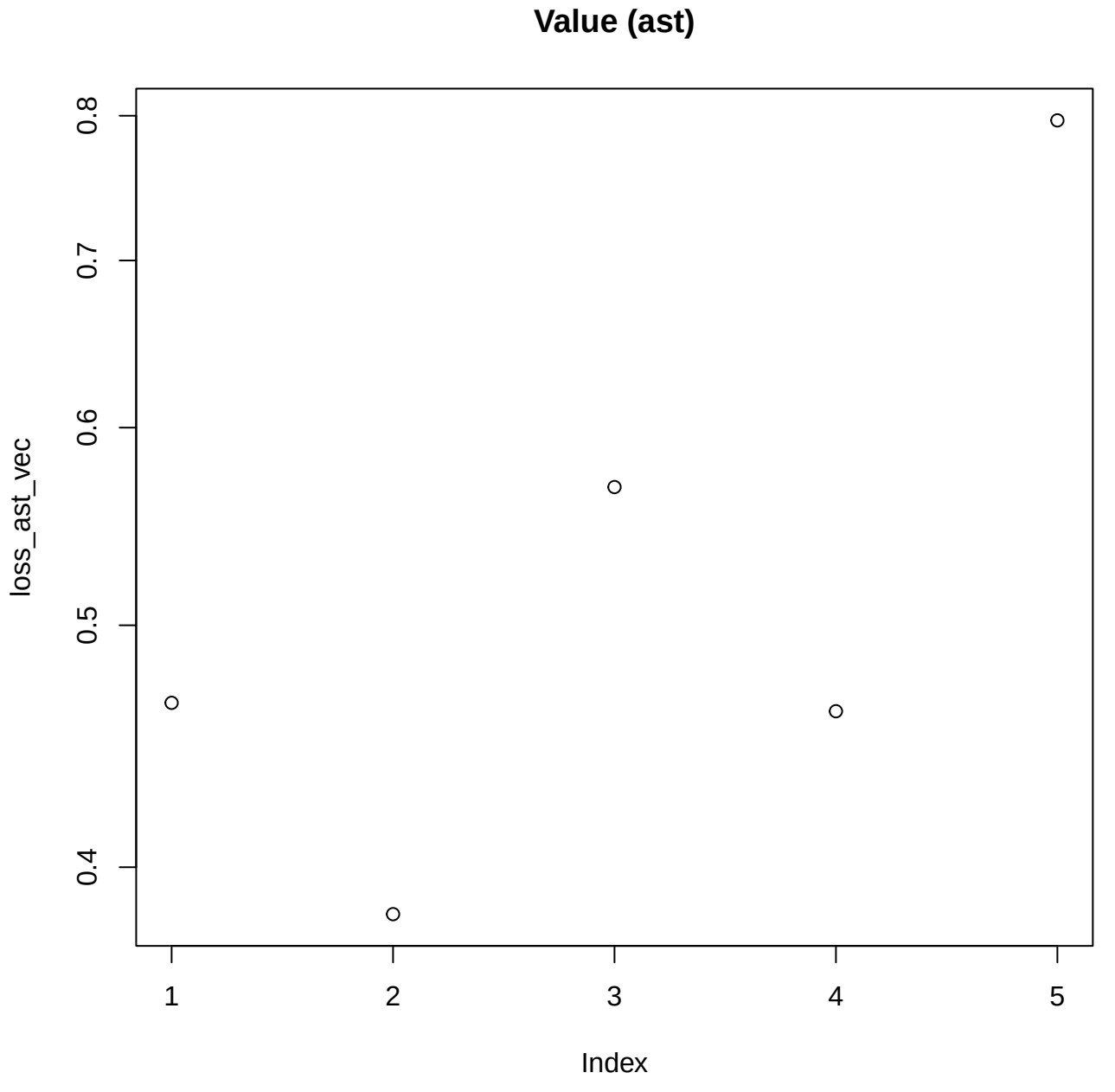

SVI ELBO Loss [98.3158]
ELBO loss
180
160
140
120
100
80
5
10
15
20
Iteration

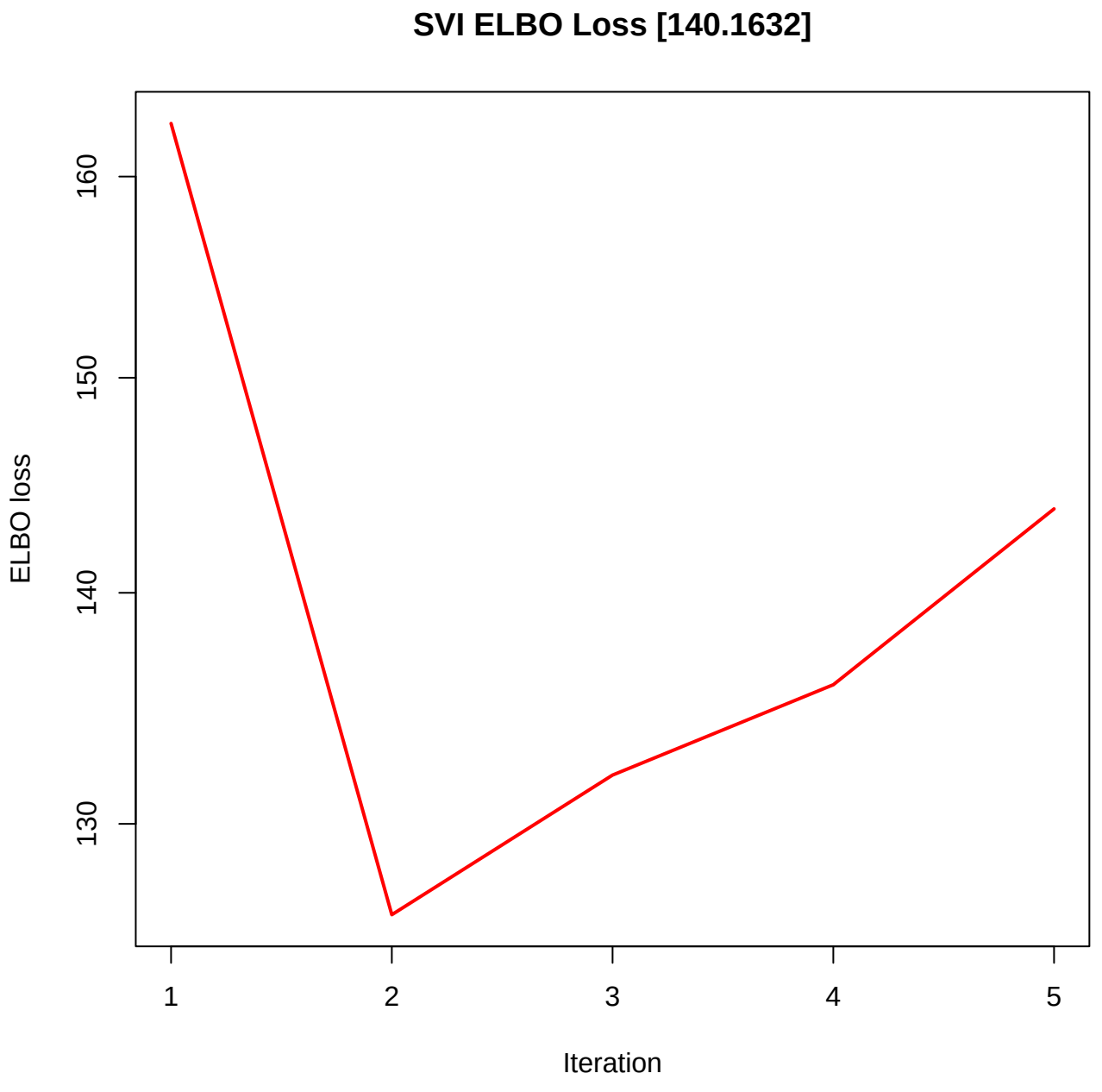

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

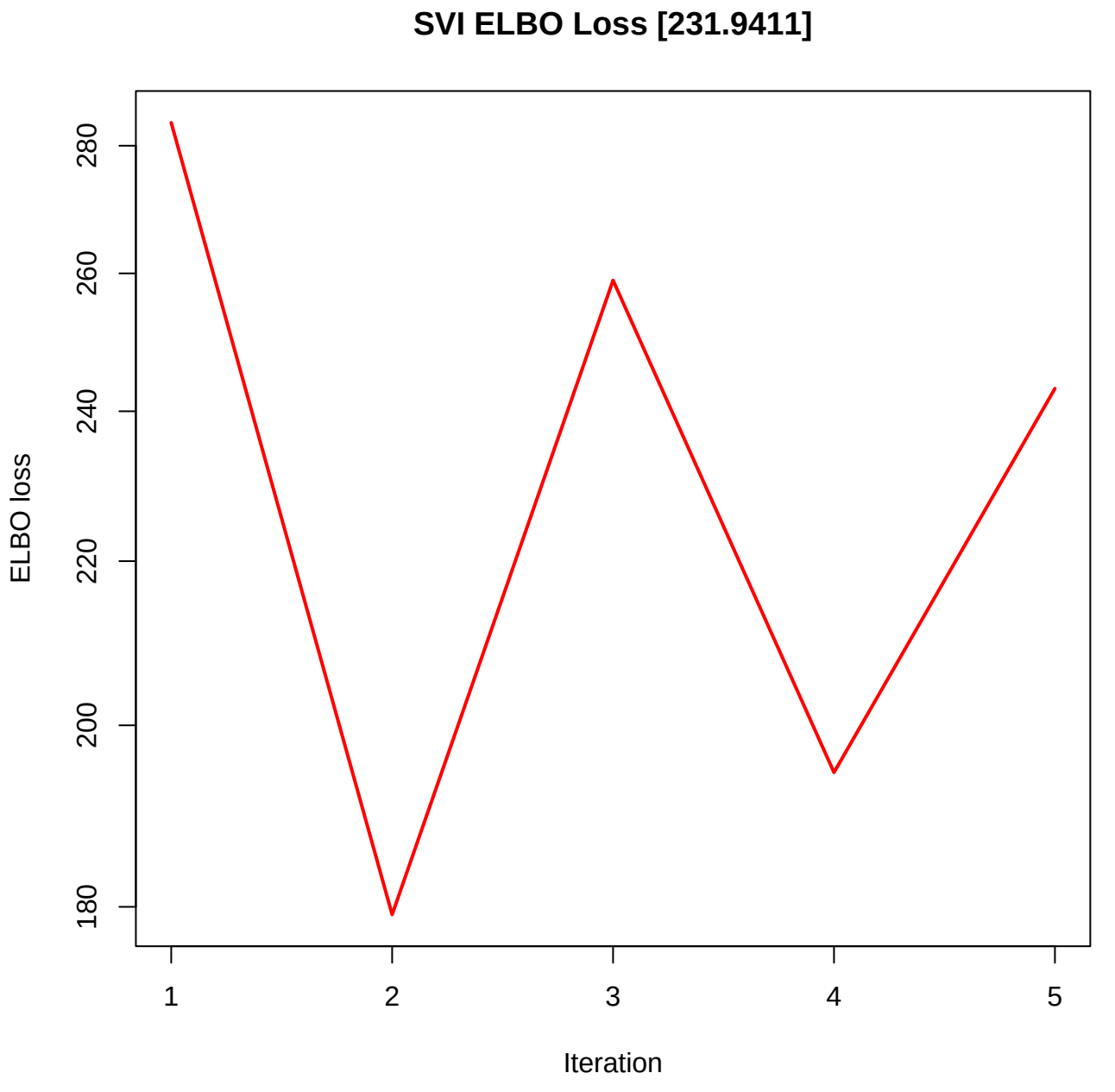

Value (ast)
loss_ast_vec
0.2
0.3
0.4
0.5
0.6
0.7
1
2
3
4
5
Index

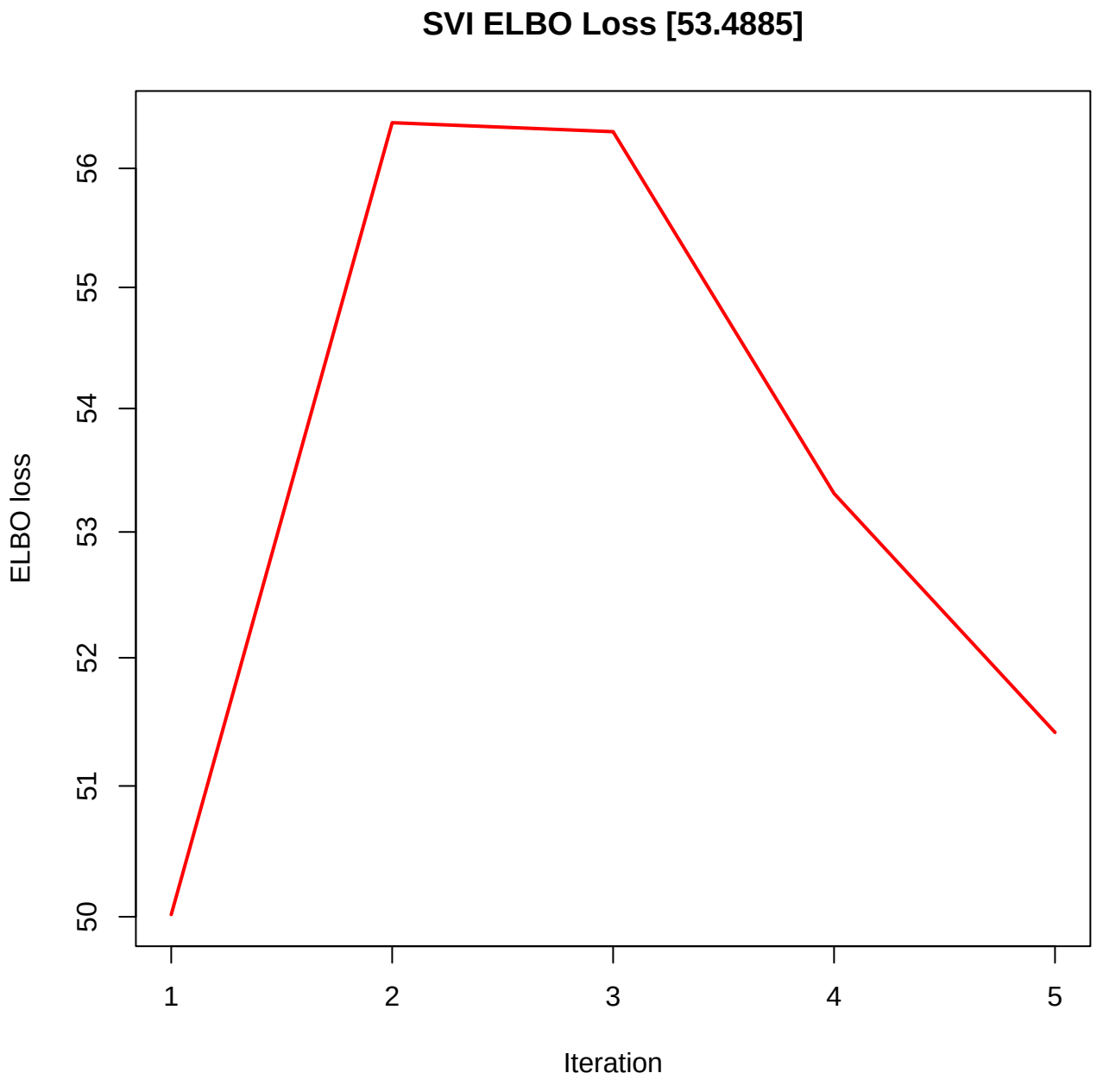

SVI ELBO Loss [75.3299]
ELBO loss
85
80
75
70
65
1
2
3
4
5
Iteration

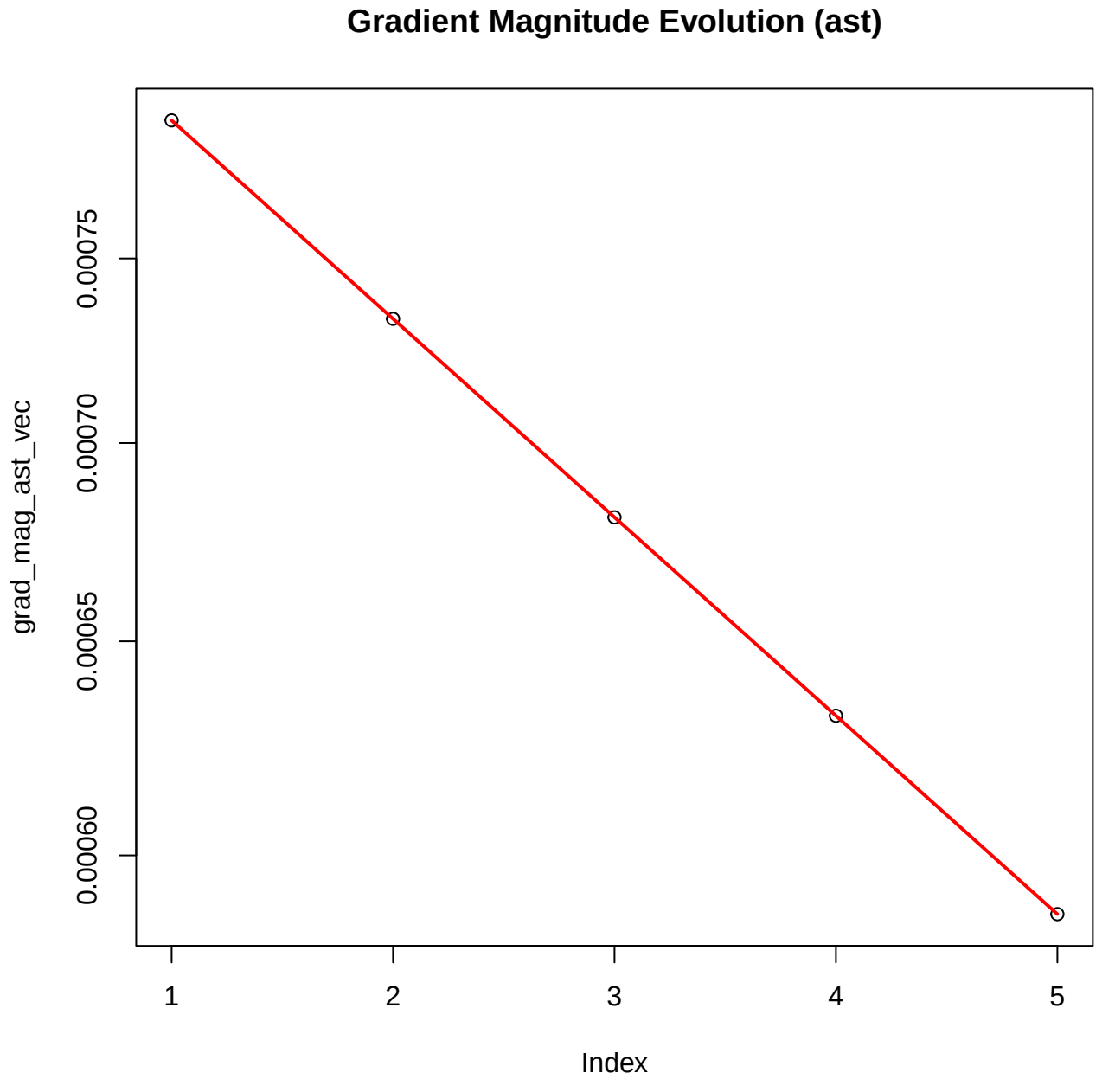

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00055
0.00060
0.00065
0.00070
1
2
3
4
5
Index

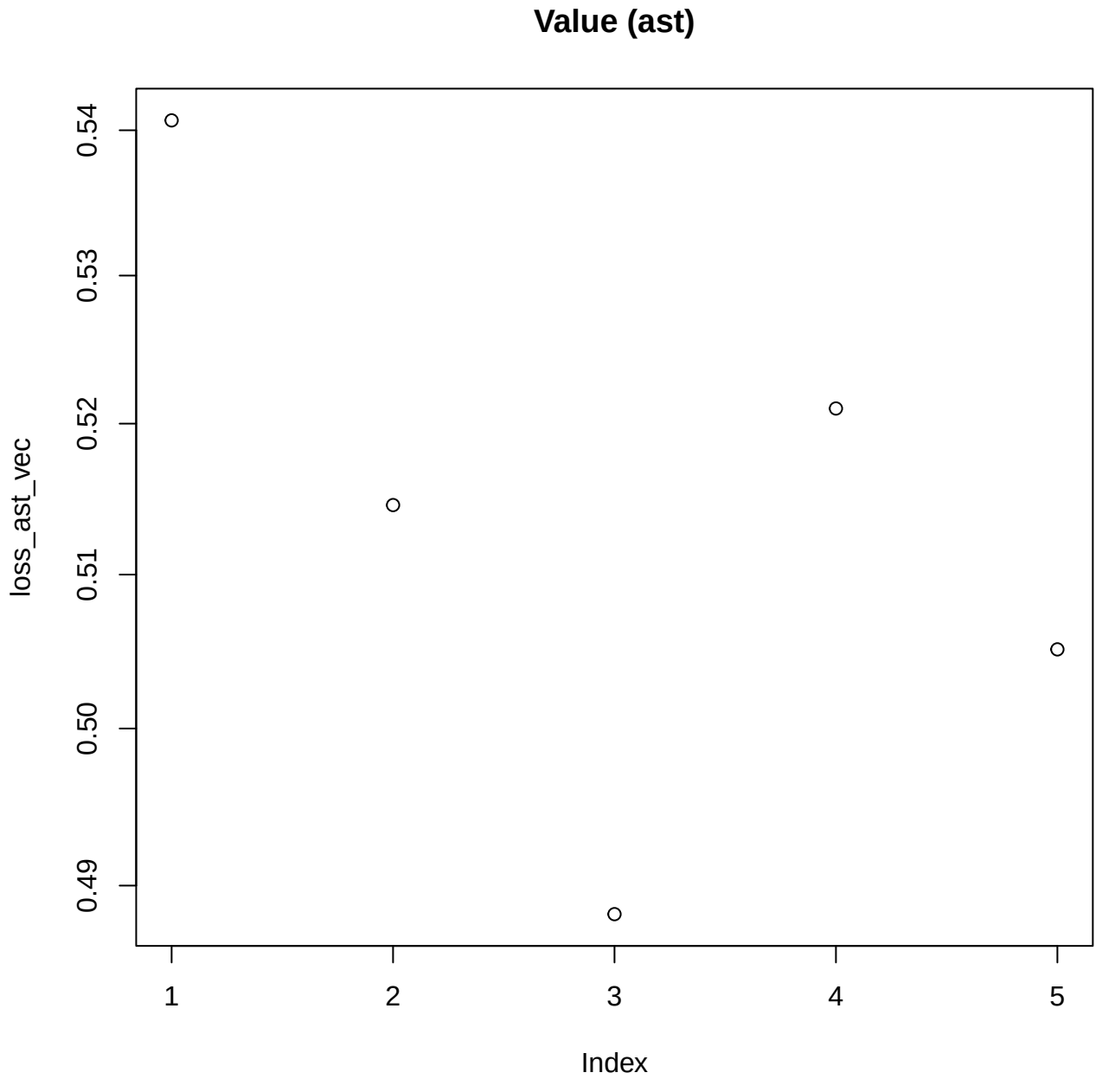

Value (ast)
loss_ast_vec
0.4
0.5
0.6
0.7
0.8
1
2
3
4
5
Index

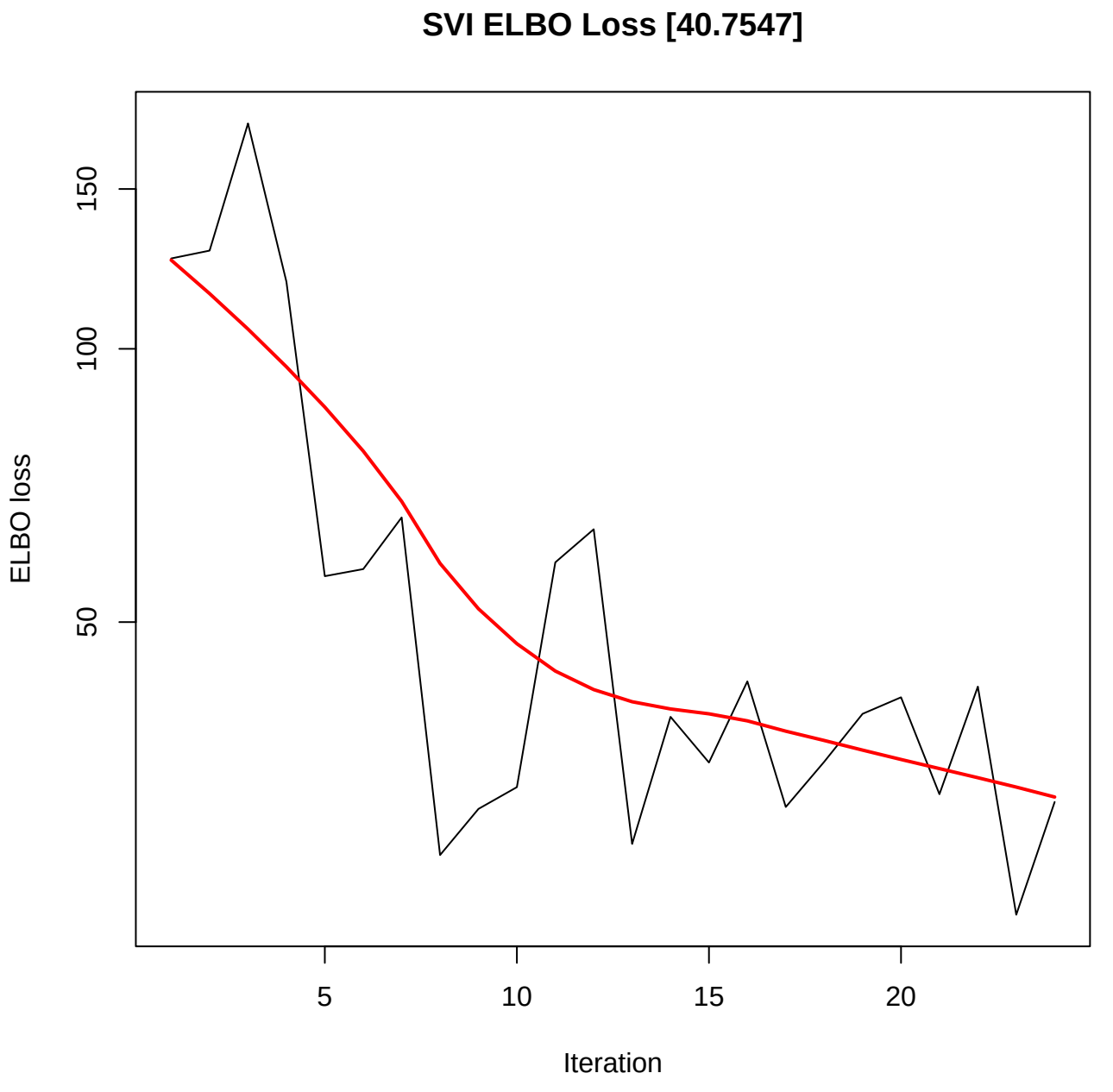

SVI ELBO Loss [140.1632]
ELBO loss
160
150
140
130
1
2
3
4
5
Iteration

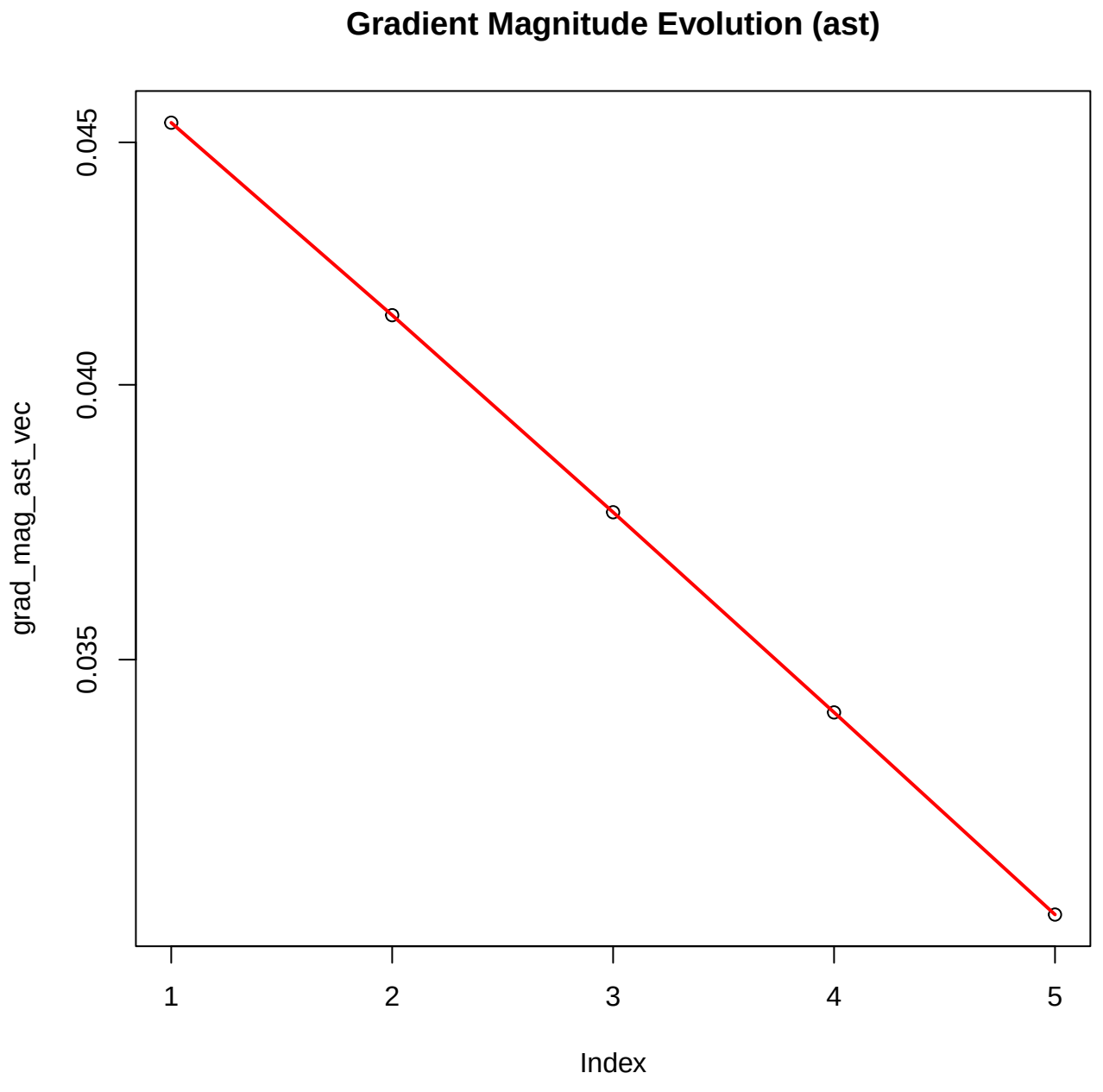

SVI ELBO Loss [231.9411]
ELBO loss
280
260
240
220
200
180
1
2
3
4
5
Iteration

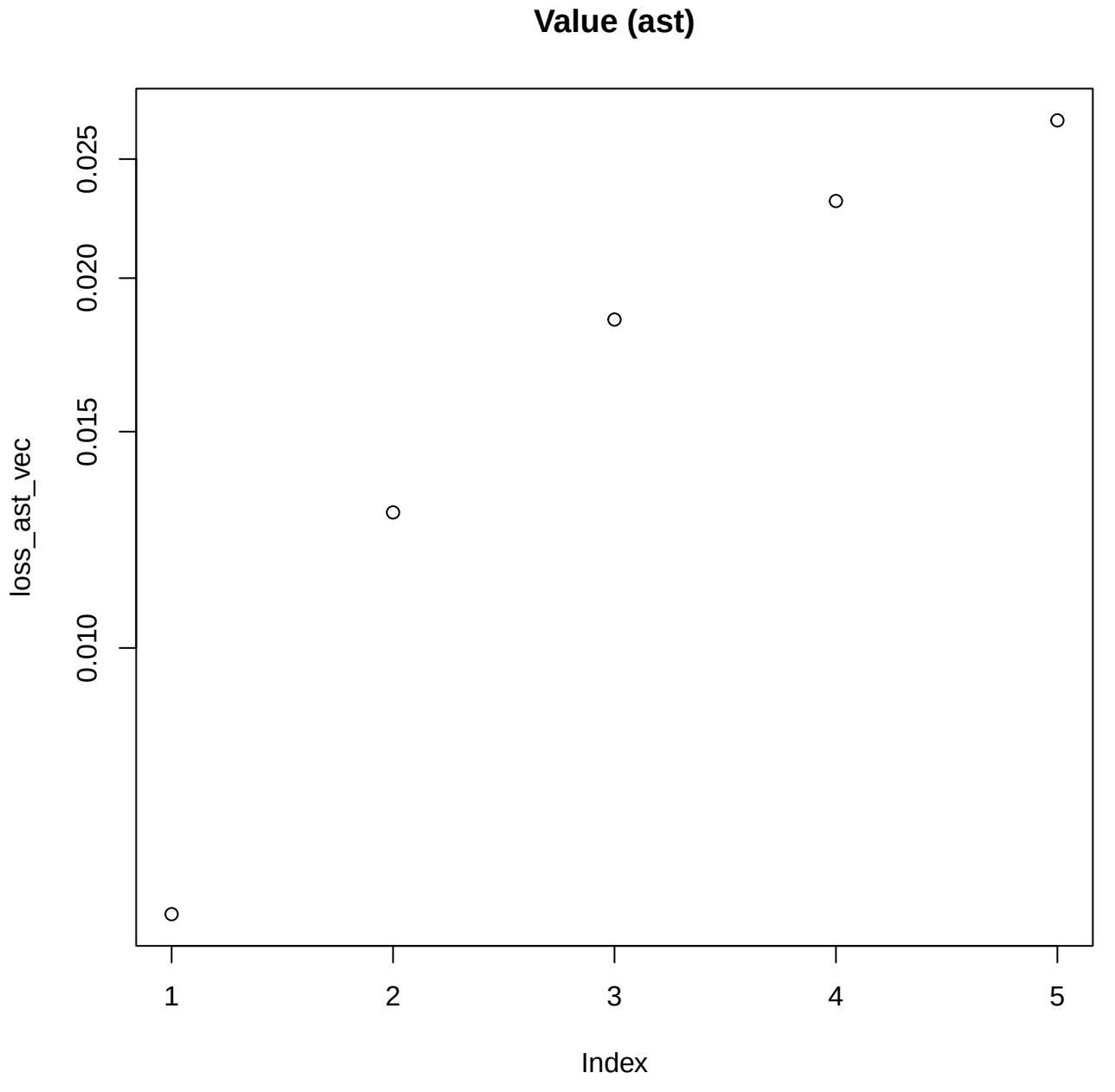

SVI ELBO Loss [53.4885]
ELBO loss
50
51
52
53
54
55
56
1
2
3
4
5
Iteration

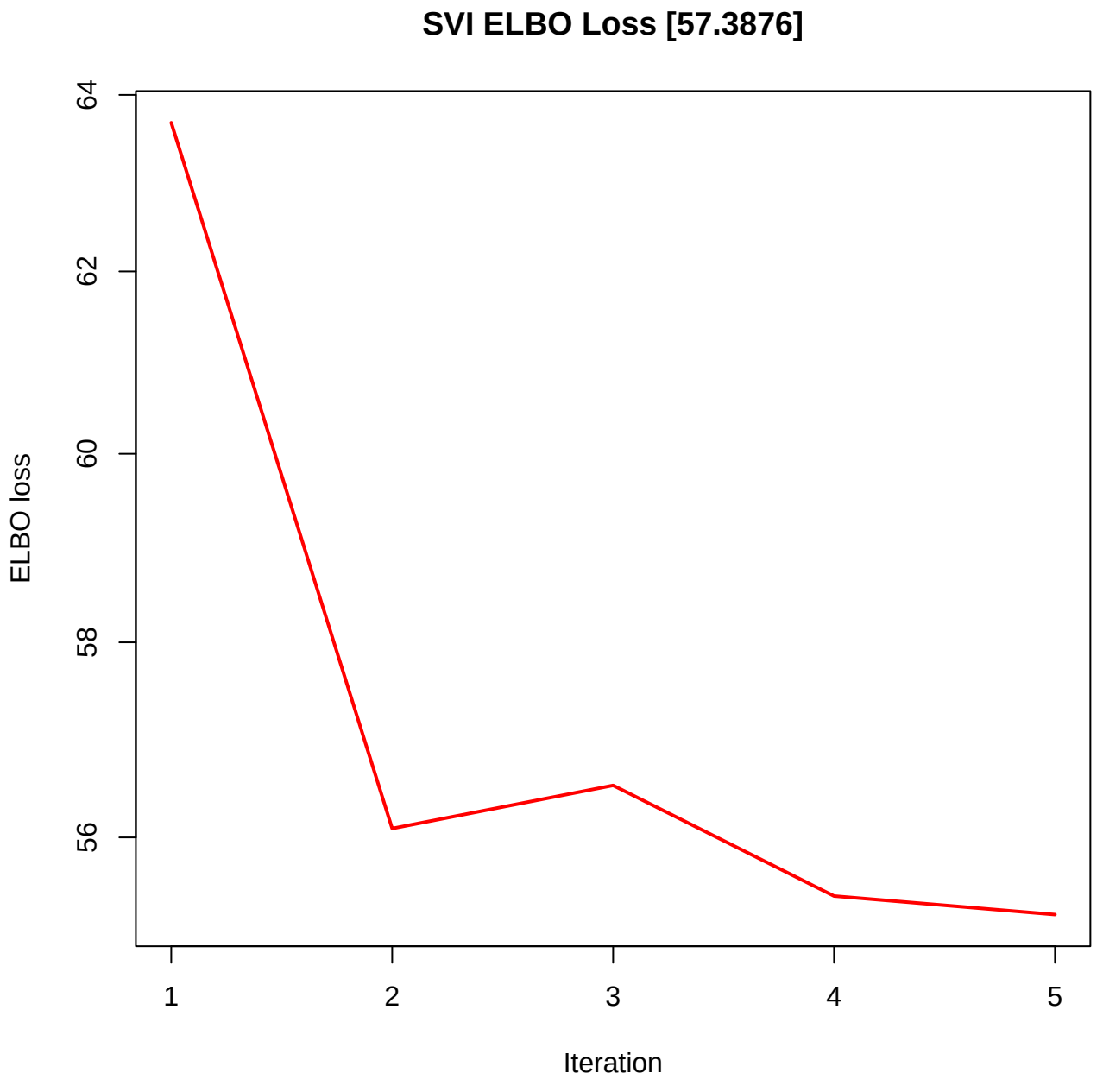

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

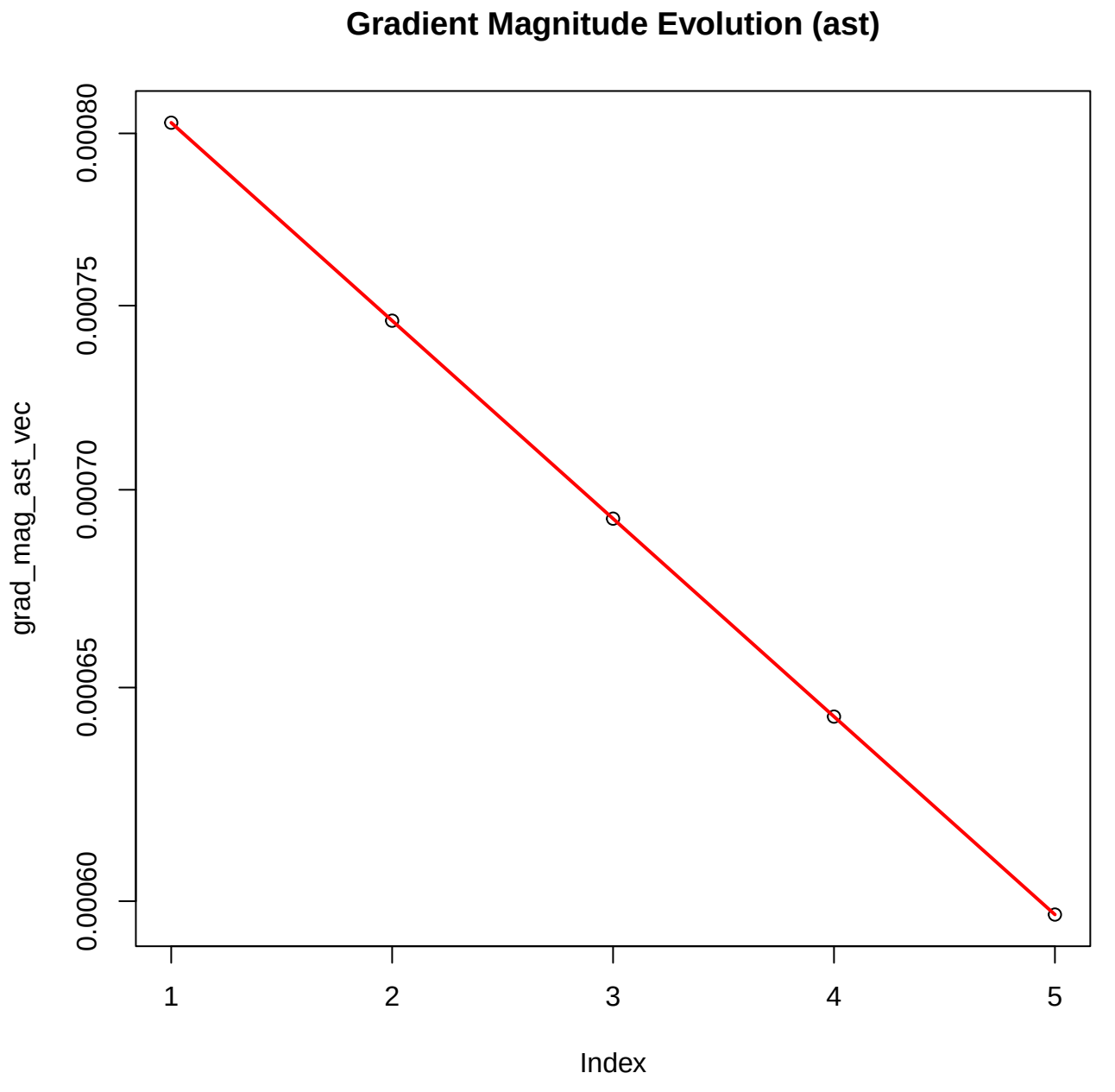

Value (ast)
loss_ast_vec
0.49
0.50
0.51
0.52
0.53
0.54
1
2
3
4
5
Index

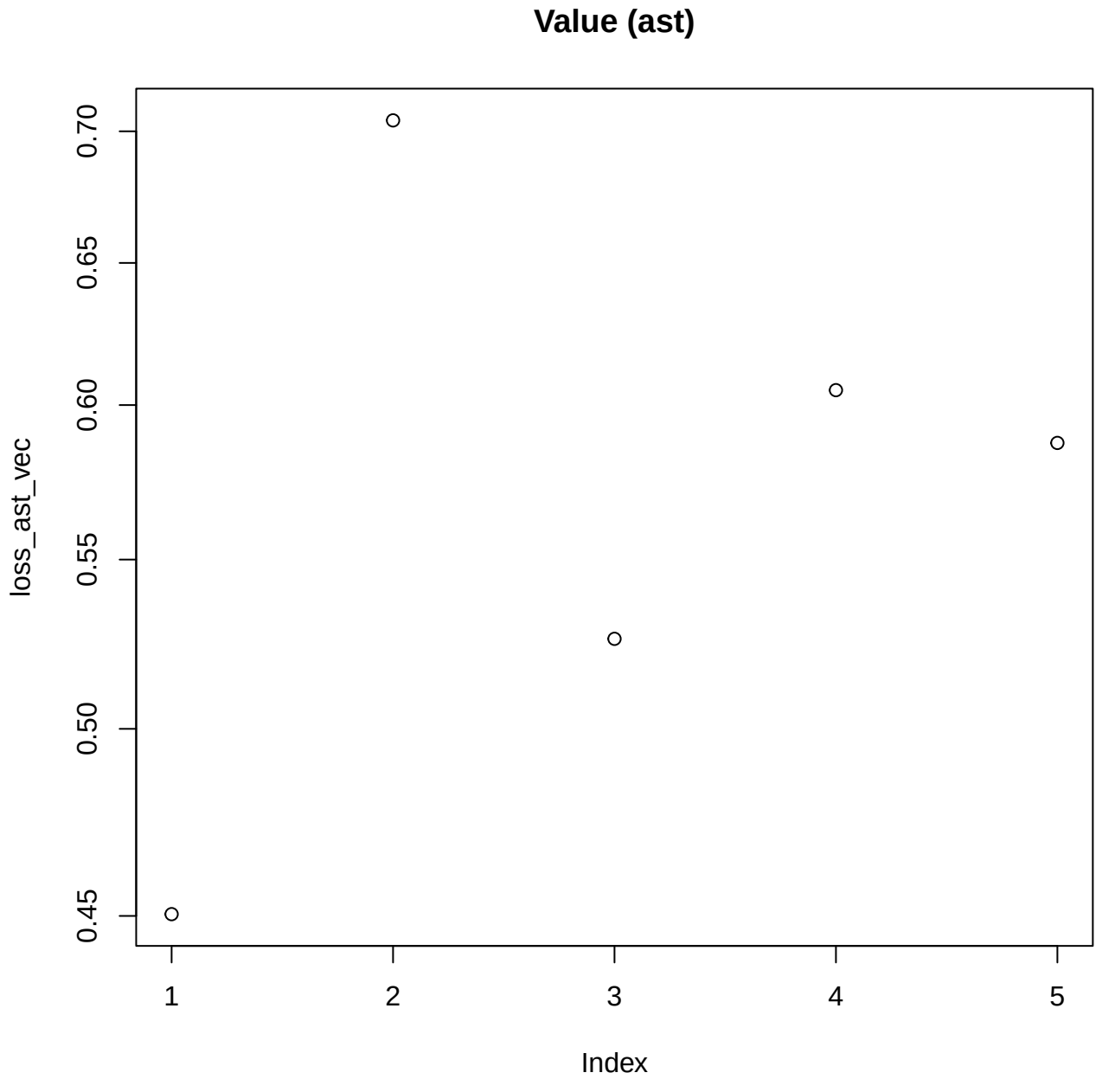

SVI ELBO Loss [40.7547]
ELBO loss
150
100
50
5
10
15
20
Iteration

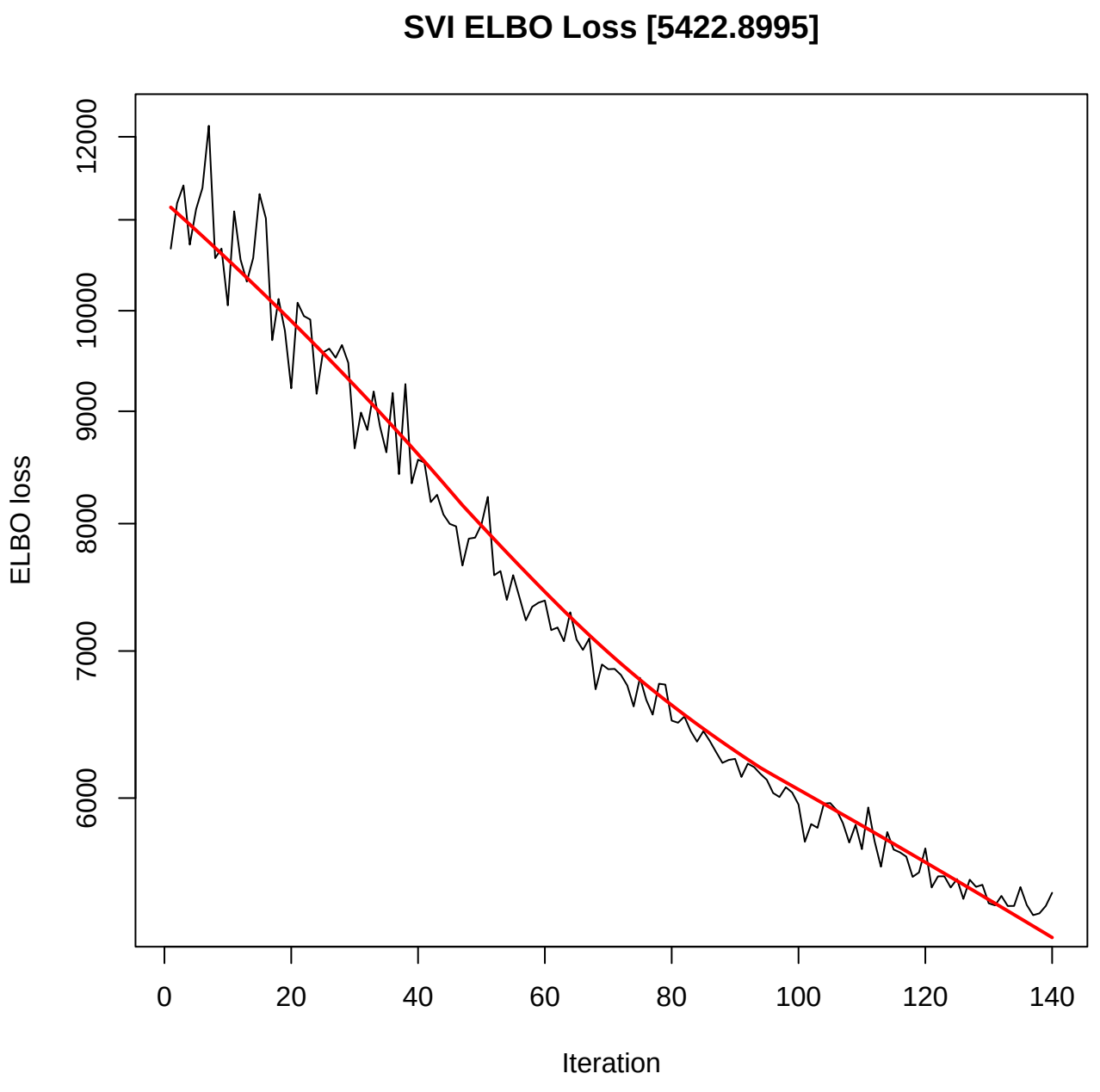

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.035
0.040
0.045
1
2
3
4
5
Index

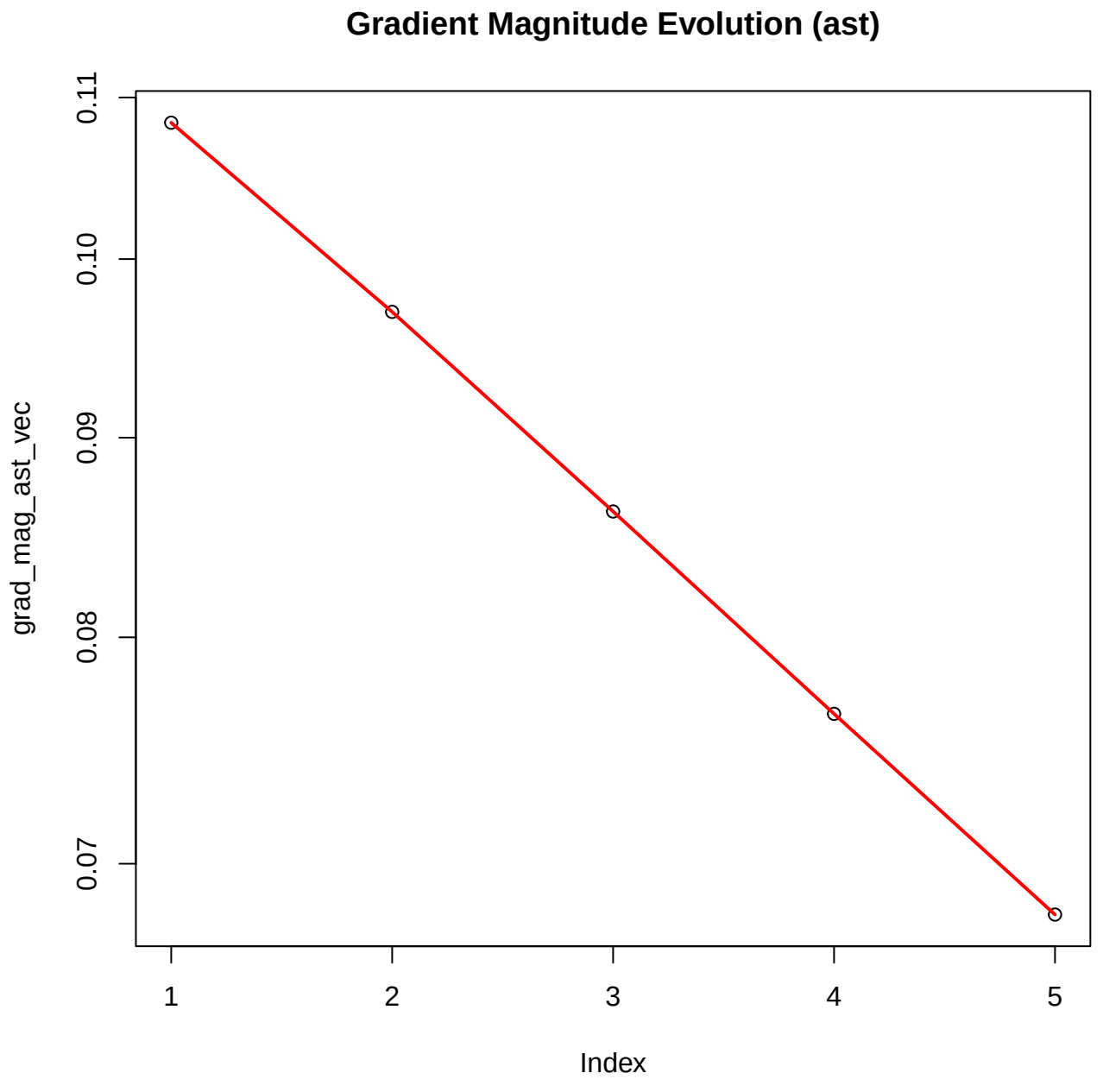

Value (ast)
loss_ast_vec
0.010
0.015
0.020
0.025
1
2
3
4
5
Index

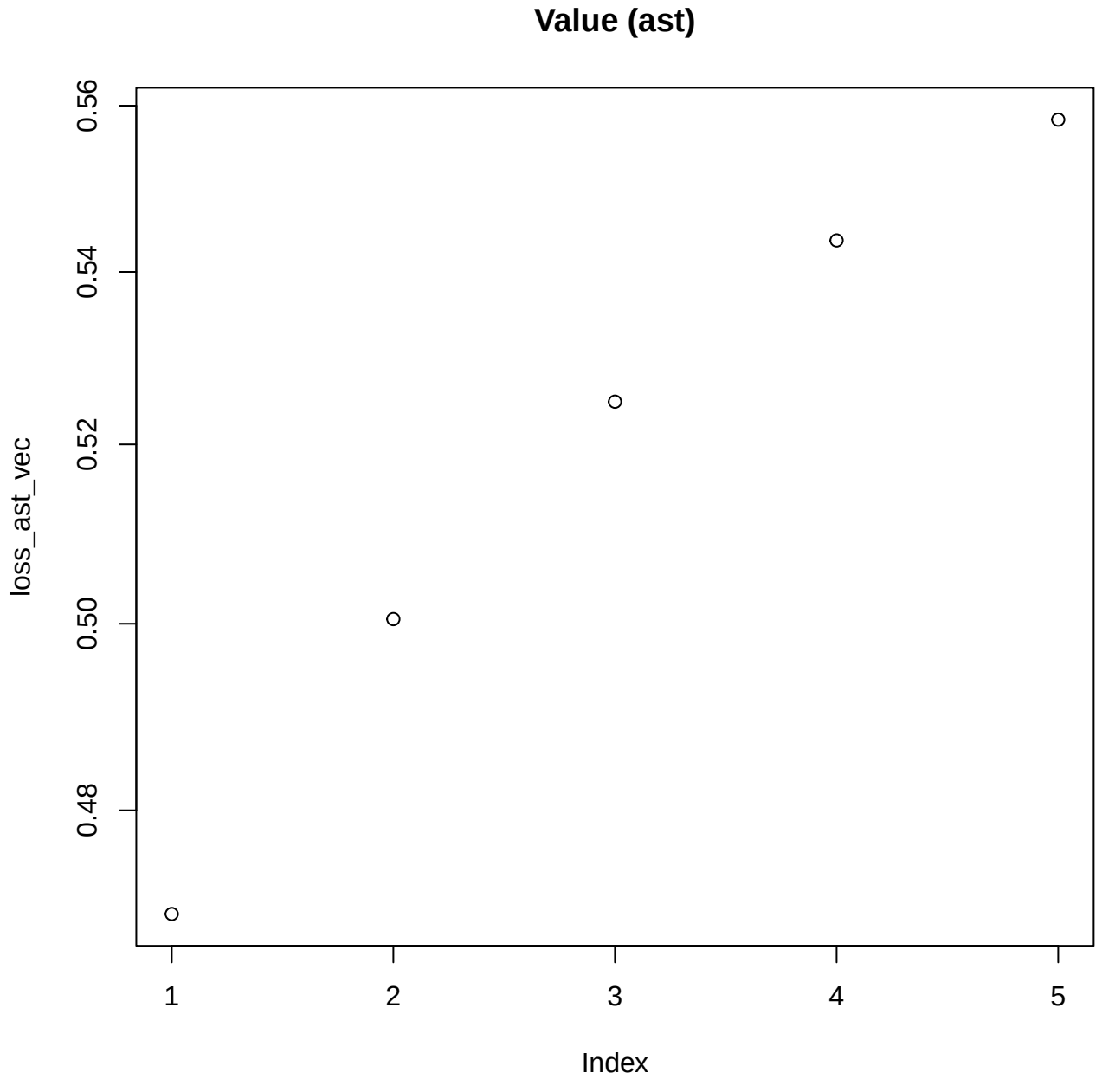

SVI ELBO Loss [57.3876]
ELBO loss
64
62
60
58
56
1
2
3
4
5
Iteration

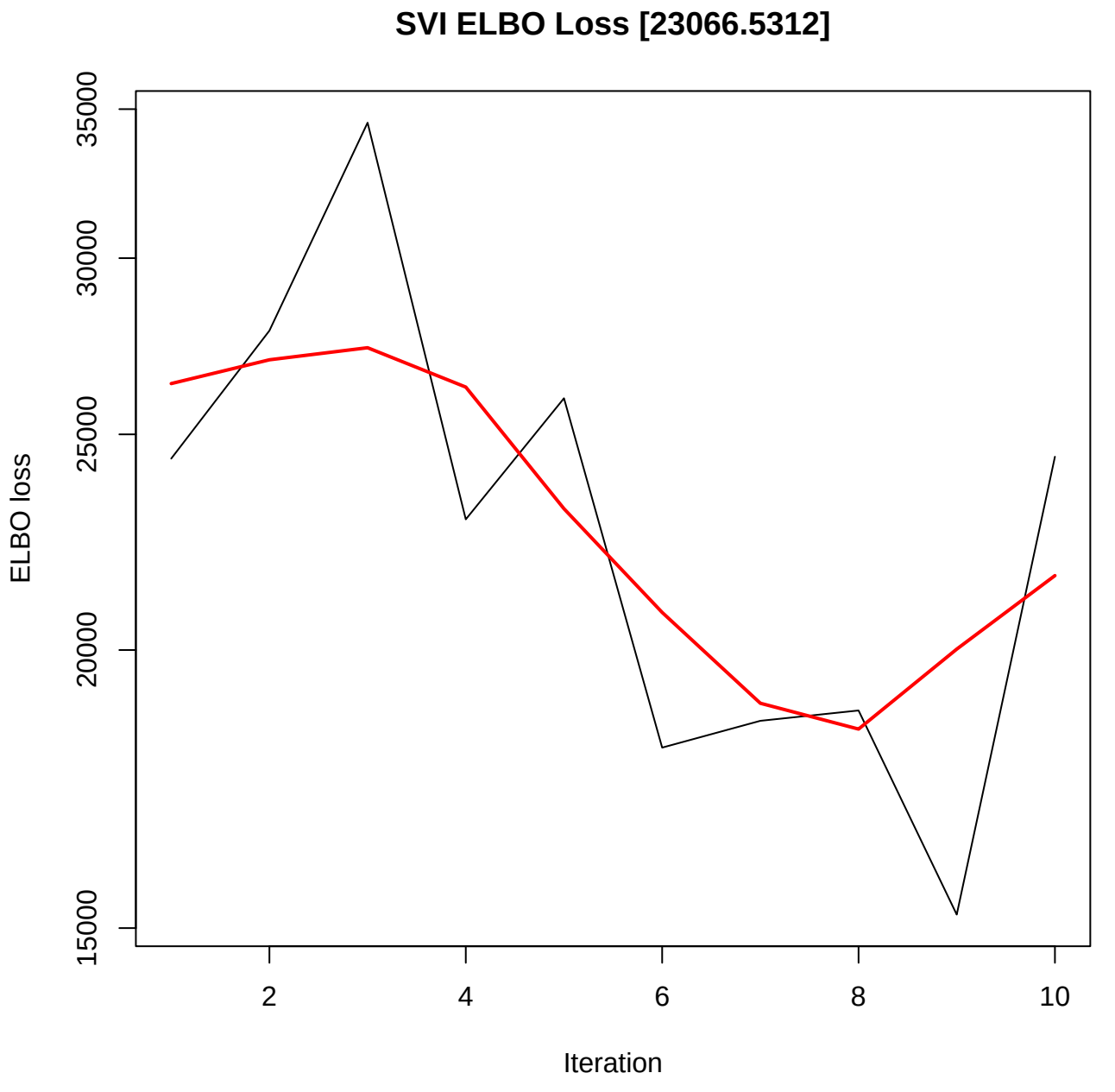

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
0.00080
1
2
3
4
5
Index

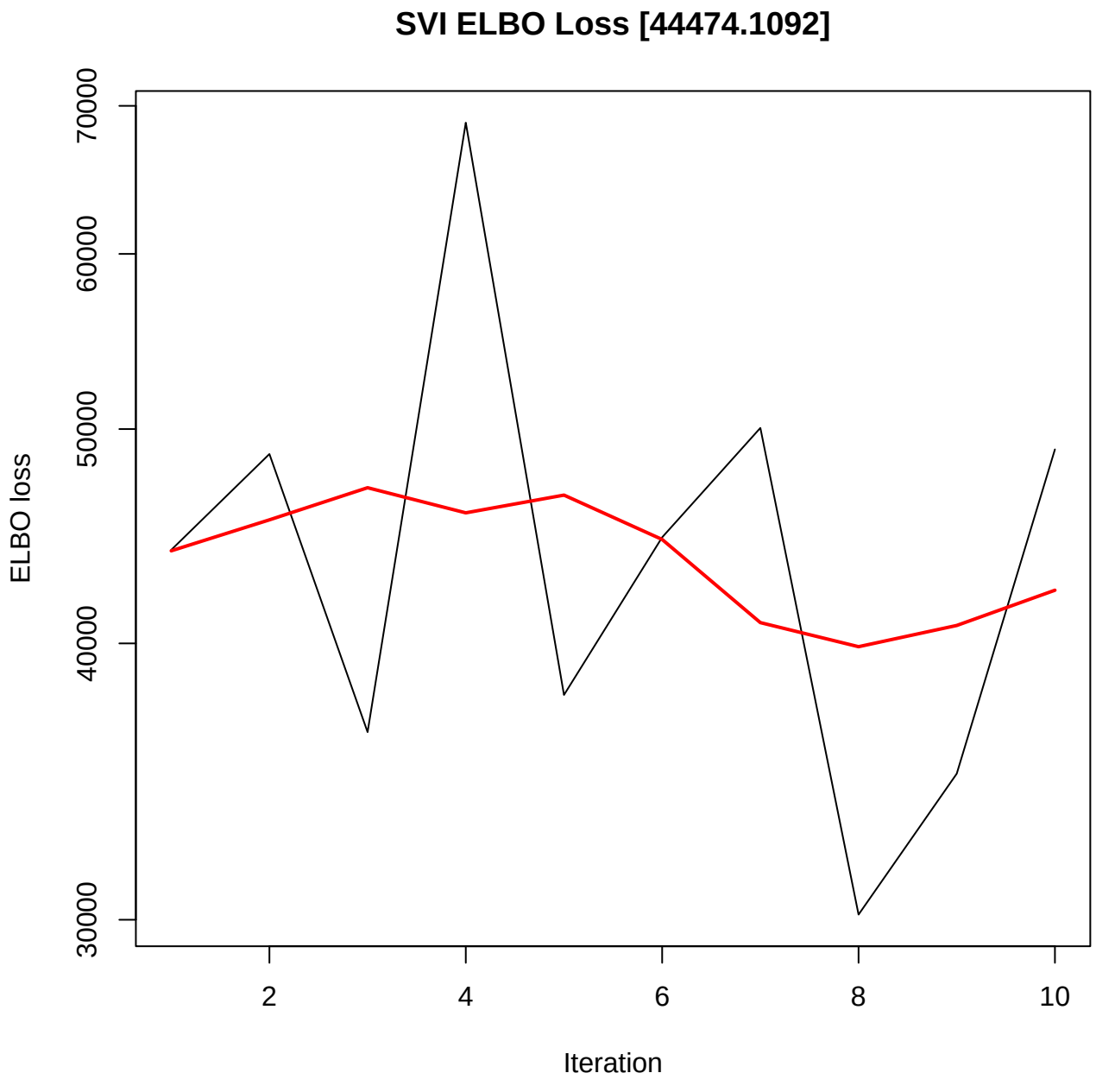

Value (ast)
loss_ast_vec
0.45
0.50
0.55
0.60
0.65
0.70
1
2
3
4
5
Index

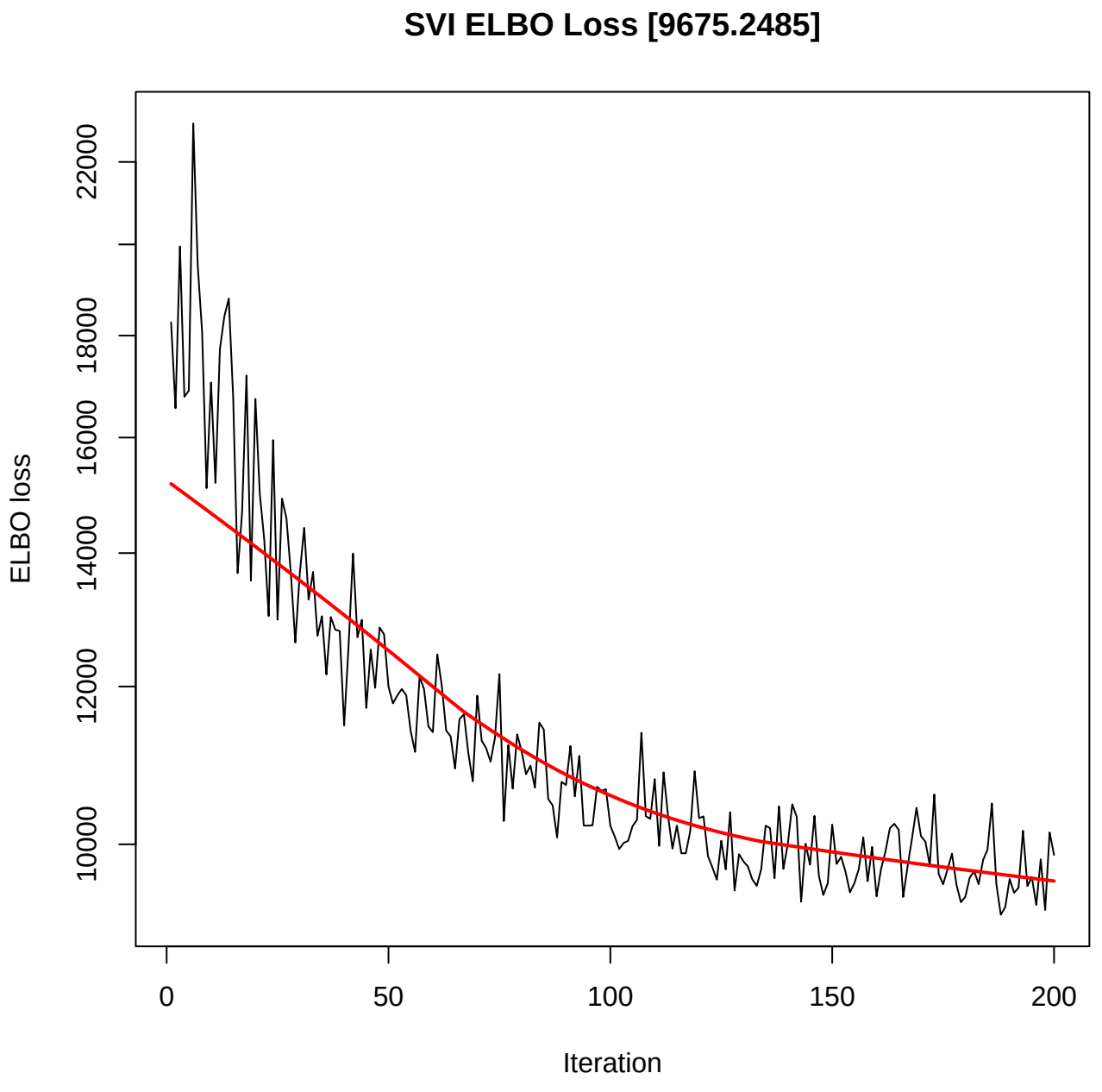

SVI ELBO Loss [5422.8995]
ELBO loss
12000
10000
9000
8000
7000
6000
0
20
40
60
80
100
120
140
Iteration

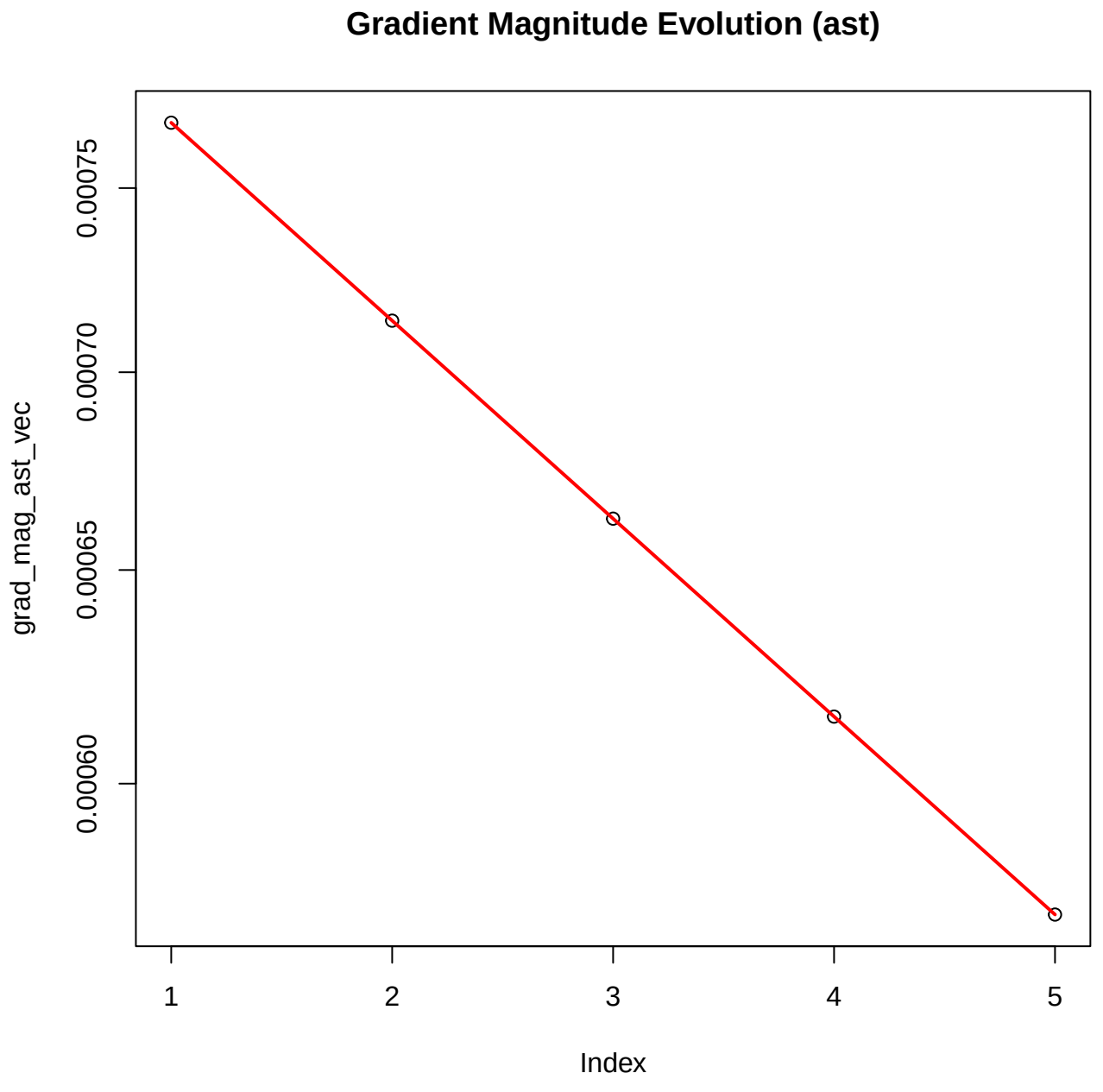

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.07
0.08
0.09
0.10
0.11
1
2
3
4
5
Index

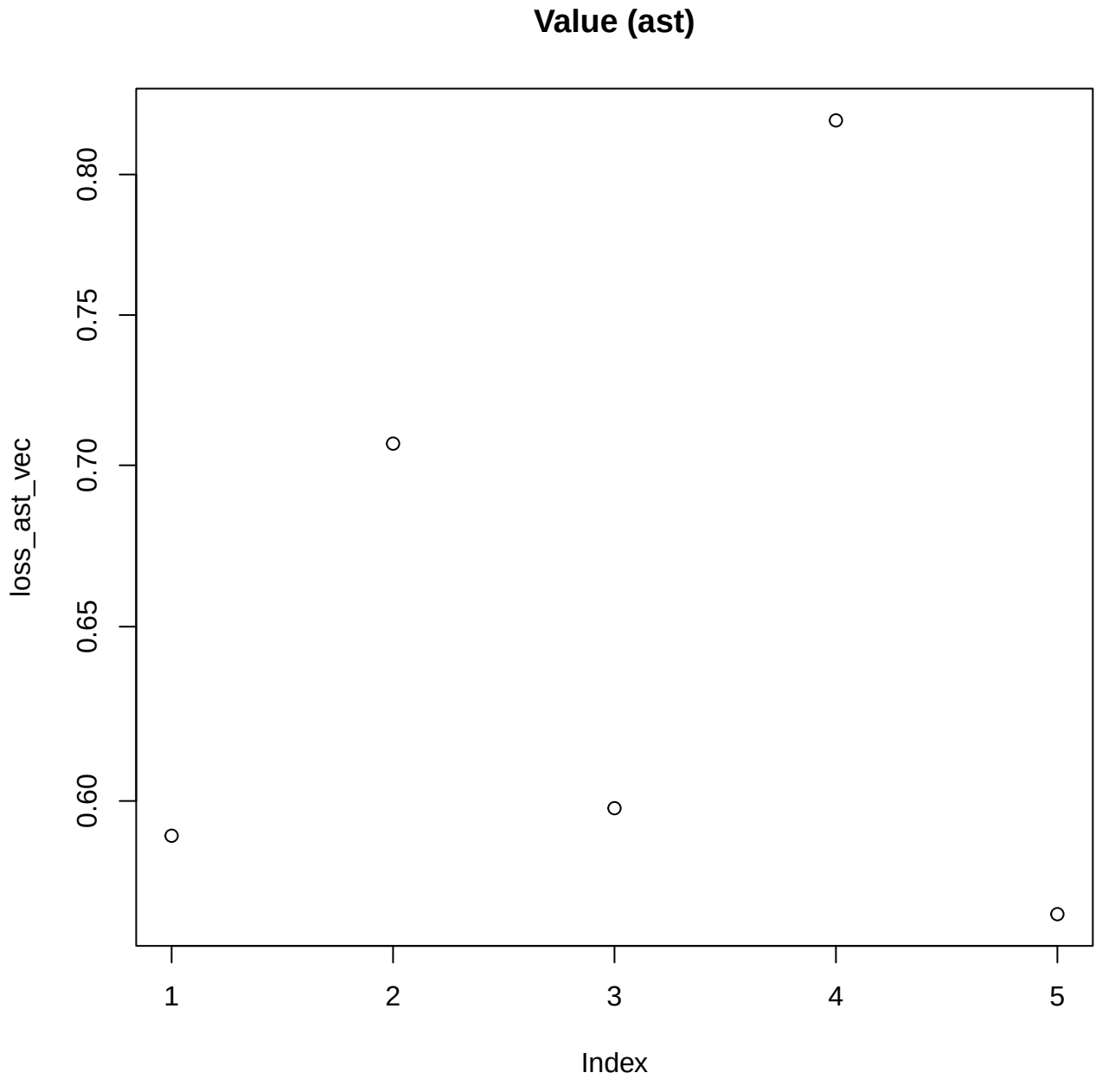

Value (ast)
loss_ast_vec
0.48
0.50
0.52
0.54
0.56
1
2
3
4
5
Index

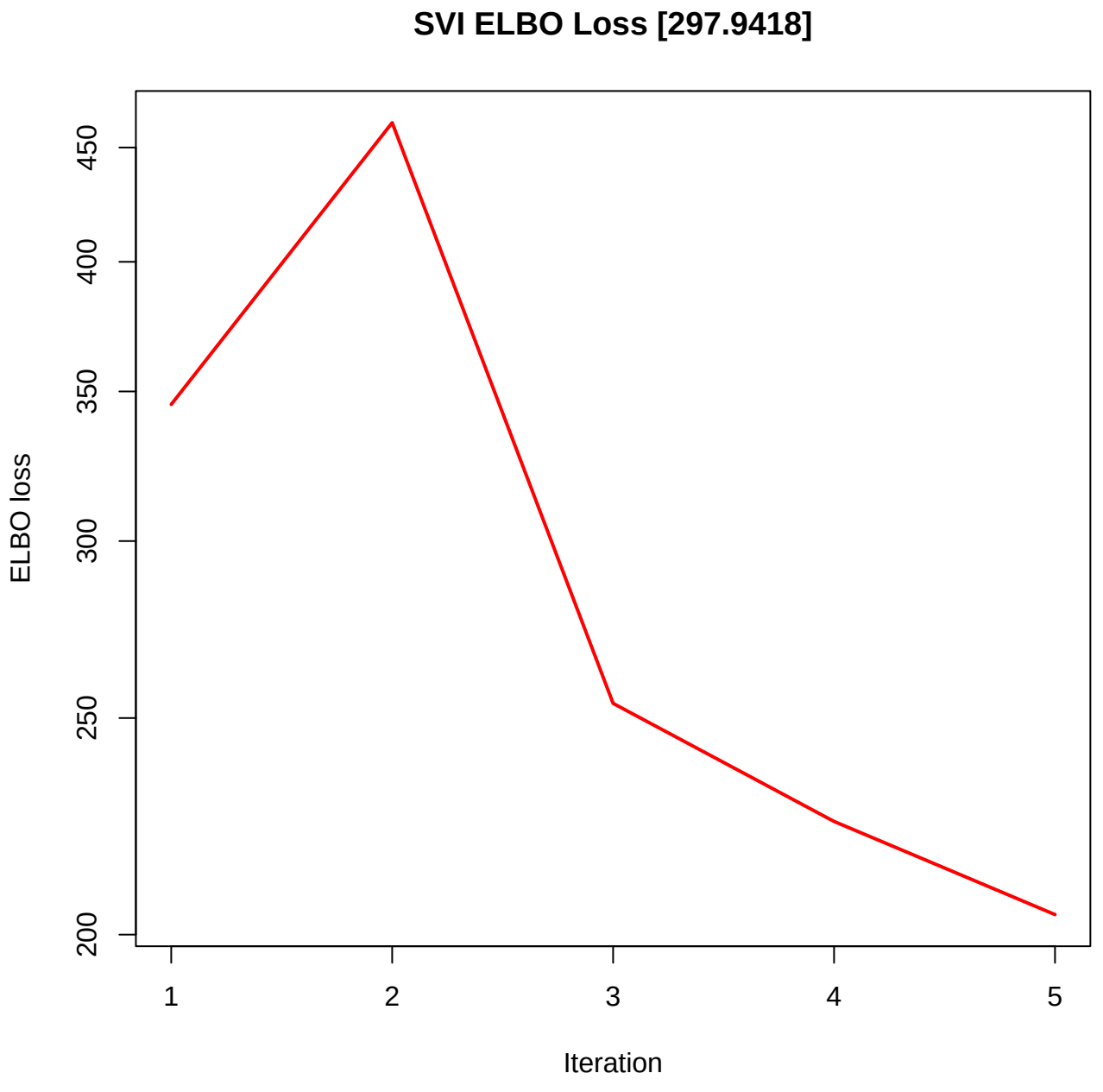

SVI ELBO Loss [23066.5312]
ELBO loss
35000
30000
25000
20000
15000
2
4
6
8
10
Iteration

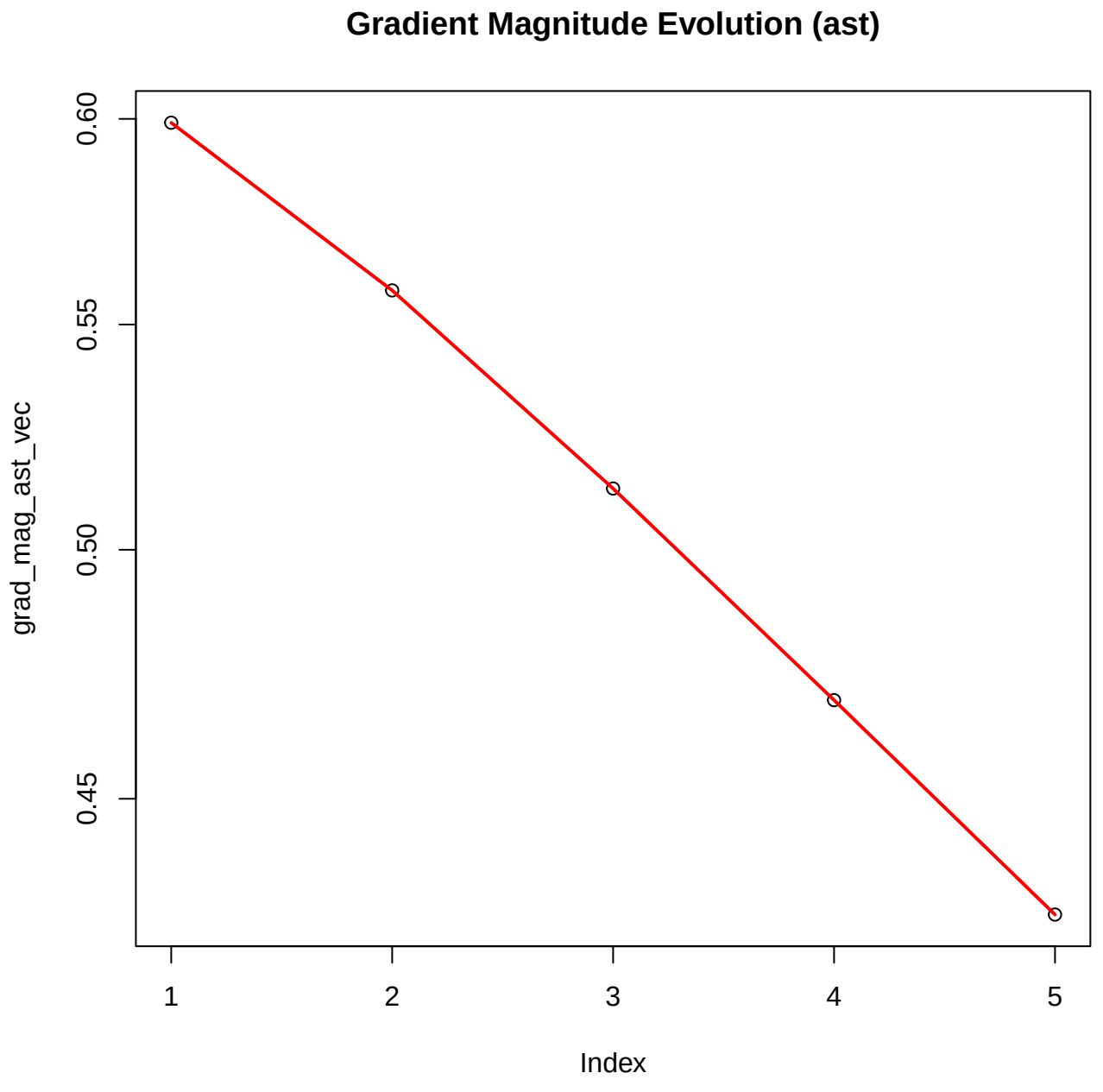

SVI ELBO Loss [44474.1092]
ELBO loss
70000
60000
50000
40000
30000
2
4
6
8
10
Iteration

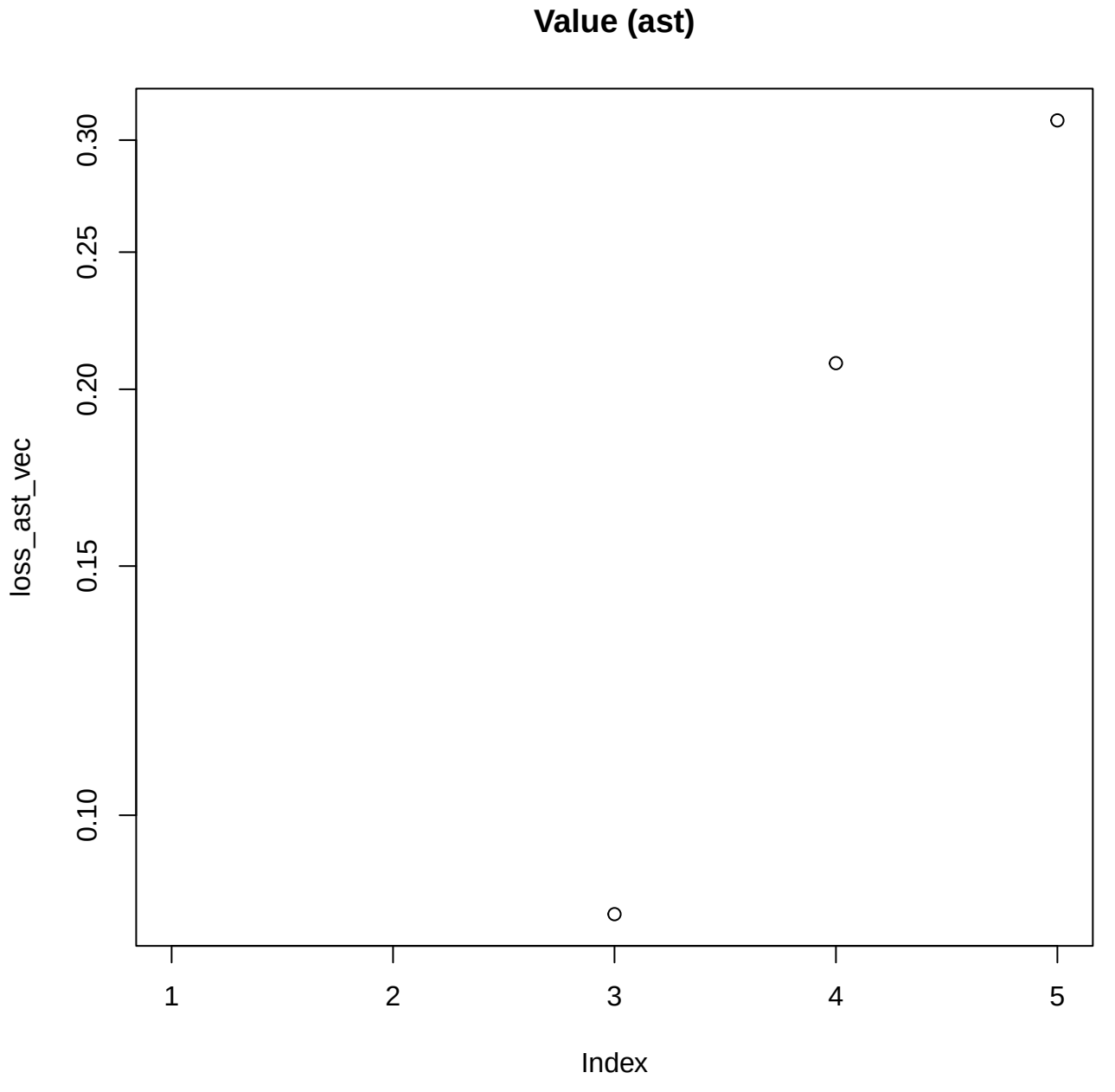

SVI ELBO Loss [9675.2485]
ELBO loss
10000
12000
14000
16000
18000
22000
0
50
100
150
200
Iteration

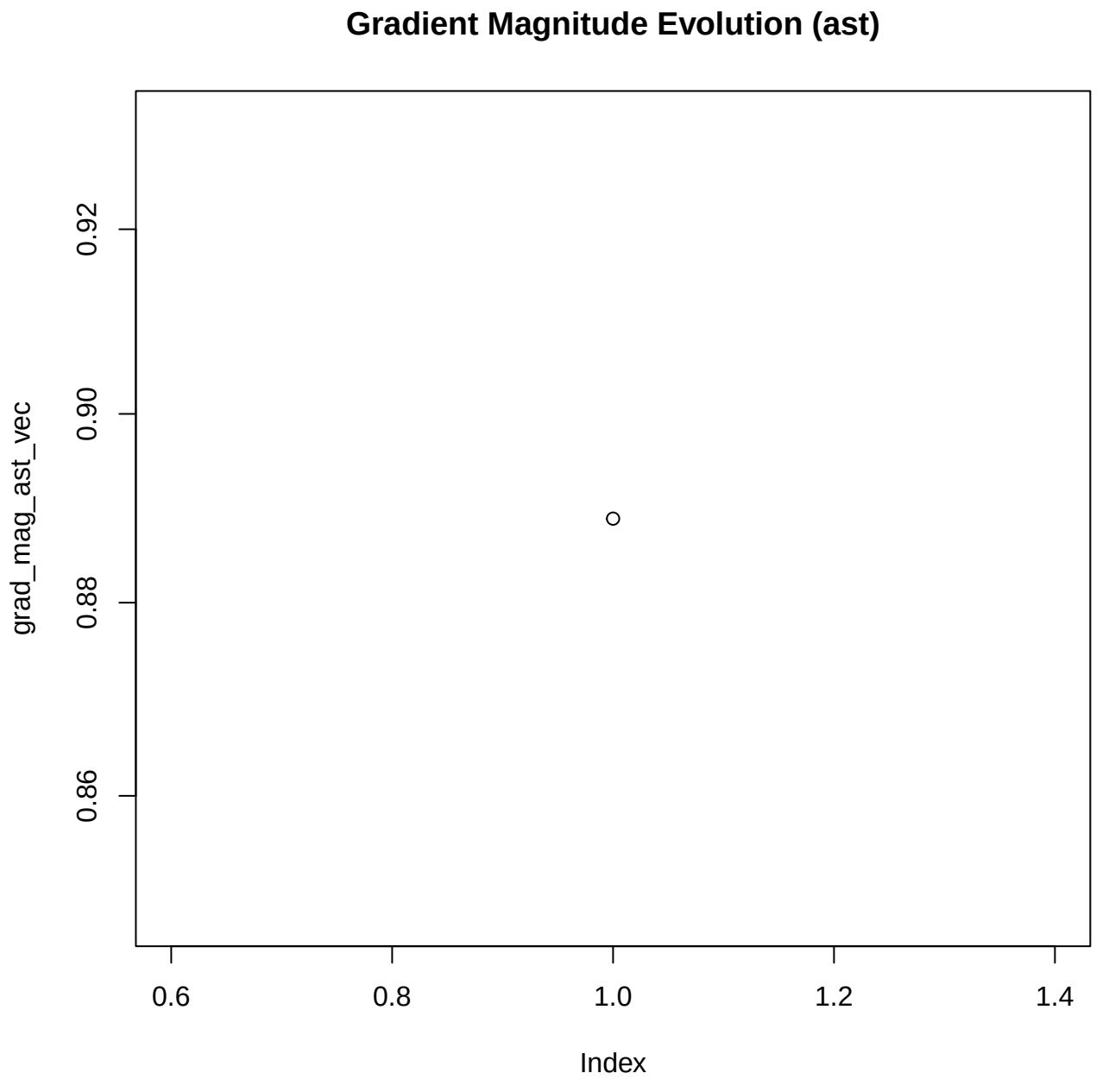

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.00060
0.00065
0.00070
0.00075
1
2
3
4
5
Index

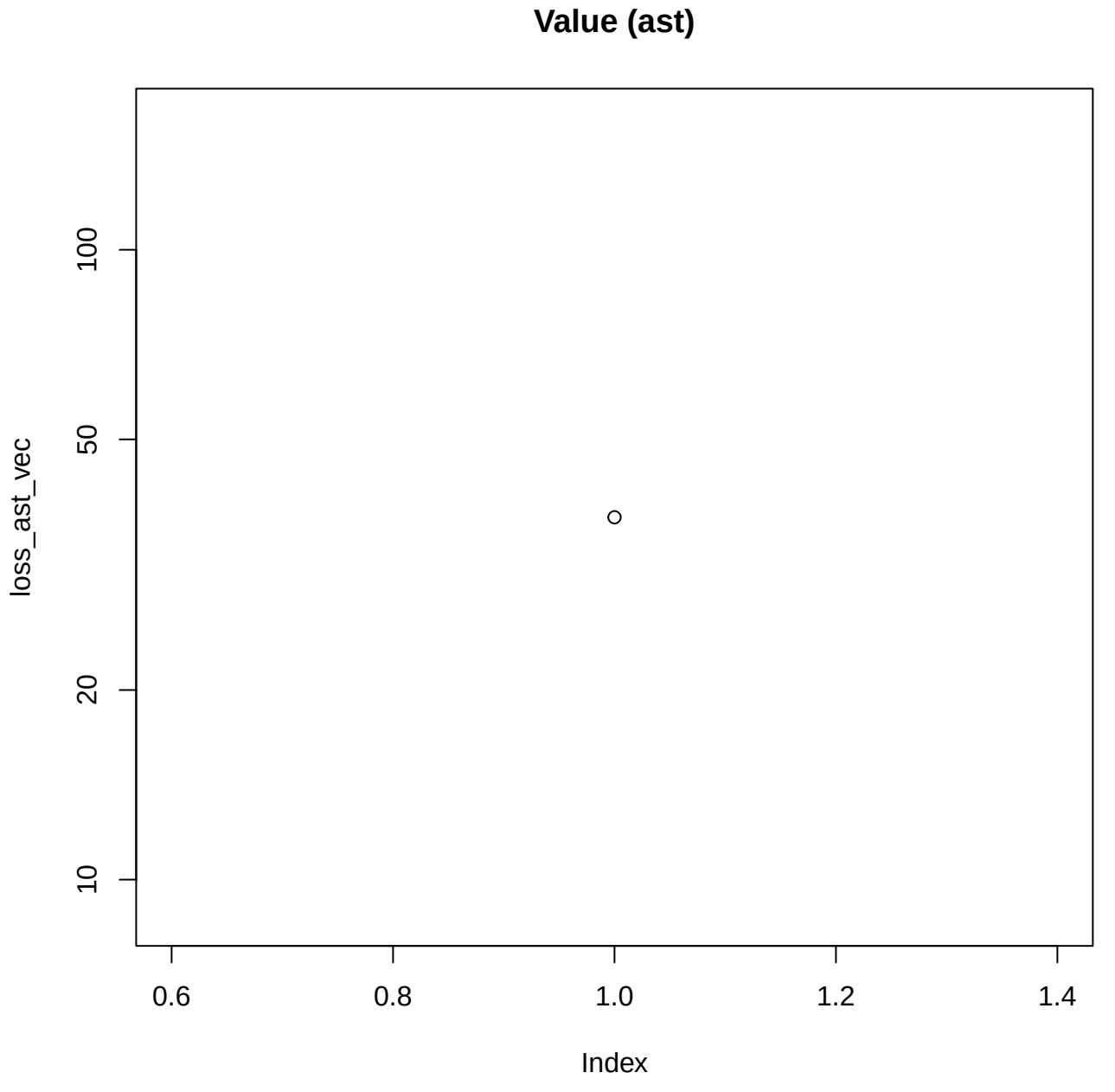

Value (ast)
loss_ast_vec
0.60
0.65
0.70
0.75
0.80
1
2
3
4
5
Index

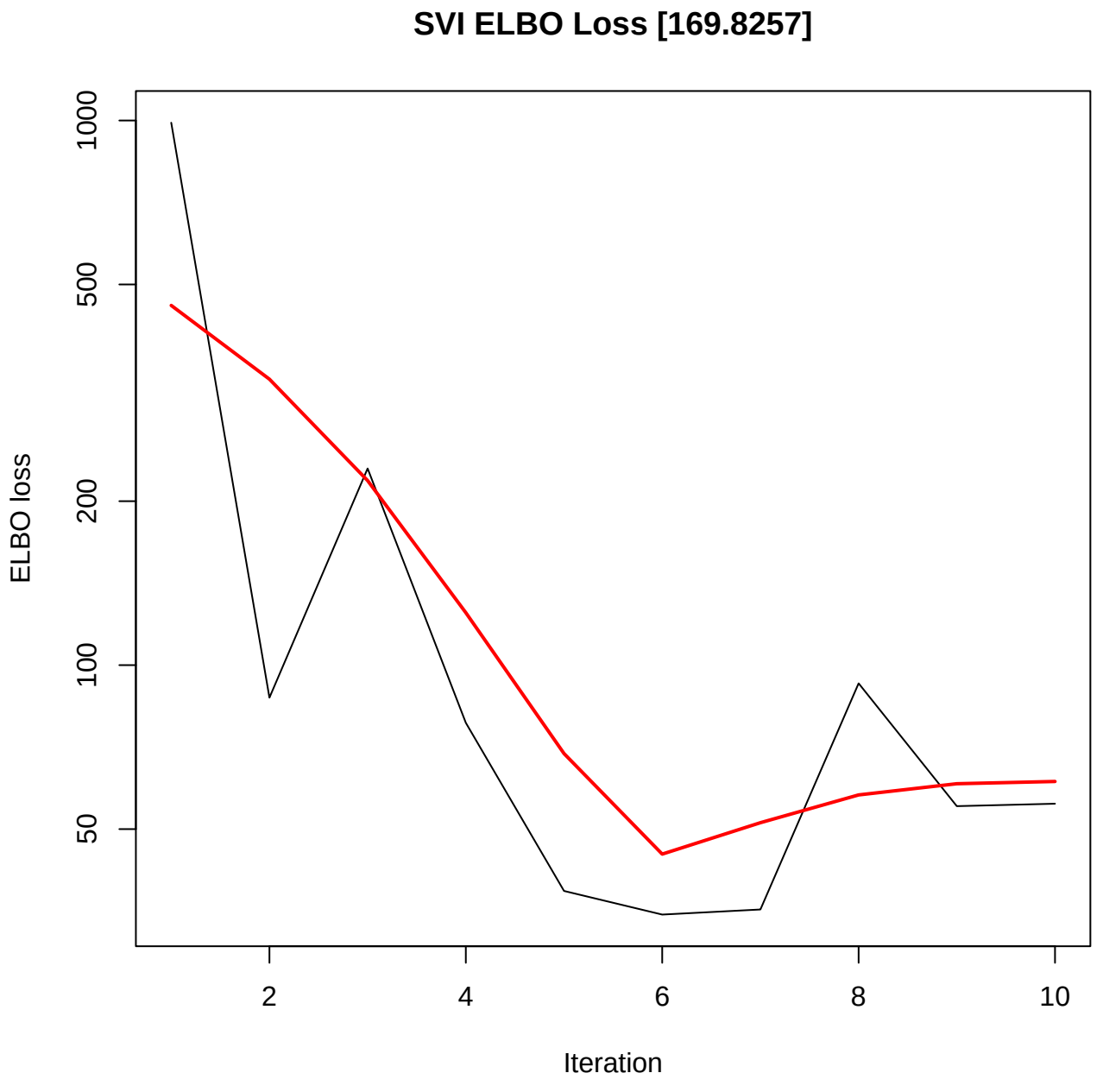

SVI ELBO Loss [297.9418]
ELBO loss
450
400
350
300
250
200
1
2
3
4
5
Iteration

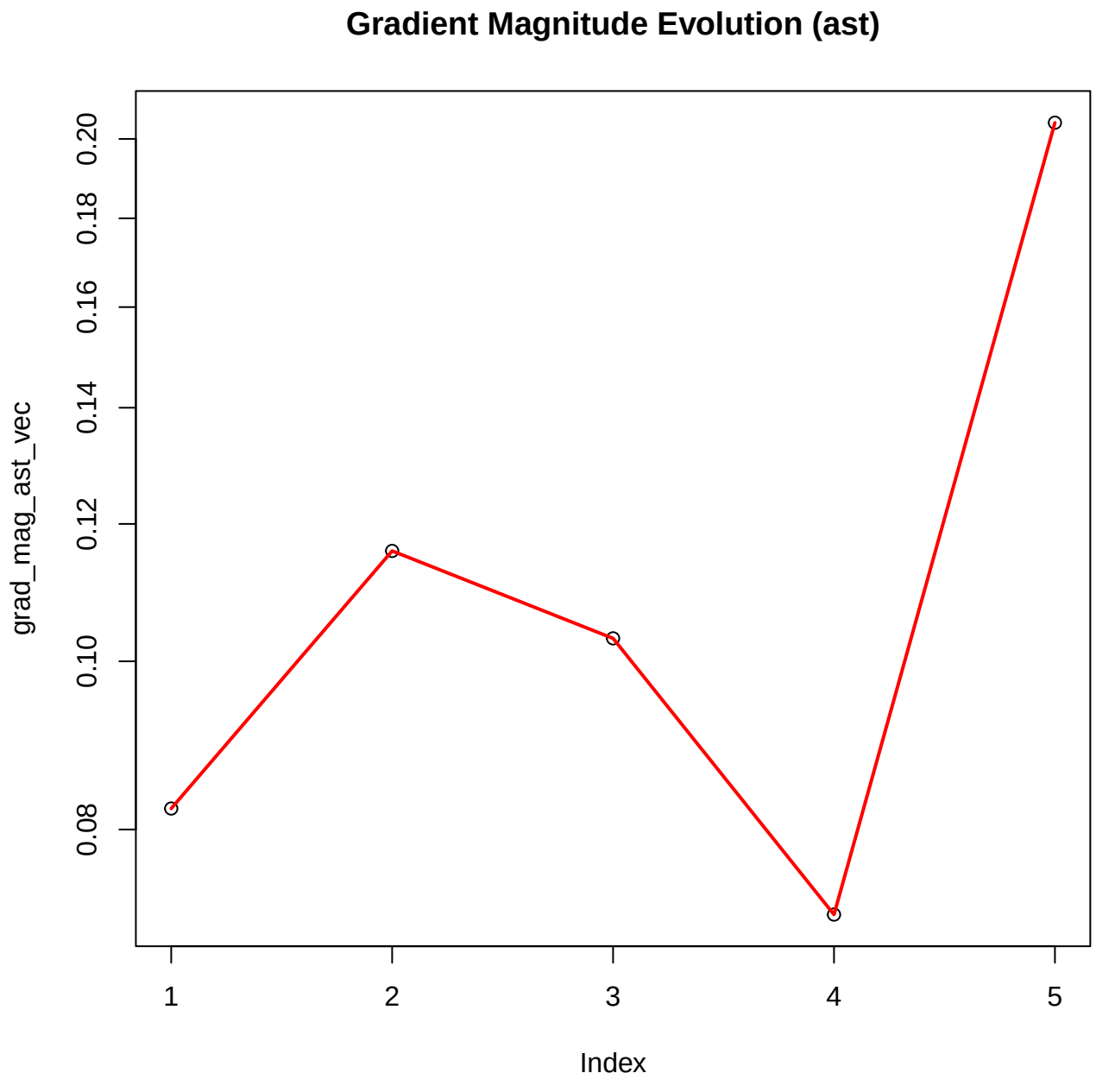

Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.45
0.50
0.55
0.60
1
2
3
4
5
Index

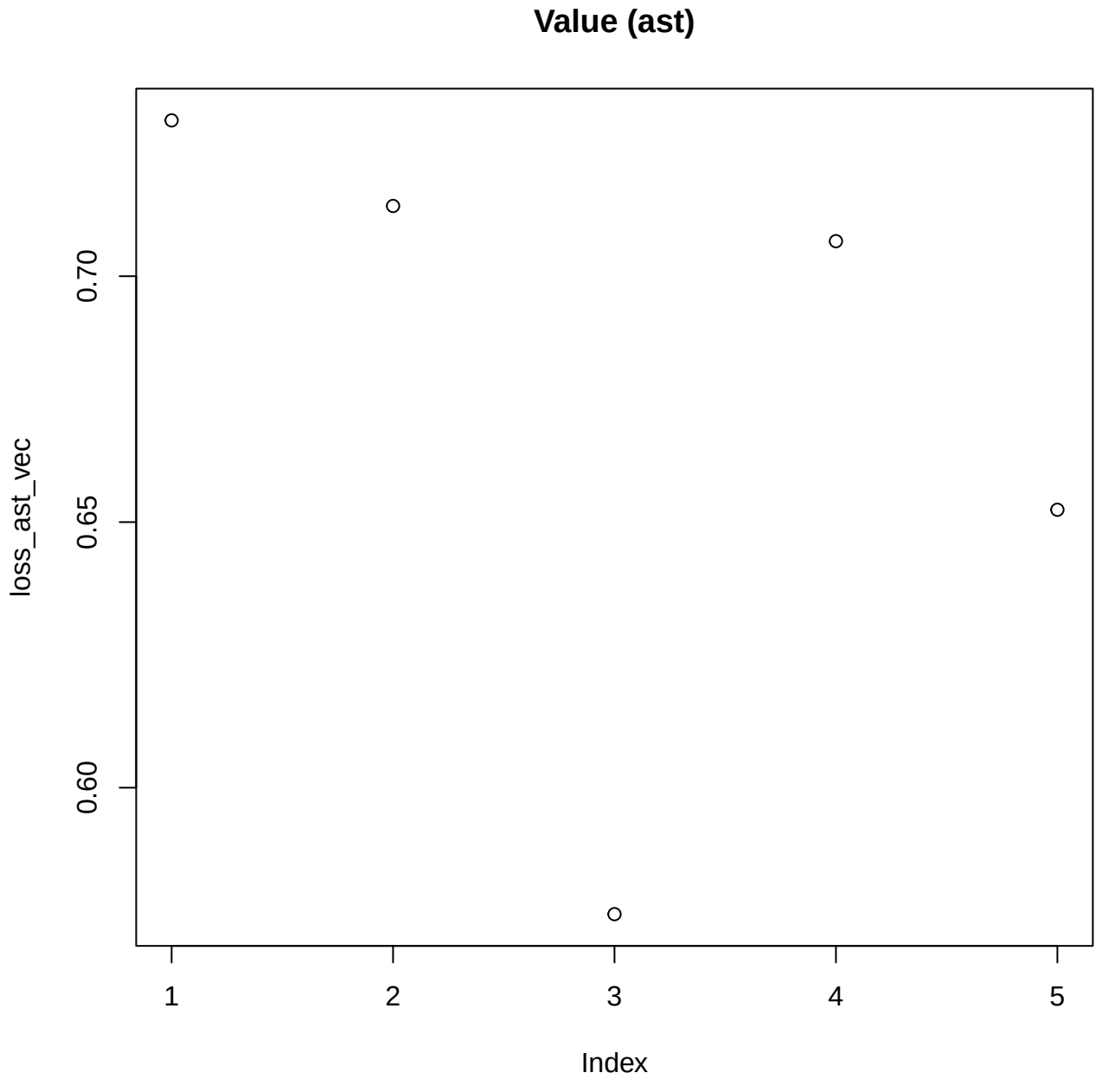

Value (ast)
loss_ast_vec
0.10
0.15
0.20
0.25
0.30
1
2
3
4
5
Index


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.86
0.88
0.90
0.92
0.6
0.8
1.0
1.2
1.4
Index


Value (ast)
loss_ast_vec
100
50
20
10
0.6
0.8
1.0
1.2
1.4
Index


SVI ELBO Loss [169.8257]
ELBO loss
1000
500
200
100
50
2
4
6
8
10
Iteration


Gradient Magnitude Evolution (ast)
grad_mag_ast_vec
0.08
0.10
0.12
0.14
0.16
0.18
0.20
1
2
3
4
5
Index


Value (ast)
loss_ast_vec
0.60
0.65
0.70
1
2
3
4
5
Index